GOOD CHARTS

UPDATED + EXPANDED

SCOTT BERINATO

The HBR Guide to Making Smarter, More Persuasive Data Visualizations

HARVARD BUSINESS REVIEW PRESS

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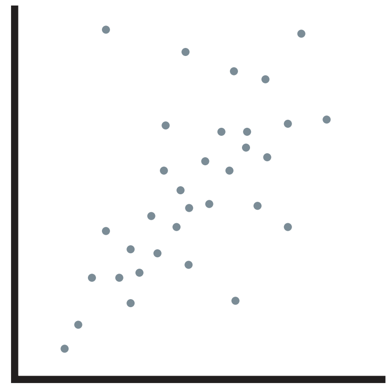

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AUTHOR'S NOTE

HELLO.

More than five years after the publication of *Good Charts*, everything has changed. And nothing has as well.

When I say everything has changed, I'm of course talking about, well, everything. The world has endured a pandemic, and many feel a great sense of unease. Information—data—was everywhere five years ago, a fact I mentioned in the introduction to *Good Charts*, but it has moved beyond that now. In the past half-decade, data and data visualization have grown exponentially. They've been used to great effect. They're connected, for example, to creative breakthroughs in medicine and the creation of entirely new business sectors pouring massive value into the economy. They've been used to educate a grieving, locked-down public about the dangers of a virus. Data and visualization have transformed worlds big and small, from health care to fantasy sports. From agriculture to daily exercise. From sports to finance to education, and much more.

Of course, it hasn't all been good. We've seen data and dataviz deployed to cover up corporate malfeasance and spark public debates over the meaning of facts and truth. But for the most part, we are more data visually literate than we were when I wrote *Good Charts*. And for the most part, visual discourse has improved things.

I looked for a suitable word to describe what's happened and I couldn't find one. So now I paw at descriptions of this sense that we've reached a level of supersaturation with information that is both awe-inspiring and also feels like a lowering presence. Data feels like a hyperstimulus. Information has become a kind of megacosm, a universe in itself that we created and now must inhabit.

When I say nothing has changed, I'm of course talking about our need to embrace, understand, and use data visualization to make things better. To help us *ascend* from the low place of the early 2020s. I'm talking about the need for broad datavisual literacy. The need to learn to positively wield data visualization's power.

I'm optimistic about this. In the five years since *Good Charts* was published, I've spoken to and worked with thousands of people about the power of effective visuals and superior information design. Nearly all are open and eager to learn the skills that can help them get better at communicating visually.

And the overall quality of the data visualization I see is improving for many reasons, including a spate of excellent books and manuals from several tremendous authors, a thriving internet community, and new and improving tools that make it easier to generate good visual information. I've seen a kind of virtuous cycle emerge in which people who experience a good chart are inspired to want more like it and to make better ones themselves.

Most of all, I believe dataviz is getting better because of people's recognition that *they can do this*. That pleases me, too, because I wrote *Good Charts* for precisely that reason. I wanted to show people that by learning a little, they can change a lot. I wanted to remove the intimidation many felt (and many still feel) about visualization, thinking of it as the domain of a few specialist masters when, in fact, it's for everyone.

Early on when I was writing the book, a colleague said to me, skeptically, "Why should you be the person to write a book about this?" My answer surprised her: "Because I'm not an expert. Most people who have to do data visualizations aren't either. Basically, I'm the reader."

I don't know if my colleague was convinced, but I do know that when I say to an audience now, "You don't have to be a designer or a data scientist to make good visual communication—you just need a few simple strategies," faces physically relax. A palpable wave of relief washes over the room. It's amazing to see and feel.

I've learned so much from the audiences I've interacted with in person and, recently, on screen, over the past five years. This updated edition of *Good Charts* is my way to send out into the world what I've learned from them.

Speak to enough people for enough time and themes emerge. I'm updating and expanding this book to address the two most common questions I get after finishing a presentation or during a workshop:

What tools do you use to make charts?
How do I get buy-in to make people realize this is worth the time and investment?

You may suspect that the answers to these questions aren't nearly as tidy as you want them to be, and that's true. The tools environment is evolving rapidly; the information in the original edition of this book is frankly not good enough anymore. But more than a call for a list of potential software, the tools question gets at something more fundamental about the nature of dataviz work, which has been left to individuals, some of whom are unprepared for it, or plain uninterested in it. So, this edition will not only update the tools information but also explore how to put together *teams* to operationalize visualization—a step that will in some ways answer the tools question.

The second question about buy-in dovetails with this, for in the effort to make dataviz an operational team sport, you prove its value and get the buy-in that can sometimes feel elusive. An entirely new chapter called "A Return to Teamwork," which is based on a successful *Harvard Business Review* article, will explore both questions and provide a framework for moving forward.

"Facts and Truth," the chapter on chart manipulation (chapter 6) is also significantly updated. I still address that blurry line between persuasion and manipulation, but in a more holistic way to look at the nature of facts and truth, their similarities and differences, and to explore the relationship between visualizations and our emotions about them.

There are other, more prosaic reasons to update *Good Charts*. Time-series data ages, of course, so new charts with recent data replace older ones to keep things fresh. It also affords the opportunity to improve on charts I was never quite happy with in the first place. And I've included even more examples that didn't exist five years ago. The pandemic, for example, was fertile ground for effective and innovative visualization.

This edition is updated and expanded, but it does not bulldoze the original. Remaining from the first edition are all the core frameworks and the design and decision-making principles that so many of you have told me have improved your ability to create good charts. Some of the handy reference material remains as well, including the glossary of chart types and their use cases. Just as with the world around us, in *Good Charts* everything has changed, and nothing has as well.

I hope this book will become a well-worn, dog-eared companion. (You will see this sentence again in the introduction, where it originally appeared.) I hope you mark up *Good Charts*, highlight passages, plant sticky notes, and sketch improvements to the charts I've created in ways that make it uniquely yours.

I'm thrilled when someone who saw me speak, or who read the book, tells me how they had learned to transform a visual to great effect, and how their data visualization *changed something*—an attitude, a strategy, their career. They sometimes tell me this as if it felt like a magic trick.

But there is no magic to it. As I say in the introduction to *Good Charts*, data visualization is neither art nor science; it's the amalgam of the two. The beauty of the thing, the art, is found in its technical effectiveness, the science. Data visualization is a craft. Like cabinetmaking. Anyone can learn it. Well-crafted things take skill, and skill takes learning. I hope this book is part of what helps you learn, practice, and hone your craft.

INTRODUCTION

A NECESSARY CRAFT

"... for there is nothing either good or bad, but thinking makes it so."

—*Shakespeare*

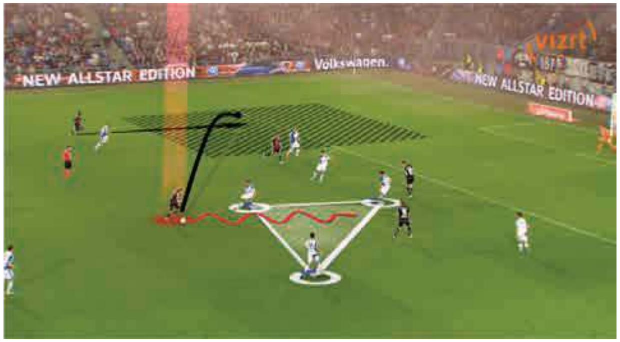
IN A WORLD governed by data, in knowledge economies where ideas are currency, visualization has emerged as our shared language. Charts, graphs, maps, diagrams, interactive visuals, dashboards—even animated GIFs and emojis—all transcend text, spoken languages, and cultures to help people understand one another and connect. This visual language is used everywhere in the world, every day.

Dashboard maps in car navigation systems help commuters avoid the thick red lines of heavy traffic and find the kelly green routes where traffic is light. Weather apps use iconography and rolling trend lines to make forecasts accessible at a glance. Real-time visualization of the markets is lifeblood for investors. Fitness-tracking apps default to simple charts that show steps taken, sleep patterns, eating habits, and more, some designed well enough to make sense on a watch face. Utility company bills include charts so consumers can compare their energy use with their neighbors'. Newspapers, magazines, and websites all use visualization to attract audiences and tell complex stories. Public debate is fueled by charts. The social web teems with data visualizations—some practical, some terrible, some rich with insight, some simply fun to look at—all vying to go viral. Until that metaphor stopped being cute, and a virus stopped the world. And then data visualization became crucial to understanding the demographics of a pandemic, the horrifying pain of exponential growth, and the efficacy of vaccines.

Sports broadcasts superimpose visual data on live action, from first-down lines on a football field to more-sophisticated pitch-sequence diagrams and spray charts that show a baseball's trajectory and expose pitching and hitting trends. Behind the scenes, data analysis in sports has transformed the games and become a model industry for showing how effective visualization creates competitive advantages—literal winners and losers.

You may not notice all the ways in which dataviz has seeped into your daily life, but you have come to expect it. Even if you think you can't *speak* this language, you hear it and understand it every day.

It's time to learn to speak it, too. Just as the consumerization of technology adoption and the widespread use of social media changed business, the ubiquity of dataviz in our lives is driving demand for good charts in unit meetings, sales presentations, customer research reports, consumer-facing advertising, performance reviews, entrepreneurs'



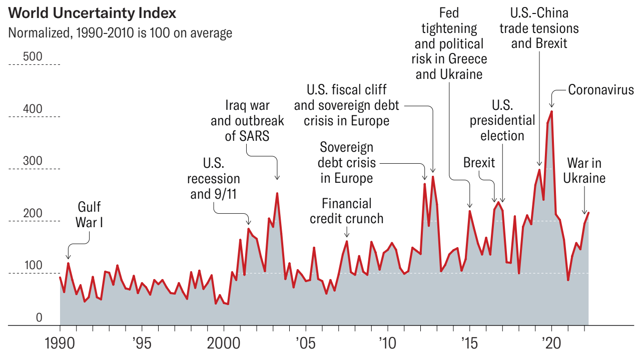
SOURCE: ROBIN STEWART, WEATHERSTRIP.APP

Economic Uncertainty Is Rising

The World Uncertainty Index, a measure of economic uncertainty, has been consistently rising since 2016, with its biggest spike during the beginning of Covid-19.

World Uncertainty Index

Normalized, 1990-2010 is 100 on average



SOURCE: HITES AHIR, NICHOLAS BLOOM, AND DAVIDE FURCERI, "WORLD UNCERTAINTY INDEX," STANFORD MIMEO, 2018

Data visualization is deeply embedded in our lives, from live sports broadcasts to weather apps to fitness apps to the news. We all "hear" and understand this visual language every day. Now we must learn to speak it.

pitches, performance metrics dashboards, and the boardroom.[1] Increasingly, when an executive sees a line chart that's been spit out of Excel and pasted into a presentation, she wonders why it doesn't look more like the simple, beautiful charts on her fitness-tracker app. When a manager spends time trying to parse pie charts and donut charts and multiple trend lines on a company dashboard, he wonders why they don't look as nice or *feel* as easily understood as his weather app.

BUSINESS'S LINGUA FRANCA

Speaking this language requires us to adopt a new way of thinking—*visual thinking*—that is evolving quickly in business. Making good charts isn't a special or a nice-to-have skill anymore; it's a must-have skill. If all you ever do is click a button in Excel or Google Charts to generate a basic chart from some data set, you can be sure that some of your colleagues are doing more and getting noticed for it. No company today would hire a manager who can't negotiate the basics of a spreadsheet; no company tomorrow will hire one who can't think visually and produce good charts.

Dataviz has become a competitive imperative for companies. Those that don't have a critical mass of managers capable of thinking visually will lag behind the ones that do. Vincent Lebunetel was the vice president of innovation at Carlson Wagonlit Travel, where he invested in hiring and training information designers. He's now cofounder and CEO of an AI-driven skills planning and mapping company that helps HR predict and plan for the skills it needs. He says that business managers and leaders who can't create clear visualizations are just less valuable: "If you're not able to make your message simple and accessible, you probably don't own your topic well enough. And visualization is probably the best way to help people grasp information efficiently."

After a group at Accenture Technology Labs produced visualizations of NBA team shooting patterns that went viral, its consultants started asking the group for help producing charts that would produce a similar visceral reaction in their own clients.[2] So Accenture built an online and in-person "visual literacy curriculum" for them. The VLC has been so effective internally that Accenture made the curriculum a client service and developed a visualization career track for its consultants.

Daryl Morey, the president of basketball operations for the NBA's Philadelphia 76ers, puts it plainly: "Everyone in our business knows they need to visualize data, but it's easy to do it poorly. We invest in it. We're excited if we can use it right while they use it wrong."

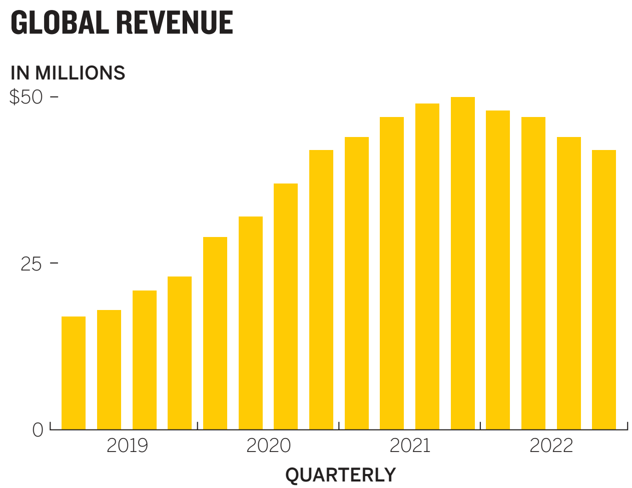
Sports in general—and basketball in particular—should serve as inspiration or a warning, depending on how you want to look at it. Over the past two decades in sports, the organizations that are best at using data win more. They see opportunities to exploit and hazards to avoid. They find novel solutions to problems, sometimes problems they didn't know they had. Most teams in a sport have roughly the same access to data, so what are the best teams doing better? They're using visualization to find the signals in the noise and then using it again to communicate their ideas clearly and effectively to decision makers. They're winning with visualization.

WHAT'S A GOOD CHART?

The rise of visualization has generated numerous opinions about how to do it right—and harsh judgment of charts that get it wrong. Missing from most attempts to establish rules are an overarching view of what it means to think visually and a framework and repeatable process for constructing good charts.

To build fluency in this new language, to tap into this vehicle for professional growth, and to give your organization a competitive edge, you first need to recognize a good chart when you see one.

How about this Global Revenue chart? Is it a good chart?



SOURCE: COMPANY RESEARCH

Ultimately, when you create a visualization, that's what you need to know. Is it good? Is it effective? Are you helping people see an idea and learn from it? Are you making your case?

So, is this one good?

It certainly looks smart. It's labeled well. It eschews needless ornamentation. It uses color judiciously. And it tells a clear, simple story: After years of healthy growth, revenue peaked and then started to taper. If we held this chart up to the rules and principles proffered by data visualization experts and authors such as Edward Tufte, Stephen Few, and Dona Wong, it would probably pass most of their tests.[3]

But does that mean it's good?

Data tools such as Excel can create charts instantly, but does that mean they're good charts?

You might think it looks better than what you could produce quickly in an Excel doc or analytics package like Power BI or even a visualization tool like Tableau—some of managers' go-to dataviz tools. You could turn a row of data into a chart there with a single click. And if you needed to present to the CEO, or to shareholders, you might play with some of Excel's preset options to make it look fancier and more dynamic. Many people love the 3-D option; it seems to draw the eye. That seems like a good thing. So does that make a chart good?

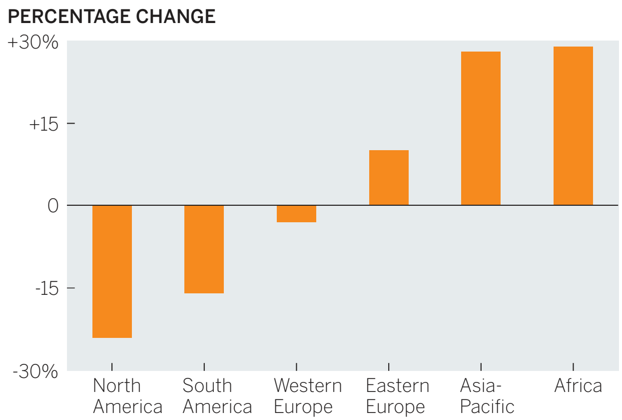
These tools are right there with our data, and they're very easy to use. But as dataviz becomes a *thing*, and we constantly encounter more well-designed, thoughtful, persuasive, and inspiring charts and graphs, we recognize that charts like these fall short, even if we can't yet say exactly why. As most managers use it, Excel visualizes data cells automatically, unthinkingly. The result beats looking at a spreadsheet—but that's a low bar.

So this chart isn't as good as the first one, but the question remains: Is that first chart good?

The answer is that we don't know. Without context, no one—not me, not you, not a professional designer or data scientist, not Tufte or Few or Wong—can say whether that chart is *good*. In the absence of context, a chart is neither good nor bad. It's only well built or poorly built. To judge a chart's value, you need to know more—much more—than whether you used the right chart type, picked good colors, or labeled axes correctly. Those things can help make charts good, but in the absence of context they're academic considerations. It's far more important to know *Who will see this? What do they want? What do they need? What idea do I want to convey? What could I show? What should I show?* Then, after all that, *How will I show it?*

If you're presenting to the board, that simple bar chart may not be a good chart. The directors know the quarterly revenues; they're going to tune you out, check their phones, or, worse, get annoyed that you've wasted their time. Maybe they're looking for markets to invest in to reverse the revenue trend. In that case, a breakdown of changes in the global distribution of revenue might make a good chart:

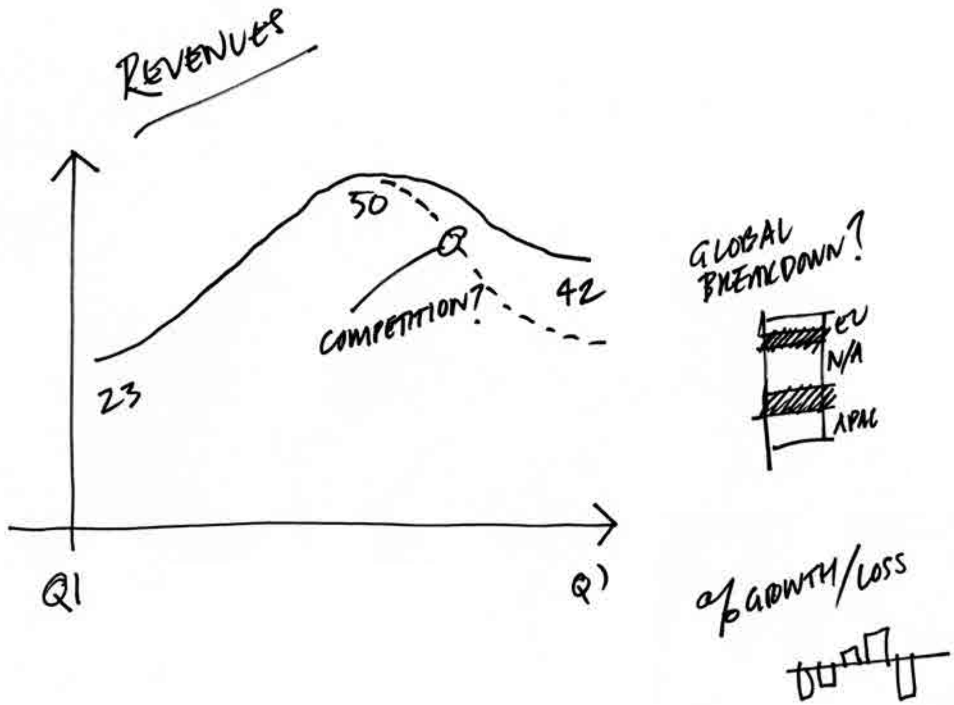
REGIONAL REVENUE TRENDS, Q1 '19–Q4 '22



SOURCE: COMPANY RESEARCH

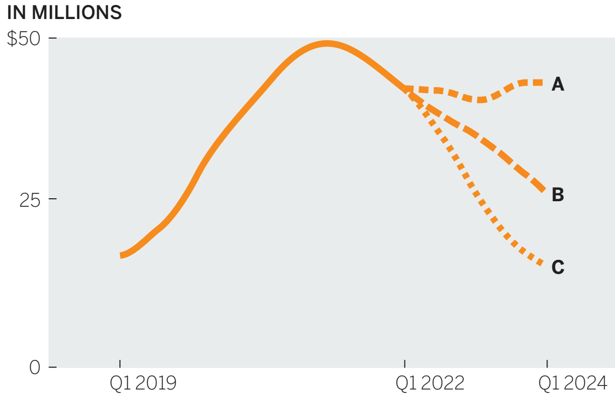
Same data set. Completely different chart.

If the boss has said, "Let's talk about revenue trends in our next one-on-one," this isn't a bad chart per se, but it may be overkill. In that scenario, the time spent refining the chart to presentation-level refinement might be better used exploring ideas around the revenue data on a whiteboard, which has the advantage of being an interactive space, ready to be marked up with ideas about what you'd want to see in that space, such as comparison to competition or a proportional breakdown:



Then again, if it's for a strategy off-site with the executive committee where future scenarios will be discussed, these probably aren't good charts. How can you talk about the future with a chart that only shows the past? A good chart in that context would reflect multiple future scenarios, as seen on the Revenue Projections chart below.

REVENUE PROJECTIONS—THREE SCENARIOS



Then again, if you're meeting with a new manager who needs to understand basic facts about the company, then yes, the original chart is a good chart.

We could go on with this indefinitely. Just notice that we're not changing our data so much as we are the view of it. Context is everything.

BEYOND RULES AND PLATITUDES

This simple example should liberate you from the idea that the value of a chart comes primarily from its execution (it doesn't) and that its quality can be measured by how well it follows the rules of presentation (it can't). Just as reading Strunk and White's *The Elements of Style* doesn't ensure you'll write well, learning visual grammar doesn't guarantee that you'll create good charts.

In his excellent writing manual *Style: Toward Clarity and Grace*, Joseph M. Williams explains why grammar rule books fall short:

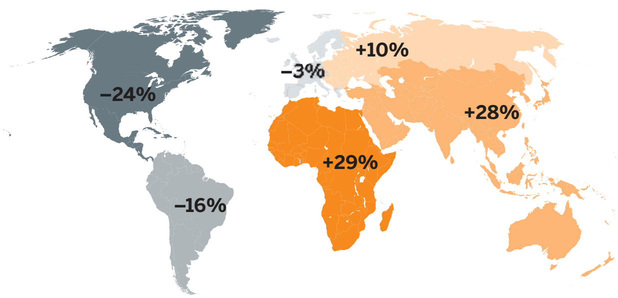
> Telling me to "Be clear" is like telling me to "Hit the ball squarely." I know that. What I don't know is how to do it. To explain how to write clearly, I have to go beyond platitudes.
>
> I want you to understand this matter—to understand why some prose seems clear, other prose not, and why two readers might disagree about it; why a passive verb can be a better choice than an active verb; why so many truisms about style are either incomplete or wrong. More important, I want that understanding to consist not of anecdotal bits and pieces, but of a coherent system of principles more useful than "Write short sentences."[4]

What Williams says about writing is just as true for dataviz. You need to get beyond rules and understand what's happening when you encounter visualization. Why do you like some charts and not others? Why do some *seem* clear, and others muddled?

How do you know, say, when to use a map instead of a line chart? One rule book for building charts states unequivocally, "No mapping unless geography is relevant."[5] That's like telling you to "hit the ball squarely." How do you know whether geography is relevant? What does *relevant* even mean? Geography could be considered the most relevant factor in your chart showing regional revenue growth for the board. Should you map it instead? Let's take a look:

REGIONAL REVENUE TRENDS, 2019–2022

PERCENTAGE CHANGE



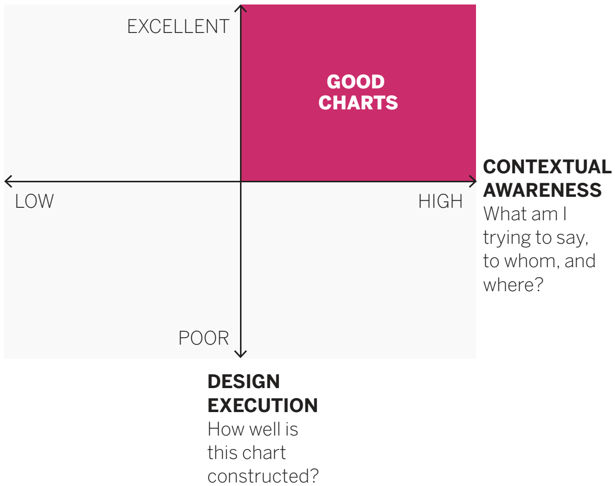
SOURCE: COMPANY RESEARCH

I could make a strong argument that this is not as good a chart for some of the contexts we shared above. Does the map make the point about regional revenues better than a chart does? Would it help you persuade the board that regional revenues matter? Are you even trying to do that? Would mapping this data geographically be worth the extra effort it takes to produce? These questions are seeking the context that simple rules can't address.

My point here is not to suggest that rules for crafting good visualizations aren't necessary or useful. They're both. But rules are open to interpretation and sometimes arbitrary or even counterproductive when it comes to producing good visualizations. They're for responding to context, not setting it.

Instead of worrying about whether a chart is "right" or "wrong," focus on whether it's *good*. You need, as Williams says, principles that help you understand *why* you'd choose a bar chart or a line chart, or no chart at all. A perfectly relevant visualization that breaks a few presentation rules is far more valuable—it's *better*—than a perfectly executed, beautiful chart that contains the wrong data, communicates the wrong message, or fails to engage its audience. Even within the following pages, you will encounter charts that don't slavishly follow the advice I'm giving for improving your data visualization. I will make any number of recommendations on color, labeling, axes, and so on and then you'll see a chart that doesn't follow that recommendation. For example, I suggest that chart titles can (and often should) be used to do more than just provide generic descriptions of the chart's structure. And yet, I've put charts in this book with titles that only describe the structure. That's okay. That particular recommendation is not a "rule." And in those cases, it just wasn't relevant to the context to do more with the title. You're seeking relevance. The more relevant a data visualization is to its context, the more forgiving, to a point, we can be about its execution.

THE GOOD CHARTS MATRIX



The charts you make should fall into the top-right zone in the Good Charts Matrix, shown above. Learning to think visually in order to produce good charts that fall in this range is the subject of this book.

THE COMPETITIVE IMPERATIVE

Three interrelated trends are driving the need to learn and practice visual thinking. The first is the permeation of visualization into our lives. The more-sophisticated, higher-quality dataviz in products and media we see now has raised expectations for the charts that others provide us, both in our consumer lives and in our business lives.

The second trend is data itself: both its sheer volume and the velocity at which it's hurtling at us. So much information hitting us so fast demands a new way of communicating that simplifies and helps us cope.

At Boeing, for example, engineers wanted to increase the operational efficiency of the Osprey—a plane that takes off and lands like a helicopter. The plane's sensors produce a terabyte of data to analyze on each takeoff and landing. Ten Osprey flights produce as much data as the entire print collection of the Library of Congress.[6] The idea of scouring that data in any raw format is inconceivably absurd, but they tried—a team of five worked on it for seven months, looking without success for ways to improve efficiency.

Then Boeing switched to visual analysis to find signals in the noise. Within two weeks a pair of data scientists had identified inefficiencies and maintenance failures. But it wasn't enough to find the signals; they had to communicate them to the decision makers. Their complex visualizations were translated into simpler ones for the management team, which approved changes to the Osprey's maintenance code. Operations improved. "It's hard to tell this kind of story," says David Kasik, a technical fellow at Boeing who worked on the Osprey project. "Ultimately we have to provide a form for telling our story in a way that others can in fact comprehend." That form is visual.

Do not presume that these are problems relegated to specialized, technical data like Boeing's. Even common business data such as financials and marketing analytics, which companies generate as a matter of course, has become so deep and complex now that we humans can't effectively deal with it in raw, nonvisual form.

The third trend: Everybody's doing it. Historically, some technologies have enjoyed a democratizing moment, when the innovation becomes cheap enough to buy and easy enough to use that anyone can try it. Examples of this shift are legion. Synthesizers were once highly technical room-size machines that only a few people in the world could operate. Synthesizer pioneer Moog made it so anyone could use them—and they did. Professional music used to require a team of engineers and expensive studio space as table stakes. In the 2010s, the singer Billie Eilish built a Grammy-winning music career on a MacBook and a few hundred dollars in software. Aldus PageMaker, the first word processor, and hypertext markup language (HTML) each in its own way made everyone a potential publisher. Dan Bricklin, a cocreator of VisiCalc, the first spreadsheet, once said that his democratizing software "took 20 hours of work per week for some people and turned it out in 15 minutes and let them become much more creative."[7]

When ownership of the technology suddenly shifts from a small group of experts to the masses, experimentation flourishes, for better and worse. (HTML led to garish GeoCities websites, but also to Google.) Dataviz is no different. What was once a niche discipline owned by a few highly skilled cartographers, data scientists, designers, programmers, and academics is now enjoying a noisy experimentation phase with the rest of us. For the first time, the tools used to visualize data are both affordable (sometimes free) and easy to use (sometimes drag-and-drop). Scores of websites have emerged that allow you to upload a data set and get bespoke visualizations kicked out in seconds. Many programs are aiming to become no less than the word processor of data visualization, guiding you to a good chart type and managing your "visual grammar" and design, the way Microsoft Word informs you that you've forgotten a comma or tells you to change the passive voice.

Meanwhile, vast reserves of the fuel that feeds visualization—data—have been made freely or cheaply available through the internet. It costs virtually nothing to try to visualize data, so millions are trying. But abundant raw material and drag-and-drop software can't ensure good charts any more than a lot of dirt and a hoe can ensure a farm. Learning to think visually now will help managers use these burgeoning tools to their full potential.

A SIMPLE APPROACH TO AN ACCESSIBLE CRAFT

The best news of all is that this is not a hard language to learn, even if it seems intimidating. Mastering a simple process will have an outsize impact on the quality and effectiveness of your visual communication. You may have heard people refer to the "art" of visualization, or the "science" of it. A better term for what this book presents is *craft*, a word that suggests both art and science. Think of a cabinetmaker, who may understand some art and some science but who ultimately builds something functional.

An apprentice cabinetmaker might start learning his craft by understanding cabinets—their history, how people use them, the materials and tools needed to make them. Then he'd learn a system for building good cabinets, and he'd probably build a hell of a lot of them. He'd also install them and learn how cabinets work in different types of spaces and with different types of customers. Eventually his skills would be deep enough to add his own artistic and clever functional details.

Learning how to build good charts isn't unlike learning how to build good cabinets, so this book will proceed in the same way. Part one—**Understand**—provides a brief history of visualization and a high-level summary of the art and science behind charts. It leans on (and sometimes also challenges) the wisdom of experts and academics in visual perception science, design thinking, and other fields to illuminate what visualization is and what happens when a chart hits our eyes. In addition to providing an intellectual foundation, this brief section should assuage your fears about learning a whole new discipline. You don't have to become a professional designer or data scientist to reach a new level in your chart making.

With a foundation of knowledge in place, you can start making better charts. Part two—**Create**—is the practical core of the book. It lays out a simple framework for improving your charts. You'll learn what tools and skills you must develop (or hire) to succeed with each of four basic types of visualization. You'll learn how to think through what you want to show and then draft an approach. The process requires less effort than you might suspect. In as little as an hour, you can vastly improve those basic charts you're used to spitting out of Excel. You may protest that because you're not a visual learner by nature, this will be harder for you. That's probably not true. Research suggests that although we clearly identify ourselves as either visual or verbal thinkers, that distinction may not exist.[8] Research also shows that anyone can improve basic visual fluency, just as anyone can learn enough fundamentals to communicate in a new language without mastering it. This section will offer a framework for evaluating yours and others' output to help you improve it. Learning this constructive criticism is meant to provide an antidote to the burgeoning and frankly intimidating chart criticism that's carried out daily online and in Twitter feeds, wherein a community of dataviz enthusiasts takes it upon itself to judge visualizations publicly.[9]

Part three—**Refine**—turns to the important skill of rendering a soundly structured chart as a polished and artful visual, both impressive and persuasive. Rather than present a list of design dos and don'ts, it connects design techniques to the feelings they create. What techniques can you employ to make a chart feel *clean*, or so simple that viewers get it instantly? This section shows how to craft charts that don't just convey some facts clearly but change minds and impel people to action. It also explores the limits of persuasion and why certain techniques can drift across a blurry line into dishonest manipulation. It looks into the role of visualization in understanding facts and truth—they are different—and why every chart is a manipulation and what that means to us as chart makers.

Finally, part four—**The Last Mile**—shows you how to operationalize your chart making for maximum value when it comes time to present your visual information. Too often, data visualization is left to individuals who happen to be holding the data, and who may or may not be equipped by themselves to produce good charts in a specific context. It wasn't always this way, and it shouldn't be now. In addition to sharing team designs and workflows, it will show you how to make charts even more effective by controlling how they're presented and using storytelling to get your dataviz beyond eyes and into minds.

Good Charts is structured as a single argument, but each of its four parts can also stand alone as a reference for information and inspiration, depending on your specific need, and a brief recap of key concepts is included at the end of each chapter. When your challenge is an upcoming presentation that will include charts, dive right into the **The Last Mile** section. If you're looking to think through some visual challenges with your team, use the **Create** section. I hope this book will become a well-worn, dog-eared companion.

Two final points: First, the subject of data itself—finding it, collecting it, structuring it, cleaning it, messing with it—fills entire books. To focus on the process of visualizing, I begin after the data has been collected and assume that readers understand and use spreadsheets and other data manipulation tools regularly. For more-complicated data analysis and manipulation, I recommend working with experts—we'll discuss this in part four.

Second, most of the charts in this book, and their narrative context, are based on real-life situations and real data. In some cases, the data, the subject of a chart, names, or other attributes have been altered to protect identities and proprietary information.

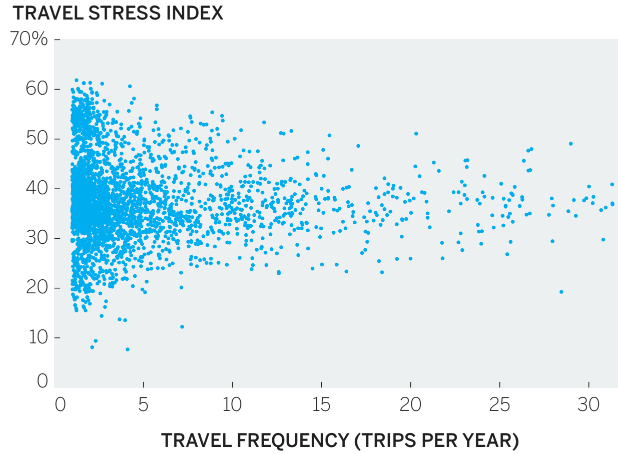
A GOOD CHART

Before we get started, take inspiration from Catalin Ciobanu. Once, he was a physics PhD brand-new to the business world, hired as a manager at Carlson Wagonlit Travel, where he worked with Vincent Lebunetel. Later he cofounded an AI software company with Lebunetel and became its chief technology officer. As a physicist, Ciobanu had learned to think visually; analyzing the massive data sets physicists use demanded it. “I had used many visual tools for analysis in science,” he says, “and when I moved to business, I found everything based in Excel. I felt very limited in the amount of insights I could convey from this. Greatly limited.”

Ciobanu was preparing for an event in Paris at which he’d present data to clients regarding what Carlson Wagonlit was learning about business travel and stress. The clients, he knew, were well versed in the aggregate figures on travel spending and the stress of business travel. But Ciobanu wanted them to see more. “What I wanted to convey wasn’t in the Excel file,” he says. “I wanted to convey this idea that travel stress is personal. It’s about people.”

After thinking through his challenge, Ciobanu produced this scatter plot:

WHO SUFFERS MOST FROM TRAVEL STRESS?



SOURCE: CARLSON WAGONLIT TRAVEL (CWT) SOLUTIONS GROUP, TRAVEL STRESS INDEX RESEARCH (2013)

When he put this chart up during his presentation, its effect was immediate and visceral. The dots created a sense of individuality that a table of percentages or trend lines couldn’t. Ciobanu focused on individuality by plotting *everyone*, not categories of people combined in bars representing some aggregate level of travel frequency. “Every point here is somebody,” Ciobanu says. “We found ourselves talking about people, not chunks of data.” Even the title, with its use of *who*, stressed the humanness of the challenge.

On the spot, clients began forming new insights from this visualization. They had assumed that stress rose with frequency of travel along a steady slope—a positive correlation that goes up and to the right: As trips increase, stress increases. This chart, though, shows that stress can either increase or decrease with more-frequent travel. It *normalizes* and becomes more predictable. Infrequent travelers, though, show wild variability in the amount of stress they experience.

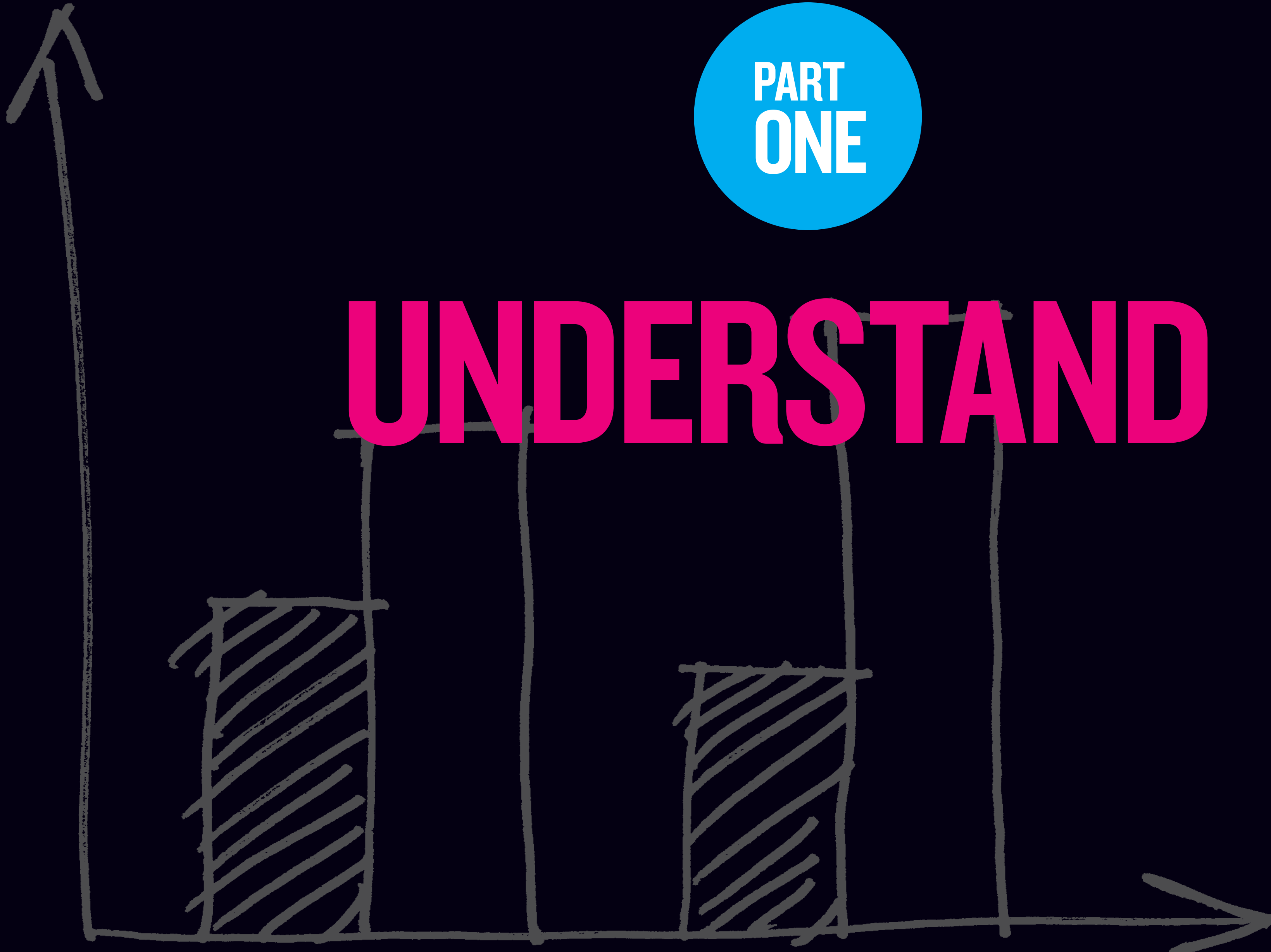
The client group eagerly discussed why that might be. Maybe some people who rarely travel view any trip as a treat and don't let delays or cramped economy-class seating bother them. Or some travelers may have to coordinate home and work schedules while they're away without executive assistance, creating the greater stress of holding down the fort while hitting the road. (Both these hypotheses were borne out by further research.) The clients discussed how programs and services could be adjusted on the basis of this one chart's shape alone.

"The conversation got passionate," Ciobanu remembers. "There were powerful outcomes in terms of re-sign rates and engagement." His colleagues and bosses were impressed, too; he gained respect for his visualization. "Following this," he says, "executives were coming to me asking me how we could show some data set of theirs or asking if I could help them make their charts better. Personally, this was one of those moments where I hit the mark."

It was a good chart.

PART ONE

UNDERSTAND



CHAPTER I

A BRIEF HISTORY OF DATA VISUALIZATION

THE ART AND SCIENCE THAT BUILT A NEW LANGUAGE

HERE'S A BREAKNECK SYNOPSIS of data visualization's development from simple communication tool to burgeoning cross-disciplinary science.

ANTECEDENTS

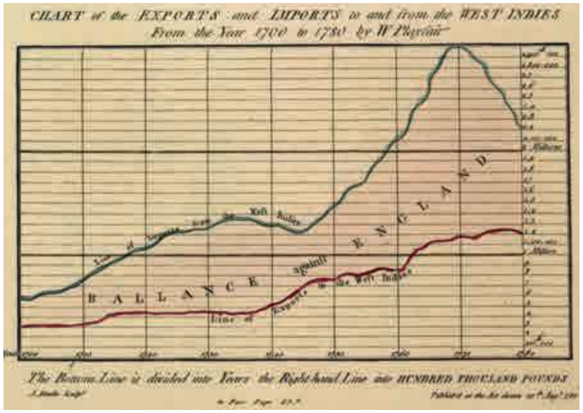
The first data visualization was probably drawn in the dirt with a stick, when one hunter-gatherer scratched out a map for another hunter-gatherer to show where they could find food. If data is information about the world, and if communication is conveying information from one person to another, and if people use five senses to communicate, and if, of those five senses, sight accounts for more than half our brain activity, then visualization must have been a survival tactic.[1] Far from being a new trend, it's primal.

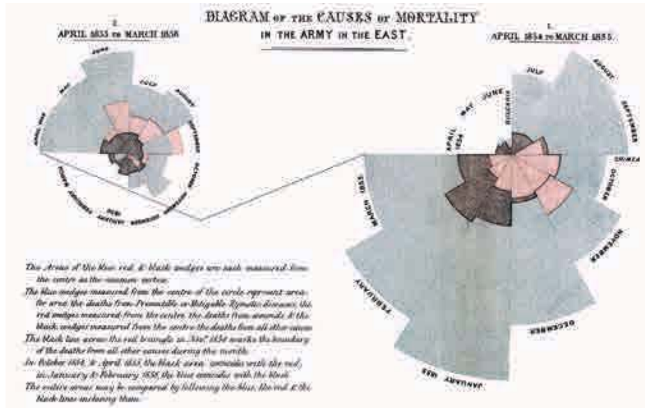
For a long time, visualization was probably limited to cave paintings and simple counting; eventually, maps, calendars, networks (for example, genealogies), musical notation, and structural diagrams emerged. In a sense, an abacus provides a visualization of data. No matter, I'm flying forward: Tables arrived in the late seventeenth or early eighteenth century and created spatial regularity that made reading many data points much less taxing. Ledgers were born. For two centuries, tables dominated information design.

What we think of as data visualization today—charts and graphs—dates to the late 1700s and a man named William Playfair, who in 1786 published *The Commercial and Political Atlas,* which was full of line charts and bar charts. He later added pie charts. Histories of infographics often start with a celebrated 1861 diagram by Charles Minard that shows the decimation of Napoleon's army during his doomed Russian campaign. Praise also goes to Florence Nightingale's "coxcomb diagrams" of British casualties in the Crimean War, published about the same time as Minard's well-known chart. Nightingale's work is credited with improving sanitation in hospitals because it showed how disease, above all, was what killed soldiers.

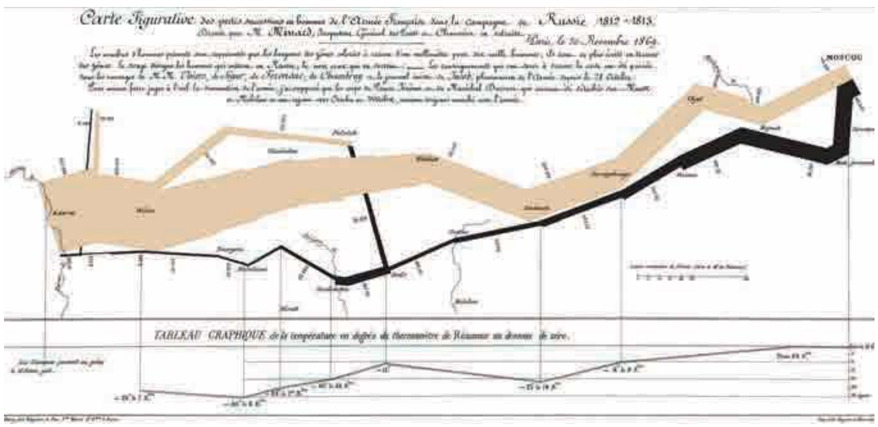
BRINTON TO BERTIN TO TUKEY TO TUFTE

It's no accident that charting began to take off with the Industrial Revolution. Visualization is an abstraction, a way to reduce complexity, and industrialization brought unprecedented complexity to human life. The railroad companies were charting pioneers. They created some of the first organizational charts and plotted operational





William Playfair, Florence Nightingale, and Charles Minard, the big three of early modern charting.



Willard Brinton's *Graphic Methods for Presenting Facts* provided advice to chart makers and critiques of charts from the early twentieth century.

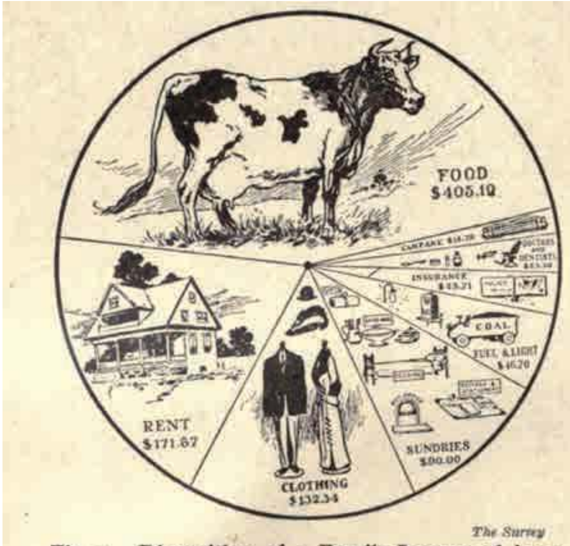


Fig. 3. Disposition of a Family Income of from $900 to $1000

This cut shows an attempt to put figures in popular form. The eye is likely to judge by the size of the pictures rather than by the angles of the sectors

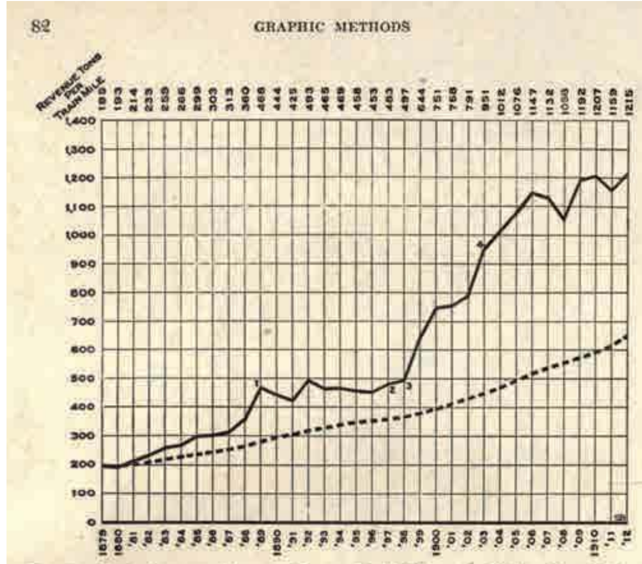


Fig. 80. Yearly Average of Revenue Tons per Train Mile on the Pittsburgh and Lake Erie Railroad. The Slanting Line Shows a Progressive Average

Here we have the data of Fig. 79 plotted in a curve which can be interpreted easily and accurately. This chart may be considered a model of good practice in curve plotting. All of the work, including the lettering, has been done by hand, thus insuring better results than can usually be obtained from printing

data such as "revenue-tons per train mile" (line chart) and "freight car-floats at a railroad terminal" (dual-axis timeline).[2] The work of their skilled teams of draftsmen (alas, they were all men) was a prime inspiration for what can be considered the first business book about data visualization: *Graphic Methods for Presenting Facts*, by Willard C. Brinton, published in 1914.

Brinton parses railroad companies' charts (and many others) and suggests improvements. He documents some rules for presenting data and gives examples of chart types to use and types to avoid. Some of his work is delightfully archaic—he expounds, for example, on the best kind of pushpin for maps and how to prepare piano wire for use as a pin connector ("heated in a gas flame so as to remove some of the spring temper").

Then again, many of his ideas were in the vanguard. Brinton lays out the case for using small multiples (he doesn't call them that), currently a popular way to show a series of simple graphs with the same axes, rather than piling lines on top of one another in a single graph. He shows examples of bump charts and slope graphs, styles many people assume are more modern inventions. He looks askance at spider graphs (they should be "banished to the scrap heap"), and he questions the efficacy of pie charts a century ahead of today's gurus.

Eventually, Brinton lays out a system for creating "curves for the executive" which can "tell the complete story [of the business] in every detail if placed in proper graphic form."

By midcentury, the U.S. government had become a complex and data-driven enterprise that demanded abstraction in unprecedented volume. Fortunately for the feds, they employed Mary Eleanor Spear, a charting pioneer who worked for dozens of government agencies and taught at American University. She produced two books in the spare, directive prose of someone who has a lot of work to do and not a lot of time to explain. *Charting Statistics* (1952) arose as a response to "problems encountered during years of analyzing and presenting data" in government. *Practical Charting Techniques* (1969) was an update and expansion on the previous, advocating for the power of data visualization: "The eye absorbs written statistics, but only slowly does the brain receive the message hidden behind the written words and numbers. The correct graph, however, reveals that message briefly and simply."

Spear's books, like Brinton's, are filled with smart, commonsensical advice, along with some now-obsolete passages of her own (she expertly lays out how to apply various cross-hatching patterns to distinguish variables on black-and-white charts; the resulting material is beautifully executed). And she engaged in some ahead-of-her-time thinking—in 1952 she included tips and techniques for presenting charts on color TV.

Jacques Bertin, a cartographer, wanted to ground all this practical advice about chart making in some theoretical foundation. So he formed a theory of information visualization in his watershed 1967 book, *Sémiologie Graphique*. Rather than focus on which chart types to use and how to use them, Bertin describes an elemental system that still frames and provides the vocabulary for contemporary dataviz theory. He broadly defines seven "visual variables" with which we *encode* data: position, size, shape, color, brightness, orientation, and texture.[3]

Bertin also established two ideas that remain deeply influential to this day. The first is the *principle of expressiveness*: Say everything you want to say—no more, no less—and don't mislead. This is a reasonably universal idea: It's editing. Writers, composers, directors, cooks, people in any creative pursuit, strive (okay, struggle) to pare down their work to the essential.

The second is the *principle of effectiveness*: Use the best method available for showing your data. That is, choose the visual form that will most efficiently and most accurately convey the data's meaning. If position is the best way to show your data, use that. If color is more effective, use that. This second principle is obviously trickier, because even today, determining the "best" or "most appropriate" method isn't easy. Often, what's best comes down to convention, or taste, or what's readily available. We're still learning, scientifically, what's best, and the process is complicated by the fact that in a world of digital interactivity and animation, what's best may change from page to screen, or even from screen to screen.

Bertin was followed in the 1970s by John Tukey, a statistician and scientist who was making 3-D scatter plots way back in the mainframe era. Tukey can be credited with popularizing the concepts of *exploratory* and *confirmatory* visualization—terms I'll borrow and use later in this book. Roughly, exploratory visualization is used to find patterns you don't know are there, while confirmatory visualization is used to show what you know is there.

Jock Mackinlay built on Bertin's work in his influential 1986 PhD thesis.[4] Mackinlay focused on automatically encoding data with software so that people could spend more time exploring what emerged in the visuals and less time thinking about how to create them. He also added an eighth variable to Bertin's list: motion. Working in computer science at the dawn of the PC era, he could see animation's powerful application for communicating data.

If Brinton is modern data visualization's first apostle, and Spear and Bertin its early disciples, Edward Tufte is its current pope. With disciplined design principles and a persuasive voice, Tufte created an enduring theory of information design

THAT'S A GOOD CHART

RED ZONE, 1912

Willard Brinton's classic book from 1914 shows us that less is new in visualization than we think. That includes sports visualization. Witness the beautiful relic on the next page, a visualization of the second half of the 1912 Harvard-Yale football game, from the *Boston Globe*.

As a historical artifact, this viz is fascinating and fun. Back then, the forward pass was so insignificant that it didn't merit a visual distinction, just the meek notation "F.P." (I count two completions, two incompletions, and one interception.) Modern fans exhausted by the officious nature of today's games will note also that only four penalties were called in the half.

One of cheapest tricks on social media is to post pictures of antediluvian technology or obsolete inventions solely for us to collectively mock them. *Ha ha, look at those simpletons with their giant calculators. They really were dumb, weren't they?* I don't present this Harvard-Yale game chart to snicker at it, but to praise it. It's an impressive feat. In 1912, for people who weren't at Harvard-Yale—the Super Bowl of its day—there was no replay, no highlights, not even radio to help them get closer to the game. The only way for most people to experience the game action was through the newspaper. That puts a lot of pressure on the game story and this visual to serve its audience well. And remember that "J.C.S."—whoever he or she was—was working on tight deadline and visualizing by hand. The chart had to be made fast, kept clean, and include the right—and the right amount of—information.

The Harvard-Yale game chart reminds us that making visual sense of games is hardly a recent innovation. Today, dataviz is deeply enmeshed in sports and layered into live sports and highlights in ways we take for granted. Think of the yellow first-down lines in football, pitch trackers in baseball, the world record pace lines that scroll across a pool in swimming races. Golf broadcasts display the trajectory of a ball, its speed and

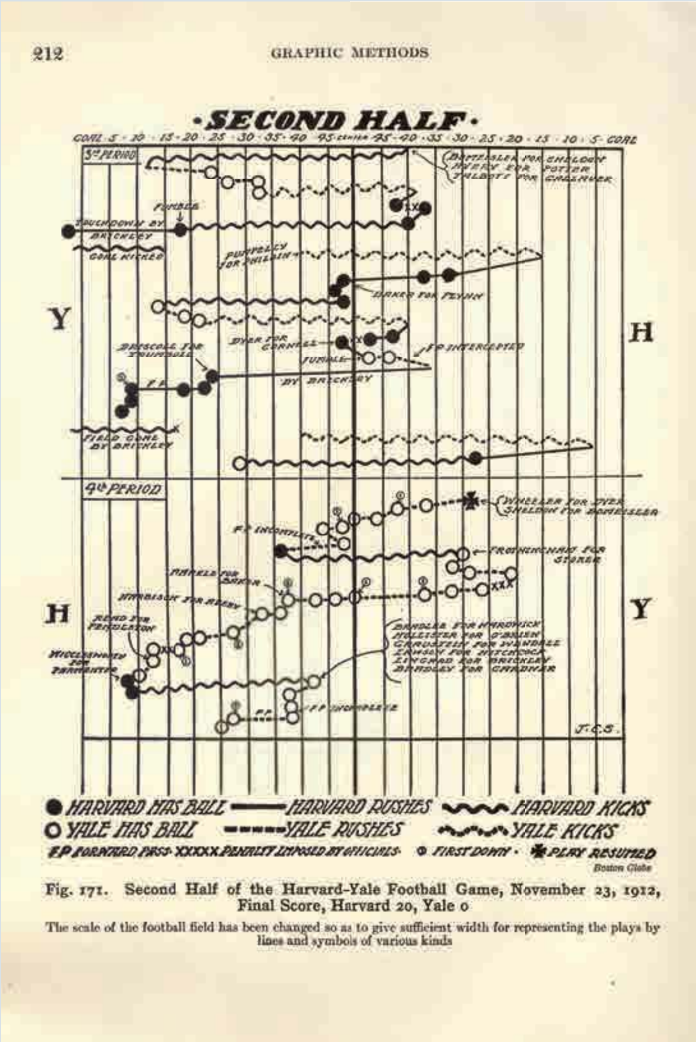


Fig. 171. Second Half of the Harvard-Yale Football Game, November 23, 1912, Final Score, Harvard 20, Yale 0

The scale of the football field has been changed so as to give sufficient width for representing the plays by lines and symbols of various kinds

height at its apex as it's happening. Greens are turned into topographical maps.

But I still see so much to admire in the 1912 chart. It creates a clear sense of progression; it communicates the fits and starts of the game, and I can immediately see Yale's fruitless domination of the fourth quarter. I found myself immersed in this.

Later, when I went to check on my own fantasy football team, I noticed a feature called Game Center that tracked plays visually during a game. It followed a remarkably similar concept to this Harvard-Yale chart. I still might like the 1912 version better.

in *The Visual Display of Quantitative Information* (1983) and ensuing tomes. For some, *Display* is visualization gospel, its famous commandments oft repeated. For example: "Above all else show the data" and "Chartjunk can turn bores into disasters, but it can never rescue a thin data set." Even though his work was rooted in scientific precision, Tufte is to the design-driven tradition what Bertin was to the scientific. A generation of designers and data-driven journalists grew up under the influence of Tufte's minimalist approach.[5]

EARLY EVIDENCE

While Tufte was declaring the best ways to create beautiful, effective charts, researchers were learning how people read them. In 1984 William S. Cleveland and Robert McGill took on "graphic perception" by testing how well people could decipher simple charts.[6] Pie charts have seemingly been under assault as long as they've existed, but Cleveland and McGill provided the first evidence that people find the curved area of pie slices more difficult to parse than other proportional forms. The two instigated a decade-plus of research aimed at understanding how we read charts and applying the results to a burgeoning visual grammar.[7] They felt duty-bound to challenge accepted wisdom: "If progress is to be made in graphics," they concluded, "we must be prepared to set aside old procedures when better ones are developed, just as is done in other areas of science." A few old procedures were set aside; a few new ones were developed.[8] This research deeply influenced the rapidly developing computer science community. Foundational texts that emerged from this era were Cleveland's *The Elements of Graphing Data* (1985) and *The Grammar of Graphics* (1999) by Leland Wilkinson.

Viz communities grew apart. Computer scientists increasingly focused on automation and new ways to see complex data, scientific visualization using 3-D modeling, and other highly specialized techniques. They were comfortable with visualizations that didn't *look great*. (In some ways this was unavoidable; computers weren't very good at graphics yet.) Meanwhile, designers and journalists focused on capturing the mass market with eye-catching, dramatic, and decorated charts and information graphics.

Wedged between these two worlds was Chart Wizard, the Microsoft innovation in its Excel spreadsheet program that married the automation of computer-generated visualization with some design options built in—albeit design options much maligned for their superfluous ineffectiveness. From extraneous three-dimensionality to limited and unintuitive color palettes, Excel charts have become an immediately identifiable trope.

Still, Excel was a democratizing moment that put dataviz in the hands of millions, and the effect of that can't be understated.

The internet happened and messed up everything.

REFORMATION

Tufte couldn't have anticipated when he published *Display* that the PC, which debuted about the same time as his book, would, along with the internet that runs through it, ultimately overwhelm his restrained, efficient approach to dataviz. This century has brought broad access to digital visualization tools, mass experimentation, and ubiquitous publishing and sharing.[9]

The early twenty-first century's explosion of infoviz—good and bad—has spurred a kind of reformation. The two traditions have dozens of offshoots. The followers of Tufte are just one sect now, Catholics surrounded by so many Protestant denominations, each practicing in its own way, sometimes flouting what they consider stale principles from an academic, paper-and-ink world.

Some offshoots have mastered design-driven visualization in which delight and attractiveness are as valuable as precision.[10] Others view dataviz as an art form in which embellishment and aesthetics create an emotional response that supersedes numerical understanding.[11] There are new storytellers and journalists who use visualization to bolster reporting and to lure and engage audiences.[12] Some use it as a means of persuasion, in which accuracy or restraint may be counterproductive.[13]

No one owns the idea of what data visualization is or *should be* anymore, because everyone does.

This transfer of ownership from experts to everyone has diminished the influence of scientific research from the 1980s and 1990s. Cleveland and McGill's results are sound, but most of their work focused on learning how people see static, mostly black-and-white charts, and it was limited to simple tasks such as identifying larger and smaller values. In a full-color, digital, interactive world, new research is needed.

Additionally, two assumptions were embedded in that early research: The first is that chart makers already have the undivided attention of the person decoding the chart. They don't. You need only look at a Twitter feed, or at all the faces staring down at smartphones during presentations, to know that every chart must fight to be seen. Early research didn't test how charts gain attention in the first place, which requires different and possibly conflicting techniques from the ones that show data most effectively. For example, complexity and color catch the eye; they're captivating. They can also make it harder to extract meaning from a chart.

The second assumption is that the most efficient and effective transfer of the encoded data is always our primary goal when creating a visualization. It's not. Our judgments may not be as precise with pie charts as they are with bar charts, but they may be *accurate enough*. If one chart type is *most* effective, that doesn't mean others are *in*effective. Managers know they must make trade-offs: Maybe the resources required to use the *best* chart type aren't worth the time or effort. Maybe a colleague just seems to respond more positively to pie charts. Context matters.

EMERGING SCIENCE

The next key moment in the history of dataviz is now. This disruptive, democratizing moment has fractured data visualization into a thousand different ideas, with little agreed-upon science to help put it back together. But a group of active, mostly young researchers have flocked to the field to try. While honoring the work of the 1980s and 1990s, they're also moving past it, attempting to understand dataviz as a physiological and psychological phenomenon. They're borrowing from contemporary research in visual perception, neuroscience, cognitive psychology, and even behavioral economics.

Here are some important findings from this new school of researchers:

Chartjunk may not be so bad. *Chartjunk* is Tufte's term for embellishment or manipulation—such as 3-D bars, icons, and illustrations—that doesn't add to data's meaning or clarity. It has long been scoffed at, but new research suggests that it can make some charts more memorable.[14] This does not suggest that overloading a data visualization with adornment is necessarily a good idea—most professionals know the value of restraint. It only suggests that an absolute dictum against chartjunk may be officious. Even if you're not adding to the meaning, you may be drawing someone's attention, or you may be giving them a memorable visual cue.

Other studies are evaluating the role of aesthetics, persuasiveness, and memorability in chart effectiveness. The findings aren't yet definitive, but they won't all align with the long-held design principles of the past. Some research even suggests that if you have only a few categories of information, a pie chart is probably fine.[15]

A chart's effectiveness is not an absolute consideration. Of course, reality is turning out to be far more complicated than "Don't use pie charts" or "Line charts work best for trends." Personality type, gender, display media, even the mood you're in when you see the chart—all will change your perception of the visualization and its effectiveness.[16] There may even be times to forgo visualization altogether.[17] Research shows that charts help people see and correct their factual misperceptions when they're uncertain or lack strong opinions about a topic. But when we understand a topic well or feel deep opposition to the idea being presented, visuals don't persuade us. Charts that present ideas counter to our strongly held beliefs threaten our sense of identity; when that happens, simply presenting more and more visuals to prove a point seems to backfire. (The research goes on to suggest that what's more persuasive in those situations is affirmation—being reminded that we're good, thoughtful people.[18])

Our visual systems are quite good at math. In some cases we can process multiple visual cues simultaneously (say, color, size, and position), and when we're looking at charts with multiple variables, our ability to identify average values and variability is more precise than when we're looking at numbers. That is, show me many numbers in a spreadsheet and ask me to estimate their average, or how much change occurs within them, and I won't do as well as if you show me, say, a scatter plot and ask me to do the same. Ronald Rensink at the University of British Columbia and, later, Lane Harrison at Tufts University have also shown that we can sense correlation in charts in a predictable way, and how effective that sense is varies from chart type to chart type—allowing us to rank order the effectiveness of certain visual forms for showing correlation (more on this in the next chapter).

All of this suggests that visual representation is even more powerful than we know and sometimes a more intuitive and human way to understand values than statistics is.[19]

Visualization literacy can be measured. Some researchers are attempting to create standard visual literacy levels. Early results suggest that most people test just below what could be considered "dataviz literate," but that they can be taught to become proficient or even fluent with charts and graphs.[20] This research also shows that we don't trust our judgments of charts as much as we should: Even when we correctly identify the idea a chart conveys, we want to check whether we're right. Helen Kennedy, a professor and researcher at Leeds University, has done groundbreaking work here on defining what seems to matter with datavisual literacy and our confidence. Many of the findings are expected—we need to be confident in our math abilities and our familiarity with visual forms. But others are surprising; for example, emotions play a large role in how people respond to visualization. (More to come on this topic in chapter 6.)

In just over a century, data visualization has evolved from manuals of simple visual grammar to frameworks for understanding the practice to, now, more-sophisticated discussions about visualization's role in the world. Whereas Brinton and Spear were concerned with simply helping people get their cross-hatching right, today entire books have been written on visualization and misinformation, and data and feminism. Kennedy herself has researched understanding visualization's role in public discourse, diversity, and how we live. She coedited *Data Visualization in Society* in 2020, a collection of scholarly articles that aims to do no less than create a philosophy of visualization, attempting to answer questions such as: Can visualizations be objective? What is the value of beauty in data visualization? How do charts affect policies and institutions? What role do emotions play in dataviz?[21]

You may look askance at these questions; you're just here to learn how to make some good charts.

Don't worry, this book is focused on those practical skills and techniques. Still, even as you're learning the practical skills, you'll find these questions naturally emerge as you try to find visual solutions to presenting your data, and as you observe others' work. Such questions emerged during the pandemic as charts became a key force in informing and debating trends. It's worth considering such questions.

A RETURN TO CRAFT

The science of visualization and information design is hurtling forward, but it will not stamp out the art of it. If anything, the science has demonstrated the need for the art. We know, empirically, that skillful design plays a role in effective visual communication. Humans have subjective *feelings* about data visualizations that can't be ignored in the process of creating them.

And so, the two broad communities in the visualization world—the computer-driven science community and the design-driven creative community, the Tukeys and the Tuftes—that were cleaved in the late twentieth century have drifted back toward each other.

This is mostly out of necessity. The volume of data we have demands automation and machine processing; at the same time, the tools we have to turn these massive pools of data into visual information still aren't good at understanding (never mind setting) the human context the data will be used in.

So on the one side, the technology is limited in its ability to intuit human needs and desires. For example, no computer program can ever know the needs of my audience and what part of my data is most important to them. If I generate a line chart with five lines on it, the software treats them all equally—each gets a unique color, they are all the same thickness, and none looks more important than another. But usually, one is more important than the others for my audience. Usually, I want them to focus on *this* information while using *that* plot as a reference point to make the primary data make more sense. I need to design my intention into the visualization. Only a person, not a computer, can know how to group the variables, or how to change the range of the axis to create a certain focus, or overlay information that's not in the data set to bring an idea into high relief.

On the other side, to find signals in the noise, I need to process and visualize hundreds, thousands, millions of data points. I need to make hypotheses and test them by generating quick visuals to see what's in the data. I need to be able to react to dynamic, changing data sources. This is the work of the machines; no human can manage this.

We need both the science and the art. And as we'll see later, we need *teams* to marry objective data with the needs of human beings in specific situations. It's the combination of tools and people that makes *effective* visualization.

As the grammar of graphics evolves (and it will continue to evolve, just as linguistic grammar does), visualization will remain what it always has been—an intermingling of the scientific and design traditions. It will be a mash-up of art and science, of taste and proof. But even if the grammar were already fully developed, understanding it alone wouldn't ensure good charts, just as knowing the rules for prepositions and the passive voice doesn't ensure good writing. The task at hand remains the same: We must learn to think visually, to understand the context, and to design charts that communicate ideas, not data sets.

And the best way to start learning how to produce good charts is to understand how people consume them. That starts by understanding some of the basics of visual perception.

RECAP

A BRIEF HISTORY OF DATA VISUALIZATION

Visual communication is primal, but what we now think of as data visualization started just two centuries ago. The history of visualization provides a foundation for learning and helps dispel several misconceptions about the practice. Above all, it allows us to dismiss the myth that dataviz is a fully formed science with rules that must be obeyed. In fact, dataviz is a craft that relies on both art and science, in which experimentation and innovation should be rewarded, not punished.

A TIMELINE OF SOME KEY MOMENTS:

Late 1700s
William Playfair produces what are often considered the first modern charts, including line charts, bar charts, pie charts, and timelines.

1858
Florence Nightingale produces "coxcomb diagrams" that show the devastating effect of disease on the British army.

1861
Charles Minard publishes a diagram showing the toll taken on Napoleon's army by his march on Russia.

1914
Willard Brinton publishes *Graphic Methods for Presenting Facts*, the first book about visualization for business.

1952
Mary Eleanor Spear publishes *Charting Statistics*, a book of chart-making best practices based on decades of work with many groups in the U.S. government.

1967
Jacques Bertin publishes *Sémiologie Graphique*, the first overarching theory of visualization, and one that remains deeply influential. Bertin describes seven "visual variables": position, size, shape, color, brightness, orientation, and texture. He also establishes two core principles: the *principle of expressiveness* (show what you need to; no more, no less) and the *principle of effectiveness* (use the most efficient method available to visualize your information).

1970s

John Tukey pioneers the use of visualization with computers and popularizes the concepts of *exploratory visualization* (finding patterns in data that you don't know are there) and *confirmatory visualization* (showing patterns in the data that you know are there).

1981

Microsoft introduces Chart Wizard into its Excel spreadsheet program, allowing millions to create fast, quasi-designed visualizations.

1983

Edward Tufte publishes *The Visual Display of Quantitative Information*, combining statistical rigor with clear, clean design principles and inspiring two generations of information designers and data journalists.

1984

William Cleveland and Robert McGill publish the first of several research papers that attempt to measure "graphic perception," setting off two decades of research into what makes visualizations effective.

1986

Jock Mackinlay publishes his highly influential PhD thesis, which carries Jacques Bertin's work into the digital age.

1990s–2000s

The computer-driven, scientific visualization community and the design-driven, journalistic visualization community diverge in their approaches to dataviz.

2010

Ronald Rensink publishes research suggesting that our perception of correlation in a scatter plot follows what's known as Weber's law and, for the first time, that a method for calculating a chart type's effectiveness may exist.

2010s

The social internet, cheap and easy-to-use software, and massive volumes of data democratize the practice of visualization, creating mass experimentation. Viz is no longer the province of a small community of experts; it's an internet phenomenon.

2014

Lane Harrison replicates Rensink's findings and applies them to additional chart types. He creates a ranking of chart-type effectiveness for showing correlation. Harrison's work is part of a new generation of research into establishing science around graphic perception, which draws on many other disciplines, including psychology, neuroscience, and economics.

2016

Helen Kennedy publishes an influential paper on visual literacy and the critical role emotions and "feelings of numbers" play in helping us make sense of data. She will push her research in the coming years into new territory of understanding the role of visualization in influencing people, affecting equality, diversity, and the effectiveness of institutions.

Today

Experimentation continues across a broad spectrum of disciplines. Tools for visualizing increasingly improve. They create better charts faster and allow for interactivity and dynamic updating of visuals. Social media is rife with visualization by professionals and amateurs. The discipline is a mass phenomenon.

CHAPTER 2

WHEN A CHART HITS OUR EYES

SOME SCIENCE OF HOW WE SEE

I'VE COMPARED THE PROCESS of learning dataviz to learning how to write and to learning a new language. Maybe the best analog, though, is music. Everyone hears music and forms opinions about it, and most do this without taking courses in music theory. We may sense *something* about the music we like—its "texture," or that it sounds "brooding," without knowing that we're describing syncopation, or a minor key.

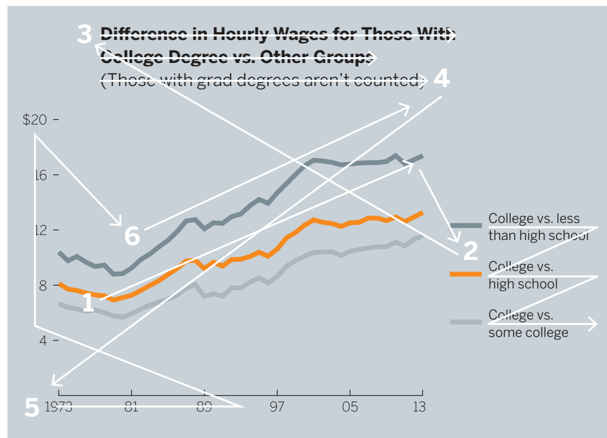
Similarly, everyone sees charts and decides whether they're good or bad without a degree in visual perception theory. You may sense *something* about a chart you like—you may even be able to describe it as "clear" or "revealing" without understanding that you're describing elementary encodings or perceptual salience.

If you wanted to compose, you might learn some music theory. Now that you've decided to make good charts, it's helpful to learn a little bit about how we see. Unlike music, though, the theory of data visualization is new and changing. It draws on multiple disciplines, including perception science, neuroscience, and psychology.[1] We're not here to become experts in all this; you just need seven broadly applicable ideas to understand what we see when we see a chart.

There's an unspoken contract between writer and reader about how readers will proceed through text. No such contract exists with visualizations.

SEVEN IDEAS TO KNOW

1. We don't go in order. In the tacit contract between a reader and a writer, the writer agrees to use words to communicate. The words will be strung together as sentences, the sentences as paragraphs, and the paragraphs as stories that will be presented "in order," which in the West means starting on the left, then moving left to right and



top to bottom on the page. Different cultures read in different orders.[2] In all cases, though, reading is done sequentially and at a reasonably even tempo.

With visualizations, no such contract exists between a producer and a consumer. We know that a reader often won't start with the title at the top of a chart until well after she has started scanning the visual middle. She may jump around. She may read halfway across an axis and then move on to something else—or skip some parts of the chart entirely.

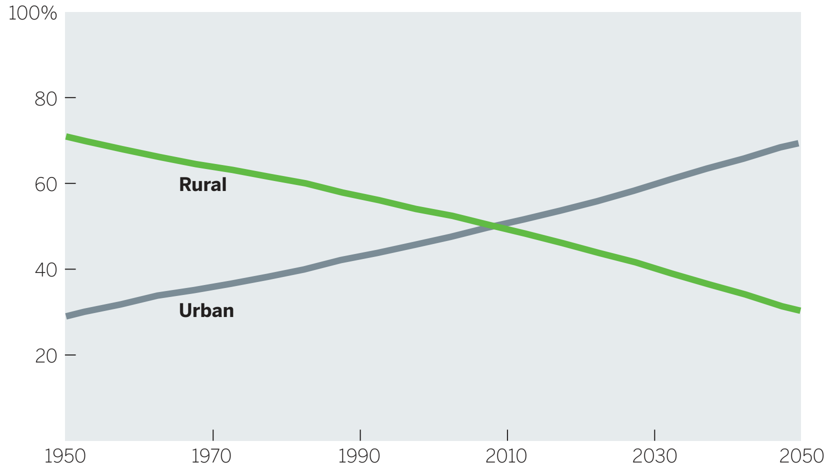
Pacing, too, is completely different. Reading a book is like running a marathon, taking a steady pace along a linear path. Parsing a chart is more like playing hockey, with fast bursts across space interspersed with intense action in concentrated zones. We go where our eyes are stimulated to go. There's no agreed-upon convention.

The order in which people look at charts varies because of many variables: chart type, who's looking, how much time they have, and more. Some research suggests that people with expertise in the subject matter of a chart or with practice using a certain chart type will read through it differently (and more efficiently) than others.[3] Although the challenges of producing good visual communication—to achieve the proper focus and clarity—are in some ways no different from those of producing any other kind of communication, they're in other ways more distinct and more difficult.

2. We see first what stands out. Our eyes go right to change and difference—peaks, valleys, intersections, dominant colors, outliers. Many successful charts—often the ones that please us the most and are shared and talked about—exploit this inclination by showing a single salient point so clearly that we feel we understand the chart's meaning without even trying. Like this:

WHERE PEOPLE LIVE

PERCENTAGE OF WORLD POPULATION
LIVING IN URBAN AND RURAL AREAS



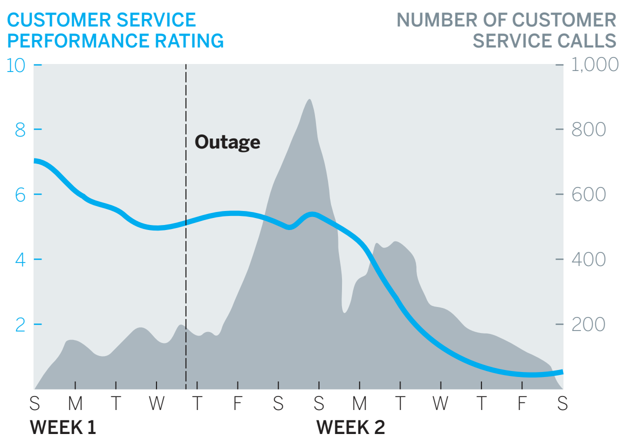
SOURCE: UNITED NATIONS, DEPARTMENT OF ECONOMIC AND SOCIAL AFFAIRS, POPULATION DIVISION (2014)

Here, the crossover is the inescapable visual your eyes can't resist. You probably didn't even read the axes before you began to know what this meant. Most likely you saw the crossover, then checked the labels, and glanced at the title, in roughly that order, nearly instantaneously, and you got the point: The world has flipped from mostly rural to mostly urban.

Note that you don't choose this path. Your eyes and your brain always notice more dynamic visual information first and fastest. The implicit lesson is to make the idea you want people to see stand out. Conversely, make sure you're not helping people see something that either doesn't help convey your idea or actively fights against it.

This is more easily illustrated with a slightly more complex chart. What are the first three things you see here?

CUSTOMER SERVICE CALLS VS. PERFORMANCE



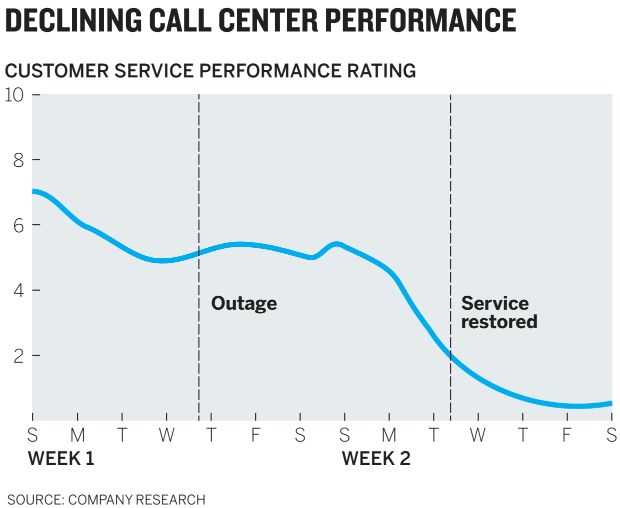
SOURCE: COMPANY RESEARCH

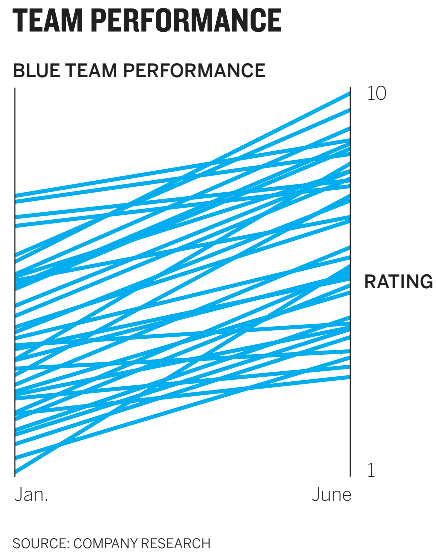
Without choosing to, most people will first see the blue line, the steep gray mountain, and the "outage" line. If the manager who presents this chart wants to communicate the relationship between an outage, customer service calls, and customer service performance, this chart rightly calls attention to those three points.

But what if he's concerned that customer service's issues are systemic rather than a result of the outage? What if he wants to convey to his boss that even after customer service calls returned to pre-outage levels, customer service performance continued to decline?

If we work at it, we can find that trend in the chart, but it's not what we notice first. It doesn't stand out. Our eyes have been drawn to something else. How might the manager make his idea what we see first?

Including the number of customer service calls in the first chart made our eyes go straight to data that this manager thinks is *not* the issue. His new chart (shown on the left of the facing page) eliminates that, thereby removing a distracting message. The addition of the "service restored" marker provides important context that highlights the continuing downward trend even after that point. And we may not have noticed it before, but now we see that the downward trend started *before* the outage.

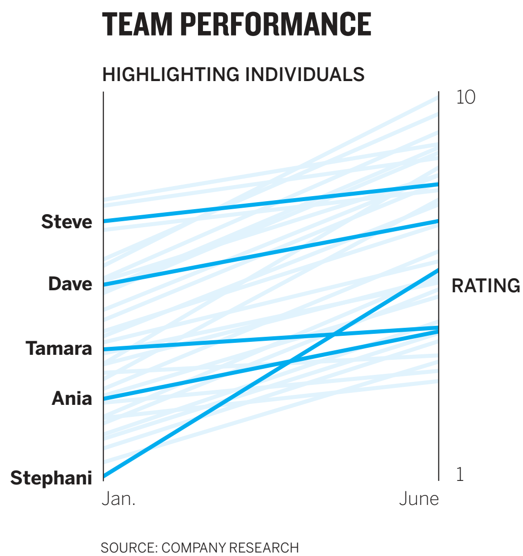




The capper is the new title: "Declining Call Center Performance." Remember we don't have an implicit contract with chart readers that they'll start with the title. And despite their position, titles aren't usually the first thing a chart reader sees. Rather, they're clues to help us find the meaning that started to emerge when we looked at the picture. Here, the word *declining* confirms the chart's message and purpose.

If this manager had given his boss the first chart, he'd have to fight her inclination to focus on what stands out in it—that peak in customer service calls—and get her to see the trend he cares about. Now he can start a conversation about performance overall.

3. We see only a few things at once. The more data that's plotted in a visualization, the less individual data will matter to the viewer, and the more singular the chart's meaning becomes. To build on the last example, to see a simple before-and-after representation of call center employees' performance, the manager might produce a slope graph. In the top chart to the right, he has plotted January and June ratings for a few dozen employees.



Despite the fact there are 60 or so data points plotted here for 30 or so variables, someone looking at this chart only sees one data point, a collective trend. "Things are going up." It's impossible to have a conversation about individuals or even subsets within this group with this chart. The boss sees only generally rising performance in a thick band.

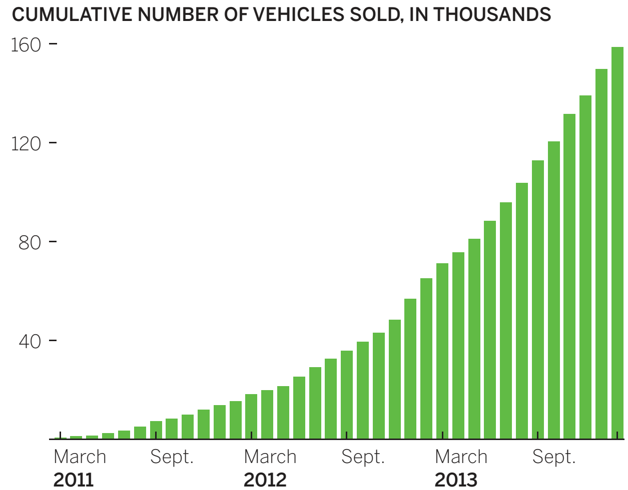
But the manager knows his boss will be making decisions about individual employees' performance. How few points of data should he show so that she can do that?

The threshold at which individual data points melt into aggregate trends is surprisingly low. It varies according to chart type and task. For example, experts think that we can't distinguish more than eight colors at a time, at most.[4] A good guide is that with more than five to ten variables, individual meaning begins to fade into the aggregate.

The manager's boss can judge individuals' work in the second chart, but even this hints at the limits of showing multiple individual data points together. It takes a moment to separate the pickup sticks before we can start to see a singular pattern in their performance. If the manager needs to convey the individual performance of hundreds of people, he has a challenge ahead of him with a slope graph. He may have to consider other techniques.

A bar chart is more effective than a slope or a trend in a line chart at getting us to focus on each discrete category of data—each bar. But even bars create singular shapes when enough of them are plotted and they're snug against each other. What's the first thing you see in the Plug-In Vehicles chart—30 plus separate values or a slope?

PLUG-IN VEHICLES: THE FIRST THREE YEARS



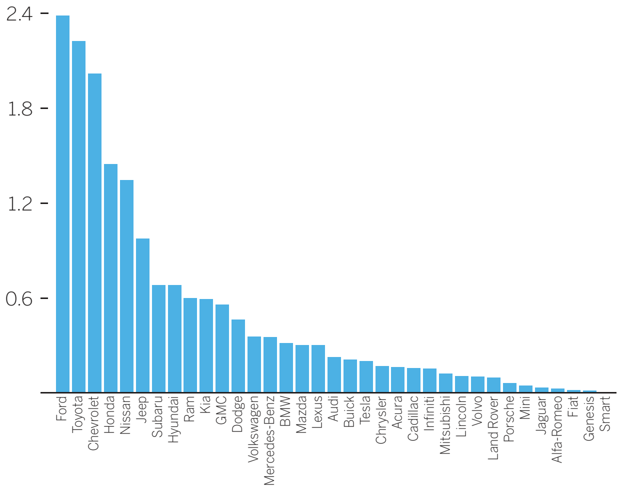
SOURCE: ADAPTED FROM PLOT.LY PLOT BASED ON DATA COMPILED BY BRETT WILLIAMS AND CHARTED AT FIGSHARE.COM

Again, if the point is the trend, this will do just fine (though since it is a trend, a line chart would work as well and avoid any confusion). But if we need to compare discrete values within this series for our discussion, or some subset of this data matters most, or there are other variables we need to make our point (such as, say, profit per vehicle), then we're not there yet, because we still just see one overarching trend.

This melding into an aggregate becomes problematic when bars make us sense a trend when there isn't one.

TOTAL VEHICLE SALES BY BRAND

VEHICLES SOLD, IN MILLIONS



SOURCE: EVADOPTION.COM

Here we clearly see a steeply declining trend; there's no trend because each bar represents a different company, not the same value (units sold, dollars, etc.). Think of it this way: We see a trend, which means it could be a trend line, but that wouldn't make sense. Kia's sales don't "go to" GMCs. They're not connected. They're discrete.

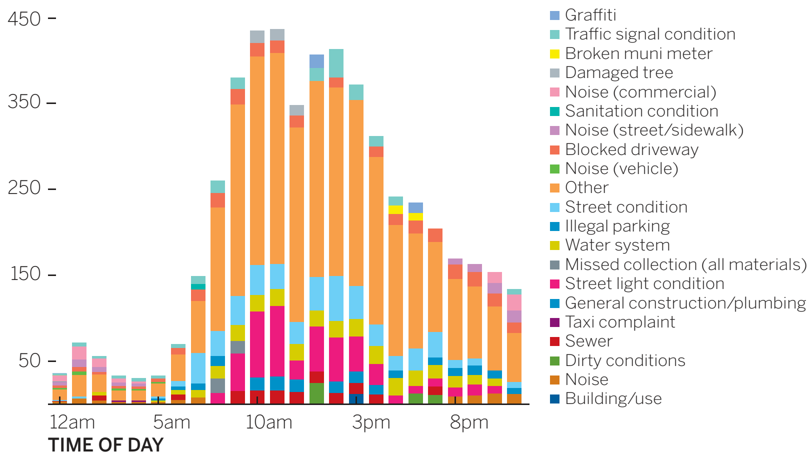
The previous examples examine generally well-constructed charts that create a single view in the mind because the mind can only process a few variables at once. But there's another kind of bad complexity in which too much information doesn't melt into a singular meaning; it just baffles us.

You've probably encountered such charts or made a few yourself. They use color, callouts, statistical labels, legends, and other devices to draw our attention in several directions at once. People find these charts taxing and frustrating. Sometimes people blame themselves for not getting meaning out of them, but it's not them. It's because we only see a few things at once.

The Most Common 311 Complaints chart plots 21 discrete categories across 24 tightly packed hours.

THE MOST COMMON 311 COMPLAINTS IN NYC

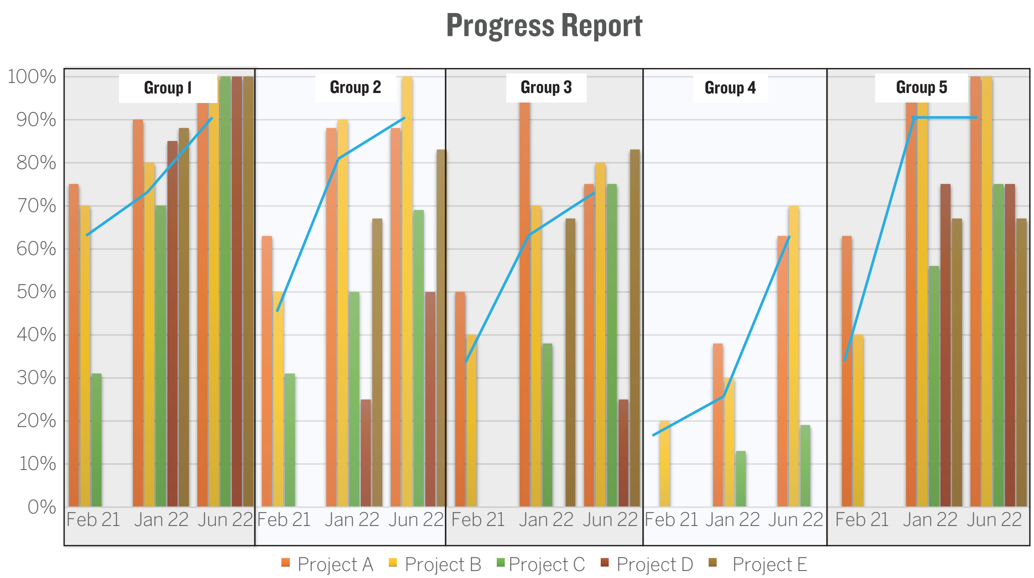
NUMBER OF COMPLAINTS (IN THOUSANDS)



SOURCE: PLOT.LY

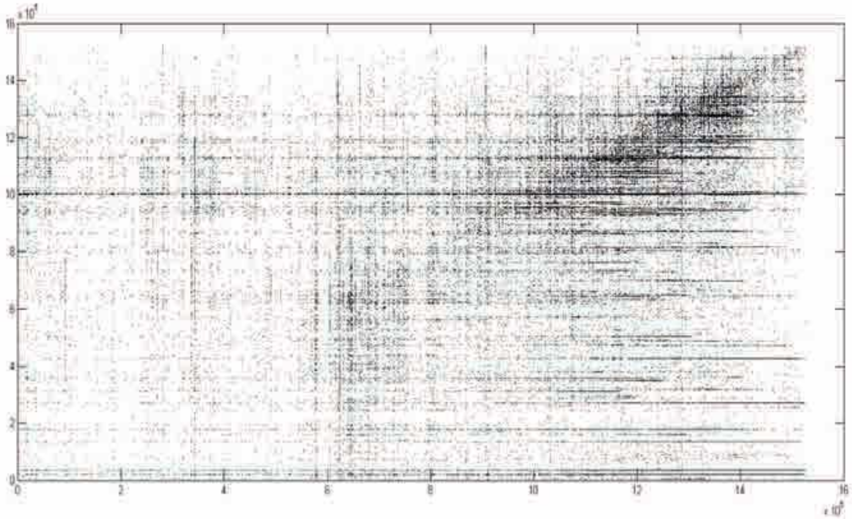
Some of the categories' values are so small as to become barely distinguishable slivers. For example, try following changes to "illegal parking" over the course of the day. The color choices aren't systematic—in fact, different complaints share like colors ("graffiti" and "damaged tree" look remarkably similar). The lengthy legend disconnects the y-axis values from the bars. And not every variable is found in every bar. What stands out here? We might argue that a general middle bump is discernible—that all the complexity forces us to look for something simpler and that's what we can grab onto—but if that's what needs to be shown, all those categories and colors are distracting from it.

In truth, this kind of chart is common, and I've seen and worked with much more challenging examples. Here's a reproduction of a project I worked on. No need to belabor the point, just notice that as you look at it, your brain is working hard to make simple sense of a complex visual. (We fixed this—transformed it, really—in a matter of a couple of hours. More on the process we used to do that in the coming chapters.)



Bad complexity neither elucidates important salient points nor shows coherent broader trends. It will obfuscate, frustrate, tax the mind, and ultimately convey trendlessness and confusion to the viewer.

Good complexity, in contrast, emerges from visualizations that use more data than humans can reasonably process to form a few salient points. Here's an extreme example:



This is a scatter plot of *10 million* data points that charts the social connections between stock traders on a social trading platform. Despite the overwhelming amount of data displayed, we see just a few things to focus on: the dense black spot, a correlative upward-right increase in density, and some striation, especially to the right.[5] That's all we can talk about here.

We can process these visualizations at the "blurry level," as one researcher puts it, and estimate the values they represent reasonably well.[6] When deeply complex charts work, we find them effective and beautiful, just as we find a symphony beautiful, which is another marvelously complex arrangement of millions of data points that we experience as a coherent whole.

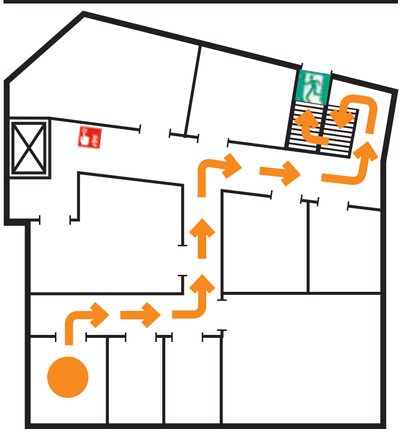
4. We seek meaning and make connections.
Once we see what stands out, we try to make sense of it immediately and incessantly. When you looked at the complex scatter plot to the left, you may have thought, *Why is it smudged black in the top right?* Sometimes we even vocalize the impulse to make meaning, with a "Hmmm," or a "What's that about?"

Even as we ask ourselves such questions, we're generating a narrative. With the original Customer Service Calls chart, for example, it doesn't take long to string the first three points we see into a simple story: An *outage* led to a *spike in service calls* and then a *performance decline.* With the Team Performance slope chart, we quickly translate the angle and density of the lines as *Performance is improving in general, but most people are lower-performing to begin with*.

Once again, we're not choosing to do this. It's what happens, automatically, when a chart hits our eyes. Understanding this will help you anticipate how people will automatically react and help you plan your visual in a way that honors and exploits how their minds work.

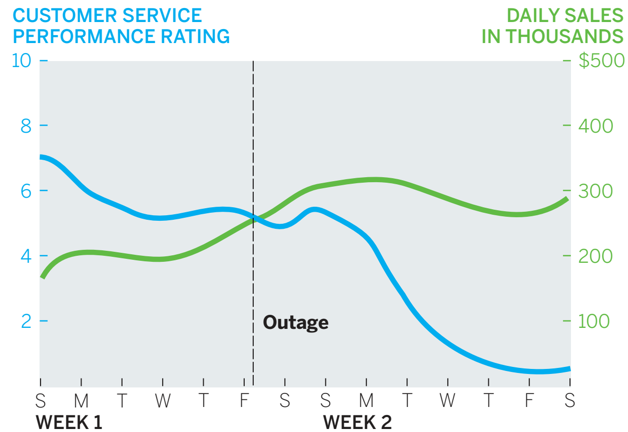
Exit this room. Turn right and walk 10 feet to the end of the hallway, where you'll be facing a large conference room. Turn left and walk another 12 feet until you come to the end of that hallway. To your left is a fire alarm, near the elevator. To your right at the end of the hall is a stairwell. Do not go to the elevator. Turn right and walk another 12 feet to the end of the hall, turn left and enter the stairwell. Go down two flights of stairs and exit the building at the door at the bottom of the stairs.

FIRE ESCAPE PLAN



Seeking sense this way has obvious benefits. For one, we process visual information thousands of times more efficiently than we do text. Some of our processing is even "pre-attentive"—it happens before we're aware we've done it—so we can grasp visual information more clearly with less effort. For example, imagine your office building is on fire. As smoke fills the room, you rush to the door, where you see the emergency exit placard to the left.

CUSTOMER SERVICE PERFORMANCE VS. REVENUE



SOURCE: COMPANY RESEARCH

In the room next to yours, someone rushes to the door and sees the Fire Escape Plan map instead. Who do you think makes it to the first exit faster?

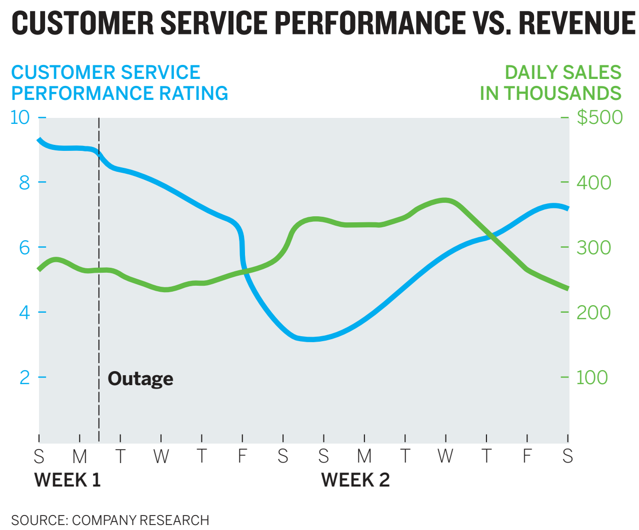
The ability to find meaning so efficiently may be a blessing in a fire, but it can also lead us to construct false narratives from data visualizations. What if the customer service manager showed his boss the chart to the right comparing customer service ratings to revenue when she asked for some data to review the effect of the outage?

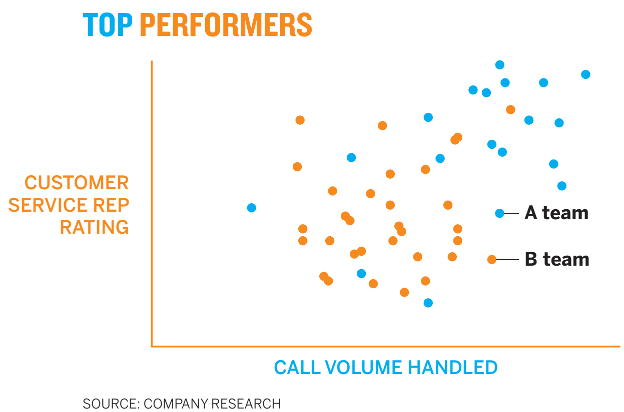
We can't help making connections in what we're presented with. Anything that stands out becomes part of the narrative we're trying to form, so what's presented becomes a crucial factor in the success of the chart—its ability to convey the idea the chart maker wants it to. The manager's boss, seeking meaning, may reasonably conclude from this chart that revenue is steady despite the outage. The narrative she's forming may convince her that she can deprioritize a proposed customer service overhaul. After all, revenue is unharmed.

But before she can do that, the manager shows her a chart from a previous outage that extends the length of time plotted, shown on the right.

In this version she sees a different story: Revenue dropped, but not until nine or ten days after the outage.

Of course, correlation does not equal causation. Did the outage and customer service's performance eventually affect revenue? Or was the drop related to something not included in the chart? The manager and his boss don't know, but he, knowing that she will seek meaning and make connections, has produced a better chart with which to start the discussion. Good visual communication should be used not just to produce better answers but also to generate better conversations. In this case, the two can wait a few days to see whether revenue starts dropping.

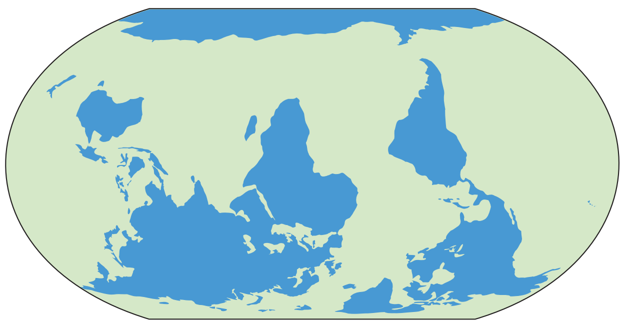




This need to make sense of what we notice is so powerful that it extends to the subconscious. In the Top Performers chart the bold orange and blue headline is one of those instantly noticeable cues. It makes us immediately connect the colors to other like colors on the chart. Somehow, our brains say, they want me to put together "performers," "rating," and "B team." And they want me to see "top," "call volume," and "A team" as connected. The colors mean *something*.

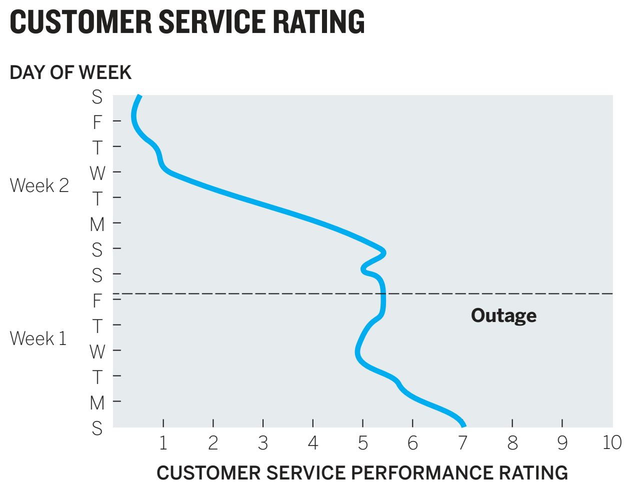
In fact, you can't stop your mind from finding meaning in this. Research has shown that our visual system will subconsciously create cohesion among the orange items while tuning out other colors and information in order to increase its focus on the dominant color, in this case orange.[7] Without realizing it, we've prioritized the color relationship over other information. That's unfortunate, because the color connection here is meaningless—just a suboptimal design decision. In fact, the blue team is higher performing.

5. We rely on conventions and metaphors.
It's not just how we're wired to see the world that defines how we see charts. It's also how we're *taught* to see the world. In this case, literally:



Is this map wrong? No. We think it's "upside down" because we've learned that "north is up," even though there is no up or down for a ball spinning around in space. You also may not have seen the world right away. In your mind, land is green and water is blue, even though that's not actually true.

Likewise, the Customer Service Rating chart below is accurately plotted, but most of us would still say it's "wrong" because time *doesn't go up*. Once we look at the axes, we find ourselves doing some cognitive gymnastics, expending significant mental energy trying to twist the lines back into a form we're used to seeing. You even may have tilted your head to the right in an effort to make the time axis horizontal, only to realize that even then, it's still "backward" because time *doesn't go right to left*.



SOURCE: COMPANY RESEARCH

In fact, time visualizations can move in any spatial direction and remain factually accurate. But we've learned to *think* of time as moving horizontally left to right on a page or a screen, and back to forward in three-dimensional space.

Moving time to the y-axis creates another perception problem. It generates a line that, literally, goes down as performance goes up. The highest performance is found at the lowest point. Again, that messes with our learned expectations: "High" performance shouldn't be spatially "low."

Conventions are a form of expectation, and our brains use experience and expectation as cognitive shortcuts so that we don't have to process everything anew every time we see it. In fact, as the neuropsychiatrist Jon Lieff points out, "The over-arching analysis of visual signals depends on what is expected . . . the influence of the brain and expectation are far greater than the raw data."[8]

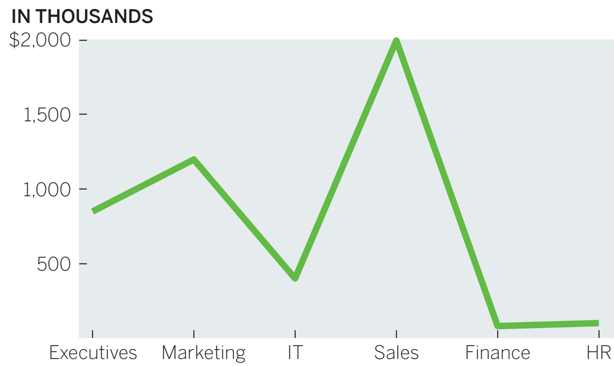
The point here is don't fight it. Your cognitive shortcuts, many developed culturally, are generally useful (though some do lead to bias; awareness of those situations and retraining the brain, and shifting the culture to eliminate those cognitive shortcuts, are essential). There are innumerable cognitive shortcuts we use when a chart hits our eyes: Up is usually good, down is bad. North is up, south is down. Researchers have found that we even connect those metaphors to value judgments.[9] For example, because south is "down," we think it's easier to go in that direction than to go north, which requires us to go "up."

Red is negative, green positive. But *red* sometimes also means "hot" or "active" (which can be thought of as positive), and in those cases, *blue* means "cold" or "inactive." Blue is water. Green is land. Hierarchies move from the top down. Lighter color shades are "emptier" or lower than darker ones. Gray things are less important than color things. Steep curves mean volatility and flat lines seem steady, or safe. These are just a few of many heuristics we use every day.

Anytime conventions like these are flouted, confusion, uncertainty, and frustration will weaken a chart's effectiveness. Some of the heuristics are so powerful and obvious that we rarely see them violated. Virtually no one maps the world "upside down." Desert temperatures aren't shown as a deep blue. Imagine a CEO announcing to her employees, "We're going to take this company into the future!" as she points behind her.

Or consider the Travel Expenses chart below. Conventionally, we connect data points only when there's a relationship from one value to the next. But here each value is an unchanging category. "Sales" doesn't change as a value; there's no inherent

TRAVEL EXPENSES BY DEPARTMENT



SOURCE: COMPANY RESEARCH

VIRTUOUS CHAOS

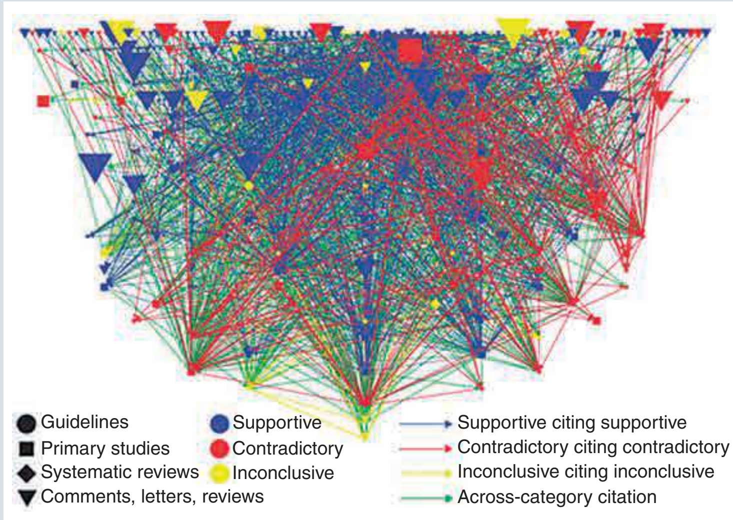
Whoa! That's terrible, right? It flouts nearly every principle in this chapter about what happens when a chart hits our eyes. Primary colors fight for attention. A dense skein of links makes it impossible to make sense of the network connections. Twelve possible dimensions to four variables make for 48 possible types of network connections, not including the size differences in nodes. It's chaos!

All valid critiques, but let's not start by focusing on these details. Start instead with the idea the authors wanted to convey. Ludovic Trinquart and colleagues created this chart for a meta-analysis of research papers on salt and its health effects.[10] Their goal was "to try to lay bare and begin to unravel the challenges underlying the dispute" over whether salt is bad for you.

Given that context, I'd argue this rainbow thicket works beautifully. It's a kind of virtuous chaos in which flouting the principles forces us to see just one inescapable idea: The mess is the thing. The colors fight with each other? So do conclusions about salt. Is it hard to make sense of how all the studies, comments, and guidelines connect? Exactly. It's challenging to find a coherent message about salt's health effects in the chart. That's reality.

The authors included several clear, orderly charts in the paper as well—after they "laid bare the challenges," they wanted to begin to unravel them after all—so it's unlikely this one is chaotic by accident. Now, imagine if the chart were honed with sound design principles. Actually, don't imagine, because a lovely chart on salt research also appeared in *Fortune*, shown on the facing page.[11]

I'm still getting that a lot has been written about salt, but conflict doesn't hit me as forcefully here. I don't feel the underlying confusion; I sense organized, passive disagreement. Even the title uses the word "patterns," which suggests some order. I see some people on one side, more on the other side, and a lot in the background, fading away in that

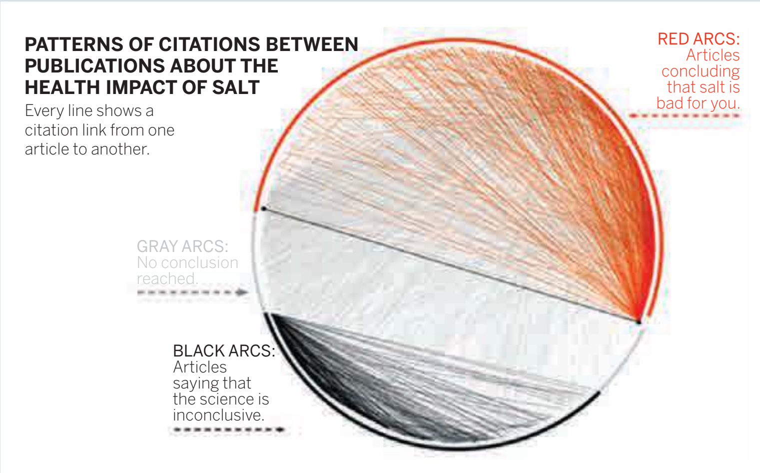


barely there gray area, even though those hard-to-see lines represent "no conclusion reached," which may be the most important group to highlight, not something to relegate to the background.

I like the idea of using virtuous chaos in visualizations in three scenarios:

- To convey the singular idea that "things are complicated and difficult to understand." Just don't expect the chart to do more than establish that one idea, no matter how much design work you put into it.

- To comment on negative complexity. Our inclination is to seek order and make meaning. Messy views can persuade us of the need for



change to something more orderly. If you want to streamline operations, show a chaotic workflow chart that maps out how things work today.

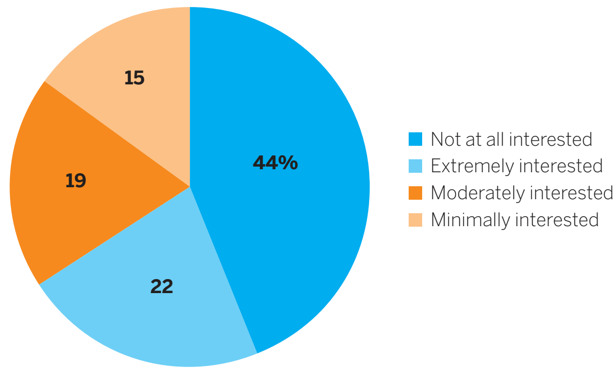
- To provide a storm before the calm. Using chaos as a "before" view will heighten the dramatic effect of the clean "after" view.

Finally, use virtuous chaos sparingly. Don't force it. Overdesigning messiness into an orderly data set is a cheap trick that people will see through if it's used inauthentically.

connection between executives' and marketing's expenses. One value doesn't *go to* the next as this line does. Each value here would be better served being plotted discretely, say, with a bar.

The real challenge with conventions comes from subtler violations of our expectations. Here's a re-creation of a published chart I encountered:[12]

HOW INTERESTED ARE YOU IN THIS PRODUCT?



SOURCE: COMPANY RESEARCH

More is happening here than may first appear. Without thinking about it, we access three conventions in our minds to help explain the meaning of the chart:

- Like colors mean like items—the blue things go together.
- Color saturation indicates higher and lower values—lighter colors have lower values than darker ones.

- Categories are arranged and plotted from one extreme to another—we can read this in order from most to least interested.

We're making meaning before we know it: There are two groups of people here with varying levels of interest, and the blue group is bigger than the orange group. But a closer look shows just how far off we are.

Like colors mean like items. You probably assumed that the blues are a pair and so are the oranges. But the key shows that the blue pieces represent diametrically opposed viewpoints (no interest, high interest), and the orange sections represent middle viewpoints (some interest, little interest). Our expectation is that "not at all" and "minimally" will be in one color because they represent the pessimists, while "moderately" and "extremely" describe another group, the optimists.

Color saturation indicates a progression of values. We expect light-color values to be lower than dark-color values, but here light blue has a higher value (22%) than dark orange (19%). If we match hues to actual numerical values, descending order *should be* rich blue, pale blue, rich orange, pale orange. Here the color groupings aren't in order either. Optimists are pale blue and rich orange, while pessimists are rich blue and pale orange. The color differences provide no guidance here. They only generate confusion.

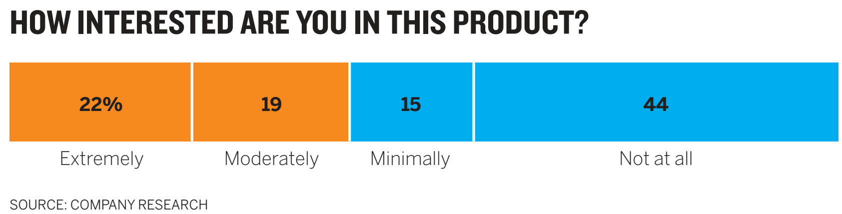
Categories are arranged and plotted from one extreme to another. Our minds want information to be arranged in order. But the key here lists categories "out of order." If we think of "extremely interested" as category one and "not at all interested" as category four, then this key is arranged four, one, two, three.

What at first glance appeared to be a simple, well-constructed pie chart turns out to repeatedly disrupt our expectations, forcing us to reset them and think harder about what we're looking at than we should have to. We can't take advantage of the mental shortcuts that help us get to meaning more quickly. Instead, we have to parse.

To show just how much disrupting expectations can affect viewers' ability to find meaning in a chart, look again at the pie chart for a few seconds and see if you can answer these two questions, and how quickly can you answer them:

- Which group makes up the majority, optimists or pessimists?
- Which single category represents the smallest proportion of people?

Now look at this version and see if it's easier to answer those questions.

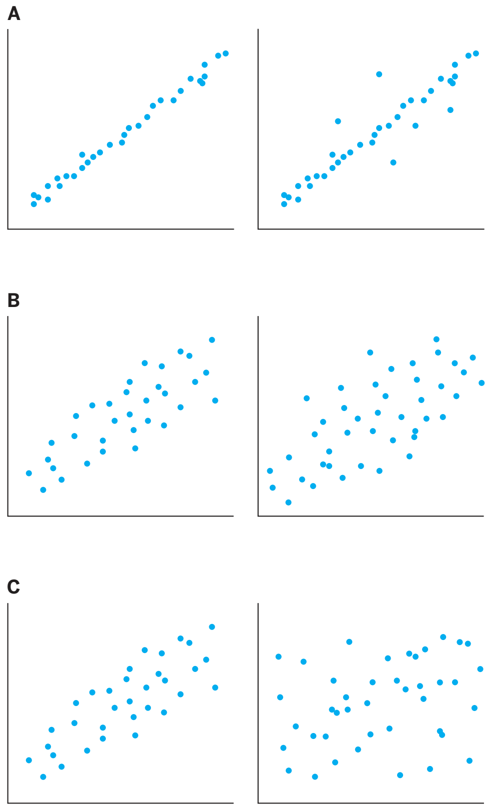


6. We sense statistical values in visuals. Another way to think of this visual perception principle is that we do math with our eyes. We can get a sense of correlation without knowing the actual values we're looking at. Our ability to detect change in charts seems to follow a fundamental rule of sensory perception known as Weber's law.

Weber's law states that "a noticeable change in stimulus is a constant ratio of the original stimulus."[13] Imagine a perfectly black room. Light a match, and you'll notice a big change in how bright the room is. But if you start with three lamps turned on, lighting a single match won't make the room seem brighter, even though statistically it is. The more light you start with, the more light you need to add to notice a change in brightness.

The key to Weber's law is that the relationship between starting state and new state is predictable and linear—twice as much original light means you need to generate twice as much new light to create a "just noticeable difference" or JND. We perceive change in the world in this linear way with light and color, scent, weight, sound, even taste.

Researchers discovered that we perceive change in correlation in scatter plots the same way.[14] For example, in scatter plot pair A on the next page, with a tight correlation near 1, you notice a big change when just a few dots are moved. But pair B has a looser correlation around 0.5, so you don't notice much change in correlation when the same number of dots as before are moved. To notice the change, you need to move twice as many dots, as shown in pair C.

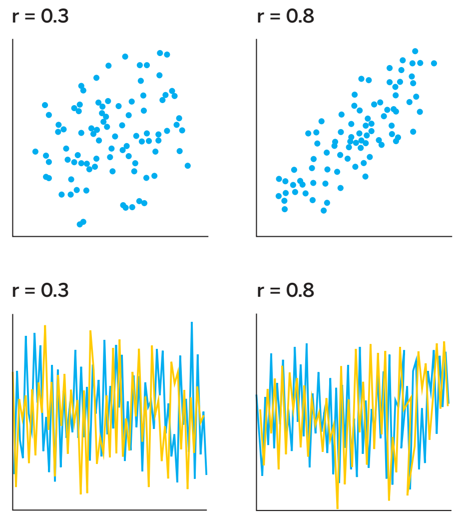


SOURCE: LANE HARRISON

More researchers have applied this to other types of charts as well. The implications are profound, for two reasons. First, if Weber's law applies to an instance of higher-order thinking, not just fundamental physical stimuli like light, then maybe we're not reading data at all, but rather doing something more fundamental in our brains with shape, angle, and space, which we then "calculate" visually to find correlation.

Second, although the relationship between perception and correlation is linear for all types of charts, the linear *rate* varies between chart types. As shown in the Perceiving Change chart, people see a difference between 0.3 and 0.8 correlations much more easily in a scatter plot than they do in, say, a line chart.[15]

PERCEIVING CHANGE

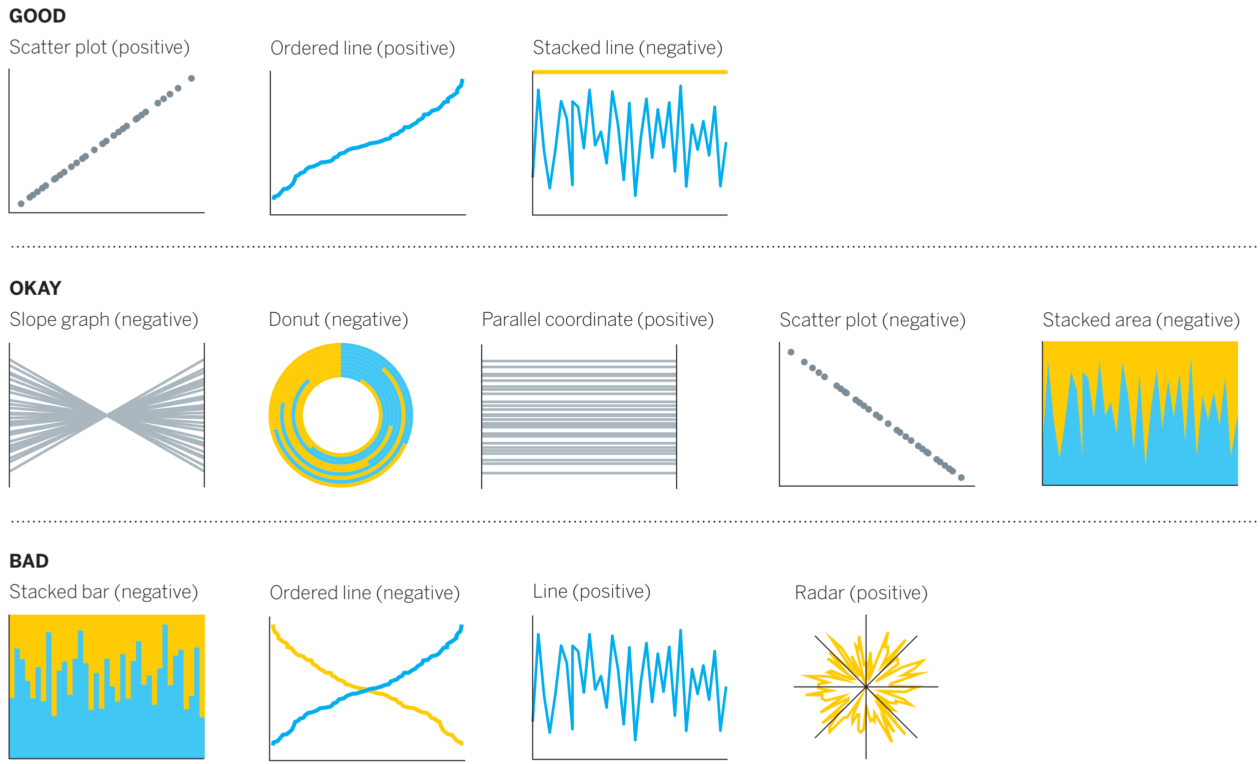


SOURCE: LANE HARRISON

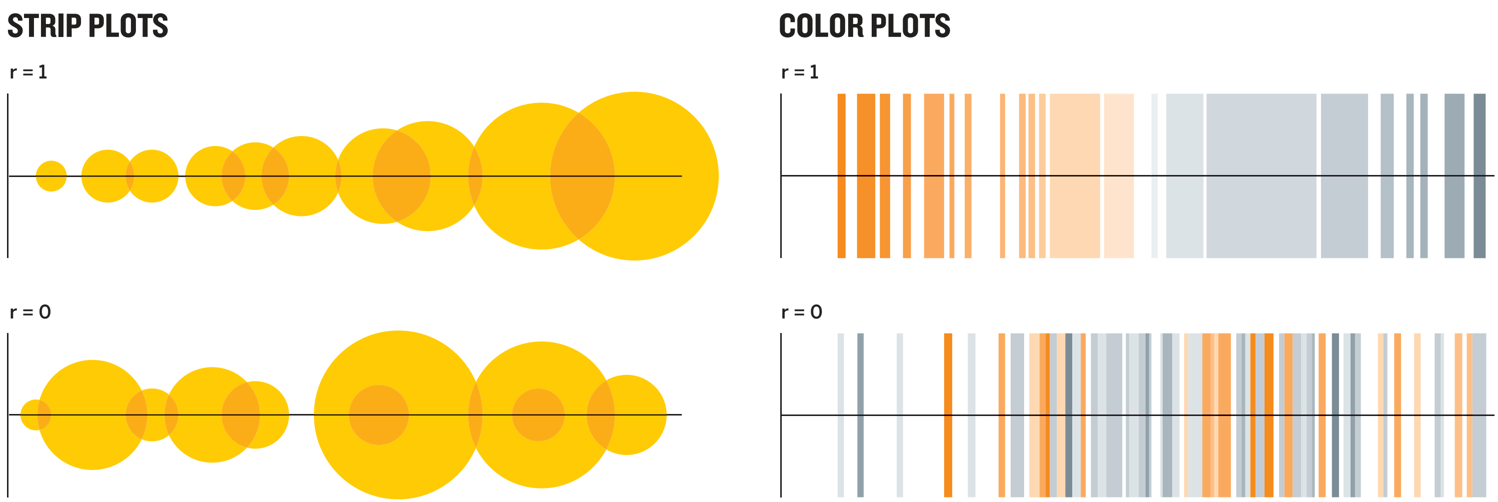
That means we can begin to measure and rank order the effectiveness of various chart types for showing correlation—which researchers have begun to do. Some of the results of this effort are found in the Ranking Methods matrix on the facing page.[16]

It also opens the possibility of more broadly being able to define what charts work best for what tasks. It could be that we're underutilizing certain chart types. For example, researchers surprisingly find that we see correlation changes in strip plots and color plots, shown on page 54, as well as or better than we do in scatter plots. This is a surprising result in a field where many believe in *space über alles*—that spatial relationships are the best way to plot data. Perhaps, as Cleveland and McGill said in their research cited in the last chapter, new procedures are being discovered, and old ones will be set aside.

RANKING METHODS TO SHOW CORRELATION



SOURCE: LANE HARRISON, MATTHEW KAY, AND JEFFREY HEER



SOURCE: RONALD RENSINK

7. We feel numbers. You may have noticed how these principles have moved from accessible ideas to esoteric ones. We started with the nearly self-evident notion that we see what stands out and moved all the way to an idea about how our eyes can do math with visuals.

Here, at the end, is the most abstruse idea of all: People *feel* data. They don't just process statistics and come to rational conclusions. They form emotions about the data visualization. We are not informed by charts; we're affected by them.

Researcher Helen Kennedy continues to do vanguard work on what she calls *feeling numbers*.[17] She has found that what sticks with people about a chart is not just what it *reports*, but how it made the user feel. "They showed emotional reactions . . . to the data itself," she recalled in a podcast. "They'd realize knife crime went up in their area and they'd feel scared."[18]

What's more, when Kennedy followed up with participants a month later, most could not recall specific data from the charts they looked at. "But they could remember the feeling they had," she reports. "You know they'd say, 'I remember feeling surprised that that number was higher than I expected.'"

Data visualizations are emotional experiences, and Kennedy believes it's important to accept that and to change how we teach statistics and data visualization to accommodate the fact. This, she says, will make our "datafied times" more inclusive. "Privileging the rational over the emotional . . . also means privileging some groups over others. This is because certain groups (often white, middle-class men) are better equipped to understand mathematical and statistical information, not because they are more naturally capable of doing so, but because they are significantly better represented in maths, science, and computing subjects at school and beyond . . . Changing how we do statistical education might mean including previously excluded groups in understandings of data and in engagements in data-driven conversations and decision-making."[19]

For our purposes, it's important to remember that not only are you bound to make people feel something, but that feeling will be even more enduring than the data itself.

CREDIBILITY AT STAKE

Understanding what people see, and what their minds do when they set eyes on a chart, is the best way to guide you in deciding what to show and how to show it. The applicability of these tenets may not yet be perfectly clear, but you will find yourself returning to these ideas repeatedly in the coming chapters. As you hone your chart-making skills, you'll see how understanding what happens to your audience when a chart hits their eyes guides you to certain solutions.

The stakes here may be higher than you suspect. Perceptual fluency research suggests that we make qualitative judgments about information based on its presentation.[20] If something is hard to perceive, people not only struggle to find the right meaning, but judge it less favorably.

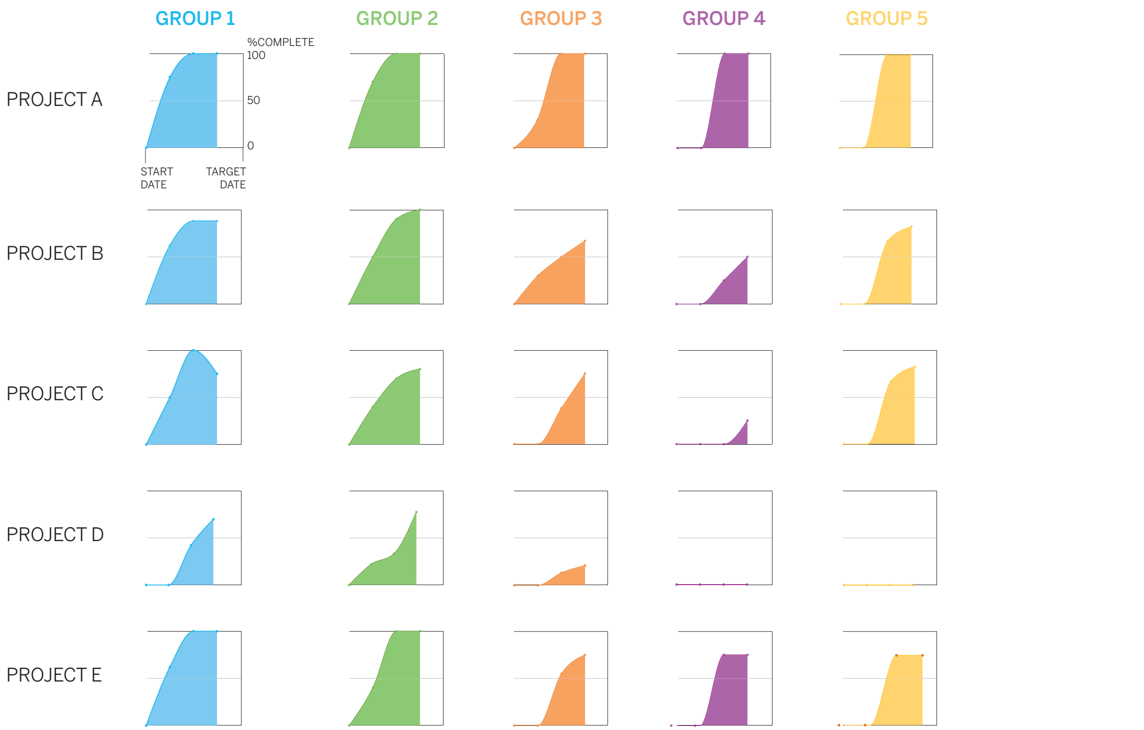
There's a fine point here that mustn't be lost: It's not the *chart* that they'll judge harshly if the meaning is hard to find; it's the *information itself*. They'll consider it less credible. The emotions they have about it (and will remember) will be more negative.

With that in mind, jump back for a moment to page 42 and take another look at the chart there that was part of a transformation project I worked on. The chart on the next page is *the same data* represented on that previous chart, only now reimagined as a series of small multiples that can be navigated either horizontally or vertically.

How we made this transformation will be the subject of the coming chapters. For now, just think about what happens when these two charts hit your eyes. What do you feel? If you were presenting these charts, which do you think will earn you more credibility? If you were at a meeting where others were presenting these charts, which speaker is earning your trust? Which chart is more usable? Which will you remember?

This is the kind of transformation you're after.

PROGRESS REPORT



That's enough theory to make you an amateur composer, and to know that if your charts don't make what's important stand out, if complex data doesn't coalesce into a few clear ideas, if the information visualized fosters a false narrative, if unconventional visual techniques confuse your viewers, then you've promised music but delivered noise.

RECAP

WHEN A CHART HITS OUR EYES

Unlike text, visual communication is governed less by an agreed-upon convention between “writer” and “reader” than by how our visual systems react to stimuli, often before we’re aware of it. And just as composers use music theory to create music that produces certain predictable effects on an audience, chart makers can use visual perception theory to make more-effective visualizations with similarly predictable effects.

Understanding these is crucial, because if users of your dataviz find it hard to understand, they will judge your data less credibly. Seven high-level, mostly agreed-upon principles can guide you:

1. We don’t go in order.
Visuals aren’t read in a predictable, linear way, as text is. Instead, we look first at the visual and then scan the chart for contextual clues about what is important.

What this means:
Whereas we write sequentially (in the West, left to right and top to bottom), we should “write” charts spatially, from the visual outward, using other elements to provide clues to the visual’s meaning.

2. We see first what stands out.
Our eyes go directly to change and difference, such as unique colors, steep curves, clusters, or outliers.

What this means:
Whatever stands out should match or support the idea being conveyed. If it doesn’t, it will distract from and fight for attention with the main idea.

3. We see only a few things at once.
The more data that’s plotted in a chart, the more singular the idea it conveys. If a visual contains dozens, hundreds, or thousands of plotted data points, people will see a forest instead of individual trees.

What this means:
If we need to focus on individual data points, we should plot as few as possible so that the visuals don’t disappear into an aggregate view.

4. We seek meaning and make connections.
Our minds incessantly try to assign meaning to a visual and make causal connections between the elements presented, regardless of whether any real connections exist.

What this means:
If visual elements are presented together, they should be related in a meaningful way; otherwise,

viewers will construct false narratives about the relationships between them.

5. We rely on conventions and metaphors.
We use learned shortcuts to assign meaning to visual cues. For example, green is good and red is bad; north is up and south is down; time moves left to right in two dimensions or back to forward in three dimensions.

What this means:
Embrace deeply ingrained conventions and metaphors when creating visuals. Flouting them creates confusion, uncertainty, and frustration, which will weaken or eliminate a chart's effectiveness.

6. We sense statistical values in visuals.
Our eyes are quite good at estimating changes to statistical information like correlation within a visual, and the ability for us to do this is predictable and measurable.

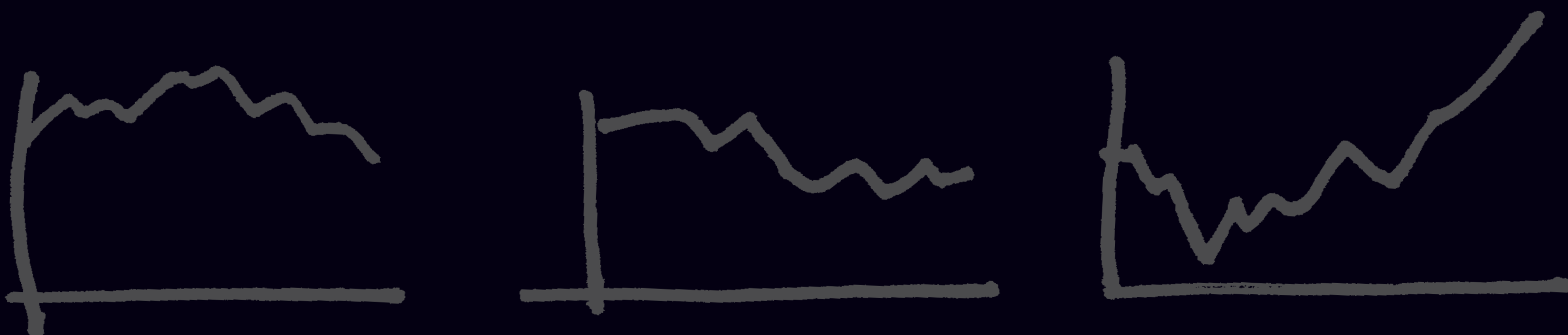
What this means:
Specific data points matter less in a visual than the overall sense of the data, and people are better at estimating values in visuals than we think. We can trust visuals to convey the data they need to.

7. We feel numbers.
People experience emotional reactions to charts, and those emotional reactions endure beyond the actual statistical information conveyed.

What this means:
When creating a chart, you're not just informing minds but affecting hearts. As you're making your chart, think about the emotions you want the audience to feel.

CREATE

PART TWO



CHAPTER 3

TWO QUESTIONS → FOUR TYPES

A SIMPLE TYPOLOGY FOR CHART MAKING

IF A FRIEND SAID TO YOU, "Pack your bags, we're going on a trip," what would you do next? Here's what you wouldn't do: You wouldn't say "Okay, great," grab a suitcase, and start filling it with clothes. How could you? You have so many questions: *Where are we going? For how long? How are we getting there? Why are we taking this trip? Where will we stay when we get there?* You can't pack until you know what you're packing for.

But when it comes to dataviz, many of us follow our impulse to unthinkingly choose a chart type and click a button to create it. In some ways software has made it so easy to create any chart that we opt for that convenience over a more deliberate approach.

I'm advocating for a more deliberate approach that resists the impulse to "click-and-viz." It starts with understanding what data visualization is—it's actually not just one thing but a collection of activities—and the resources, skills, and mindset you'll need to make good charts. This simple framework provides you with a foundation for building your charting skills. You're going on a trip; this will make packing easier later on.

THE TWO QUESTIONS

To start thinking visually, consider two questions about the nature and purpose of your visualization:

1. Is the information *conceptual* or *data-driven*?
2. Am I *declaring something* or *exploring something*?

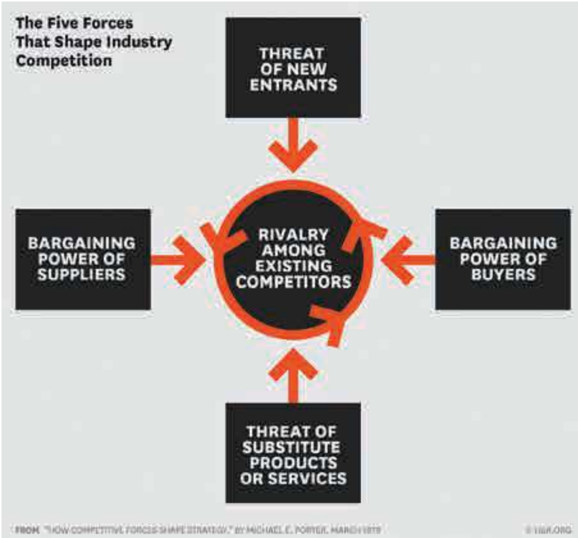
If you know, generally, the answers to these two questions, you can plan what resources and tools you'll need and begin to define the type of visualization you may finally settle on using.

CONCEPTUAL OR DATA-DRIVEN?

	CONCEPTUAL	DATA-DRIVEN
Focus	Ideas	Statistics
Goals	Simplify, teach "Here's how our organization is structured."	Inform, enlighten "Here are our revenues for the past two years."

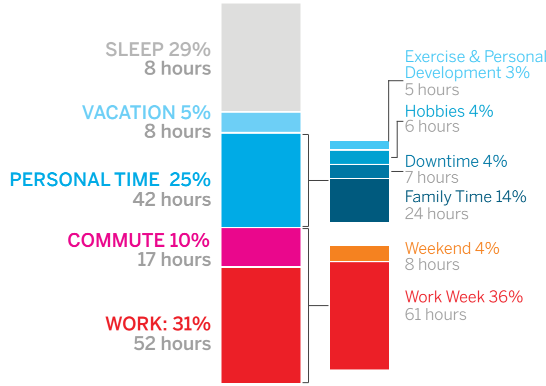
The first is the simpler of the two questions, and usually the answer is obvious. Either you're visualizing concepts and qualitative information or you're plotting data and information. But notice that the question is about the *information itself*, not the forms that might ultimately be used to show it.

The first two examples on the facing page are clearly identifiable as a conceptual chart about the five forces of strategy and a data chart about time management. The chart with the axes and a line is actually conceptual, despite using a statistical form. Conversely, the map is just a map with some logos on it. And yet, it represents data. Sometimes a data-driven chart will take on a conceptual form, and



MEASURING 13 WEEKS OF A CEO's TIME

We had 27 CEOs track their time, 24/7, for 13 weeks. That's 2,184 total hours. Here's how the average CEO spent their time.



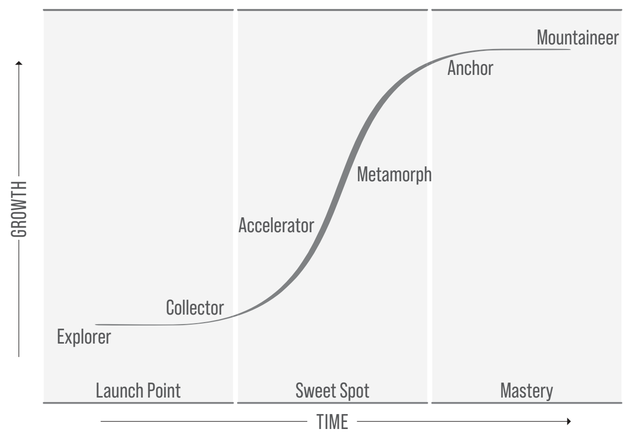
SOURCE: MICHAEL E. PORTER AND NITIN NOHRIA, HARVARD BUSINESS REVIEW, JULY-AUGUST 2018

MOST SEARCHED CAR BRANDS BY COUNTRY

Darker color represents higher average monthly search volume

SOURCE: TOPSPEED.COM ANALYSIS OF GOOGLE SEARCH DATA

The Stages of Growth



SOURCE: FROM *SMART GROWTH*, BY WHITNEY L. JOHNSON, 2022, © 2022, DISRUPTION ADVISORS, LLC.

vice versa. Make sure that when you answer this question, you're thinking about the information, not the form.

DECLARATIVE OR EXPLORATORY?

	DECLARATIVE	EXPLORATORY
Focus	Documenting, designing	Prototyping, iterating, interacting, automating
Goals	Affirm: "Here are our revenues over the past five years."	Corroborate: "Let's see if marketing investments contributed to rising profits."
		Discover: "What would we see if we visualized customer purchases by gender, location, and purchase amount in real time?"

If the first question identifies what you *have*, the second question elicits what you're *doing*. It's a more complicated one to answer, because it's not a binary proposition. Within this question, you could choose three broad categories of purpose—declarative, confirmatory, and exploratory—the second two of which are related.

Most often we work with declarative visualizations. These make a statement to an audience—usually in a formal setting—a presentation, a report, or a tweet. They tend to be well-designed, finished products. That doesn't mean they're unassailable. Declarative viz shouldn't preclude conversation about the idea presented; a good one will generate discussion. Still, what you think of dataviz usually falls into this category. If you have a spreadsheet workbook full of sales data and you're using that data to show quarterly sales or sales by region in a presentation—your purpose is *declarative*.

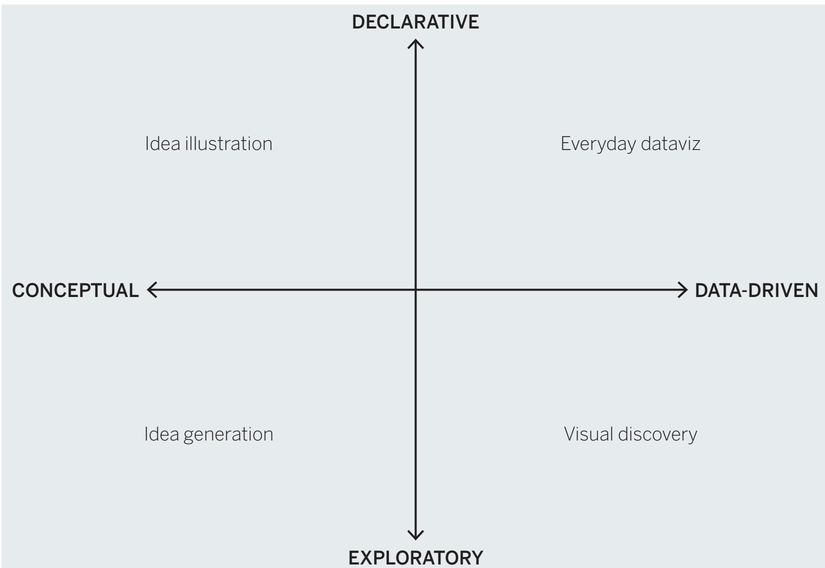
But let's say your boss wants to understand why the sales team's performance has been lagging lately. You suspect that seasonal cycles have caused the dip, but you're not sure. Now your purpose is *confirmatory*, and you'll dip into the same data to create visuals to learn whether or not your hypothesis holds. Charts used to confirm are less formal, and designed well enough to be interpreted, but they don't always have to be presentation worthy. The audience is yourself or a small team, not others. If your hypothesis is confirmed, it may well lead to a declarative visualization you present to the boss, saying, "Here's what's happening to sales." If it turns out that seasonality isn't the culprit, you may form another hypothesis and do another round of confirmatory work.

Or maybe you don't know what you're looking for in the sales data. Instead, you want to mine this workbook to see what patterns, trends, and anomalies emerge. What will you see, for example, when you measure sales performance in relation to the size of the region a salesperson must manage? What happens if you compare seasonal trends in the Northern and Southern hemispheres? How does weather affect sales? This is *exploratory* work—rougher still in design, usually iterative, sometimes interactive. Most of us don't do as much exploratory work as we do declarative and confirmatory; we should do more. It's a kind of data brainstorming that can expose hidden insights waiting to be found. Big strategic questions—*Why are revenues falling? Where can we find efficiencies? How do customers interact with us?*—can benefit from exploratory viz. Once, a colleague wondered aloud about how often the high temperature was in the 70s in Boston. That led to several hours of exploratory visualization in which some insights about the weather in the city were uncovered.

Other ways to ask the purpose question that may help you organize your thinking: "Do I need to give the answers, to check my answers, or to look for answers?" Or "Am I presenting ideas, researching ideas, or seeking ideas?"

Answering these questions can help you anticipate the kind of work ahead of you. For example, as you move from the declarative toward exploratory, certainty about what you know tends to decrease, and the complexity of your information tends to increase. Also, when your purpose is declarative, you're more likely to be able to work alone and quickly. As you move along the spectrum, you're increasingly likely to work in a team, lean on experts, and invest much more time in the process.

THE FOUR TYPES

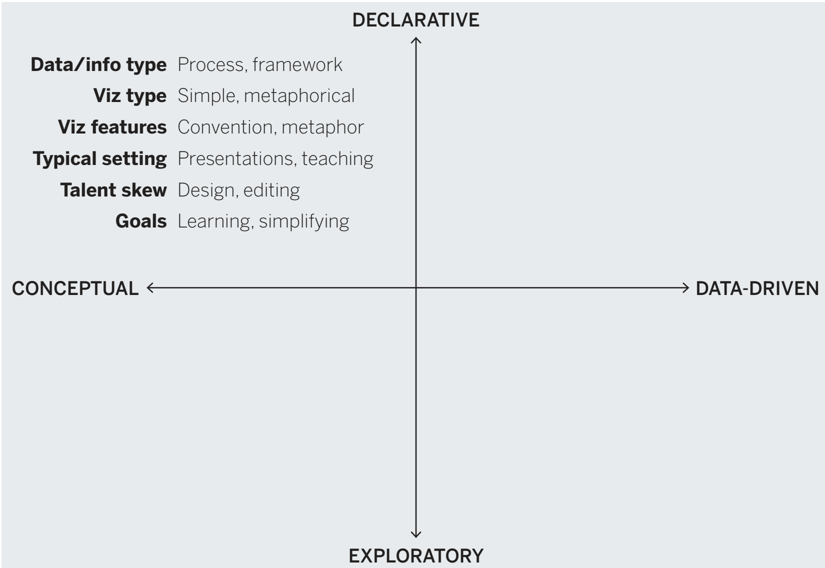


THE FOUR TYPES

The nature and purpose questions combine in a classic 2 × 2 matrix to create four potential types of visualizations that you'll use.

Knowing which quadrant you're working in will help you make good decisions about the forms you'll use, the time you'll need, and the skills you'll call on. Let's start at the top left of this 2 × 2 and proceed counterclockwise.

IDEA ILLUSTRATION: CONCEPTUAL, DECLARATIVE VISUALIZATIONS

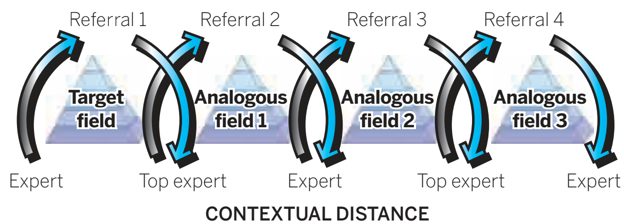


Idea illustration. We might call this the "consultants' corner," given that consultants can't resist process diagrams, cycle diagrams, and other idea illustrations—sometimes to deleterious effect. (Gardiner Morse, a former editor at HBR, has coined a term for these sorts of overwrought diagrams: "crap circles."[1]) But at their best, declarative, conceptual visualizations simplify complex ideas by drawing on people's ability to understand metaphors (trees, bridges) and simple conventions (circles, hierarchies). Org charts, decision trees, and cycle diagrams are classic examples of idea illustration. So is the 2 × 2 that frames this chapter.

Idea illustrations demand clear and simple design, but they often lack it. They don't face the constraints imposed by axes and accurately plotted data. Their reliance on metaphors invites unnecessary adornment aimed at reinforcing the metaphor. If your idea is "funneling customers," for example, the impulse may be to show a literal funnel, but literalness can lead to unfortunate design decisions. Because the discipline and boundaries of data aren't built into idea illustration, they must be self-imposed. Focus on clear communication, structure, and the logic of the ideas. The skills required here are similar to what a text editor brings to a manuscript, channeling the creative impulse into the clearest, simplest thing.

Say a company hires two consultants to help its R&D group find inspiration in other industries. They will use a technique called the *pyramid search*.[2] But how does a pyramid search work? The consultants have to sell it to the company's R&D leaders. They present something like this:

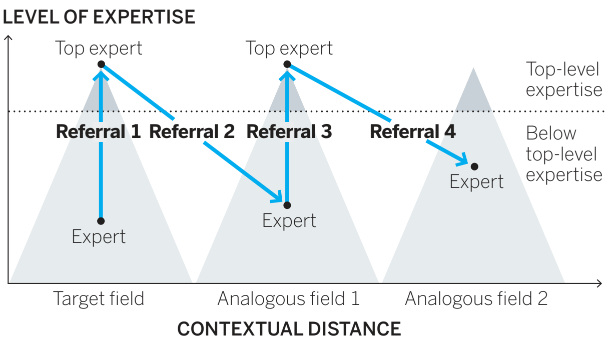
HOW A PYRAMID SEARCH WORKS



This idea illustration suffers from overdesign. The color gradient, arrows with drop shadows, and the sectioned, 3-D pyramids dominate, drawing our eyes away from the idea and toward the decoration. Stylization like this is a red flag. Additionally, the consultants haven't effectively channeled the metaphor. They're selling a pyramid search, but they present interlocking cycles; the pyramids are simply imagery doing little work. This is confusing. They have also put experts and top experts on the same plane (and the top experts are at the *bottom* of the diagram—another missed metaphor) instead of using height to convey relative status.

They'd be better off presenting something like this:

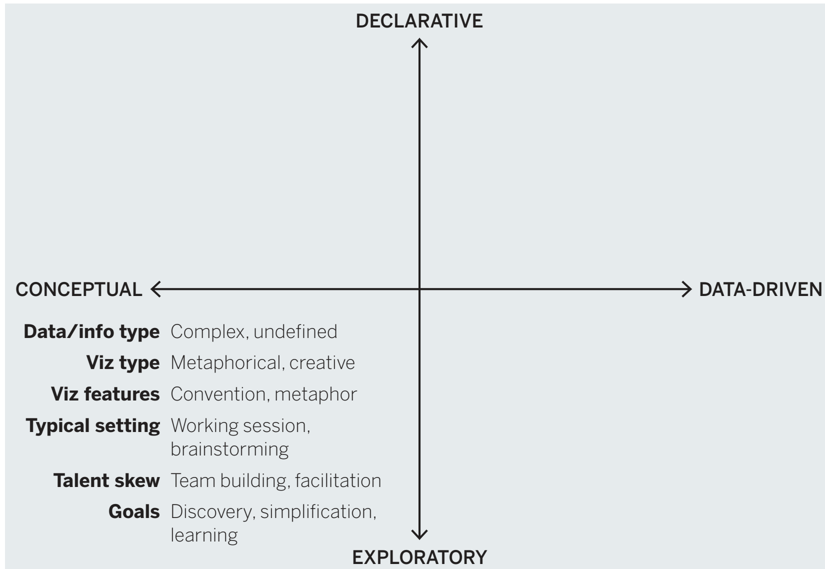
CLIMBING PYRAMIDS IN SEARCH OF IDEAS



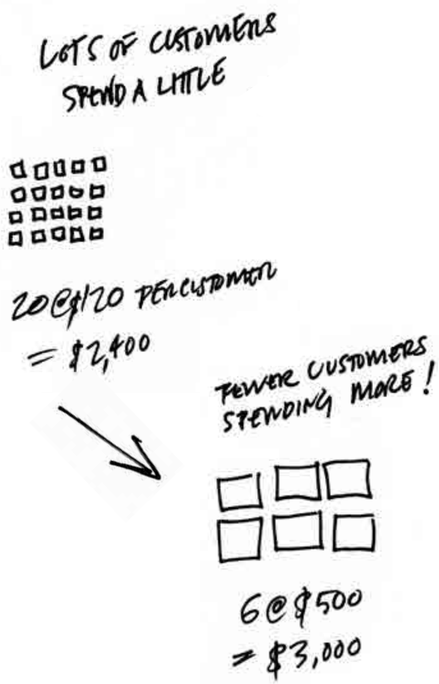
SOURCE: MARION POETZ AND REINHARD PRÜGL, *JOURNAL OF PRODUCT INNOVATION MANAGEMENT*

Here the pyramid metaphor fits the visual representation. What's more, the axes use conventions that viewers can grasp immediately—near-to-far industries on the x-axis and low-to-high expertise on the y-axis. The pyramid shape itself serves a useful purpose, showing the relative rarity of top experts compared with lower-level ones. The title words help, too—*climbing* and *pyramids* both help us grasp the idea quickly. Finally, they don't succumb to the temptation to decorate. The pyramids, for example, aren't three-dimensional or sandstone-colored or placed against a photo of the desert.

IDEA GENERATION: CONCEPTUAL, EXPLORATORY VISUALIZATIONS



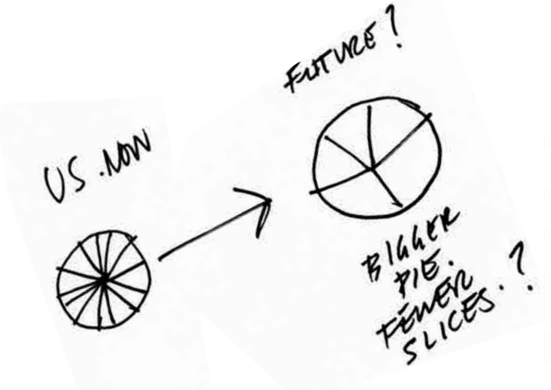
Idea generation. For many people, this quadrant is the least intuitive. When would you ever produce nondata visuals to explore ideas? The very notion of clarifying complex concepts seems to run counter to exploration, in which ideas aren't yet

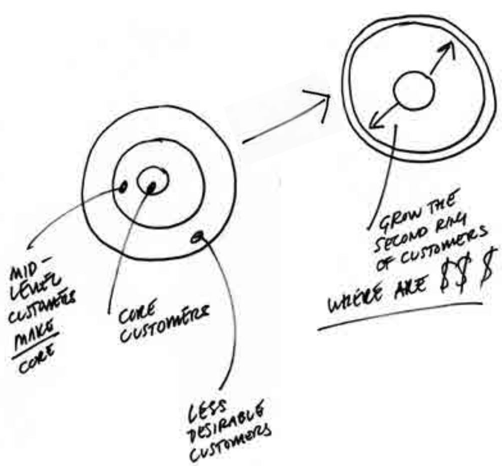


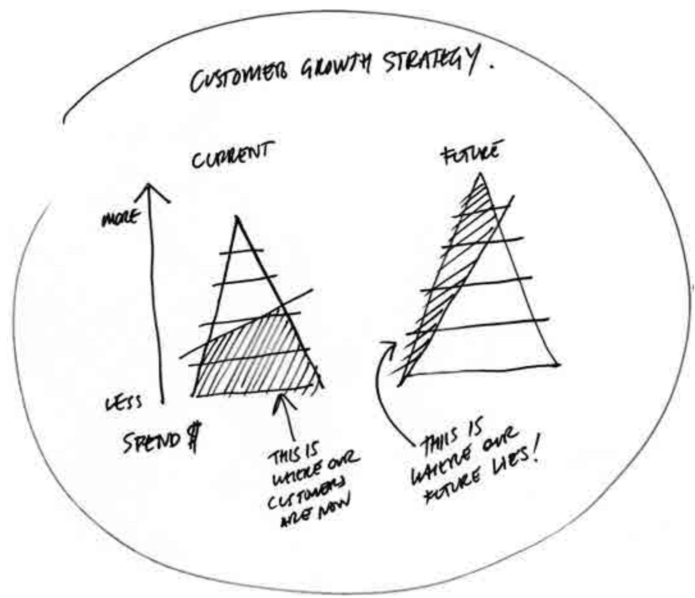
well defined. It differs in setting and media from the other three visualization types, and managers may not think of it as visualization, but they use it often. It happens at a whiteboard or, classically in entrepreneurial circles, on the back of a napkin.

Like idea illustration, it relies on conceptual metaphors and conventions, but it takes place in more-informal settings, such as off-sites, strategy sessions, and early-phase innovation projects. It's used to find answers to nondata challenges: restructuring an organization, coming up with a new business process, codifying a system for making decisions.

Idea exploration can be done alone, but it benefits from collaboration and borrows on design thinking processes: gathering as many diverse points of view and visual approaches as possible before homing in on one and refining it. Jon Kolko, the founder and director of Austin Center for Design and the author of *Well-Designed: How to Use Empathy to Create Products People Love,* fills his office with conceptual, exploratory visualizations strewn across whiteboard walls. "It's our go-to method for thinking through complexity," he says. "Sketching is this effort to work through ambiguity and muddiness and come to crispness." Managers who are good at leading teams, facilitating







brainstorming sessions, and capturing creative thinking will do well in this quadrant.

Suppose a marketing team is holding an off-site. The team members need to come up with a way to show executives their proposed strategy for going upmarket. An hour-long whiteboard session yields several approaches and ideas (none of which are erased) for showing their transition strategy. Ultimately, one approach gains purchase with the team, which thinks it best captures the key points of its strategy: get fewer customers to spend much more.

The facing page shows rough sketches of a whiteboard at the end of the idea generation session. Of course, visuals that emerge from idea exploration will often become more formally designed for presentation.

Visual discovery. This is the most complicated category, because in truth it's actually two categories. Remember that the purpose question led to three possible types of tasks: declarative, confirmatory, and exploratory. I left confirmatory out of the 2×2 to keep the basic framework simple and clear. Now, while we focus on this quadrant, I will add in that information, as seen on the adapted 2×2 on the next page.

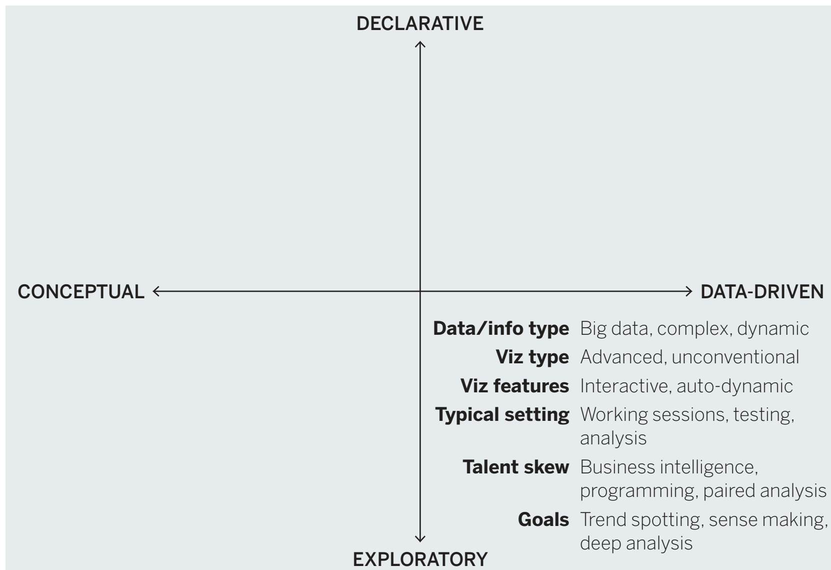
Note that confirmatory applies only to data-driven charts. A hypothesis can't be confirmed or disproved without data. Also, the division is shown as a dotted line because it's a soft distinction.

Confirmation is a kind of focused exploration, whereas true exploration is more open-ended. The bigger and more complex the data, and the less you know going in, the more exploratory the work. If confirmation is hiking a new trail, exploration is blazing one.

Visual confirmation. You're answering one of two questions with this kind of project:

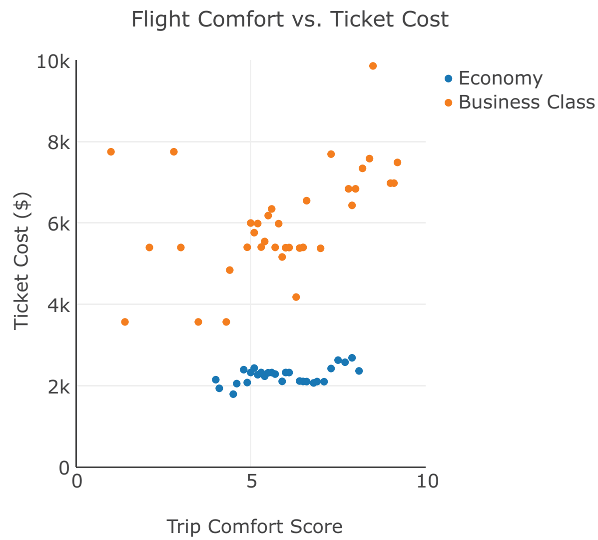
1. Is what I suspect is true actually true?
2. What are some other ways of looking at this idea?

VISUAL DISCOVERY: DATA-DRIVEN, EXPLORATORY VISUALIZATIONS



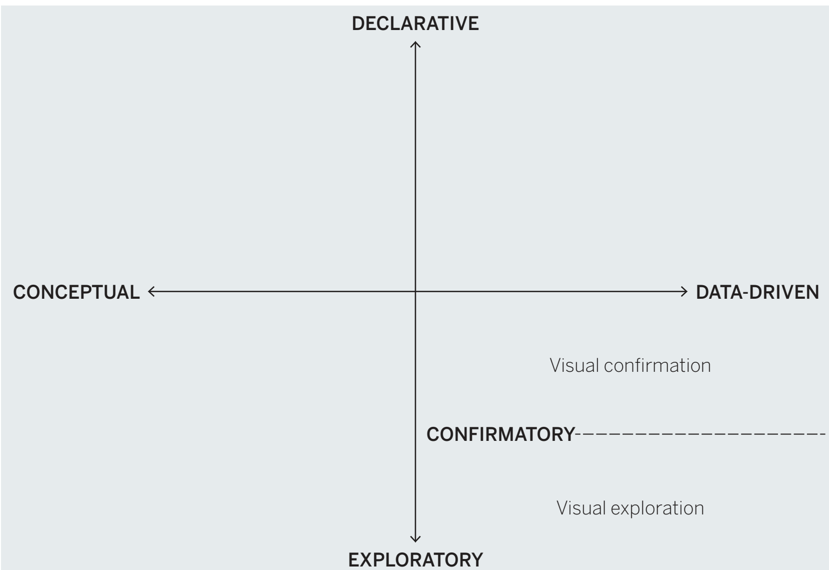
In hypothesis setting and confirmation, the scope of the data is manageable, and the chart types you're likely to use are more common ones—although when you're trying to see things in new ways, you may venture into some less common types. Confirmation usually doesn't happen in a formal setting; it's the work you do to find the charts you want to create for presentations. That means your time will shift away from design and toward prototyping that allows you to iterate on the data and rapidly visualize and revisualize.

Suppose a manager in charge of travel services wants to research whether the plane tickets the company buys are worth the investment. She goes into her visual confirmation project hypothesizing that comfort increases with ticket cost. She pulls data on cost versus comfort for both economy and business class flights and quickly generates a scatter plot. She's expecting to see correlation—dots splayed up and to the right.



Notice that the chart she creates, above, is a prototype. The manager hasn't spent much time refining the design or refining the axes and titles. It's more important for her to see if her idea is right than to make it look great. Immediately she sees that the relationship between cost and her other variables is relatively weak. There is an upward trend in comfort on business class, but it's not strong. She's startled to find her hypothesis doesn't hold. Higher cost of flights may not be worth it. So, she thinks about what other ideas to test before making any decisions.

VISUAL CONFIRMATION AND VISUAL EXPLORATION



Visual exploration. Exploratory, data-driven visualizations tend to be the province of data scientists and business intelligence analysts, although new tools have begun to engage more of us in visual exploration. It's an exciting kind of visualization to try because it often produces insights that can't be gleaned any other way.

Since we don't know what we're looking for, these visuals tend to be more inclusive in the data they plot. In extreme cases, this kind of project may combine multiple data sets or even dynamic, real-time data that is continuously updating. It may even venture beyond real-world data. David Sparks, a political scientist and statistical analyst who now works for the NBA's Boston Celtics does visual exploration. But he refers to his work as "model visualization." In Sparks's world, *data visualization* focuses on real, existing data. *Model visualization* passes data through statistical models to see what *would* happen under certain circumstances.

Exploration lends itself to interactivity—allowing a manager to adjust parameters, inject data sources, and continually revisualize. Complex data sometimes also lends itself to specialized and unusual visualization types, such as force-directed network diagrams that show how networks cluster, or topographical plots that give a third dimension to data.

Function far outweighs form here: Software, programming, data management, and business intelligence skills are more crucial than the ability to create presentable charts. This quadrant is where a manager is most likely to call in experts and advanced tools to help create the visualizations.

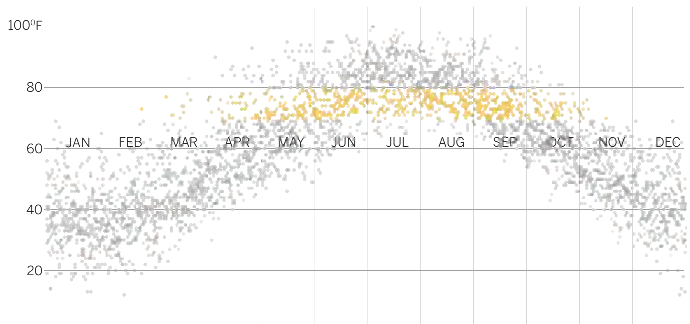
A manager at a social media company has been asked to look for new markets for its technology. He wants to find opportunities that others won't see. He connects with a data scientist who tells him how semantic analysis can be used to map thousands of companies in multiple industries according to the similarity of their text communications.

The manager loves the idea but can't do it himself. He hires the data scientist, who develops and adjusts the data set with the manager until they can generate a rough visual that links companies that are similar according to semantic analysis; the more similar the companies, the "stronger" the link and the more closely they're mapped. This results

BEAUTIFUL DAYS

DAYS WITH HIGH TEMPERATURES IN THE 70s BOSTON, 2007–2017

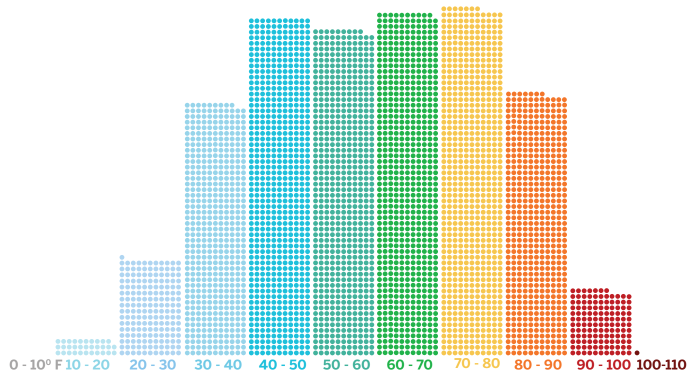
TOTAL DAYS CHARTED: 4,018



DATA: WEATHER UNDERGROUND

DISTRIBUTION OF HIGH TEMPERATURES BOSTON, 2007–2017

TOTAL DAYS CHARTED: 4,018



DATA: WEATHER UNDERGROUND

"What a perfect day!" my colleague said arriving at the office. It was 73 degrees Fahrenheit and clear. "There aren't enough days like this in Boston."

Was that true? How common was a high temperature in the 70s in our city? And what was the most common high temperature? I grabbed 11 years of high temperatures and got to work.

It turned out to be a classic visual discovery project. First, we did some visual confirmation. We set our hypothesis ("High temps in the 70s are uncommon here") and then plotted to see if that's what we found. A scatter plot of all those 4,000 days showed a simple arc with lower temps early in the year steadily rising to a July peak and then steadily falling through December. Made sense, but we couldn't see much about days in the 70s because in the initial plot, dots were color coded to the year of their observation—not useful to us.

So, after some color manipulation, we were able to highlight only those days in the 70s. A narrow band emerged that looked like quite a few days, but certainly "that doesn't look like a lot of days" my colleague said.

So, we switched forms to a histogram that showed the distribution of high temperatures in 10-degree bands. Surprise: The 70s is the most common high temperature range in Boston, by just a little. And the second most common range was the 60s, which I argued was an equally pleasant temperature range. My colleague agreed. Our hypothesis was wrong, and more often than not you will experience a comfortable high temperature in Boston.

The charts you see here are well-designed final products that came out of a more rapid confirmation process over a few hours in which the designs were similar, but not nearly so neat or considered. When doing visual discovery, the working charts don't need to be presentation worthy.

We also continued into more open-ended visual exploration to see what else we could discover. For example, we banded the temperatures by month to discover that there's far more variability in high temperatures in the winter than the summer. We whipped up a line chart in which each line represented daily highs in a year to see which years showed a wider range of temperatures and which were relatively warmer or colder. We added a sunny day/cloudy day variable to see if different seasons or temperature ranges correlated with how clear or overcast the days are.

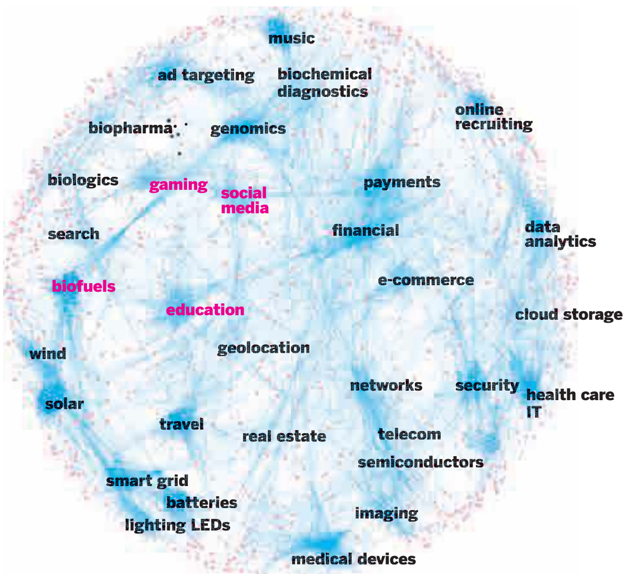
None of these charts was presentation worthy, but we were in discovery mode; they didn't need to be.

Notice in this process how we guessed at a chart type and quickly discovered it wasn't helping us confirm our hypothesis, so we tried another one. And then in discovery, we kept changing chart types as we looked for new insights. In other words, the questions we asked dictated the forms we used, not the other way around.

Once you start practicing visual discovery, you'll find it naturally leads you, as it did us, from confirmation to exploration. We had the data and found something interesting and that led to an almost irresistible impulse to wonder: What else can we find?

in this network diagram, which exposes easy-to-see industry clusters. The white space between proximate clusters represents opportunities to connect one industry to another, because although the data shows that those clusters are similar, no companies have yet emerged to fill the gap.[3]

INDUSTRY CLUSTER SEMANTIC ANALYSIS

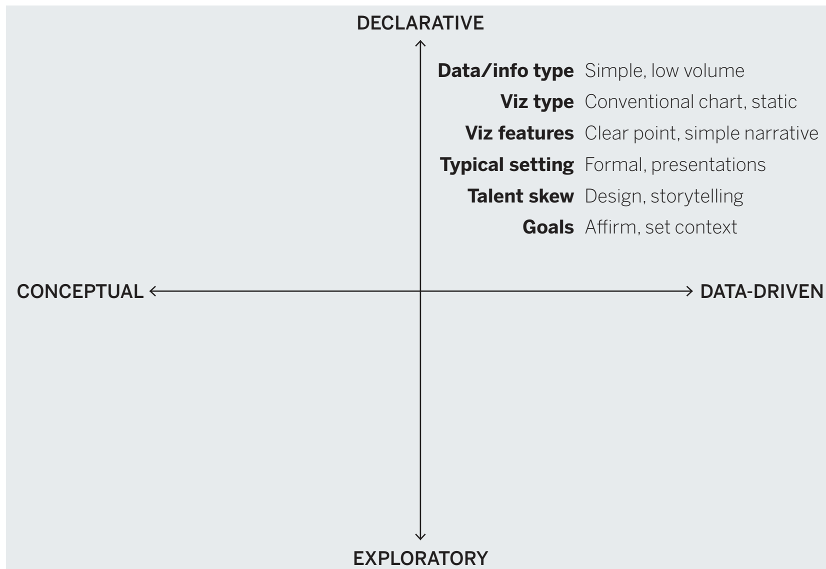


The manager is not surprised when he immediately notices that social media and gaming don't have much white space between them; he's played games on Facebook. But he does see white space between social media and other industries, such as education and biofuels—potential new markets for his technology.

Everyday dataviz. These are the basic charts and graphs you normally spit out of an Excel spreadsheet and paste into a PowerPoint. They are most often simple forms, such as line charts, bar charts, pies, and scatter plots.

The key word here is *simple*. The data sets tend to be smaller and simpler. The visualization communicates a simple idea or message, charting no more than a few variables. And the goal is simple: Give people information based on data that is, for the most part, not up for debate.

EVERYDAY DATAVIZ: DATA-DRIVEN, DECLARATIVE VISUALIZATIONS

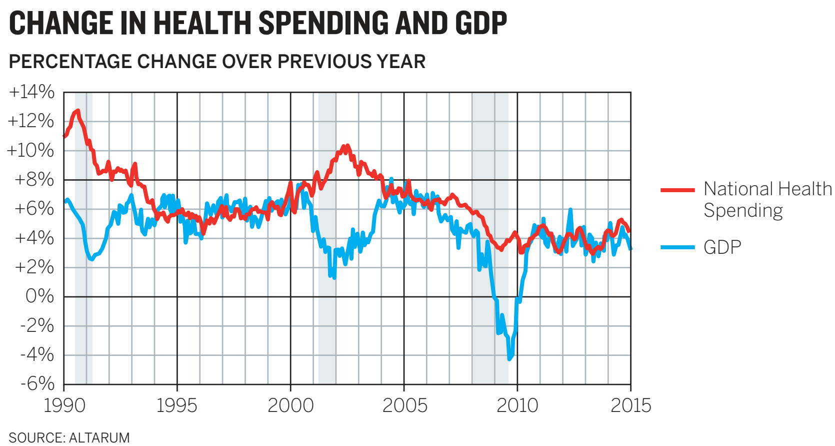


Simplicity is primarily a design challenge achieved through clarity (more on this later). Clarity and consistency make everyday dataviz most effective in the setting where they're typically used—a formal presentation or a document. Increasingly, this kind of data visualization is seen on social media or in the news, advertisements, sports broadcasts, and innumerable other settings. My car displays a real-time fuel-economy visualization!

In all these settings, time is constrained. People want to get to meaning quickly, and a poorly designed chart will waste that limited time by creating questions that either confuse the audience for the visualization or require the presenter to explain the structure of the visual or the information that's meant to come to the fore. Ideally, you should be able to present an everyday dataviz without any explanation at all. If it can't speak for itself, it has failed like a joke that needs explaining.

That's not to say that declarative charts shouldn't generate discussion. They should. But the discussion should be about the *ideas in the chart*, not the chart itself.

An HR vice president will be presenting to the rest of the executive committee about the company's health-care costs. A key message she wants to convey is that the growth of these costs has slowed significantly, giving the company an opportunity to think about what additional services it might offer. She's read an online report about the slowing growth that includes a link to some government data. So she downloads the data and then clicks on the line chart option in Excel. She has her viz in a few seconds. But since this is for a presentation, she asks a designer colleague to add even more detail from the data set about GDP and recessions, to give a more comprehensive view of the data.



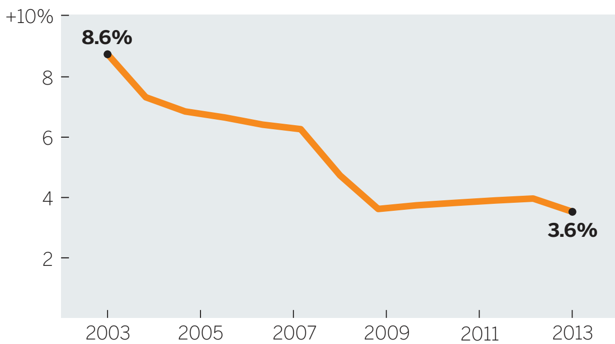
This is a well-designed, accurate chart, but it's probably not the right one for this situation. The HR exec's audience doesn't need two decades' worth of historical context to discuss the company's strategy for employee benefits investments. The only point she needs to make is that cost increases have slowed over the past few years. Does that jump out here?

In general, charts that contain enough data to take minutes, not seconds, to digest will work better on paper or a personal screen, for an individual who's not being asked to listen to a presentation while trying to take in so much information. Health-care policy makers, for example, might benefit from seeing this chart in advance of a policy hearing in which they'll discuss these long trends.

But our exec needs something simpler for her context. From the same data set, she creates the Annual Growth chart below that gets to her point simply, clearly, and quickly.

ANNUAL GROWTH IS DECLINING

ANNUAL GROWTH IN HEALTH CARE SPENDING



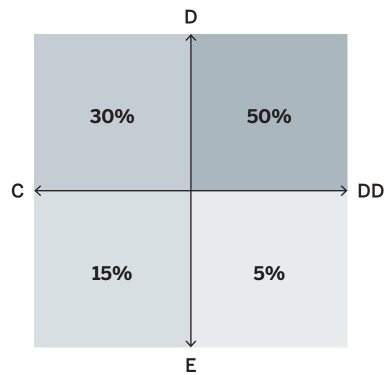
SOURCE: CENTERS FOR MEDICARE & MEDICAID SERVICES

She won't have to utter a word for the executive team to understand the trend. Clearly and without distractions, she has set the foundation for presenting her recommendations.

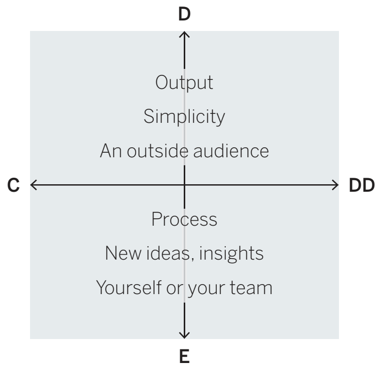
USE THE "FOUR TYPES" 2 × 2 MATRIX

The "four types" 2 × 2 is a useful framework. Just as you can layer many types of information over a basic road map—gas stations, traffic, weather—you can layer any number of ideas or pieces of information over the map of visualization types, to help you understand and plan the time, resources, and skills you'll need. Here are five examples:

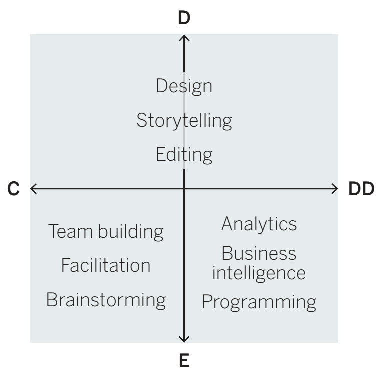
Usage frequency. Your numbers may vary. I've put in my own starting point. Most of us will spend the majority of our charting time with everyday dataviz. However, new software and online tools are making discovery and exploration much easier. I expect that number in the bottom right quadrant to grow.



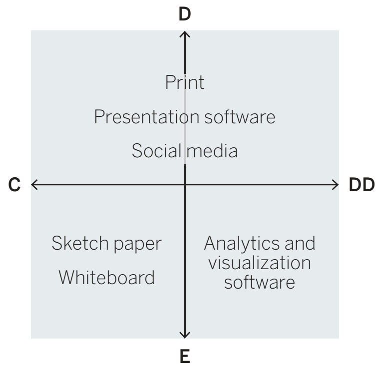
Focus. For declarative work, focus on output—creating great visuals that will move others. For exploration, worry less about how your visualizations look and more about generating ideas and allowing you and your team to learn.



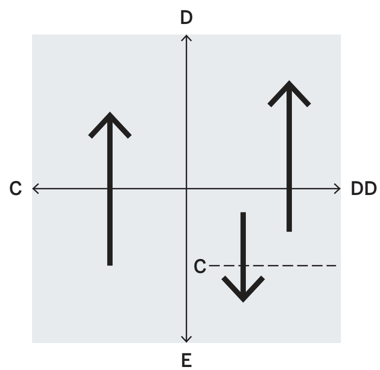
Skills. A project's importance, complexity, and deadline will dictate whether skills should be developed or hired. You're most likely to need to contract with others in the discovery quadrant and for crucial presentations, such as to the board. Idea exploration skills are worth developing whether or not they're applied to visualization.



Media. In general, tools in the exploratory half enhance your ability to interact and iterate, whereas tools for declaratives support great design. But expect more good design to be built into exploratory software tools as they're developed.



Workflows. Exploratory work often results in insights that you want to share in well-designed declarative charts for a broader audience. All the 2 × 2 idea illustrations in this chapter, for example, started as idea explorations before being designed for publication. Sometimes testing a hypothesis in



confirmatory work will produce unexpected results that you can't explain, and thus will send you into deeper exploration.

You can keep layering over the frame. You might, for example, add the names of colleagues you'll call on when doing a certain type of visualization. You might add links to the software tools you use in the various quadrants, or links to courses you want to take to improve your skills with visualization.

Data visualization isn't exactly one thing, but more a collection of related activities that vary with the task at hand. The skills you'll call on, the tools you'll use, and the media you'll visualize with can vary significantly from quadrant to quadrant. What makes an idea illustration a good chart may be different from what makes an everyday dataviz a good chart.

Spending just a few minutes asking the two questions at the beginning of this chapter—Is the information *conceptual* or *data-driven*? and Am I *declaring something* or *exploring something*?—will prepare you to visualize well. You'll have packed for the right trip.

RECAP

TWO QUESTIONS → FOUR TYPES

Visualization is a diverse craft. Different types require different skills and resources. Before making visuals, plan for them. Determine what skills and resources you'll need by defining your visual communication generally as one of four kinds. You'll put yourself in the right mindset for the project and save time by having planned ahead.

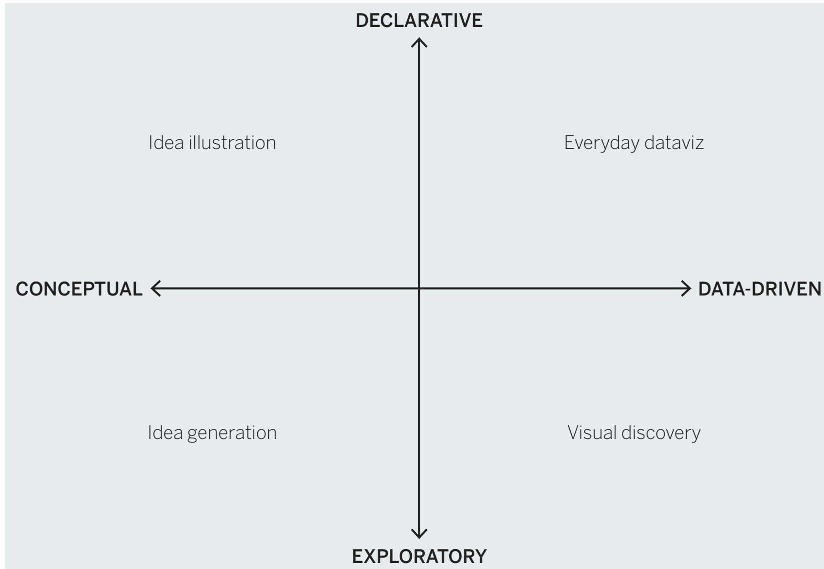
Answer two questions to learn which kind of visual communication you're about to undertake:

1. **Is my information *conceptual* or *data-driven*?**
   - Conceptual information is qualitative. Think of processes, hierarchies, cycles, and organization.
   - Data-driven information is quantitative. Think of revenues, ratings, and percentages.

2. **Are my visuals meant to be *declarative* or *exploratory*?**
   - A declarative purpose is to make a statement to an audience—to inform and affirm.
   - An exploratory purpose is to look for new ideas—to seek and discover.

Match your answers to the type of visual communication shown in the four types 2 × 2 matrix:

THE FOUR TYPES



Idea illustration
A visualization of an idea that's not connected to statistical data. Often uses metaphors, such as trees, or processes, like cycles. Examples include organizational charts, process diagrams, and this 2 × 2 matrix itself.

Idea generation

Rapidly sketched concepts for visualizing ideas not connected to statistical data. Often done in groups as brainstorming sessions, on whiteboards, or, famously, on the back of a napkin.

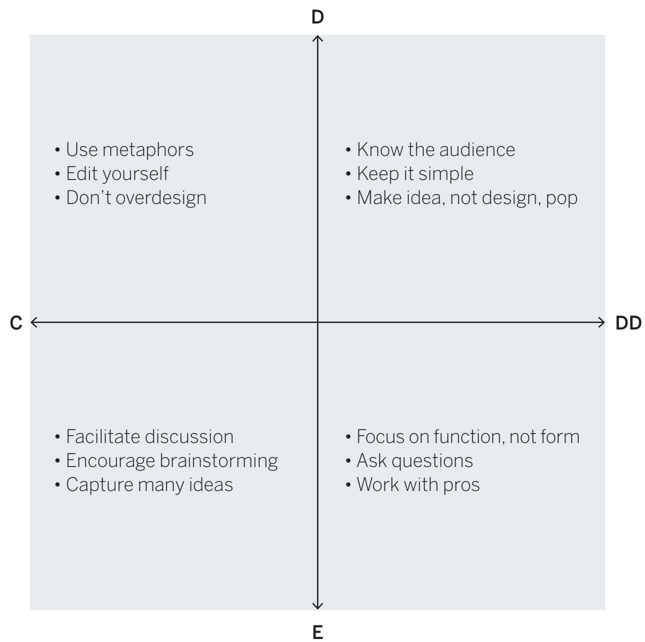
Visual discovery

Visualization in which data is used to confirm hypotheses or find patterns and trends. *Visual confirmation:* the more declarative subset of visual discovery that is generated to test a hypothesis or look at data in a new way. It's often done by an individual, usually with statistical software, such as Excel or any number of online tools. *Visual exploration:* the more exploratory portion of visual discovery, which uses data in its rawest form to see what patterns or trends emerge. Relies on large data sets and dynamic data sets that change often. Usually requires advanced software tools and data science or business analysis skills.

Everyday dataviz

Standard charts and graphs used to express an idea to an audience. Usually well designed and based on a manageable amount of data, and often used in a presentation setting.

You can use this 2 × 2 as a template to make notes about each type of visualization, the skills you want to build for developing them, the tools you'll call on, and any other hints you'll find useful whenever you start a visualization project. For example, here's a version that provides reminders of what to think about for each type:



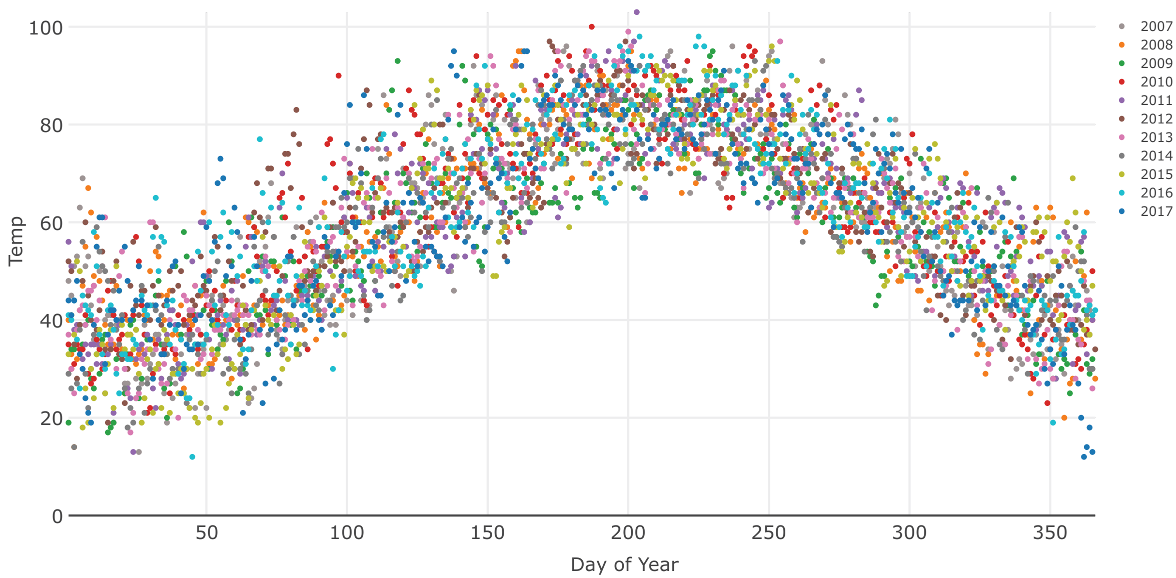
CHAPTER 4

BETTER CHARTS IN A COUPLE OF HOURS

A SIMPLE FRAMEWORK

MOST OF THE STRESS we feel about creating charts relates to picking the right kind of chart, which often amounts to scanning preset options in a software program and trying out a few until one looks right or just seems pleasing. If you have time, you might adorn it with a few more clicks—make it 3-D or change some colors. The tools make it so easy to produce a visualization that the biggest challenge in crafting good charts is overcoming the inclination—temptation, really—to just click and use what's spit out. It seems hardly worth putting more time and effort into the process.

Of course, that's flat wrong. That approach might be fast, and it might spruce up the look of a chart, but it doesn't refine the ideas that the chart conveys. Recall the Good Charts Matrix in the introduction: Good charts are a positive combination of well-understood context and some design refinement. Even if software programs automatically generated well-designed charts (most of them don't), none yet can divine the context of a chart well enough to create excellent default output. None sets context to any meaningful degree. For example, recall the weather chart from the previous chapter on page 72 in which we were exploring high temperatures in the 70s. The software initially generated this:



This is not useful for our context (and probably not useful for many contexts at all) in which we were trying to learn about days with high temperatures in the 70s. The colors are assigned to years, which don't matter at all to us. The y-axis was automatically created and doesn't include a line for 70 degrees, the range I'm interested in. This was a fine starting point, but it was only that. The software has no idea what I want to accomplish. It's incumbent on me to manipulate the visual to match my context.

Programs visualize data. People visualize ideas.

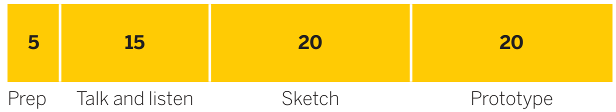
Instead of jumping right to chart types and design, you need more inputs to help define your context and identify the visual approach that will be most effective. This isn't a waste of time and effort; it's the antidote to unthinking, automatically generated charts. With just a little effort we can turn prosaic and uninformative charts into powerful, good charts.

And it doesn't take as much time or effort as you might suspect. You can make major gains in the quality of your visual communication often in under an hour. Other times you may spend an afternoon on creating a compelling visualization.

Here's how. Let's start with these steps and time frames:

BUILDING BETTER CHARTS

MINUTES SPENT AT EACH TASK



Prep time shouldn't take more than a few minutes. But as you might expect, ensuing steps' time will vary according to the type of visualization and the complexity of the project. For one or two good charts, start with this time distribution as a guide.

PREP

Cooks would call this *mise en place*—all their ingredients and their kitchen organized to prepare for cooking. Do these three things:

Create three kinds of space.

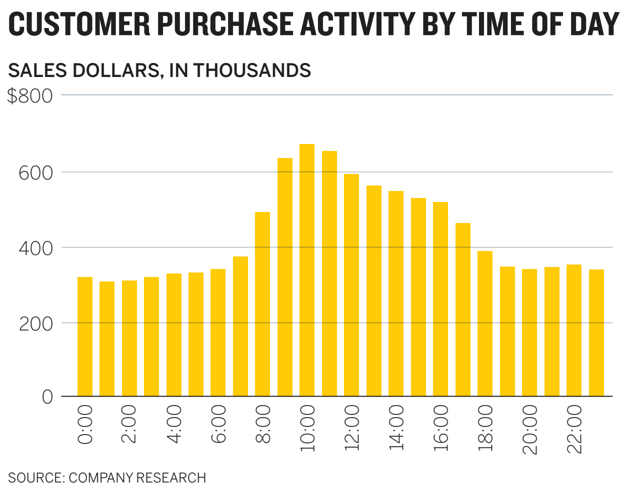
- **Mental space:** Block out time on your calendar. Turn off email and social channels. Focus.
- **Physical space:** If you're in an open-concept office, get a room. Even if you have an office, find a quiet, closed-off area away from your desk to minimize interruptions. You'll be seeking others' ideas and opinions, but you don't want random, unsolicited comments from passers-by.
- **White space:** Bring plenty of paper, whiteboards, sketch apps, whatever allows you to draw and

take notes. A rolling whiteboard will allow you to take notes back to your desk. If you can't get one, bring a phone to snap pictures of your sketches. It's helpful to have markers and pens or pencils in three or four colors.

Put aside your data. This may seem counterintuitive, but it's key to allowing for more-expansive thinking. Don't ignore the data—make sure you understand it—but don't lead with it. "When you start with the data set in mind, it limits how you think," says Jeff Heer, an associate professor of computer science who teaches data visualization. "First you need to step back and think more broadly."

Focusing on the cells of data can lead to banal results—charts that just convert tables to visual form. If you start with a more open point of view, you may discover ways to make your idea come through more strongly by introducing new data or re-crunching the data you have.

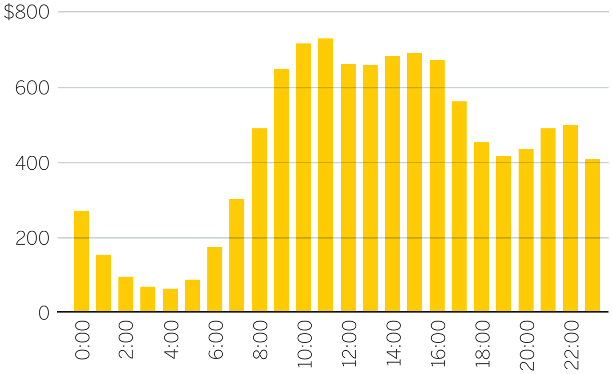
Here's a simple example. A general manager for an e-commerce site is looking at customer purchase activity by time of day. Visualizing the data from one of his spreadsheet's columns yields this:



This is not bad, and it was simple to execute. But if the manager had put aside the data and talked through what he was trying to show (a process I'll get to in a minute), he'd have realized that purchase data was set to Eastern Standard Time, in the location where the purchase was registered, not to the time in the location where the purchase was made. It would be more useful to show volume of sales by the *purchaser*'s time of day:

CUSTOMER PURCHASE ACTIVITY BY TIME OF DAY

SALES DOLLARS, IN THOUSANDS



SOURCE: COMPANY RESEARCH

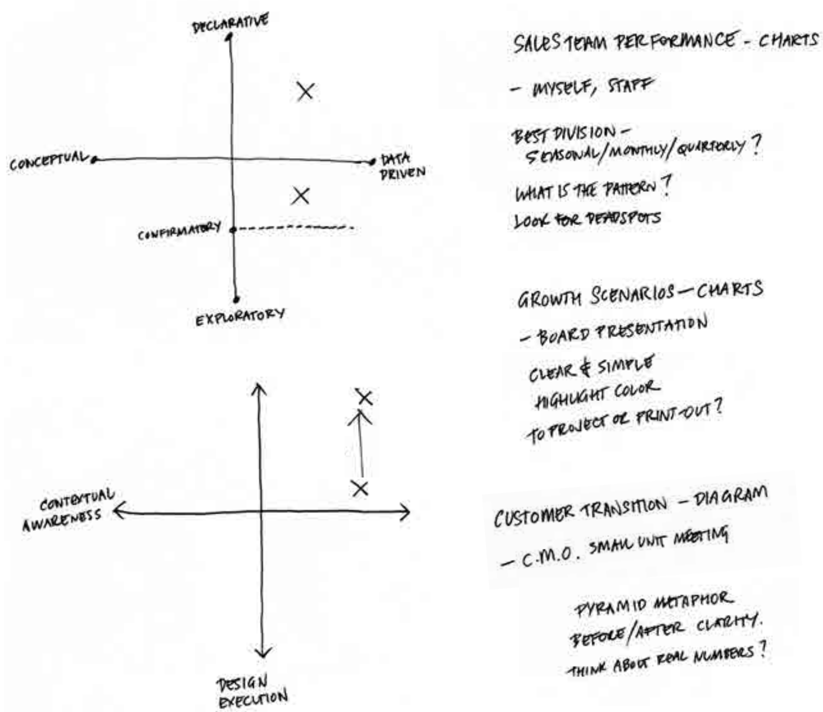
Starting from what he wanted to show rather than from what data he had on hand led to a different, more useful chart.

Write down the basics. You've created space. Now document a few key pieces of information on your paper or whiteboard to help frame your thinking. Include:

- What you'll call it
- Who it's for
- What setting it will be used in
- Which of the four types of visualization you're creating
- Where on the Good Charts Matrix between context and design you should aim

For an example, take a look at the Sales Team Performance sketches below.

Add keywords and notes as prompts and reminders. This will serve as a launching point, or as a buoy you can return to if you drift off in the talking and sketching that you're about to do (which should be encouraged; think expansively).



TALK AND LISTEN

This is the core of the context-setting that leads to good charts.

If you want your charts to get better, talk about what you're trying to show, listen to yourself, and listen to others. Conversations contain a trove of clues about the best way forward. Words and phrases will steer you to the data you need, the parts of it to focus on, and possible chart types to use.

Of all the things you do to make better charts, this will be the most revelatory, but also possibly the least natural. It takes getting used to. Practice doing these three things:

Find a colleague or friend. Although you can talk out loud to yourself or take notes, having someone to chat with works much better. Who? That depends. If you feel uncertain about how you should visualize your data, ask an outsider, someone who doesn't know much about either the data or what you're doing, whose reactions will be free from the assumptions and biases of those who are more familiar with the data and its audience. That will force you to explain even basic information, organize your ideas, and provide even more context. It will feel like brainstorming.

Conversely, if you're confident about your approach but you want to refine it or to make sure that it's sound, connect with someone closer to the project who knows more about the data and may even be part of the audience. This will feel more like a gut check.

Talk about specific questions. Don't wander into the conversation without a plan. Start with these questions:

- What am I working on?
- What am I trying to say or show (or prove or learn)?
- Why?

The first question is straightforward and factual, most useful if your counterpart is an outsider. It gives rise to necessary exposition; their ensuing questions may signal when you're making assumptions that they're not and help you notice when you're veering off topic.

Imagine starting one of these conversations this way:

> I'm working on showing the bosses we have an opportunity to invest in new HR programs—
>
> > Wait—smaller programs for the upcoming fiscal year, or more like big, long-term investments?

Even in that one exchange, you're being forced to focus more precisely on what you want to show. Make a note of it, even if it seems obvious.

The second question will vary according to whether you're in the declarative space (*What am I trying to say or show?*) or the confirmatory or exploratory space (*What am I trying to prove or learn?*). Notice that you're still explicitly avoiding your data. You don't want to ask, *What does the data say?* Even if you're reasonably certain that your viz will be a straightforward representation of the data, this is your chance to think more broadly about your approach, which may in turn lead you to seek out other data or information to incorporate into your visualization.

It will help with subsequent activities if, while you're talking, you find and jot down a short phrase or sentence that becomes the working answer to *What am I trying to say?* Here's a conversation that arrives at such a statement:

> I'm trying to show my boss that we're doing better than she thinks in terms of customer retention.
>
> Why does she think you're doing poorly?
>
> Well, our retention rate has fallen for three straight quarters. I know it looks bad right now, and everyone is panicking.
>
> So how is it better than she thinks?
>
> Well, it's not what we're doing, as far as I can tell—it's what's happening in the industry. Although our retention rate is falling, it's not falling nearly as dramatically as our two main competitors'. Something systemic is going on, I'd guess.
>
> Ah!
>
> If I can show her that, she'll see that we should focus our worry, our energy, on figuring out what's going on in the market, not on changing how we're executing as a company.

The manager starts by suggesting that what he wants to tell his boss is "We're doing better than you think." His partner recognizes that as a qualitative statement and does well to press the manager into explaining. This leads the manager to a description of what he can show to prove it to his boss: "Although our retention rate is falling, it's not falling nearly as dramatically as our two main competitors'." Without realizing it, the manager has found a visual starting point and is drifting toward a visual solution: three falling lines with the emphasis on the fact that one line—theirs—is not falling the way the others are.

The third question to keep in mind in this conversation is the most difficult and, frankly, the most annoying. Keep asking "Why?" and encourage the person you're speaking with to challenge you as well. If you become exasperated or find yourself unable to come up with a good answer, or hear yourself saying "Just because!" that's a good sign that you need to think more critically about what you're trying to show. This conversation and its litany of

"Why?" forces a manager to admit that she's not prepared to create the declarative she's proposing:

> I want to compare financial results to key productivity data like time spent on email and in meetings.
>
>> Why? What's the connection there?
>
> It just seems like there's probably a relationship between the two. Revenues are down. I ask myself, *Why?* We're in meetings so much now. We never have time to work!
>
>> But don't you get work done in meetings? Why are they the problem?
>
> I mean, I know I'm getting less done because of all this time spent on other stuff.
>
>> Why does one lead to the other, though? How can you actually prove that more meetings and emails equals lower revenue?
>
> I'm not sure, but of course there's some connection there. There has to be!
>
>> Why? What if they're helping you get work done, too?
>
> Just because! I'm sick of sitting in meetings!

If you're trying to create a declarative dataviz and you can't adequately answer the "Why?" you might want to stop, form a few hypotheses, and test them with exploratory visuals to see what emerges.

Listen and take notes. As you talk, listen to your counterpart, but listen to yourself, too. Pick out visual words and phrases that describe how you *see* the ideas and information and write them down. If, for example, you hear yourself using words like *distributed* and *spread out over*, or *different types* and *clusters*, they are clues to your potential approach. Listen for metaphors: *The money's flying out of our department*. *We saw a huge dip*. *Revenues fell off a cliff*. *It's a crazy maze of choices.* They evoke powerful imagery that could inform how you design your information.

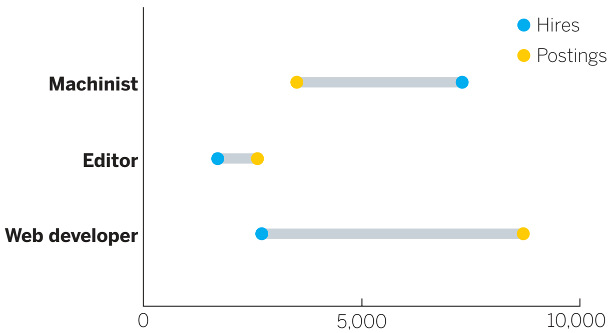
Take this statement: "I want to compare the number of job postings to hires to see what the ratio is for different types of jobs." That sentence contains enough information to suggest a strong visual approach. Here it is again, with the visual cues emphasized:

> I want to **compare the number** of job postings to hires to **see** what **the ratio** is for **different types** of jobs.

Compare the number suggests a chart that plots data points along a numbered axis. *The ratio* tells you you're comparing one number with another. *Different types* suggests that you can repeat the comparison across several categories, and maybe create subgroups. (You may also notice some of the other nouns describe potential variables: *Postings*, *hires*, and *jobs* are all important categories of data. Note these, too.)

Let's skip ahead for a moment. Pulling those keywords from that one sentence could bring the manager, eventually, to the following final visualization. Reread the sentence the manager captured: It's all reflected in the chart:

MONTHLY JOB HIRES VS. MONTHLY POSTINGS



SOURCE: ECONOMICS MODELING SPECIALISTS INT'L.

Here's another example, this one of a sales manager who wants to do some exploratory visualization of his teams' sales performance. "It's not clear," he says to his friend, "that there's any regular pattern to our sales. I'm really trying to understand how and when they make sales—how sales are happening over time. Is it mostly smooth, or are there bursts of sales with periods of nothing? Is it the same month to month or not? Are different seasons showing different sales patterns?"

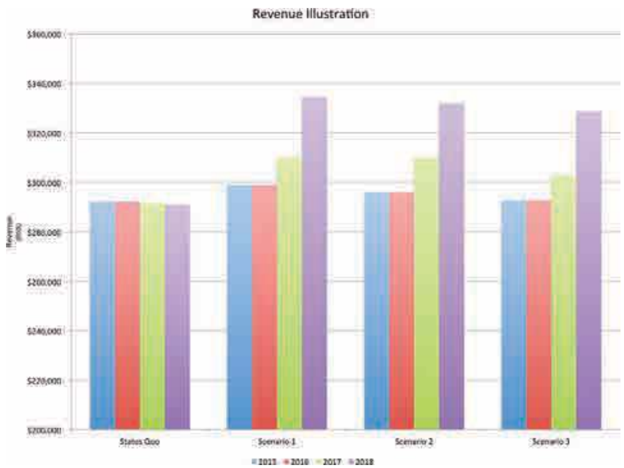
If he's been doing this for a while, he will have jotted down several keywords and phrases from his chat:

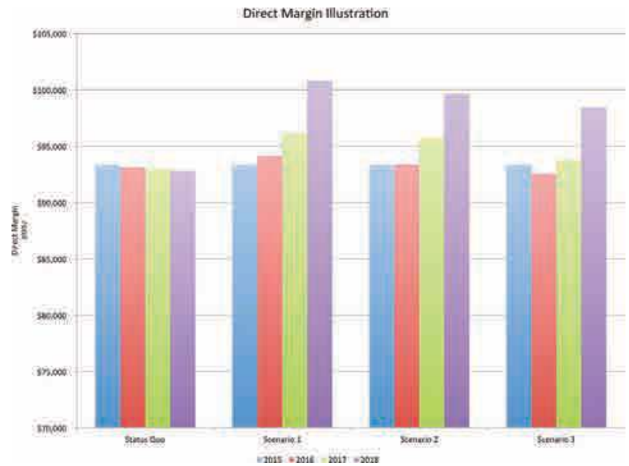
> It's not clear there's any regular **pattern** to our sales. I'm really trying to understand how and **when** they make sales—how sales are **happening over time**. Is it **mostly smooth** or are there **bursts** of sales with **periods of nothing**? Is it the same **month to month** or not? Are different **seasons** showing different sales **patterns**?

It's a bit strange at first, listening to yourself talk in such an active way, but it's undeniably valuable. Time and again I've watched people's eyes light up as someone utters a phrase that creates a *Eureka!* moment. Suddenly they realize exactly how they'll make their chart.

On the next page is one of my favorite examples. A consultant needed to transform his ineffective bar charts. These represent four possible strategic directions the client could pursue, and the consultant was asking the client to compare scenarios.

That's not easy! The type is small, so the consultant found himself explaining all of the labels and axes because his audience couldn't see them. What's more, the y-axes are not the same range, so even though he was asking his audience to compare sets of bars, that was near impossible given that the bars on the right are generally as high or higher than the ones on the left, but they represent about one-third





the value. The chart so confused the client that it scuttled the meeting.

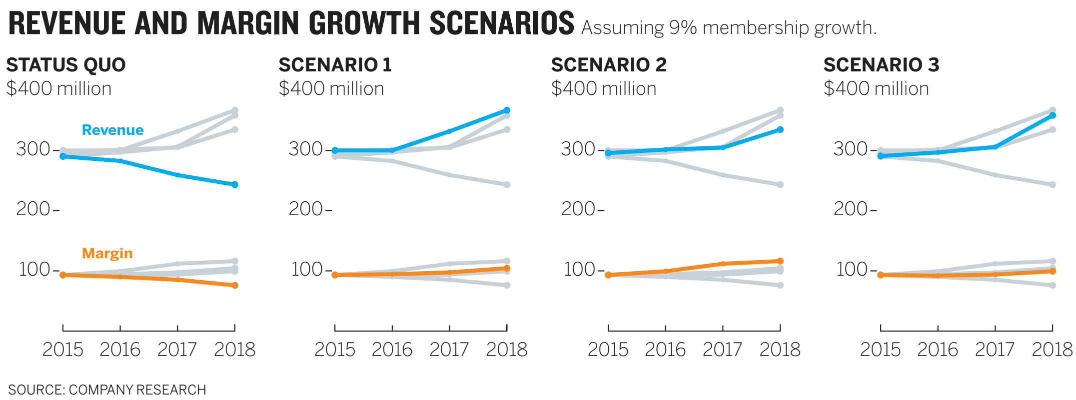
He needed something better. We began talking and trying to set his context. I asked some of the basic context-setting questions, and at one point he said, "I need the client to be able to compare revenues *and* margins across scenarios and revenues *to* margins within scenarios." I jotted down several words there: compare, across, within.

After some time, I asked him why the bars were different colors for each year. "I don't know. That's just what the software did," he said, defeated. "Well, does it matter? Do you need to distinguish between years?" I asked.

It felt like he had reached his limit with my questions, and he impatiently blurted out, "Look! The years don't matter to me. I don't care about the bars! All that matters is the trend line for each scenario. Do revenues go up or down and do margins go up or down?"

It was an amazing moment. Without realizing it, the consultant had literally blurted out that the chart type he was using, a bar chart, wasn't useful to him, and the chart type he *should* have been using, a line chart, is what mattered. That, combined with his note about comparing across and between led to a successful revision, shown on the next page.

It was a good outcome. And it's a much more pleasing, pretty chart. But here's the wonderful thing about that: It's not a good chart because it's



nicer looking. It's nicer looking because it's a good chart. That is, when you take time to set your context, design decisions are made for you. Extraneous information—color, extra variables, whatever—is eliminated because it's not important to your context.

When you're talking and listening, force yourself to answer that fundamental question out loud: *What am I trying to show or say (or learn, or prove)?* More of your answer than you may suspect lurks in a brief conversation. Once you extract those words, it's time to draw.

SKETCH

Finally, you're drawing. You should come out of this step with an approach and a rough draft that can be refined. Here's how to start:

Match keywords to approaches. The words, phrases, and notes you wrote down can now be put to use. Start drawing examples of the visual words you captured. Match those words to types of visual forms. You can match them to the types of visualizations that typically best show what they describe.

There are references you can use. Andrew Abela, the university provost and former dean of the business school at Catholic University of America who has written books about effective presentations, created a guide, on the facing page, that organizes typical charts well, but it comes with caveats.[1] For instance, not everyone will agree on which chart types should be included and which excluded. Some people will take exception, for example, with Abela's inclusion of pie charts and spider charts (or what he calls "circular area charts"), which they consider difficult or suboptimal. Others will ask why unit charts and slope graphs *aren't* included. Newly popular forms such as dot plots and lollipop charts aren't here either. And what about tables?

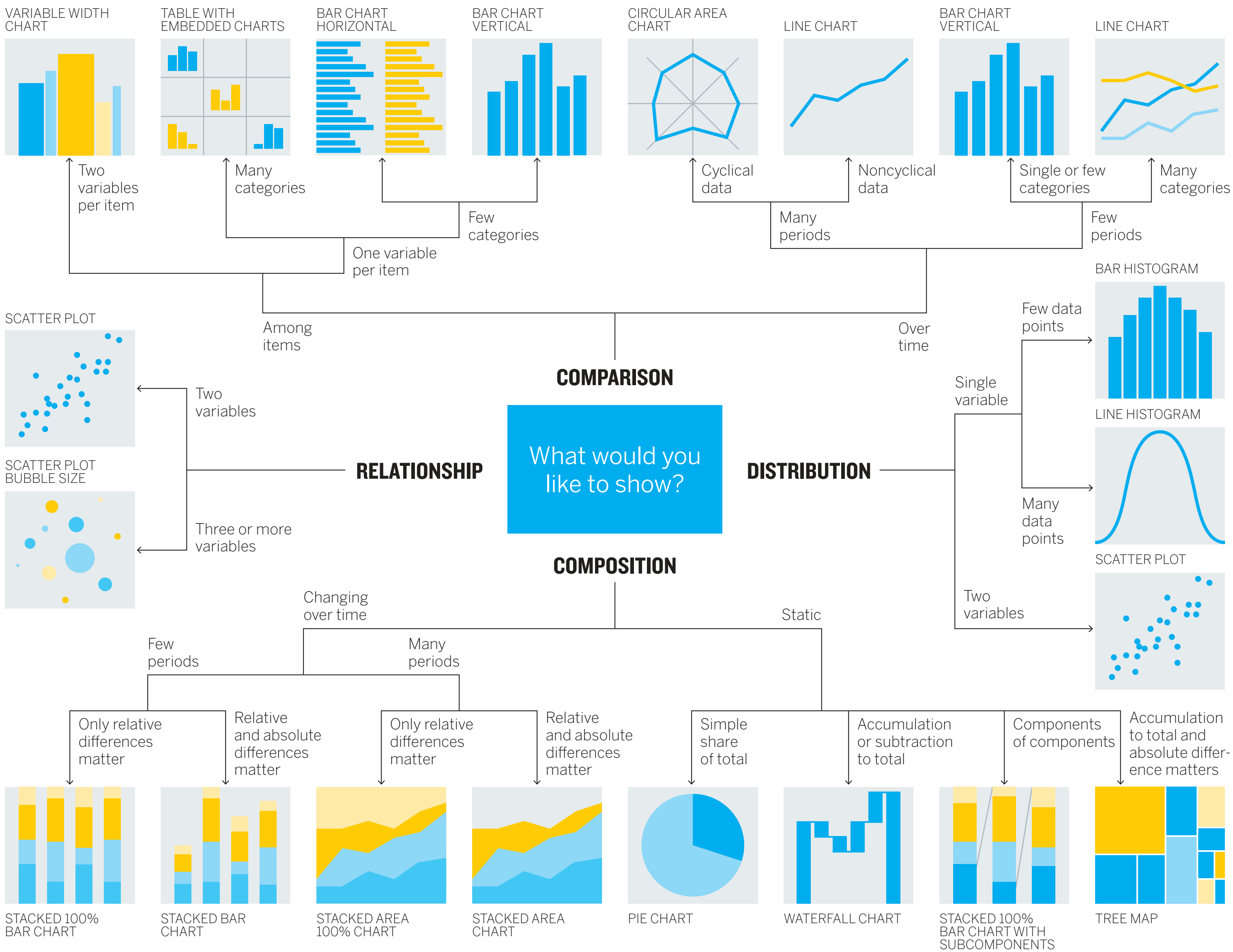
Also, a guide like this could narrow our thinking at a stage when we should be broadening it. It's something like pouring out a bucket of Legos in front of a child and then telling her she can make only the ten things in the instruction booklet. At the beginning of the sketching phase, we're better off just messing around with the Legos.

Remember that no cheat sheet will encompass all options. Under every common chart type shown here is a remarkable diversity of variations. New chart species are being spawned all the time. Trying to identify and document every variation of every chart type would be a quixotic effort at best. It's better to just learn basic categories and types and then become a collector. Look around; collect examples of chart types that appeal to you or that you find exceptionally effective. Make notes about what you think works well or caught your eye. Visit websites devoted to dataviz and follow people on Twitter who post new charts daily. (Shortcut: Make lists for #dataviz, #visualization, #viz.)

Still, Abela's guide is here for two reasons. First, it's as good as any typology out there (an online search will yield many more) at helping us understand categories of forms—comparison versus distribution, for example.

Second, I'm showing Abela's because I've adapted its main categories for a worksheet that matches typical keywords you may find yourself saying during the talk and listen stage of this process to the types of charts you might try to sketch.

This transforms Abela's decision machine into more of an inspiration guide. I've simplified the categories and types and added conceptual forms that don't appear in Abela's chart. (Ironically, the type of visualization Abela used to create his typology—a hierarchical decision tree—isn't listed *on* the typology, because he shows only data-driven forms.) To use this guide, see if your visual words match those in any of the quadrants. For instance, if you wrote down *proportion* and *a percentage of*, you might consider starting with stacked bars or a pie.



SOURCE: ANDREW V. ABELA

MATCHING KEY WORDS TO CHART TYPES

COMPARISONS

NOTES

before/after
categories
compare
contrast
over time
peaks
rank
trend
types
valleys

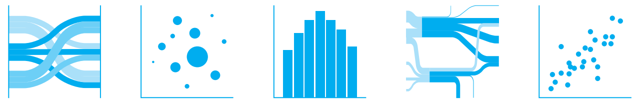
BARS | BUMP | LINES | SLOPE | SMALL MULTIPLES

DISTRIBUTIONS

NOTES

alluvial
cluster
distributed
from/to
plotted
points
spread
spread over
relative to
transfer

ALLUVIALS | BUBBLE | HISTOGRAM | SANKEY | SCATTER

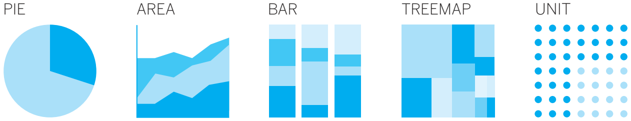


COMPOSITIONS

NOTES

components
divvied up
group
makes up
of the whole
parts
percentage
pieces
portion
proportion
slices
subsections
total

PIE | STACKED AREA | STACKED BAR | TREEMAP | UNIT

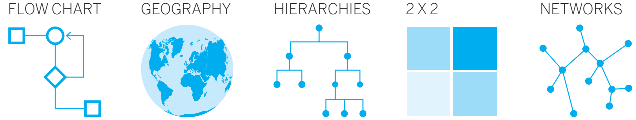


MAPS NETWORKS LOGIC

NOTES

cluster
complex
connections
group
hierarchy
if/then
network
organize
paths
places
relationships
routes
structure
space
yes/no

FLOW CHART | GEOGRAPHY | HIERARCHIES | 2 X 2 | NETWORKS



Keep in mind that this worksheet is neither complete nor definitive. It's not meant to tell you what chart type to use, only what types to play with as you start sketching. You may find that some projects, for example, can benefit from multiple chart types or hybrids (say, a bar chart overlaying a map). It's just meant to help you get started. (Also, you can use the glossary at the back of this book to quickly reference chart types and see some of their strengths and weaknesses.)

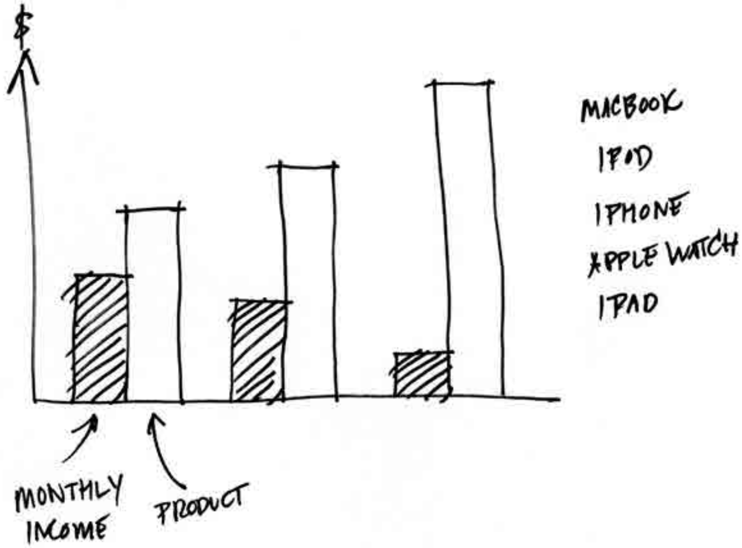
Keep in mind that bars, lines, and scatter plots are your workhorses. Those three forms alone will help you arrive at many good charts in most situations. While you shouldn't shun other forms, you also don't need to choose different ones just to be different.

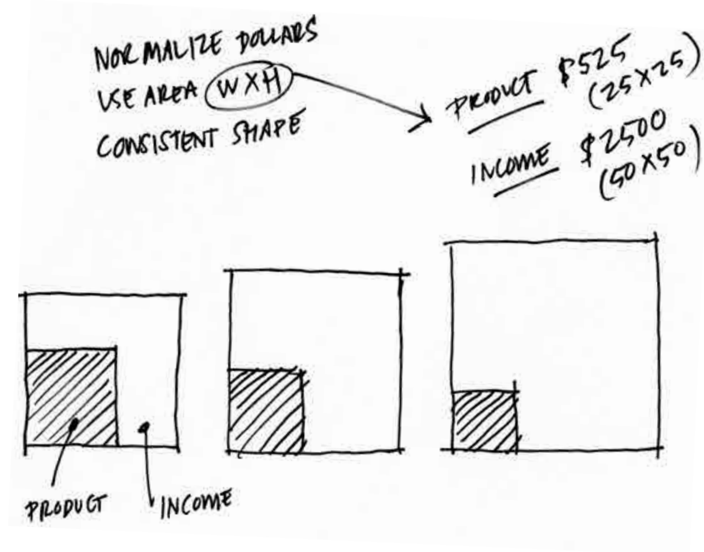
Start sketching. Sketching bridges idea and visualization. Good sketches are quick, simple, and messy.[2] Don't think too much about real values or scales or any refining details. In fact, don't think too much. Just keep in mind those keywords, the possible forms they suggest, and that overarching idea you keep coming back to, the one you wrote down in answer to *What am I trying to say (or learn)?* And draw. Create shapes, develop a sense of what you want your audience to see. Try anything.

Sometimes the form will seem so obvious that you won't feel the need to sketch a lot of alternatives. A basic comparison between categories can often result in a bar chart. Trends over time are usually plotted as line charts. Still, don't forgo the exercise altogether. Hannah Fairfield, a graphics editor behind some of the most celebrated data visualizations in the *New York Times*, always tries out at least two completely different forms to check her assumptions about the best approach and to stay creatively open.

For an article comparing the price of various Apple products to median monthly household income, my coauthor on that piece, Walter Frick, and I thought we'd show a simple bar chart, with one bar for the cost of a product and the other for income.

It would have been a natural choice because we were comparing values within categories. The bar chart is valid. But, in keeping with Fairfield's advice to always sketch a couple of options, we decided to look for other ways to show the comparison. One phrase kept recurring while we sketched: how much monthly income the cost of an Apple product would *take up*. This led us to think about the product's cost as a piece or *portion* of monthly income, rather than just a comparative value. Eventually, we settled on the less likely but arguably more effective approach of mini treemaps. Arguably, a stacked bar would have worked as well. The key was our discussion, the keywords we kept using, and sketching led us away from comparisons to proportions. Sketches of both the simple approach and the alternative are shown on the next page.

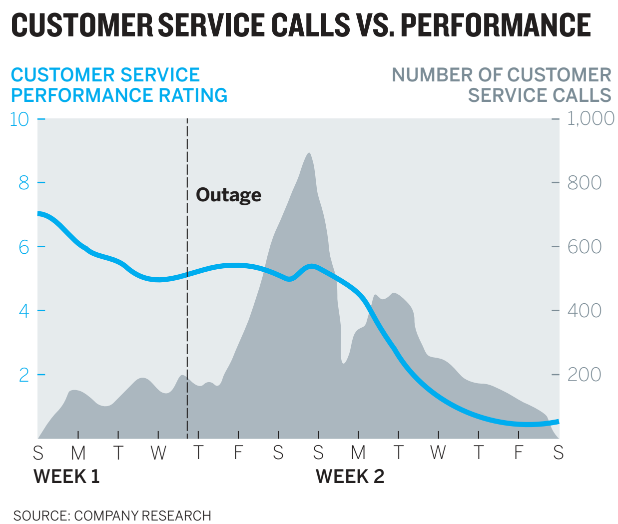




Even if you think a standard chart type will work for your visual, sketch an alternative to check your assumptions and stay creatively open. Sometimes it will lead to a better form.

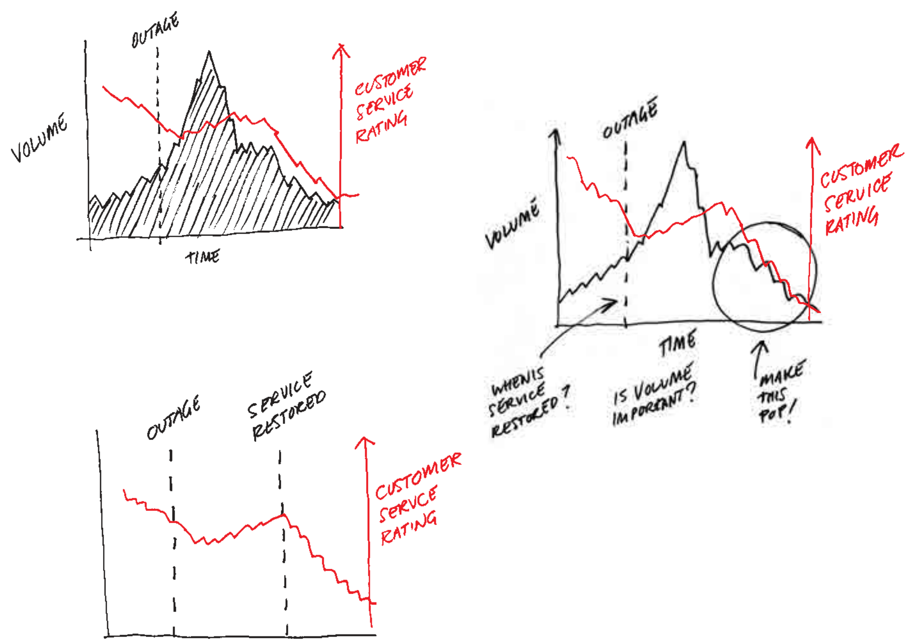
Even if you're confident that you should be using a simple bar chart, line chart, or a scatter plot, sketching these basic forms is still important. Just as rough drafts improve even staff memos and other prosaic writing, sketches will make even simple charts better.

Remember the manager in chapter 2 who wanted to show his boss that customer service performance was declining in spite of, not because of, a website outage? He could have thrown together a basic line chart showing the data he'd collected: customer service calls and customer service performance. As a reminder, that chart is shown on the right.



But when he sketched the basic chart, he saw that the dramatic shape of the call volume would probably fight for attention with the performance trend. So he spent a few minutes sketching alternatives (shown below), looking for ways to increase the focus on customer service performance. He kept referring back to the statement he had jotted down to describe what he wanted to communicate to his boss: *Even when service was restored after a website outage, customer service ratings continued to decline. And they started declining before the outage.*

The breakthrough came when he realized that his statement didn't mention customer service calls at all. The data was there, and he had plotted it unthinkingly. So he sketched a version *without* the outage data and immediately felt it was better. Then he added two key points that were reflected in his statement: *after service was restored* and *before the outage*.



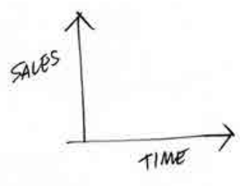
As he sketched, he didn't try to portray the data accurately; he knew the trend was downward—that was good enough for the time being. He added notes about possible treatments, such as magnifying and shading. But few final decisions were made. The most important decisions at the sketching phase are what *not* to pursue and what form to use. This is illustrative brainstorming. In 15 minutes, the manager went from visualizing some cells of data to visualizing what he wanted to say.

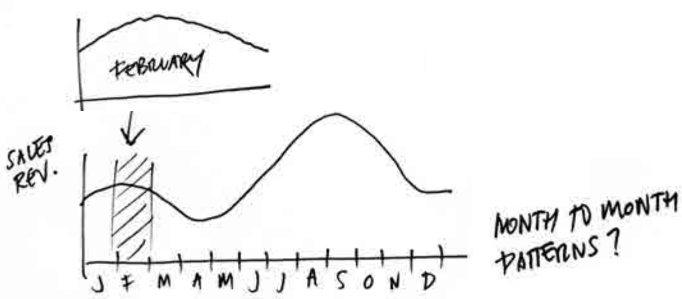
Sometimes sketching lasts longer. The sales manager from earlier in the chapter who was looking for seasonal and month-to-month patterns in his team's sales performance noted some keywords from his conversation with a friend about his project:

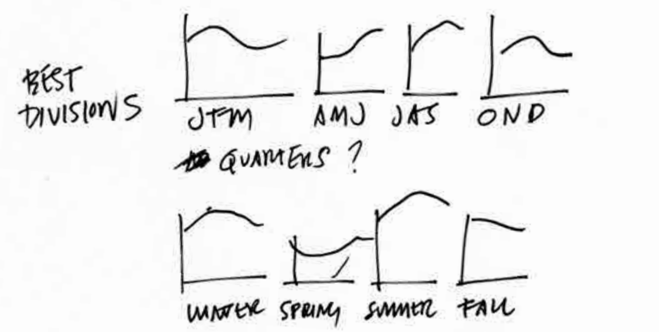
> It's not clear there's any regular **pattern** to our sales. I'm really trying to understand how and **when** they make sales—how sales are **happening over time**. Is it **mostly smooth**, or are there **bursts** of sales with **periods of nothing**? Is it the same **month to month** or not? Are different **seasons** showing different sales **patterns**?

Looking at his notes, he saw that he was really talking about two things here: patterns and time. He actually used a phrase that was *the* potential visual approach: *sales over time*. He sketched those two variables as axes and then started to think about how to use them.

Line graphs are usually a good starting point for trends. So he drafted one of those over a year. From there his sketches reflect an effort to find the right set of line graphs based on some of his words—*seasons*, *periods*, and *month to month*—which suggested ways to organize his visuals. As he proceeded, his approach came into focus; but again, his charts weren't accurate or to scale. He was just homing in on the approach.







Sketching is also useful to help us try different approaches to complex stories. Here's part of a conversation from the talk and listen stage for an economics student. The student extracted lots of keywords from her conversation:

> I'm trying to show **a lot of things**, actually. I want to see where the **greatest growth** is in jobs in the **coming decade, compare** sectors that are **strong or weak**. But also how is pay in those jobs **relative to** the **total number** of jobs that will be created? That's the tricky part, because it's easy to show a **super high growth** of jobs, but if it's growing from 10 jobs to 20, what does that **percent growth** really mean? If **high-growth** jobs are **low-paying** ones, what does that mean? What about manufacturing versus knowledge work? Could I **divide** the data that way? There's just a lot going on in the data.
>
> > Why is it important to show so many things?
>
> That's just it. Many times you see **one piece** of this data highlighted and it ignores these other factors, so it's like, "Look at all that **job growth**," but it doesn't take into account **pay** or **raw numbers** of jobs. I'm looking for a **holistic picture**, a smarter look at this.

> GROWTH IN JOBS COMING DECADE

> HOLISTIC BIG PICTURE

> COMPARING GROWTH vs RAW NUMBERS

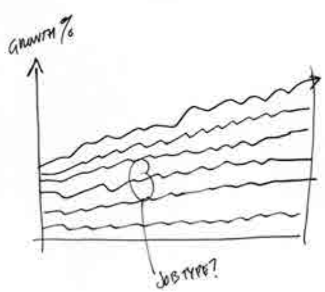
> MORE SOPHISTICATED VIEW

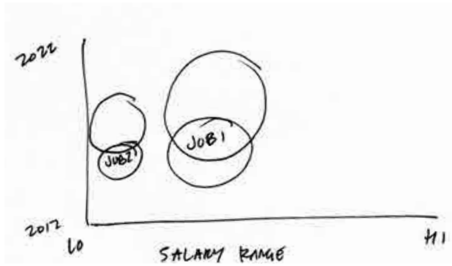
It's silly to think the student could extract one chart style or approach from this conversation. She will need multiple charts and will need to think about what information goes in which chart type. She'll need storytelling skills, which we'll explore later on. On this page, sketching is meant to explore options for organizing this student's *holistic picture*.

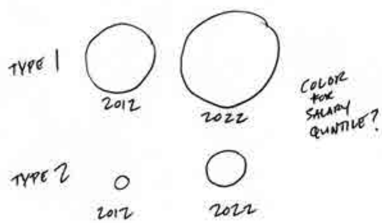
Whether it takes five minutes to confirm the approach you sensed you should use, or an hour of slogging to find a good way to organize your information, sketching is a crucial habit to form. For many professional designers and dataviz pros, it ranks at or near the top of their list of activities that improve visual communication.

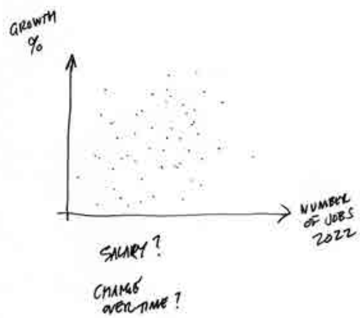
When people ask me what tools are best for making good data visualization, I tell them that paper and pencil or whiteboard and markers should top their list. You can usually get 90% of the way to a good chart just by sketching.

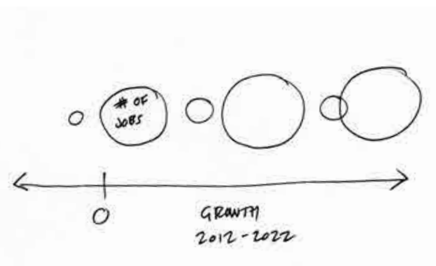
One final note. The sketches you see here are rather neat. That's because I'm trying to create a good user experience for you. But I can't stress enough how going fast and being messy is okay. It keeps your mind at a high level and open to new ideas. Just to give you a sense of what it's like in the real world, I've included here a few of my real sketches from projects.

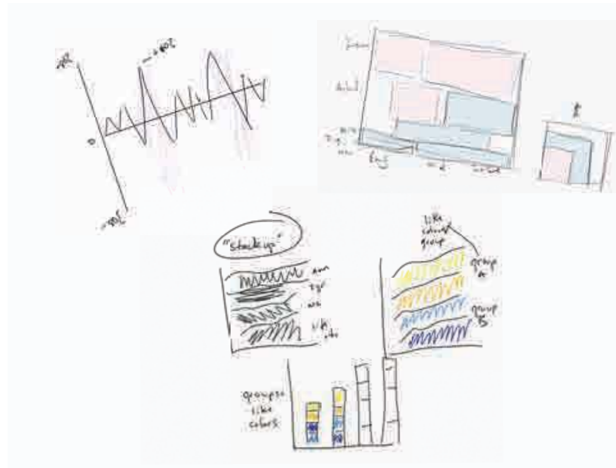








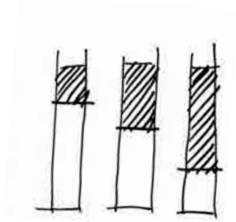


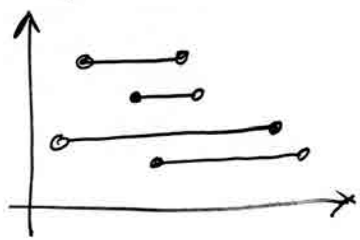


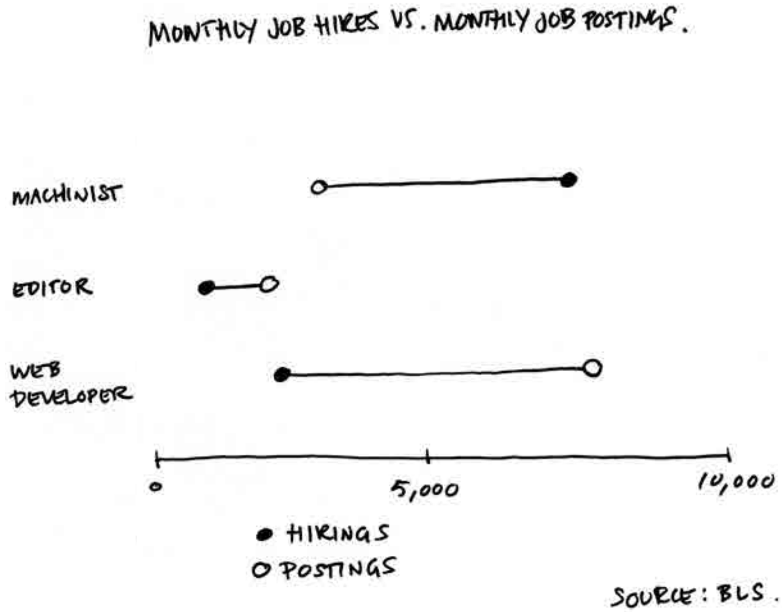
PROTOTYPE

At some point, you'll have done enough sketching and will be ready to start making more-realistic pictures. But when? Watch out for these signs that you can begin to prototype:

- Your sketches reasonably match your *What am I trying to say or show?* statement.
- Your sketches are becoming refinements of one idea, rather than broad stabs at different ideas.







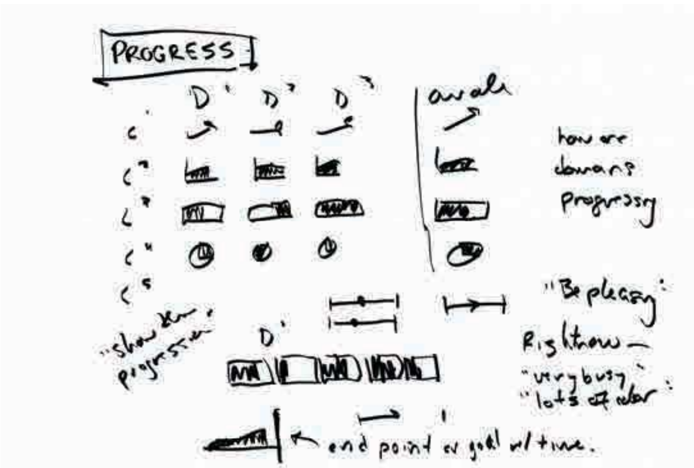
- You find yourself plugging in actual data or trying to imitate the actual values.
- You find yourself designing the charts, focusing on color, titles, and labels.
- You feel that you don't have any more ideas.

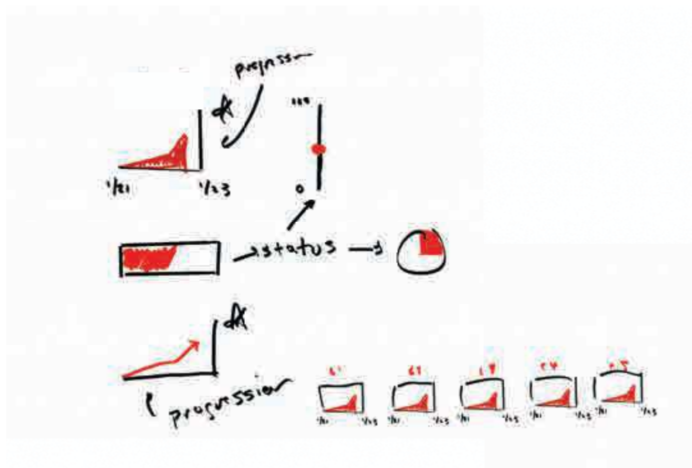
Sketching is *generative*; it's meant to bring up ideas. Prototyping is *iterative*; it's meant to hone good ones. Prototypes should incorporate real data, or *realer* data. Don't try to be perfect with your plotting but use realistic axis ranges and approximate values that give a sense of what the actual shape of the thing will be. Often, it's useful to prototype with a subset of the data to create accurate pictures without feeling the burden of having to prototype everything. The manager plotting seasonal sales data, for example, might focus on one season for his prototypes.

Prototypes should also begin to incorporate broad design decisions such as use of color, and the media you'll be building it for.

Again, to put this in the real-world perspective, recall the difficult-to-parse chart on page 42 that included many bars in five clusters. I mentioned then that we fixed that by using this process. Below is one of the sketches and the first prototype created for that project. It should be obvious which is which.

The prototype, which plots only three categories even though the final chart will catalog many more, is cleaner and more realistic than the quick sketches. It uses real labels and includes a key. It also raises questions—*Will this x-axis range work, given the data? Should color be used for categories*—that can be addressed in ensuing iterations.





Most prototyping you'll do falls into one of three categories:

- **Paper**, physically sketched on paper or a whiteboard, as with the previous examples
- **Digital**, done in software or on the web
- **Team**, done with partners who are subject-matter experts with deep knowledge of your data, or who have skills you lack, such as programming or design

Paper prototyping requires virtually no setup beyond what you've already done. Even if you plan on doing digital prototypes, a paper prototype is a good transition from sketching; a first paper prototype is like a final-draft sketch. Paper prototyping is good for simpler data sets (or subsets of larger data sets) and simpler visualizations, because it's slow. A chart with ten categories could become difficult and tedious to draw by hand. It's also harder to maintain clean plotting on paper as the amount of information piles up.

Digital prototyping is much faster than drawing and manages more information more cleanly. Here you can use tools built into the software where your data exists (such as Excel or Tableau) to quickly build visuals, or you can upload some data to a website that offers the ability to try multiple approaches. Digital prototyping is rapid prototyping. It's especially powerful for confirmatory and exploratory dataviz. When my colleague and I explored Boston's weather data, we used digital prototyping.

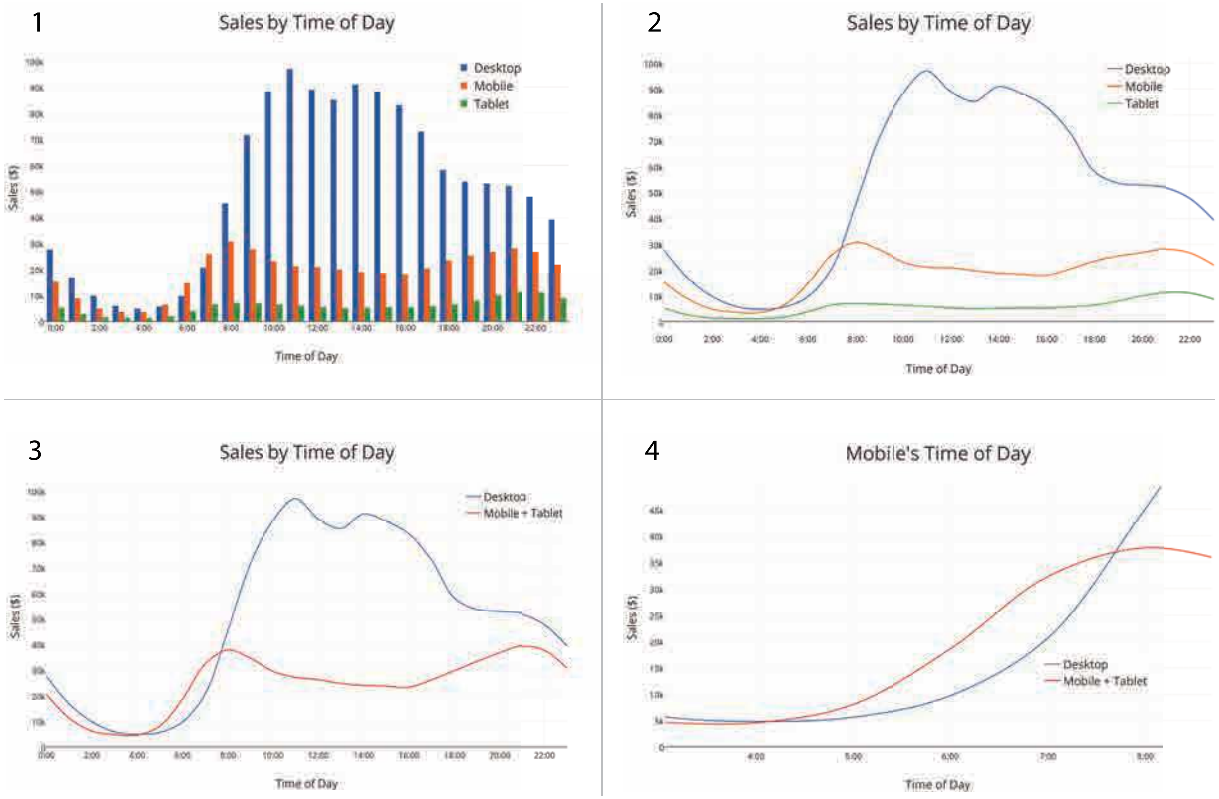
The good news for managers is that the number of tools suited to digital prototyping is growing—and the tools are improving.

More on tools in the coming chapters. Suffice it to say for now that online tools such as Plotly Chart Studio, Flourish, Infogram, Datawrapper, RAWGraphs, and others have made quick work of digital prototyping.

The four prototypes on the facing page based on when sales happen were created online in less than ten minutes. Notice they are only lightly refined—they feel as if they are in process and not like final visualizations. Even as you want prototypes to loosely reflect reality, you still don't want to use this time to refine to a finished product. You're only trying to get a sense of the final product.

That you can move a visualization prototype so far in ten minutes demonstrates the power of digital prototyping. You can almost read the manager's thoughts in the iterations: *This is way too much information crammed into a bar chart. The trend is what matters anyway, so let's try a line chart. Mobile and tablet can go together, and it's simpler to have just the two trends to compare. Now let's zoom in on this interesting slice of the data that I want to focus on.*

Digital prototyping has its limitations. For one, the tools that do it best require more training to use. The free online ones have a lower learning curve but more-sporadic feature sets. All do some



things well. None does all things well. Each has its strengths and weaknesses, so you may find yourself jumping from tool to tool depending on your project or even within a project. Digital prototyping may also be overkill for simple visualizations in which paper prototypes get you close enough to where you need to be. And few of these tools are designed to help prototype conceptual forms, which often require more sketching and prototyping than data-driven visuals. When working on conceptual graphics, paper and whiteboard are probably your best options.

Team prototyping is something else altogether. The previous techniques are defined by the tools you use. Team prototyping is defined by the way in which you work: with expert partners. The concept is based on a system of data analysis called paired analysis, which itself borrows from a software development process called extreme programming and other sources.[3] In each of these the idea is to pair a subject-matter expert with a tools expert who can manipulate data and visuals quickly, and who can suggest solutions you may be unaware are available to you.

Team prototyping speeds up development of visuals and is vital to visual exploration.

Brian Fisher and David Kasik used this method at Boeing.[4] "This turns out to be highly effective," Kasik says. "The key is to have them actually sit and work together, not throw things over the wall."

Paired analysis has proved powerful at Boeing, cutting time to visual insights dramatically since those who have and know the data aren't spending cycles figuring out how to manipulate the visualization tools. In one case, the company used it for some deep exploratory sessions in which a team of two wanted to visualize information about bird strikes on airplanes. Bird strikes are a serious safety issue (a strike by Canada geese on an Airbus A320 passenger jet caused the notable "Miracle on the Hudson" water landing in New York in 2009)—at the time of the analysis, the cost of bird strikes was estimated at anywhere from $123 million to $615 million a year, but very little was known about

THAT'S A GOOD CHART

AHA! MOMENTS

Some statistical concepts are hard to grasp without seeing them. For example, nonlinear progressions bend our brains. It doesn't seem right that eating two 8-inch pizzas is less than eating one 12-inch pizza. Or that increasing customer retention from 60% to 80% is *four times* as valuable as increasing it from 20% to 40%.

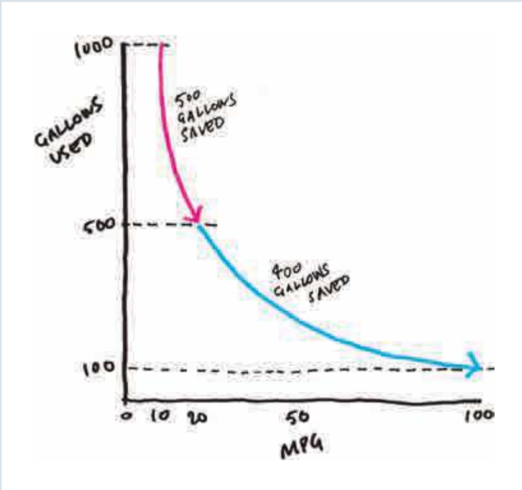
Bart deLanghe, Stefano Puntoni, and Richard Larrick wrote an excellent HBR article about this called "Linear Thinking in a Nonlinear World."[5] The article clearly lays out the pitfalls of thinking linearly (which we all do), but key to making their ideas come alive were the good charts that helped this unintuitive concept pop. Their draft manuscript included some basic Excel-generated curves, but they weren't making the idea click. I wasn't getting it. We all wanted to get the charts to the point where even at a glance, you got it.

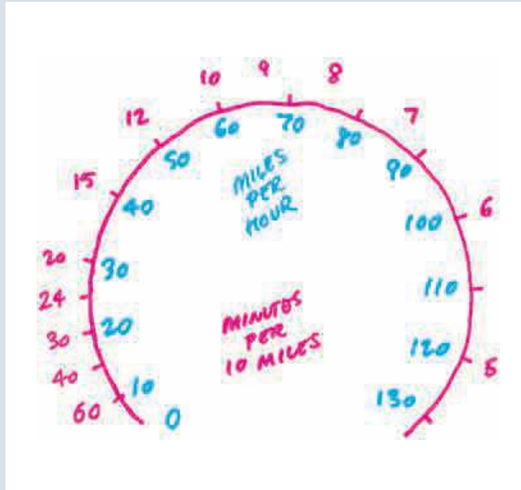
We used the talk, sketch, prototype method to make that happen. I remember our conversation generating interesting phrases like "part of the slope" and "steep then gradual" or "gradual then steep." Someone said, "When you look at the points of intersection."

Eventually we arrived at some powerful solutions. One excellent chart shows how increasing fuel efficiency from 10 to 20 mpg saves more gas than increasing it from 20 to 100 mpg.

You can see those phrases from the conversation explicitly reflected here. Notably, our paper prototype—a sketch—ended up being the basis of what was presented in print and online. Yes, this was partly an aesthetic decision but also shows that you can often reach an excellent result without much software intervention when presenting simple concepts.

But in this article, one chart captivated people more than others, the Paceometer, credited to researchers Eyal Pe'er and Eyal Gamliel.





As they note: "It will surprise most drivers that going from 40 to 65 will save you about six minutes per 10 miles but going from 65 to 90 saves only about two and a half minutes—even though you're increasing your speed 25 miles per hour in both instances." This chart instantly exposes the nonintuitive idea that the faster you're going, the less time you save by going faster, in a way that text and, I would argue, more-traditional charts do not.

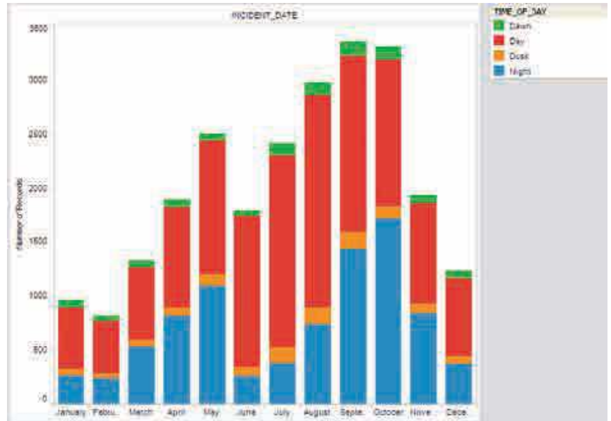
Could this have worked as a curve like the miles-per-gallon chart? Sure. But I think what captivated the audience here is that it delivers the idea in a *form they're used to using*. It also helps that they've not saddled this visual with any other information. They've only layered the relevant new information into a familiar visual form to create a powerful Aha! moment.

the patterns of bird strikes and how they might be mitigated.

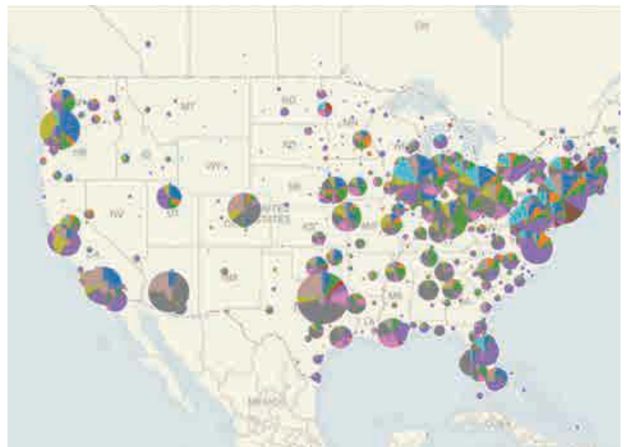
Little was known in part because the data analysis required to understand bird strikes was a tedious process of finding and reading through thousands of records from dozens of sources, correlating them, and then updating the results as new events occurred. To speed things up, Boeing paired a subject-matter expert (an aviation safety specialist) with a tools expert (in this case, an expert in both Tableau and IN-SPIRE visualization software). They worked together over several days. The following example shows the workflow. Think of the images as responses from the tools expert to requests from the subject-matter expert. Obviously, in a real-life setting they'd be discussing each of these steps in depth before the person visualizing went ahead and created charts.

Also notice how prototypes are not refined until they have to be. They work together to see just what they need to see, then change or refine based on that.

We need a way to identify and extract data on bird strikes from XYZ data sources. And once you have that system set up, we really want to see *when* bird strikes happen, by both month and time of day.

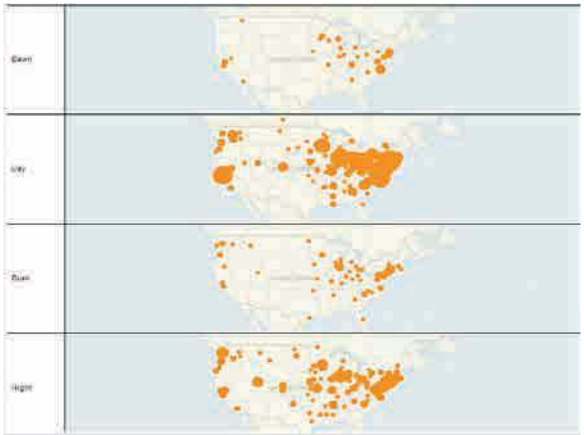


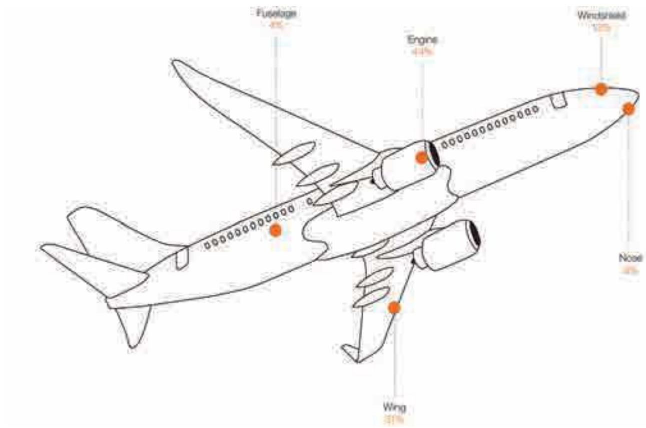
That's good, but is there a way to map this by geography? I'd like to know where the most bird strikes happen. And can each spot break down the type of bird that was involved?



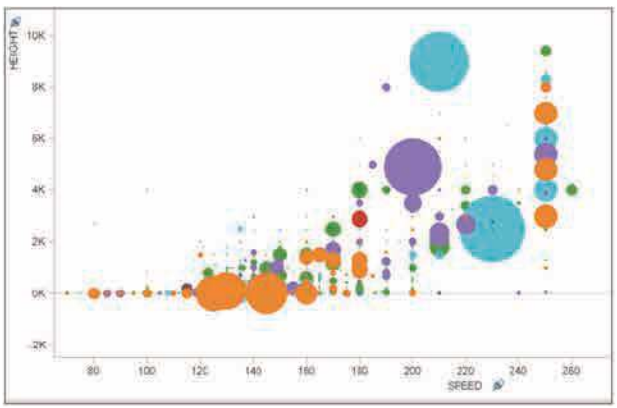
Wow. Great. Can we combine those two? Time of day and location? But less detail. I want to be able to show this to management and discuss the findings.

Great. For the presentation, we also should show frequency of where on planes birds strike. Something simple.





Can we also see altitude versus speed? Maybe see if there's any pattern there.



Obviously, this is a radically simplified and abbreviated version of the work the two put in on the project. The insights they gained along the way, for example, were put into more presentation-worthy charts that others would be able to understand. But it shows how the subject-matter expert focused on articulating the problems and explaining the context he was trying to create. The tools expert, meanwhile, drew on his knowledge of good visualization techniques and how the tools could quickly generate different views to give his partner what he needed.

Kasik says this process brought new insights faster than other methods and has led to design improvements to shield airplanes and better pilot training to recognize and react to bird strikes.[6]

You can borrow this framework to achieve similarly powerful results with your prototyping, especially on bigger and more complicated projects. Recruit a tools expert, someone with expertise in some aspect of visualization that you don't have. That could be:

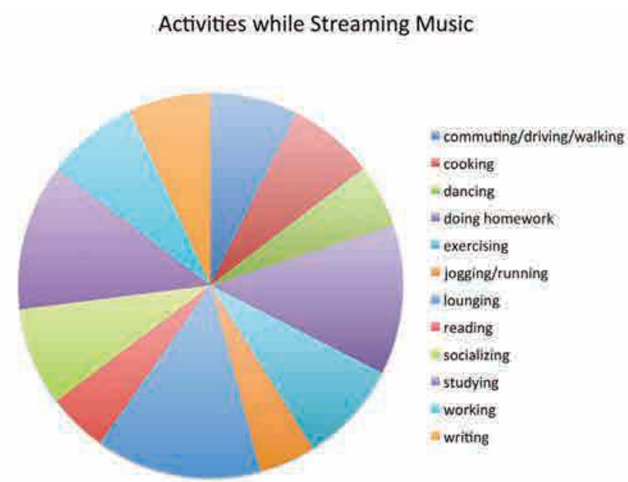
- **A developer** who can create interactivity with complex programs like D3—the most popular JavaScript visualization library for programmers
- **A designer** who can help you visualize a complex or unusual form using professional design tools like Adobe Illustrator
- **A data analyst** who knows how to find, scrape, clean, and manipulate data in business intelligence and visualization software systems like Tableau or QlikView so that you can find patterns and relationships that you'd otherwise miss.

Sit together. Describe to the expert what you're trying to achieve. Talk. (More talking!) Show the expert your sketches, the keywords you jotted down; clarify your ideas. Then begin the back-and-forth exchange. Even better, you can go through the whole development process with the expert.

IN PRACTICE, START TO FINISH

Here's an example of thinking through a visualization from beginning to end. Lisbeth is a marketing manager at a company that provides streaming music services. The company is trying to understand what other activities customers engage in while they're streaming music. Data collected by the company will help shape its multimillion-dollar marketing strategy.

Lisbeth has seen the data. She's even quickly generated a pie chart from her spreadsheet program, just to see at a high level what was there:



She knows that even a cleaned-up, well-labeled version of this pie won't be effective. She's having trouble herself extracting any meaning from it other than *users do a lot of different things while streaming music*. She blanches at the idea of presenting this to the marketing department as a visual aide to a multimillion-dollar investment decision. She decides to make it better.

Prep: 5 minutes. Lisbeth finds a small workroom with a whiteboard and a few color markers. She brings coffee for herself and a friend she has invited to help. She spends a few minutes framing her effort at the top of the whiteboard. In addition to plotting her work in the declarative, data-driven quadrant (everyday dataviz), she plots what will make this chart "good" on the Good Charts Matrix. Her sketches are shown on the following page.

Her chart should look good, but she's willing to forgo time refining the design to focus on getting the context as close to perfect as possible in the time frame she has. After all, she's presenting to her unit, where people will have deep knowledge and opinions on the topic and data. She makes a brief note that if this chart gets it right, she may have to make a better-designed one for other, more formal presentations.

Talk and listen: 20 minutes. Her friend—who's not part of this project—arrives. Lisbeth wants not only to talk through her idea but also to check her assumptions with someone who doesn't have much knowledge or bias about the project.

So, I need to show my department what people do while they're using our service. I want to be able to show any trends or dominant activities, for sure, but there are a dozen different things they do, and it all seems pretty random.

> Why can't you tell your team there's no dominant activity?

We can't market to everyone; we have to figure out who we want to target and know why. Plus I'm not convinced that it's trendless. I just think I haven't figured out the way to group the data that will expose the trends.

> Is there a category of activity, or a couple of categories you can focus on, like exercising?

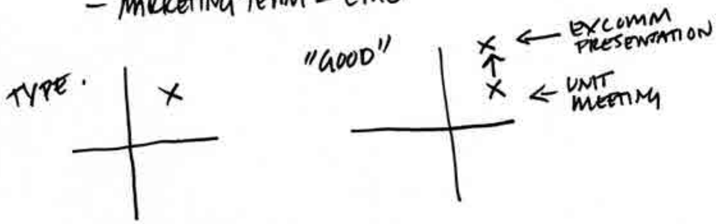
Maybe. Actually, the data wasn't grouped, but that's something to look at.

Their conversation goes on for about ten minutes. Later Lisbeth spends ten more minutes chatting up a colleague who'll be at the meeting. Here's part of what she says to this friend:

So the big meeting is coming up, and I know Tom is going to give me the "So what?" because it's not entirely clear we've found one or two dominant activities that we can focus on. Plus I get frustrated when we just put up these percentages and don't think about the individuals. You can't just market to some 60%

MARKETING STRATEGY

– MARKETING TEAM – CMO



WHAT'S MY SENTENCE?

SIMPLIFY?

FIND THE MAIN IDEA

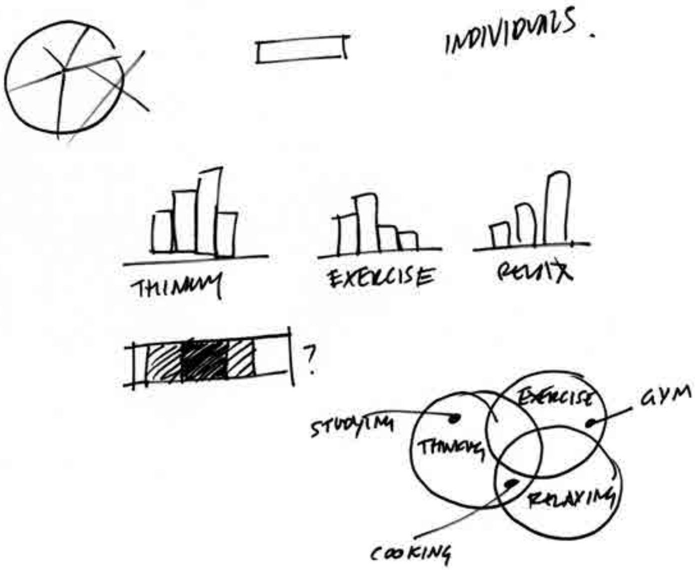
ACTIVITIES THAT DOMINATE

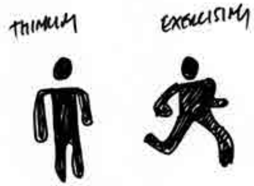
NOT PERCENTAGES – INDIVIDUALS/PEOPLE

ORGANIZE INFORMATION — TOO RANDOM

CATEGORIES

THINKING	ACTIVITY/EXERCISE	RELAXING
HOMEWORK	EXERCISE	SOCIALIZING
STUDYING	DANCE	COOKING
WRITING	COMMUTE	PARTIES
ETC.	ETC.	ETC.

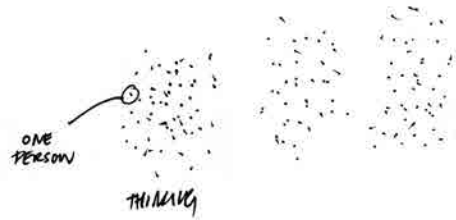




"IF TWITTER WERE 100 PEOPLE"

ACTIVITY OF 1000 / 10,000?

WHAT WOULD THAT LOOK LIKE?

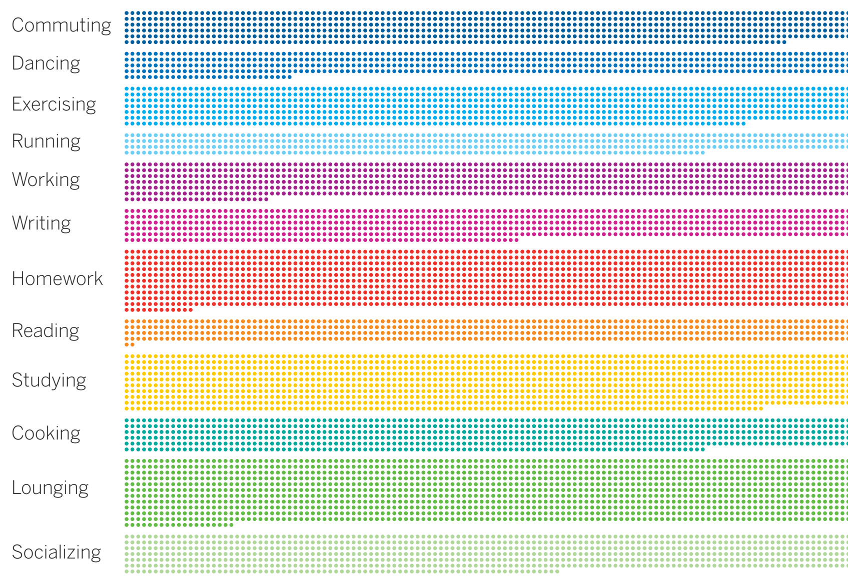


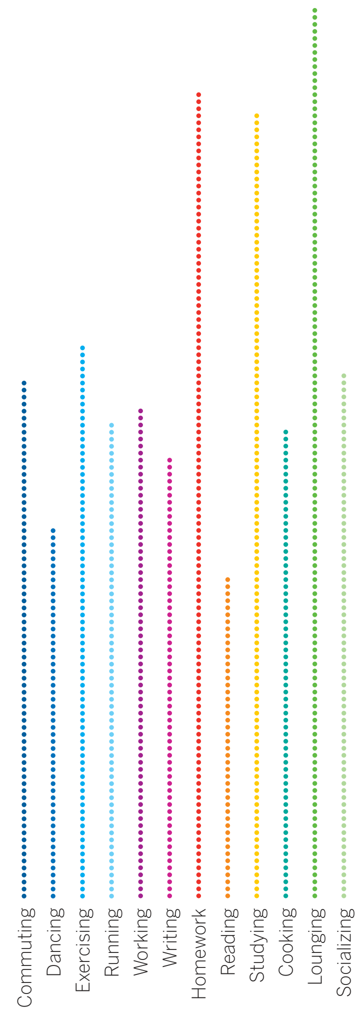
because they're all doing the same thing. You have to think about people. Anyway, I'm looking for ways to organize the information so it's not just random. But I also think it'll be more effective if we get people in a mindset to think about talking to people, not just these aggregate groups.

Sketching: 20 minutes. Even as Lisbeth captures the conversations, she starts sketching as shown on the facing page. She knows right away that grouping the activities in general categories will help make a pie chart more accessible, so she looks over the activities again and assigns each to one of three categories. Although she's pretty sure a pie chart won't work, she sketches one anyway. She sketches bars and tries out a Venn diagram, with circles for each category overlapping with some of the activities. She scribbles. The word *individuals* keeps staring back at her from the whiteboard. She really wants to make the information feel more personal and less like a generic stat. She draws a few icons of people, remembering a dataviz she found online that went viral called "If Twitter were 100 people," which used a similar technique.[7]

She writes, "Activity of 1,000/10,000? What would that look like?" And she jabs the whiteboard with dots. Could she put thousands of dots on the screen in her presentation? That form is called a unit chart—assigning visual units to some value, like a person or a dollar. It helps audiences to make that connection to the thing behind the data in a way that more abstract concepts like a percentage can't. When we see the dots, we think about the individual units, the people. When we see a bar, we think about the whole.

Prototyping: 65 minutes. Lisbeth likes the idea of a unit chart and recruits another friend who can do some light programming to create some, paired-prototyping style. In 30 minutes, they have multiple unit charts to evaluate—each showing proportional numbers of participants in activities—including the version below.

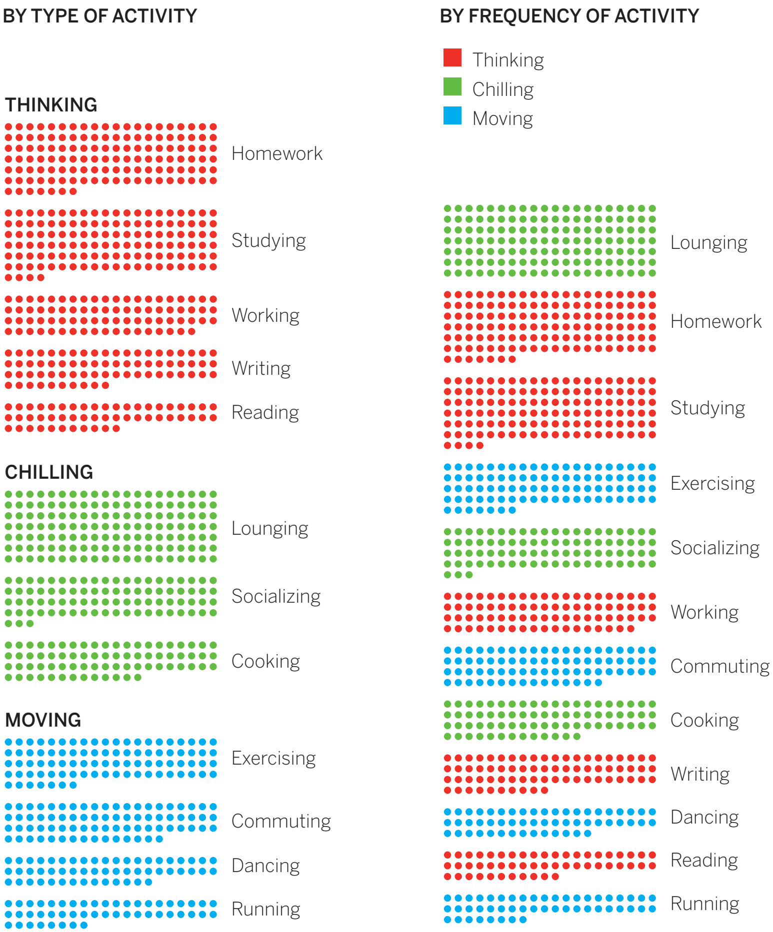




Lisbeth recognizes that 10,000 dots, although stunning, is somewhat impractical for a presentation. It's hard to see any values or differences in values in the picture. She asks her programmer to try versions with 1,000 dots. She asks him if he can "make it so the differences in value are easily seen." He iterates. They need just 15 more minutes to produce versions with 1,000 dots, including the two on this page.

Lisbeth likes the leftmost of these because the differences feel meaningful and the form feels familiar. Each group of dots evokes a group of people, but you can also easily see overall proportions within groups. It's a unit chart and a stacked bar chart at the same time. In just 20 more minutes—less than two hours after she started—Lisbeth has a presentation version of her chart, the pair on the facing page, which organizes activities both by category in one chart and by most-to-least-common in the other. She thinks, *These are visualizations we can have a conversation about.*

WHAT OUR USERS DO WHILE STREAMING



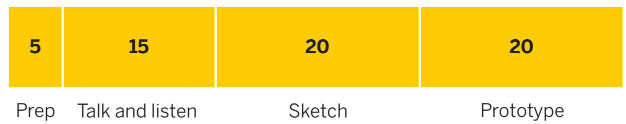
SOURCE: COMPANY RESEARCH

OVERLAPPING, NOT SEQUENTIAL

I've outlined a process that goes from one step to the next with fixed time intervals, largely because that's the easiest, most accessible way to describe the progression of activities.

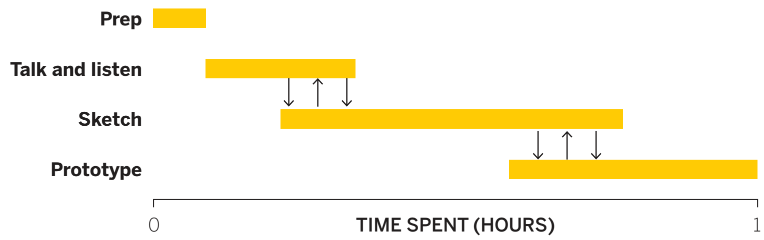
BUILDING BETTER CHARTS

MINUTES SPENT AT EACH TASK



In practice, the process won't be so linear. The steps will bleed into one another. You may find yourself sketching as you talk, for example. You should: It's hard not to start drawing as you capture keywords and talk through your challenge. Sometimes a prototype will expose a weakness in your visualization (or an opportunity you hadn't seen) that will literally send you back to the drawing board to sketch alternatives.

In short, the process may proceed something more like this, which shows how, in that typical hour, the steps might overlap:



Not every project is typical, though. When the best visual approach isn't clear, talking and sketching may dominate your efforts. Or, if you have a good idea of what you're trying to show, or if prototypes lead you to further manipulate the data to refine the idea, you may quickly settle on the visual approach and spend much more time refining prototypes. You can imagine the length of these bars stretching and shrinking, and the arrows between them shifting.

VISUAL CRIT

There's one final technique you should use as you develop your charts—constructive criticism. This can be part of your own chart development, to hone your product. It can also be done separately when you encounter others' data visualizations, so that over time, you learn about what you find appealing and effective.

Good writers are great readers. They look to others' work for ideas and borrow (okay, steal) from what inspires them. Creators in general approach their craft this way, and visualization is no exception. One of the best ways to get better at making charts is to look at, and think about, a lot of them.

Good news: There's a surplus available. It's hard to be on the internet for a hot minute without stumbling on some dataviz that's going viral. If you follow #dataviz or visit any number of visualization-heavy websites (the Upshot on the *New York Times* website, for example, or the *Economist*, which tweets many charts every day), you'll find plenty of fodder.

You can critique any chart and get something out of it. Find simple ones. Boring ones. Complex, artful ones. Ones on topics you know nothing about. Look at each one with a purpose. Do you get it? What do you like? What don't you like? Deconstruct technique. Think of ways you might have approached the chart differently. Recreate it in your own way.

This doesn't have to feel like homework. It can be done casually and quickly. Here's a way to learn from others' work or to take a fresh look at your own.

1. Make a note of the first few things you see. We see first whatever stands out. Document the first element your eyes focus on. A "spike"? "Blue bars"? It may be more impressionistic: "a long smooth line," or "pickup sticks crossing over each other all over the place." What you wouldn't see first is "interest rates going up in the past few fiscal quarters." That kind of content focus requires some parsing of the idea beyond what first hits the eyes. Here you want to get at that initial, instantaneous visual perception.

2. Make a note of the first idea that forms in your mind and then search for more. You've looked at the chart for a few seconds now. What is it trying to tell you? Here's where you might say, "It tells me interest rates are going up, and fast." Ask critical questions about this idea you've formed: "Does it match the chart's intent?" "Is the chart misleading or is something missing?" After your initial impression, study it; see if you can find deeper narratives, or if more questions arise the longer you look at it.

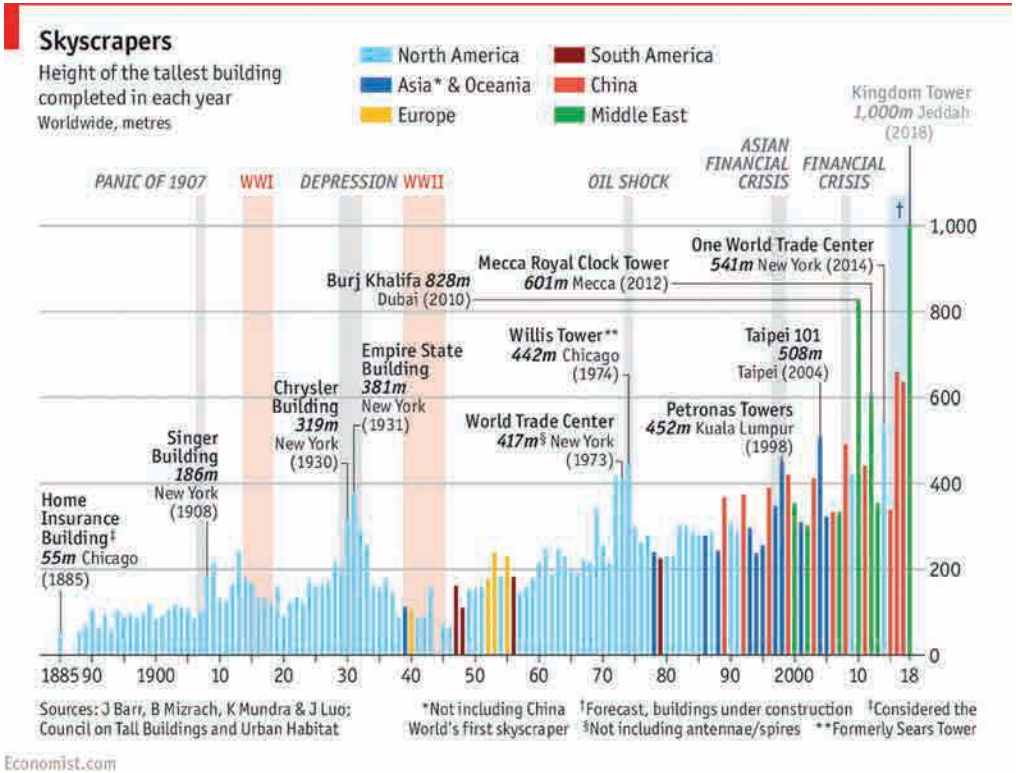
3. Make notes on likes, dislikes, and wish-I-saws. Focus on the feeling you get from elements in the chart. "I don't like all the labels." "I don't like how the y-axis is cut off." "I like how I immediately get it." "I like how they used gray for the background information." "I wish I saw this in comparison to last year." Sometimes these gut feelings signal what makes a chart successful; other times they expose what may be improved. Over time, you'll find that you react consistently to certain elements; you'll discover both common missteps and your own aesthetic.

4. Find three things you'd change and briefly say why. "Say why" is the crucial bit. Your reason should ultimately improve the chart's effectiveness. "Because I don't like blue" is thin reasoning. "Because the blue is hard to see with the yellow right next to it" is better. Limit yourself to three changes—that will force you to prioritize only the most important ones. The aim here is to focus on what will help the main ideas shine through.

5. Sketch and/or prototype your own version, and critique yourself. Revisualizing is the most powerful way to learn. The before-and-after comparison helps you see whether what you thought would make a chart better actually does. If you have a data set, great. Otherwise, create a simple spreadsheet with estimates of the key values, or just sketch and estimate the values. (If it's a conceptual visualization, you don't need anything; just start.) Value speed over precision here, as you do when you sketch and prototype your own dataviz. The self-critique will attach what you've learned about what works and what doesn't work to your effort. Try to include both positives and negatives in your self-critique.

VISUAL CRIT IN PRACTICE

Many people hesitate to do visual crits. Some may not feel qualified because they're neither a designer nor a data scientist. Others don't feel comfortable with criticism even when it's constructive. But as a consumer of the chart, you are qualified to have opinions about what it makes you see and feel. As for comfort with criticism, I tell people I do this with to avoid judgmental words about what you *like* and don't *like* and to talk instead on what you think *works* and *doesn't work*. The best way to get comfortable with visual crit is to just start doing it. Here's a simple example of one I performed using the step-by-step process we just laid out.



1. Make a note of what you see first.

- My eyes go right to the Chrysler Building and Empire State Building, maybe because I recognize them, but their bars also stand out on the left.
- I also see many stripes, with salmon-colored ones jumping out at me.
- I also see blue, until lots of colors take over when the bars shoot up.

2. Make a note of the first idea that forms in your mind and then search for more.

- All the tall buildings used to be in North America, and now they're not. I got that pretty quickly from the color. But if I'm supposed to be able to think about who's building tall buildings now, that's harder to see, because the colors are so various. It reads to me like North America and Everywhere Else, unless I work at it.
- There's an amazing surge in the height of the tallest buildings right now. It's hard to pick up, though, because heavy labels and lines and stripes denoting eras drown it out.
- Those stripes denoting important world events may be meant to tell me something about the height of buildings during that time, but the more I look at them, the more random they seem.

3. Make notes on likes, dislikes, and wish-I-saws.

Like	Dislike	Wish I saw
Thin lines, feels like skyscrapers, gives sense of great height	Era demarcations heavy and overpowering	Less stuff overall
Labeling important buildings	Pointers, y-axis grid heavy, labels redundant (year)	Catchier title?
Using color to denote location	Color choice makes it hard to quickly pick location in recent times, especially with the spatial disconnect between the key and the colors	Some point of reference for height, hugeness
Y-axis on right for easier reference of tallest buildings	Footnotes and symbols confuse me	Maybe shapes/profiles of buildings?

4. Find three things you'd change and briefly say why.

- Eliminate the demarcations for eras. It's not clear what they add, and they definitely make it harder for me to see the progression of tall buildings. It feels like it's inviting me to make a correlation that I can't find.
- Work on labels. Make them simpler so that they don't overpower the bars. No elbows in pointers. Make labels less intrusive in the visual field. Lighten the grid lines.
- Color. Find a way to make color more instructive at a glance. Combine China with Asia?

5. Sketch and/or prototype your own version, and critique yourself.

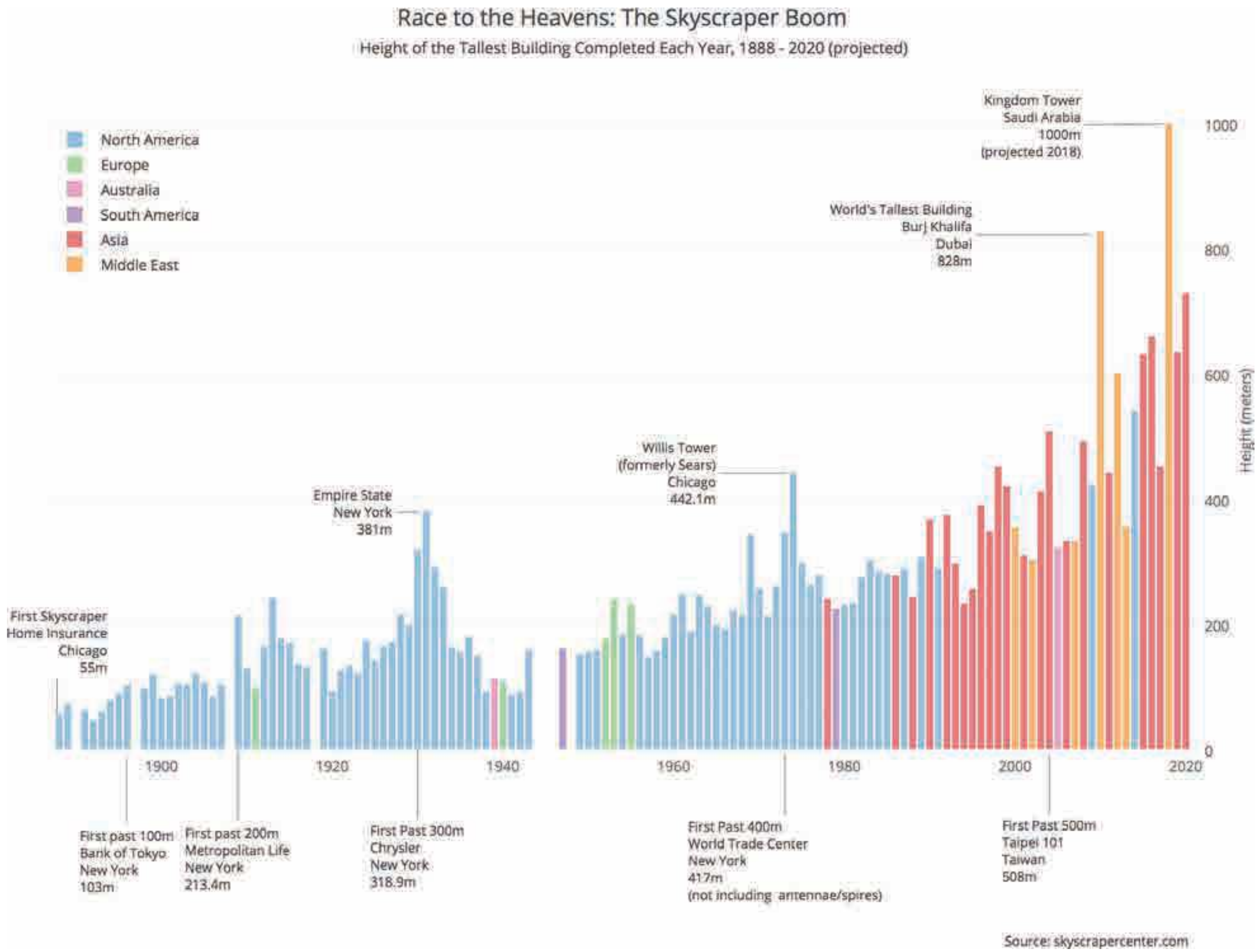
Self-critique: I like how my prototype (on the facing page) feels simpler and less cluttered. I don't miss the era demarcations. A key breakthrough was dividing the labels into new tallest-building milestones running neatly along the bottom, aligned, while well-known landmarks stay in the visual field. This helped solve the busyness of the labels. Also, the labels in the field escalate in an echo of the visual itself. Removing some belt-and-suspenders design with the labels also helped. Looking at the y-axis tick lines, I had an idea to align them to the landmark buildings rather than make them traditional, evenly divided ranges. That would be unconventional but might help add to the sense of accelerating height.

I haven't addressed the problem that some of these buildings are "projected heights" rather than completed structures. And I don't think I've solved the color puzzle at all. I want to ask a professional designer what she would do to make so many lines that require so many colors render as clearly distinct. I'd love to not have a key at all. Finally, I wish the chart included some point of reference to get a sense of just how astonishingly high 1,000 meters actually reaches. Could it be included without cluttering?

Note that this is only a prototype. My goal wasn't to create a clean, finished, good chart, but rather to learn from dissecting the original and just navigating to some ideas around a better solution. That's enough for this crit. I could keep going, but I'll stop this one here.

Inevitably when people try this process, they tell me it feels strange at first but that it does work for them. I've had some people tell me how surprised and delighted they were when they first started hearing the visual words they were saying out loud. One person told me it felt like a "cheat code" because they kept figuring out how to better show their data just by talking about it.

The talk, sketch, prototype process will get you to good charts by getting you to a better, more specific, and clearer context. A by-product of great context is good design. Yes, there are design techniques to


Race to the Heavens: The Skyscraper Boom
Height of the Tallest Building Completed Each Year, 1888 - 2020 (projected)
North America
Europe
Australia
South America
Asia
Middle East
Kingdom Tower
Saudi Arabia
1000m
(projected 2018)
World's Tallest Building
Burj Khalifa
Dubai
828m
Willis Tower
(formerly Sears)
Chicago
442.1m
Empire State
New York
381m
First Skyscraper
Home Insurance
Chicago
55m
First past 100m
Bank of Tokyo
New York
103m
First past 200m
Metropolitan Life
New York
213.4m
First Past 300m
Chrysler
New York
318.9m
First Past 400m
World Trade Center
New York
417m
(not including antennae/spires)
First Past 500m
Taipei 101
Taiwan
508m
1900
1920
1940
1960
1980
2000
2020
0
200
400
600
800
1000
Height (meters)
Source: skyscrapercenter.com

improve your charts—and we'll be covering those next—and yes, practicing visual critique will further hone your ability to make good-looking dataviz. But most of what will make an audience regard your charts as good looking and effective is not down to design skills. It's down to context. A chart that knows its context well will naturally end up looking better *because* it's showing what it needs to show and nothing else. Good context begets good design.

Good charts are only the means to a more profound end: presenting your ideas effectively. Good charts are not the product you're after. They're the way to deliver your product—insight.

RECAP

BETTER CHARTS IN A COUPLE OF HOURS

To improve visual communication, fight the impulse to go right from getting data to choosing a chart type from the preset options in a software program. First spend time creating context and thinking through the idea you want to convey. Usually, an hour or so of prepping, talking and listening, sketching, and prototyping will help produce a superior visualization.

Follow these steps to make it happen:

1. **Prep: 5 minutes**
   - Create a workspace with plenty of paper or whiteboards.
   - Put aside your data so that you can think more broadly about ideas.
   - Write down the basics as constant reminders, including who the visualization is for and what setting it will be used in.

2. **Talk and listen: 15 minutes**
   - Enlist a colleague or a friend to talk about what you're trying to say or show, or prove or learn.
   - Capture words, phrases, and statements that possibly sum up the idea you want to convey.

3. **Sketch: 20 minutes**
   - Match keywords you've captured to chart types that you may try out, using the chart on 94 as inspiration.
   - Start sketching, work quickly, and try out multiple visual approaches. Sketching is *generative*.

4. **Prototype: 20 minutes**
   - Once you have an approach you think will work, prototype it by making a more accurate and detailed sketch.
   - Refine your prototypes just enough to see what you need. Prototyping is *iterative*.
   - Use digital prototyping tools or paired-prototyping techniques if you want to iterate further.

VISUAL CRIT

Just as good writers are great readers, good chart makers are great at mining visualizations for inspiration and instruction. One of the best ways to get better at making charts is to critique them.

First pick out some charts to evaluate. Pick all different kinds. Simple ones. Boring ones. Complex ones. Ones you know nothing about. Then follow this simple process for critiquing and workshopping them:

1. Make a note of the first few things you see.
Don't think—react. What stands out? Is it a peak? A color? Lots of words?

2. Make a note of the first idea that forms in your mind and then search for more.
Decide what idea you think is being conveyed. Does it match the chart's seeming intent? Is the chart misleading? Is something missing?

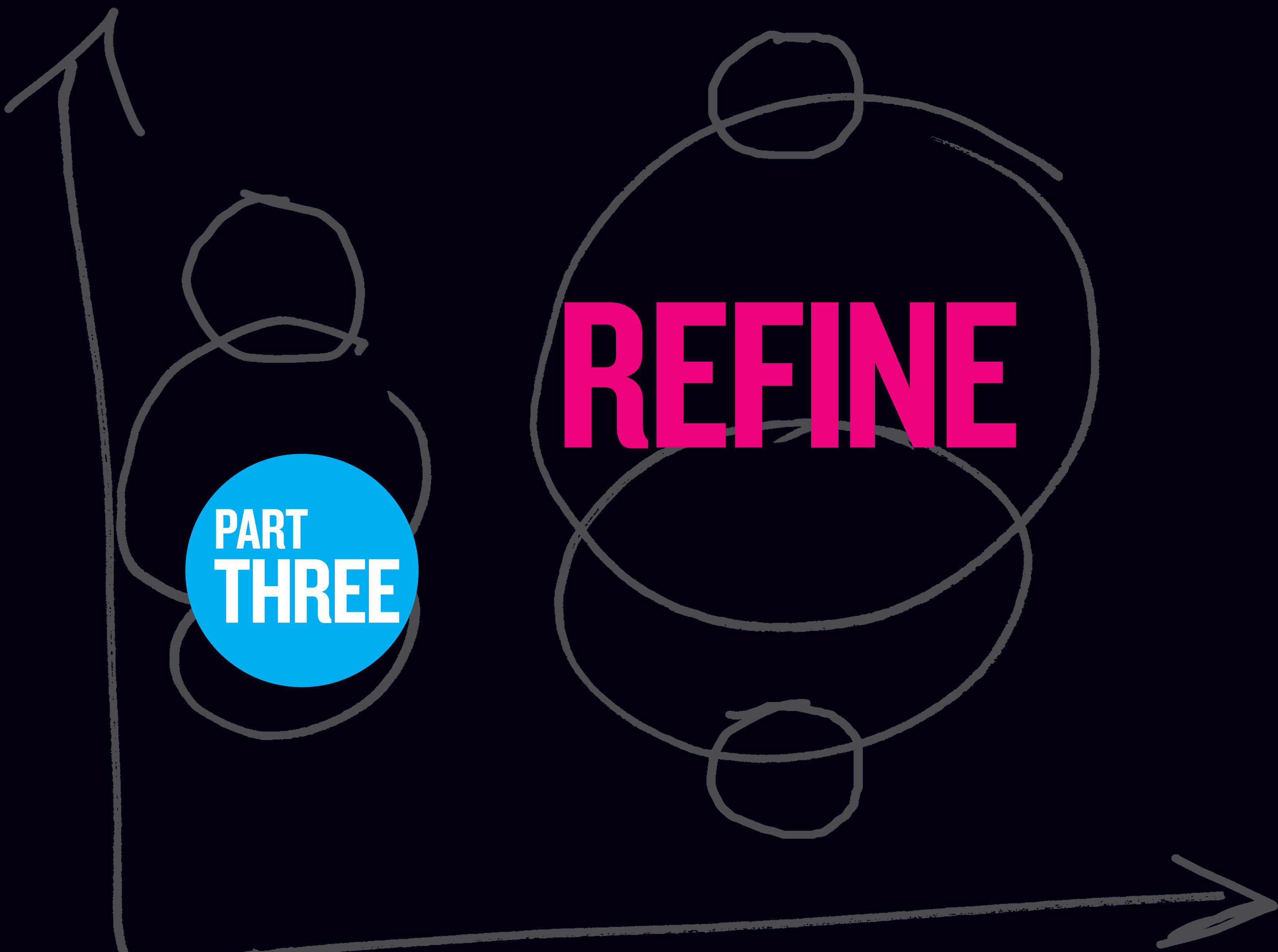
3. Make notes on likes, dislikes, and wish-I-saws.
Don't focus on what you think is *right* or *wrong*. Instead, think about your gut reaction to the visual, the feeling you get.

4. Find three things you'd change and briefly say why.
Limit them to three so that you're forced to prioritize only the most important changes. Saying "why" is key to making sure you focus on effectiveness rather than taste. "Because I don't like blue" is not a good reason to make a change. "Because it's hard to see blue next to yellow" is.

5. Sketch and/or prototype your own version, and critique yourself.
Just as when you sketch and prototype your own dataviz, value speed over precision here. Include both positives and negatives in your self-critique.

PART THREE

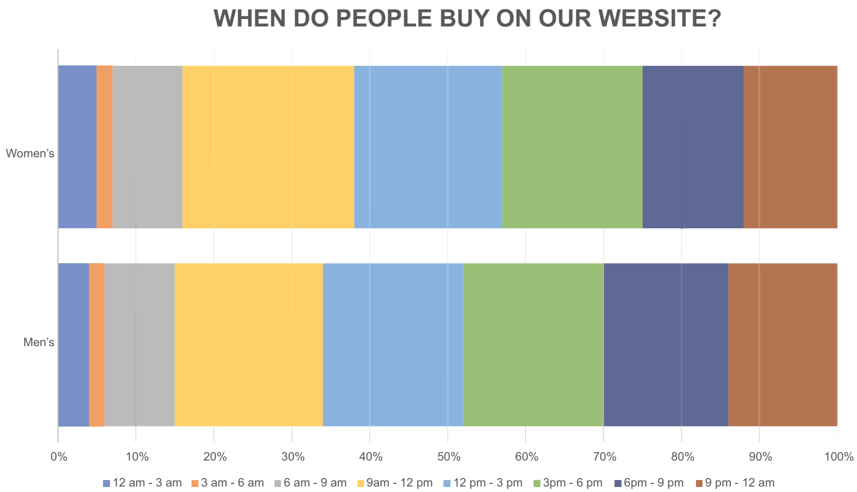
REFINE

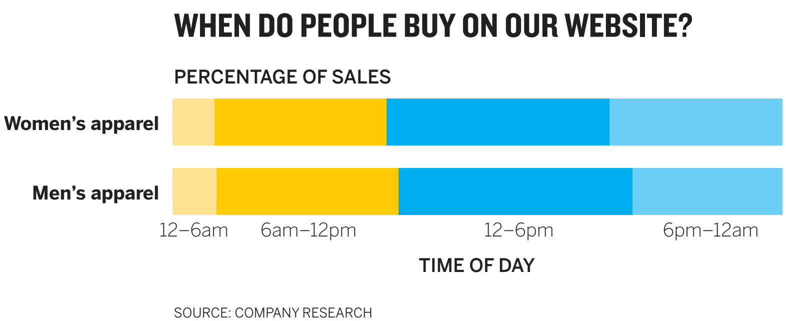


CHAPTER 5

REFINE TO IMPRESS AND PERSUADE

GETTING TO THE “FEELING BEHIND OUR EYES”





WHICH OF THESE visualizations is a prototype and which was created for a presentation to the CEO?

The top chart is obviously the prototype, generated in Excel with just a few clicks. Most of us would say the bottom chart (designed in Adobe Illustrator) looks better, is "airy" or "streamlined" or "clean," whereas the top one is "busy" or "blocky" or "messy."

In that great book on writing that I've referred to before, *Style: Toward Clarity and Grace*, Joseph Williams describes impressions of good and bad writing as "a feeling behind our eyes."[1] Charts get behind our eyes in the same way, and it's important to understand why, and what design principles and tactics lead us to have bad feelings about the first chart and good ones about the second.

It can't be reiterated enough that those good and bad feelings aren't a function of which chart is prettier. Aesthetic value is a by-product of *effectiveness*. For example, look at the charts on this page again and try to answer these questions:

- Do more people buy women's apparel before or after noon?
- Does the site get more buyers before breakfast or after dinner?

The charts are the same type and contain the same data, but the design of the second one is *easier to use*. Good design serves a more important function than simply pleasing you: It helps you access ideas. It improves your comprehension and makes the ideas more persuasive. Good design makes lesser charts good and good charts transcendent.

"THE FEELING BEHIND OUR EYES"

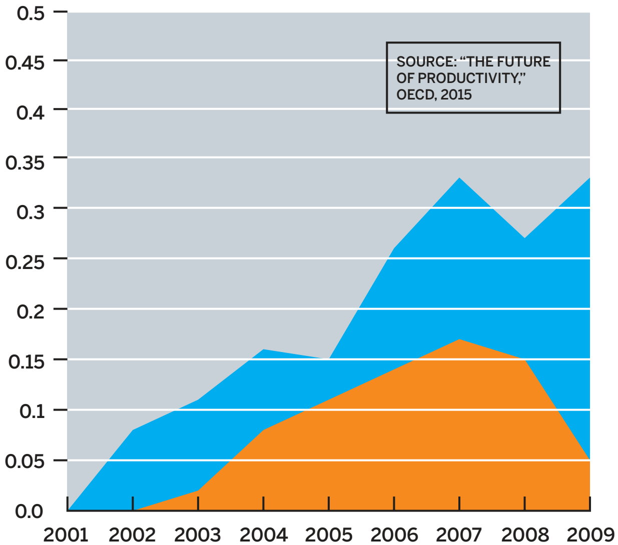
But just as Williams would tell you that following some rules (many of which are arbitrary) can't alone lead to good outcomes in writing, creating and strictly adhering to a list of rules for designing charts similarly won't work.

Another metaphor will help what we'll do in this chapter: music theory. It does not dictate what you can and can't do when you make music. Instead, it explains why you might feel what you do when music "hits your ears." It can suggest techniques that lead to typically good outcomes and why some techniques sound "bad" or "off," but it never suggests you're not allowed to do that thing that sounds off. In short, music theory is not *prescriptive,* it's *descriptive*. It gives a common language to the feelings music creates.

We'll approach dataviz design the same way. I won't be telling you what specific colors to use, the right number of tick marks for your axes, or where to put your key. The answer to all those questions is, of course, "That depends on your context."

Instead of prescribing rules and procedures, I'll try to describe the design principles that lead you to have certain feelings about charts—why it looks "clean" or "chaotic." Why you might "get it" right away, almost without thinking, or why you might feel like you don't know where to start. Why one feels persuasive and another unmoving.

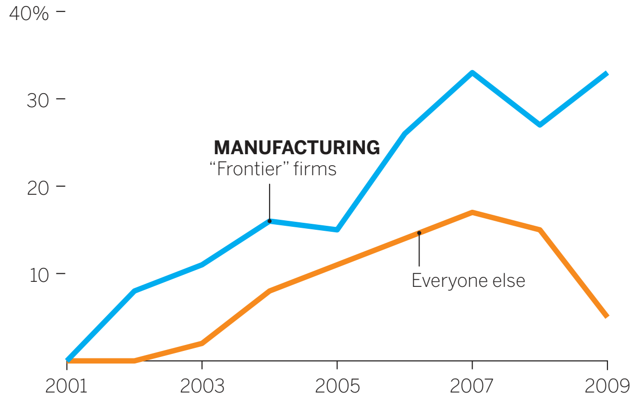
By understanding some principles around three concepts—structure, clarity, and simplicity—you'll be well on your way to upping your design game with your charts.



DIFFERENCE IN LABOR PRODUCTIVITY LEVELS FROM THEIR 2001 VALUES FOR "FRONTIER FIRMS" AND EVERYONE ELSE

THE GAP BETWEEN THE MOST PRODUCTIVE FIRMS AND THE REST IS GROWING

PERCENTAGE DIFFERENCE IN LABOR PRODUCTIVITY LEVELS FROM THEIR 2001 VALUES (INDEX, 2001=0)



SOURCE: "THE FUTURE OF PRODUCTIVITY," OECD, 2015

Structure. When the feeling behind your eyes is that a chart is "clean" or "crisp" or "orderly" or alternatively "messy" or "muddled" or "chaotic," much of that feeling is coming from how the chart maker thought about the chart's *structure*. The chart on the right on this page looks cleaner and more *professional*, even if we're not sure why. Here are the techniques that give us that impression:

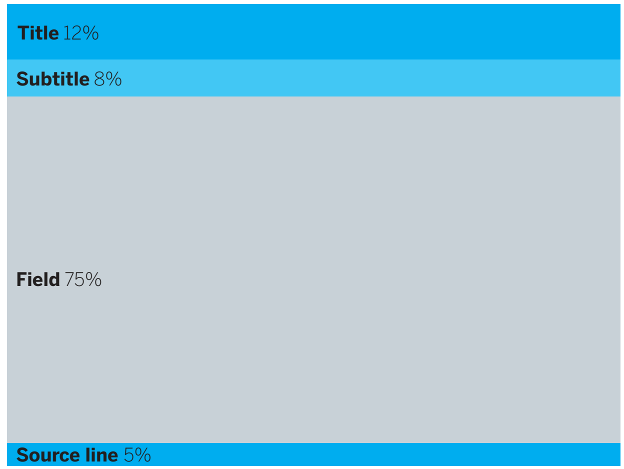
Consistent hierarchy. Generally well-structured charts include three essential elements, placed in a reasonably predictable vertical order:

Title (and sometimes a subtitle)
Visual field (visuals, axes, labels, captions, legend)
Source line

You should be able to map those elements onto any well-designed declarative chart.

How to design each of these elements and where to place them comes later. For now, just take an inventory. It may seem basic, but sticking to this consistent structure will be useful. Regularly including all those elements makes charts portable, reusable, and sharable. Your boss may want to put it in a presentation he's making for the executive committee, and he can do so with confidence that it won't raise questions he can't answer about what an unlabeled axis represents. The social team can put it on the company feed. If you want to reference this chart months or years later, its provenance won't be in doubt because you've included a source line.

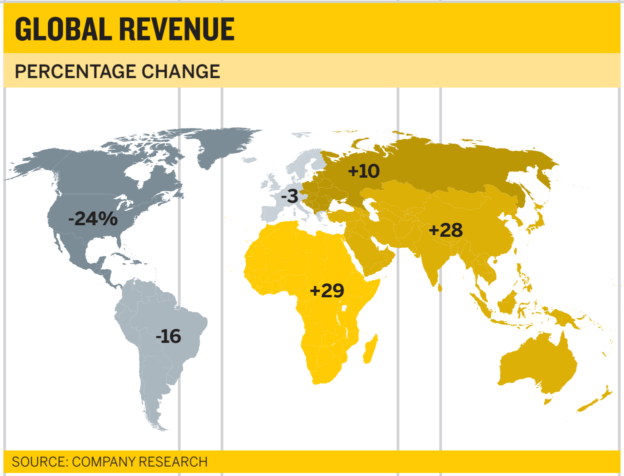
Consistent placement and weighting of elements. The structure outlined above is so common in chart making that you hardly notice it. It disappears into a convention we're all used to seeing: The title, for the most part, sits atop all, directly over the subtitle, which precedes the visual field. Sourcing is a small text line at the bottom. In the visual field, axes tend to be bottom and left, and legends often rest on the right side or in another vacant part of the field where they won't disrupt the visual. Regardless of the shape, most charts' proportions are divided up in about the same way as shown in the diagram on this page. The visual field should dominate the structure. The other elements serve the visual. Remember, we don't read charts the way we read words. Your audience's eyes go to the picture first, not the title, but you don't want to lose elements that will help your audience make sense of the visual space. Compare the first pair of charts in this chapter to see the difference between elements such as titles, keys, and labels that aren't well proportioned and ones that support and enhance the visual. Notice how you naturally move between the elements to confirm what you're seeing in the chart rather than struggle to make the connection or hold the information in your head that you need to make sense of the chart.

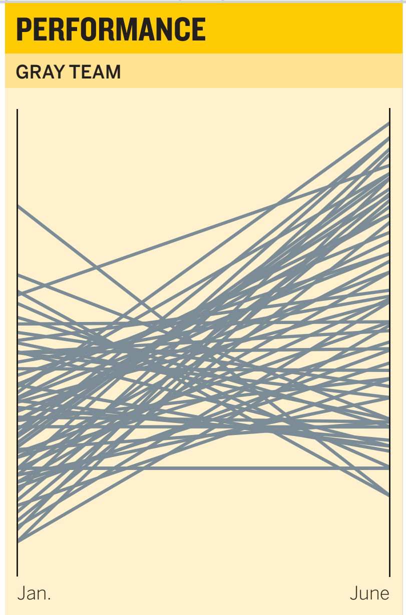


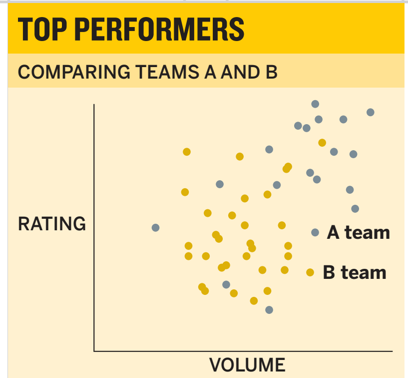
Well-designed charts we see every day are structured this way, regardless of if they're presented horizontally, as in a presentation, vertically on a phone screen, or as a square in a social media feed. (See the following page for examples of all three orientations.)

Don't go measuring charts to get your space allocation just so; use these proportions as a guideline.

This is also a good basis for building chart templates; being consistent with your sizing and placement of elements over the course of several charts adds to an audience's sense of orderliness and professionalism as well.

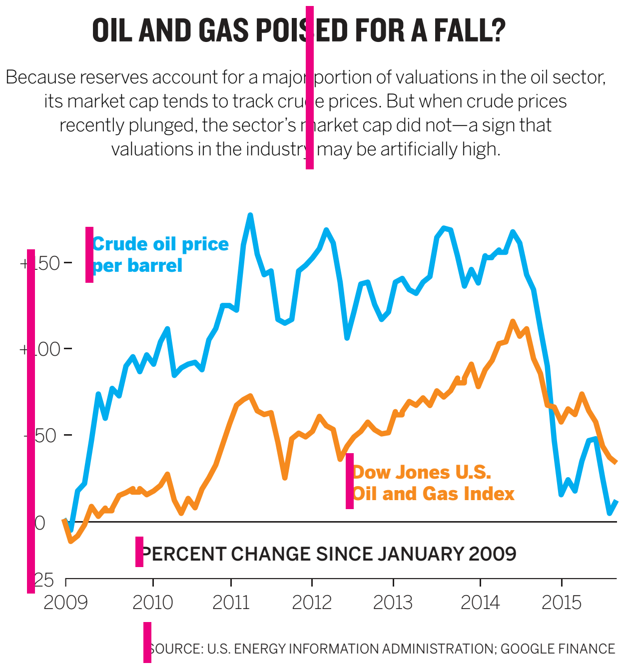


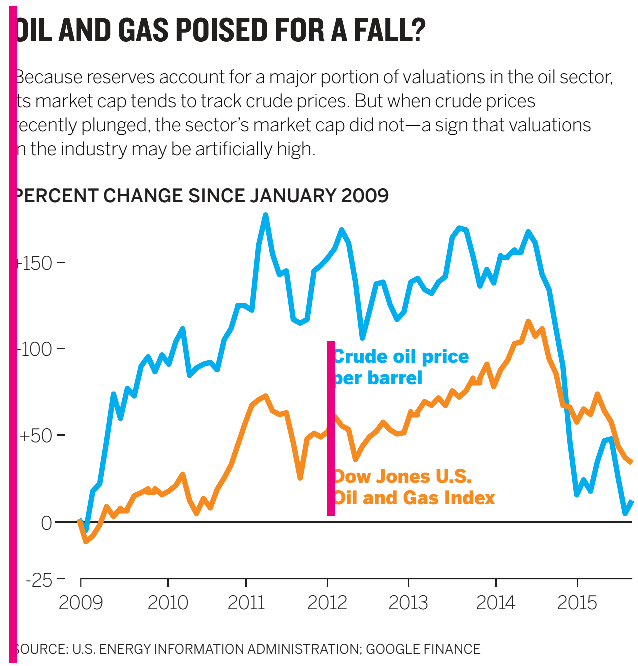




Consistent alignment. Professional designers see the world in grids. They divide their design spaces into evenly sized and evenly spaced columns and rows. When you see something that you sense is well designed or professional looking, that's *neat*, part of that feeling comes from the fact that it was designed with a grid system. This book, for example, looks smart because it's built on a grid, which is being revealed on this page.

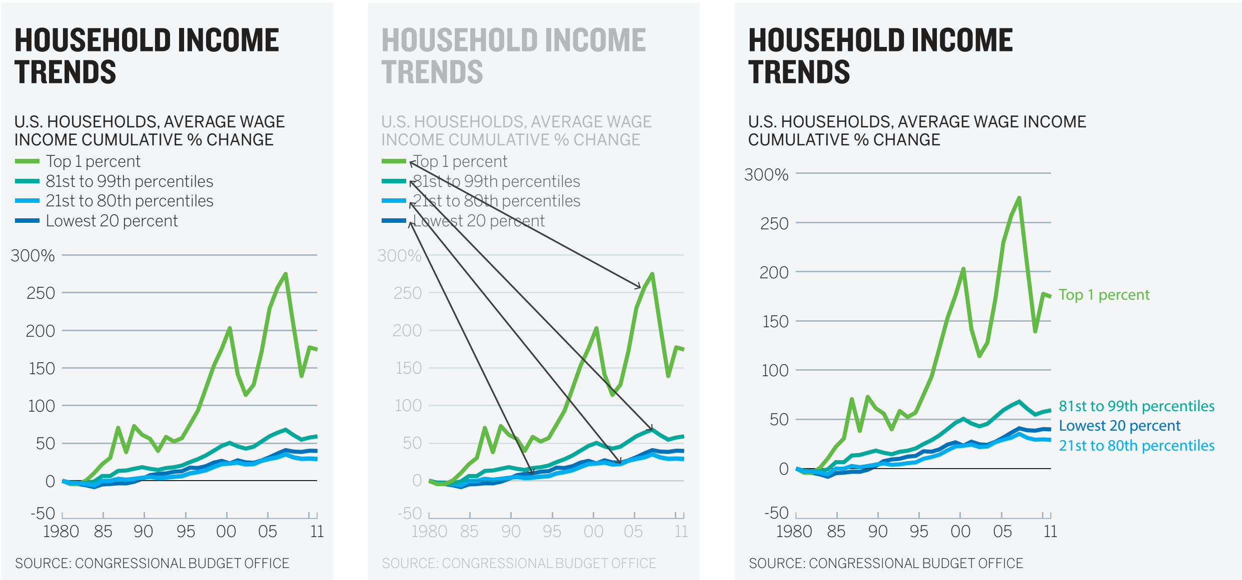
This book's grid is quite complex. Well-aligned charts don't need such sophisticated grids. They will use as few points of alignment as possible because more discrete alignments make charts feel busier. Adding center justification to a title, for example, creates multiple alignment points for elements that could share one. Unaligned labels in the visual field create a sense of haphazardness. The title, subtitle, and legend, for example, could all align to a single, left reference point. The difference in the sense of orderliness in the two Oil and Gas charts is plain, and you can see why when you mark their points of alignment. The one on the left has six; the one on the right has two.





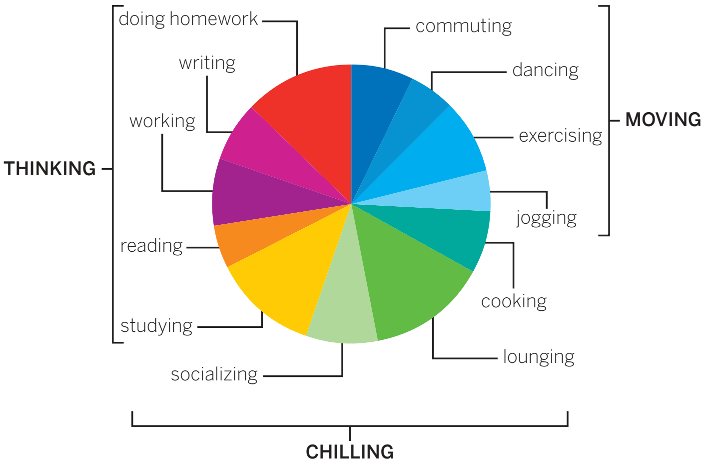
Do you need a grid system for your visuals? Many charts already have one: the axes. They are valuable guides that you can use as baselines for your labels and other elements. But it's good practice to look for elements that are floating to see if there's an opportunity to align them to other elements. And look to align elements between charts or on presentation slides as well. Many people are surprised at how much cleaner their charts look from adjusting alignment alone.

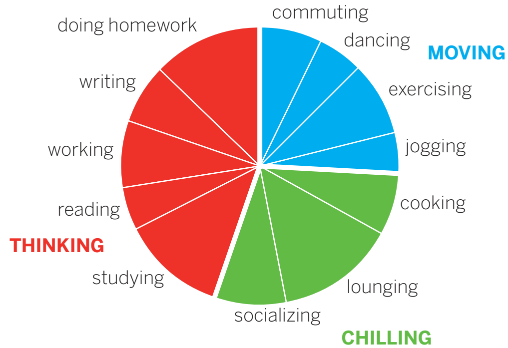
Limited eye travel. Keeping elements that work together proximate also supports a clean structure. Keys and legends, for example, can force a lot of back-and-forth eye travel to match values with visual elements. They also force the user to hold the values in their minds as they connect key to visual. Still, keys and legends are useful and sometimes necessary, but it's often best to connect values directly to their visual counterparts.



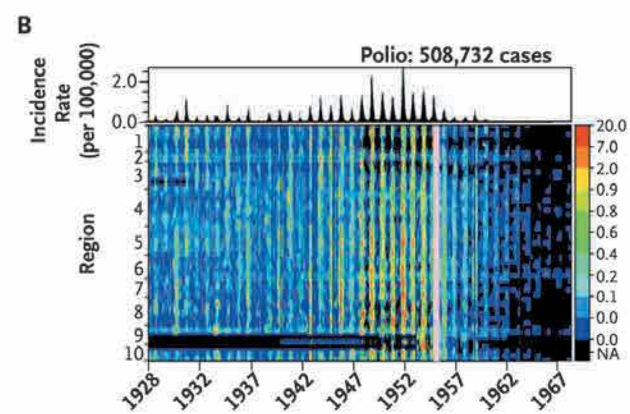
The last Household Income Trends chart *feels* simpler. Your eye travels across the visual and arrives at the label, which itself is color coded, connecting the elements more naturally than a key can. Compare that to the work it takes to try and learn the variables in the first chart, where you have to dart eyes back and forth between the key and the visual.

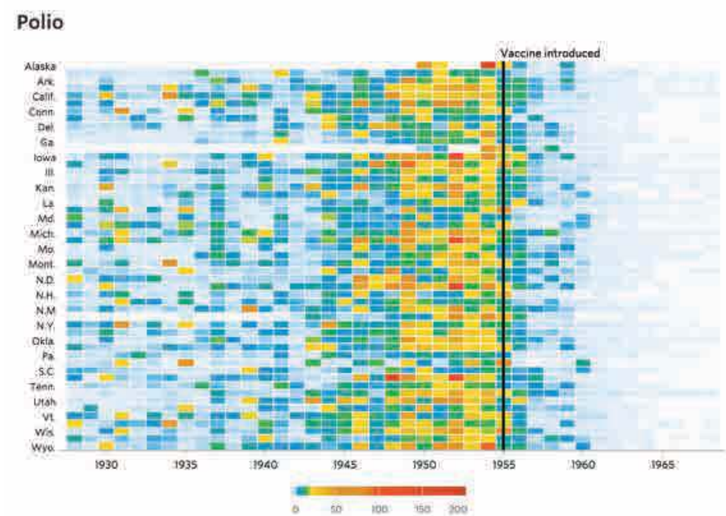
Another way to limit eye travel and keep the structure of charts neat is to make pointers and other marks as short and straight as possible, or even eliminate them altogether. Curves and elbows in lines pull your focus away from more-important elements. And the further away the label, the harder it is to connect it to its visual counterpart. Compare these two pie charts:





Clarity. Does the chart make sense to you, or are you stuck wondering what you're supposed to see? You may have experienced what the data visualization pioneer Kirk Goldsberry calls a "bliss point"—that *Aha!* moment when a visualization instantly and irresistibly delivers its meaning to you in a way that feels almost magical, as if it required no effort on your part. Such moments come from a design that achieves *clarity*. Which of these charts sparks that bliss point?





The chart on the bottom right of the previous page is an astonishing achievement in clarity, part of a set designed by Tynan DeBold, of the *Wall Street Journal*. The chart on the bottom left, which delivers the same information, was presented in the *New England Journal of Medicine* for a specialized audience. It's a good chart for its context, but does not achieve the same effect. How does DeBold's chart achieve such clarity?

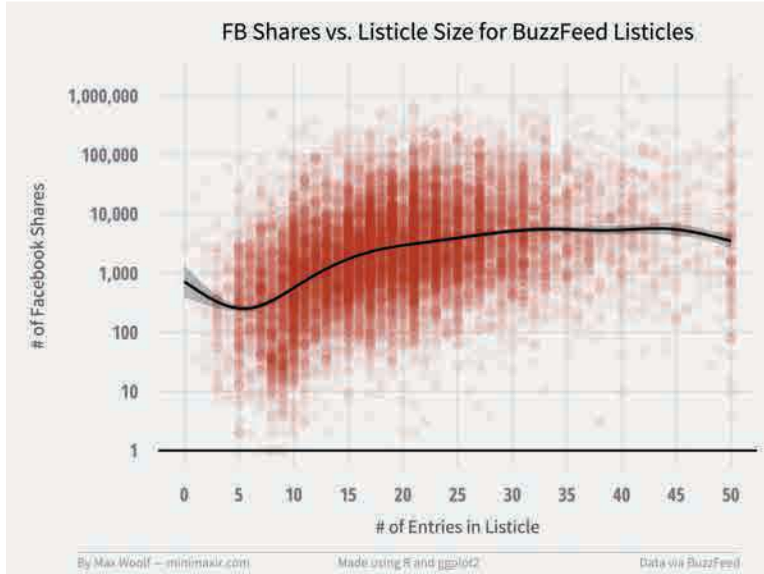
Nothing is extraneous. Other than labels, only three words accompany this visual, yet it's instantly understandable. DeBold's restraint is remarkable. For example, he doesn't add a "States" label to the y-axis, or "Year" to the x-axis, because we don't need those words to understand the labels. He even goes so far as to omit "Cases" from the title. (The chart ran as part of an article that briefly notes before a series of graphics like this that they represent "cases per 100,000 people," but even without that the meaning is clear.) Admittedly, this is an extreme example. But it serves to illustrate how clarity can be achieved by removing nonessential information.

Each element is unique and serves the visual. DeBold's chart contains seven elements: title, x-axis labels, y-axis labels, legend, visual, line of demarcation, and caption. Each one does a job that none of the others does. There's zero redundancy.

Most charts aren't so purposefully clear. They lack clarity because elements are used to describe the chart's structure rather than support the idea being conveyed. Titles or subtitles repeat axis labels. Captions describe what the visual shows. These are signs of a chart that plots data but isn't advancing an idea as well as it could, or a chart maker who lacks confidence that the visual can convey the idea on its own.

Supporting elements that have a finer purpose—that augment rather than just repeat—enhance clarity. Start by making sure elements serve to describe the chart's *idea* rather than its structure. Think of a piece of music: Which title helps you understand the *idea* behind it better: Concerto No. 4 in F Minor or *The Four Seasons: Winter*?[2]

One way to create clarity is to make the title or the subtitle the question that the visualization answers. Go back to our core question: *What am I trying to say or show?* This Facebook/BuzzFeed chart to the right is an excellent visualization. But would you more quickly understand what it's showing if its title and subtitle were, say:



Finding the Sweet Spot
How many items make listicles go viral?

The actual number of shares on the y-axis—what the original title refers to—is the data that helps show the *idea* of virality. This new title refocuses viewers on that idea, helping them get to what they're looking for more quickly. The words in the title give deliberate clues: "Sweet spot" prompts us to "find" the active region in the visual field. Convention tells us that a sweet spot will be active, positive, dense, so we make a connection between the deep red blotch we see first and the title.

If, instead, we wanted people to notice what types of articles don't go viral, we could change the text:

Viral Dead Spots
Listicles get shared less when they include too many or too few items.

Same visual, completely different effect on viewers as they check the visual and then reference the title and subtitle for context. One final note: The original title of this chart isn't always a bad idea. Sometimes you want a more objective or passive tone that simply describes the data. (This is especially true for analysts, who are meant not to make judgments on the data but only to show it.) As ever, knowing the context is key.

It's unambiguous. If you were quickly approaching the intersection where this sign is posted, and you had to get to Cambridge, would you be able to get in the correct lane in time?

Its ambiguity is paralyzing. Instead of using the sign to guide you, you have to take time to assign meaning to the sign itself. You're forced to slow down, shift your focus from driving to thinking about the sign, while trying to continue moving forward. You might feel your mind racing, or you might get panicky. Maybe people are beeping at you. It's stressful.

Ambiguity in visualizations generates a similarly stressful effect (without the beeping). We approach a visual at speed, prepared to parse it quickly, and then ambiguous elements force us to stop, refocus, and think about the visual and how it's built rather than the idea. In DeBold's polio graphic on page 133, there's no way to misinterpret any element. Compare that with the medical journal version, in which the legend is vertical, snug against the heat map. Is it an axis? What about the lavender line? It's unlabeled. What does it mean? How does the small line chart above the chart relate? Why are there seemingly *three* y-axes? We're stuck reading the sign instead of using it to get where we're going.

It doesn't flout metaphors or conventions. DeBold's polio chart uses colors in a way that our brains swiftly grasp: Red is more intense, blue is less so. He has created a low-res heat map that plots 2,250 data points (50 states by 45 years). But he's done something clever: He's tacked on a blue-to-pale-gray gradation at the low end of the scale, desaturating the blue until it's nearly colorless, or "empty," at zero. With that he has tapped into another convention we're used to: Less color saturation equals less value.

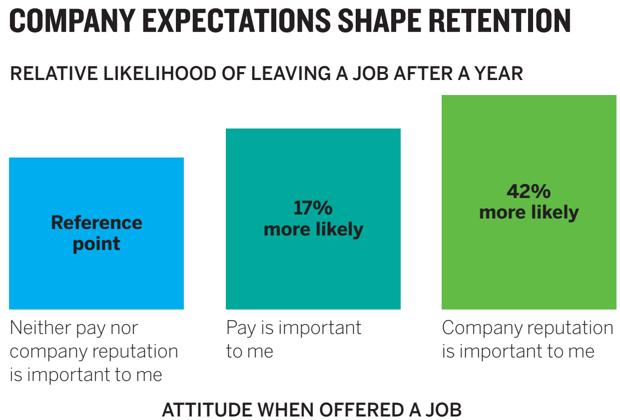
These two conventions combine to create the stunning effect of polio's literally *disappearing*. Compare this with the journal version, in which midnight blue equals zero. The disappearing effect is there, but dark blue transitioning to a darker blue doesn't feel as powerful or immediate. It could just as easily convey full saturation. The lavender line of demarcation for when the vaccine was introduced is harder to see. It doesn't elicit a before-and-after narrative as effectively.

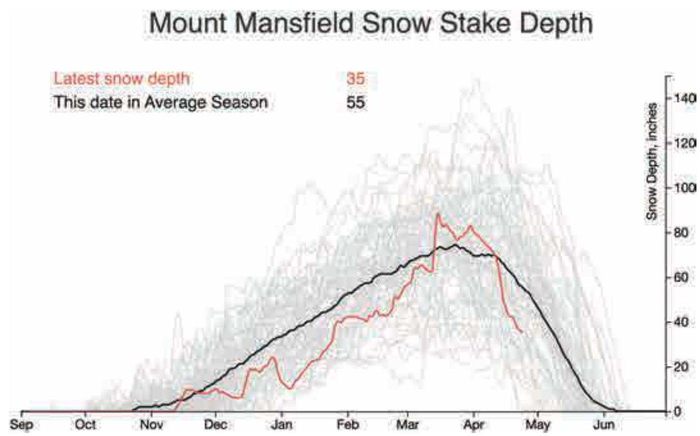
Remember all the conventions lodged in our brains as heuristics from chapter 2: North is up. Red is hot. Time goes left to right. Strong designs do not upend these conventions unless there's a good reason for it.

Simplicity. Does the chart look airy, elegant, and pleasing, or cluttered, busy, and complex? Do you naturally know where to look or do you spend time figuring out what to focus on? The sense of spaciousness, minimalism, beauty, or lyricism we may feel when we see a dataviz comes from its *simplicity*.

Clarity and simplicity are related but subtly different. Clarity concerns effective communication: Does the idea come through? Simplicity focuses on effective presentation: Are you showing only what's necessary for the idea to come through? When both are achieved, they hold together like a binary star system, serving each other. Simplicity contributes to clarity, and clarity enhances the sense of simplicity.

But simple isn't always clear, and clear doesn't have to be simple. Which of the two charts below takes longer to understand?





Although the chart on the top of the previous page is *simpler*, it probably took longer for you to understand—if you understand it at all. It's less clear. The labels fight with their visual counterparts. How much value does the first bar have? Why are there no values on the y-axis? Why are the bars different colors? If the middle bar represents 17% more likely, how can the only slightly larger third bar represent 42% more likely? (In fact, the bars represent some chance of leaving a job that we don't know, because it's not shown; the second and third labels represent the difference in height between compared to the first one.)

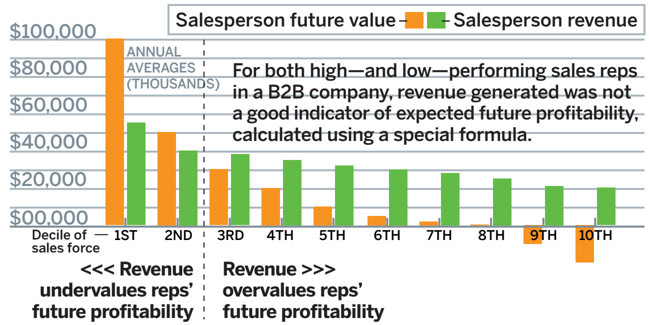
The chart on the bottom of the previous page is not nearly as simple. It plots nearly 70 trend lines each across 365 x-axis points (one for each day of the year). Still, the point of it is absolutely clear. It uses color effectively. The title and labels are unambiguous.

We tend to think of simplicity as the absence of stuff—that if we just keep taking away more and more information, we'll achieve simplicity. That's true to a point. But excessive simplicity leads to a lack of clarity. What you really need to think about is *relative simplicity*—how little you can show and still convey your idea clearly. Follow the maxim usually attributed to Einstein: "Everything should be made as simple as it can be, but not simpler."[3]

Which of the charts about sales rep performance on the right is simpler?

A REP'S PAST PERFORMANCE DOESN'T PREDICT FUTURE PROFITS

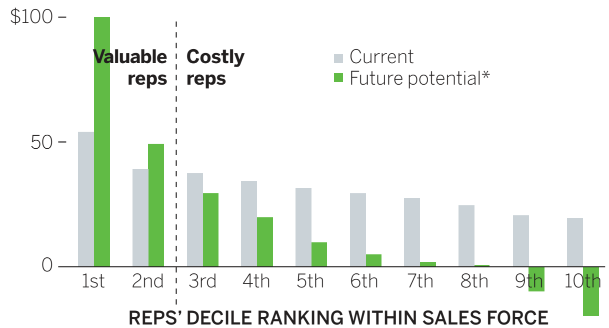
By looking not just at the revenue reps have generated but at their future profitability, you may find that your top performers are even more valuable than you thought—and your low performers even more costly.



SOURCE: V. KUMAR, SARANG SUNDER, AND ROBERT P. LEONE

A REP'S PAST PERFORMANCE DOESN'T PREDICT FUTURE PROFITS

SALES REPS' AVERAGE ANNUAL REVENUE (IN THOUSANDS)



*CALCULATED USING A PROPRIETARY FORMULA.

SOURCE: RESULTS FROM A STUDY OF ONE B2B COMPANY BY V. KUMAR, SARANG SUNDER, AND ROBERT P. LEONE

The chart on the top looks final and reasonably clear. But *simple* and *clean* probably aren't the feelings you get behind your eyes. The simplicity of the version on the bottom is impressive, given that it manages to convey the same point with so many fewer elements. What makes that version simpler?

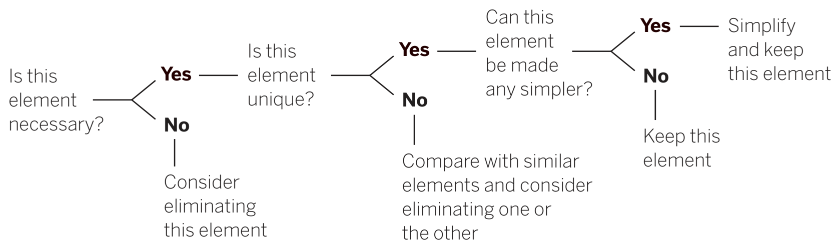
It removes stuff. Leave only what's valuable to communicating your message. Edward Tufte mathematized this idea as the "data-ink ratio"—the higher the share of ink on the page that's devoted to necessary elements, the better.[4]

Tufte's concept sounds precise, but he's really just saying don't waste ink on decoration or redundancy. In text editing, this is more colorfully referred to as "removing the deadwood." It's a sound principle. But the trouble with such aphorisms is that "necessary" is a slippery, subjective thing. What is valuable to communicating your message depends, as always, on context. Who is the visual for? Do you already have their attention? How much detail do they need? How and where will they use the visual? Do they have seconds or minutes to look at it? Are you trying to inform them or persuade them? Have they seen this kind of chart before? Are they familiar with the data? How will it be displayed? The answers to these context questions (and many more) will affect what's necessary to include.

It's also hard to edit yourself. If you didn't think some element was necessary, you probably wouldn't have included it in a prototype in the first place. It takes discipline to "kill your babies," as text editors sometimes say.

A good way to force yourself to look critically at what you've included is to evaluate the elements one by one, using this simple question flow:

WHICH ELEMENTS SHOULD YOU KEEP?



If you've been through a talking and sketching process, and your answer to *What am I trying to say or show?* is written down, you can use that to determine whether an element is necessary. The manager who created the Rep's Past Performance chart on the facing page did write down his statement: *Past sales aren't a good predictor of future performance. Highest performers are more valuable than you think, and lower performers are less valuable than you think*.

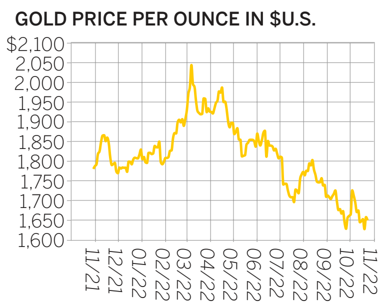
With this in mind, we can spend a few minutes applying the question flow to every element in the original version of his chart.

A **title** is nearly always necessary. But is this one unique? No. In fact, one of the captions repeats it nearly verbatim. Keep the title, kill the caption. Is there a simpler way to present the title? Not really. It can stay as is. The **subtitle** is a tougher call. It sums up the statement of purpose well. But is it really necessary? It's not unique: It recapitulates the visual. The captions below the x-axis also repeat the same idea. That's three ways to say the same thing. So, let's kill the subtitle.

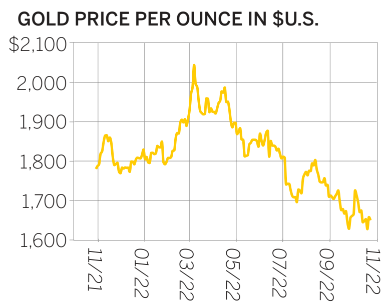
The information in the **visual field** is necessary, unique, and couldn't be made much simpler. Keep it as is. We've already decided that the **caption** is redundant, but it does contain bits of unique information, about the formula for future value and the source of the data at a B2B company. This is minor information that doesn't need to distract from the visual the way it does currently. It can be moved to the **source line**. The other two captions, about over- and underperforming, are necessary to describe the division between the two types of salespeople, which was a core idea the chart maker wrote down.

Axes are nearly always necessary on data plots, but how many demarcations they should contain is both endlessly debatable and a major factor in how simple a chart feels. The "airiness" of a simple visualization is often achieved by diminishing or removing a chart's background structure—reference lines, ticks, value intervals. Look at the three gold price charts below (we'll come back to the sales performance chart).

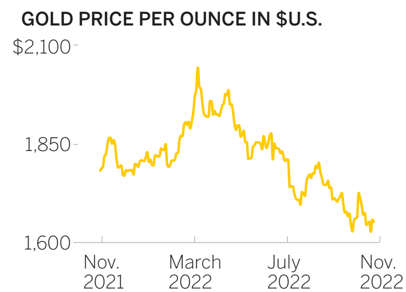
The chart without gridlines and fewer labels feels simplest, but is that kind of minimalism always a good thing? Think about display media: A chart presented on paper or on a personal screen—a format in which viewers can spend time with it—may benefit from more detail that allows the viewer to reference individual values and explore the chart in depth. But for a chart in a presentation—when you want the audience to understand the visual in seconds—fewer structural elements will reduce distractions and make it easier to focus on the broad ideas.



Context: Prototype
Use: Research, individual, informal
Media: Personal screen, paper



Context: "Let's talk about gold prices"
Use: Analysis, informal or formal, one-on-one, small group
Media: Paper, personal screen, public screen



Context: "Gold prices are dropping this year"
Use: Presentation, formal, small or large group
Media: Paper, small screen or large screen

SOURCE: BULLIONVAULT.COM

CLARITY THROUGH CONTEXT, DEPTH THROUGH DESIGN

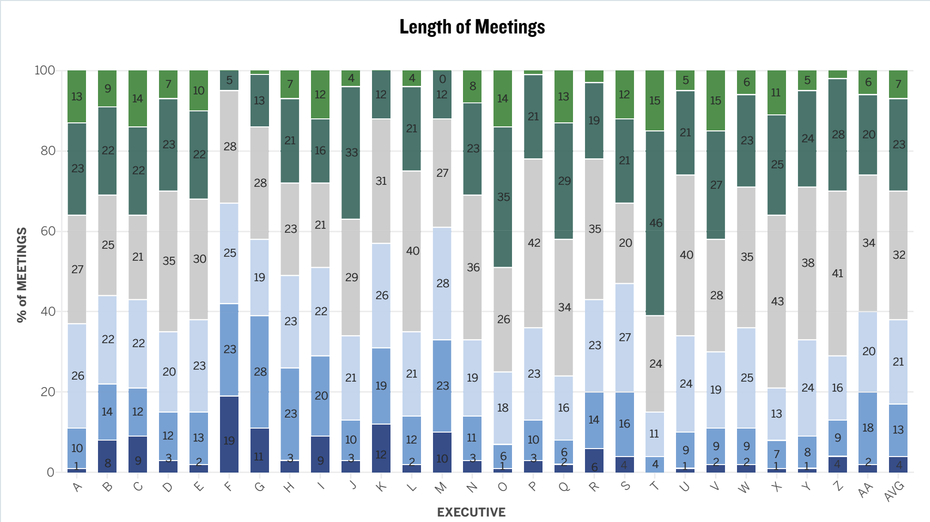
Legendary management professor Michael E. Porter for years collected data from dozens of CEOs on how they spend their time, enough data to present a compelling portrait of what CEOs really do with their time in a *Harvard Business Review* series of articles.

He provided the team at HBR the data and two dozen rough visuals; many looked something like the one below.

They weren't publication ready, but they did provide some context. Meeting size and meeting length were two key variables that came up repeatedly, for example. And each chart plotted each CEO's activities discretely, signaling a desire to make the information feel like more than just aggregate data.

On the next page is one of my favorite charts produced for this project.

This chart is based on a stacked bar chart like the first one. It's a considerable transformation, and there's much to learn from this chart about context, clarity, and design.



MICHAEL E. PORTER AND NITIN NOHRIA, HARVARD BUSINESS REVIEW, JULY-AUGUST 2018

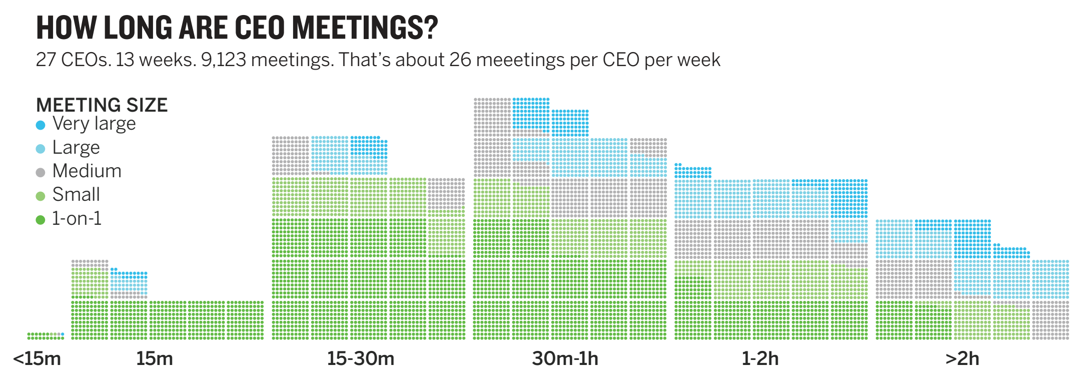
- It wasn't conceived this way. It started as a histogram of meeting length. Then the units were layered in. Then meeting size data. Good charts often go through such prototype iteration. Here we had to do some new calculations to cross-reference meeting time and size. Sometimes it doesn't work out—the new data distracts or it doesn't easily map to the visual—so you scrap it. Other times it works well.

- It's restrained. Notice that the actual percentage values are absent. The x-axis forgoes a label. The subtitle and the key both suggest that each dot represents a meeting, so a separate caption isn't needed. Notice that the key is built vertically to match the way the data is plotted and the color scheme matches a typical convention (lighter colors equal less value). This restraint is similar to DeBold's vaccine chart on page 133. It won't always work to be so minimal, but it does here because in the context setting we learned that the specific values were less important than the *sense* of proportion.

- It's multiple charts. What type of chart is this? If you thought histogram, you're right. If you thought unit chart, you're right, too. And if you see a stacked area chart, also correct. The distribution gets you thinking about meeting length. The units are a powerful way to manifest events and people: Each dot is a meeting a person must attend. If it were just presented as percentages, I lose that connection to the image encoded in the unit, the dot. And the color coding allows me to think about meeting length and size in two ways: across the whole data set and within each meeting length bin.

Despite so much going on, the chart maintains coherence, clarity, and most crucially, usability, largely due to three key factors: We spent time setting context. We iterated and tried things. And we practiced restraint.

It becomes easier to create such transformations with practice, but it all flows from setting context above all. The more context you have, the easier it will be to achieve these goals.



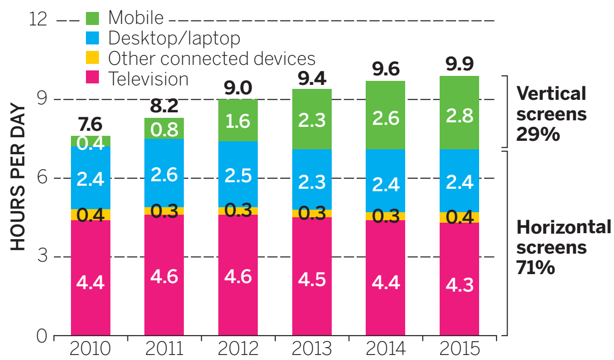
SOURCE: MICHAEL E. PORTER AND NITIN NOHRIA, *HARVARD BUSINESS REVIEW*, JULY–AUGUST 2018

Ask yourself, *What do I want viewers to do with this chart?* If the overall shape of the trend is what matters, be more aggressive taking away reference points such as grid lines and axis labels. Communicating the idea that "the price of gold is going down" probably doesn't need detailed stratification on the y-axis. But if you're hoping to have a conversation about monthly gold price trends, more reference points may be helpful so you can more easily connect months to approximate values. Imagine, for example, using the right-most chart on page 140 and saying to your audience, "Look what happened to prices in May." That's much harder to see here than on the middle chart. Then again, the prototype at left has so many dollar values on the y-axis that it's hard to follow them across the grid.

Back to the sales performance chart on page 138: The x-axis is unique and necessary—each pair of bars needs a label. But do we need more or fewer values on the y-axis? If we reduced it to just low, middle, and high values, would that adversely affect its ability to convey the idea? Probably not. The manager's statement of purpose shows that comparing the relative value between two time periods matters more than specific dollar values. The y-axis can be simplified.

In general, though, labels present another challenge to simplicity. A common technique for many managers is to label every visual element on the page with its specific value:

TIME SPENT ON SCREENS BY ORIENTATION, U.S.



SOURCE: MARY MEEKER'S INTERNET TRENDS REPORT

The labels begin to overtake the visual. But why are they there? Are we meant to focus on the specific values, or on the overall shape of the thing we're looking at? Are we meant to look at the data or read it? A visualization is an abstraction. Labeling every value is a concretization. If you feel that it's necessary to show every value, and for your audience to have access to all specific values, a table may be a better option:

TIME SPENT ON SCREENS BY ORIENTATION, U.S.

HOURS PER DAY SPENT ON SCREENS, U.S.

	2010	2011	2012	2013	2014	2015
Television	4.4	4.6	4.6	4.5	4.4	4.3
Desktop/laptop/other	2.8	2.9	2.8	2.6	2.7	2.8
Mobile	0.4	0.8	1.6	2.3	2.6	2.8
Total	**7.6**	**8.3**	**9.0**	**9.4**	**9.7**	**9.9**
% Horizontal screens	95	90	82	76	73	71
% Vertical screens	5	10	18	24	27	29

The manager who made the chart on the previous page may argue that the table isn't as effective because it doesn't provide instant recognition of an upward trend and the growing share of mobile screen use. That manager is correct and has unwittingly argued against her labeling every value in the chart: If the trend and the growing share are most important, the specific values shouldn't be put there to steal our attention from the overall trend.

The manager needs to ask, *Is each individual value important to expressing my idea?* and *Do specific data points have to be available to discuss the idea?* If the answer to either question is yes, a table should be made available. The manager can provide a visual as well, but he's now free to make the chart much simpler. Compare the original chart with the suite of three below, which make every value available and give viewers at-a-glance trends:

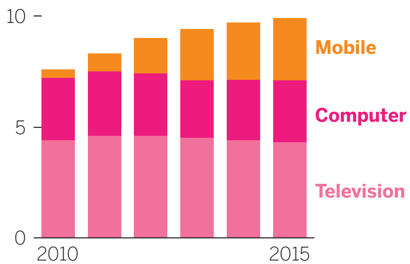
HOURS SPENT ON SCREENS

HOURS PER DAY SPENT ON SCREENS, U.S.

	2010	2015
Television	4.4	4.3
Desktop/laptop/other	2.8	2.8
Mobile	0.4	2.8
Total	**7.6**	**9.9**
% Horizontal screens	95	71
% Vertical screens	5	29

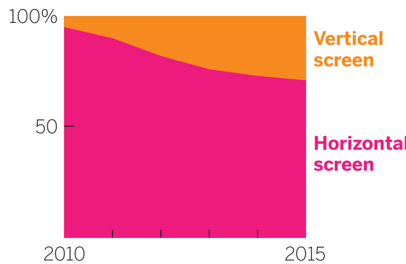
SCREEN TIME IN THE U.S.

NUMBER OF HOURS PER DAY



TIME BY ORIENTATION

PERCENTAGE SHARE



There's no right answer here without knowing the context. But it's true that more labels will reduce simplicity and demand that the viewer make decisions about what's important. Overall, be aggressive in your efforts to reduce marks on the page. You can almost always take away more than you think—and more than you want to. Test very sparse versions of your chart on colleagues; you may be surprised at how little you need to include to convey your idea.

It’s not redundant. Removing repetitive elements, as we just did, helps simplify, but so does removing redundant design within elements. Here are a title and subtitle for a chart:

WHAT IS MIDDLE CLASS?
Family income by city, 2013

This is clear and crisp text. But design-wise, the title is highly redundant. To make it stand out, it’s been given *five* special treatments: size, boldface, underline, color, and all caps. Does it catch your eye? Yes. Does it need so many signals that it’s special? No.

The subtitle has two distinguishing elements: size and italics. But if the text is smaller and appears right below the title, it must be the subtitle. Italics are superfluous here.

This is called belt-and-suspenders design. You don’t need both to hold up your pants, so pick one. In general, a design will feel simpler if you apply as few unique attributes as possible. Here’s the same title and subtitle but with only one difference assigned in each case—size, weight, or color:

What Is Middle Class?	**What Is Middle Class?**	What Is Middle Class?
Family income by city, 2013	Family income by city, 2013	Family income by city, 2013

You might even argue that the line space between the two levels of information is redundant. If you want more space for your visual, you could put the title and subtitle on the same line and still achieve the proper relationship between them:

What Is Middle Class? Family income by city, 2013

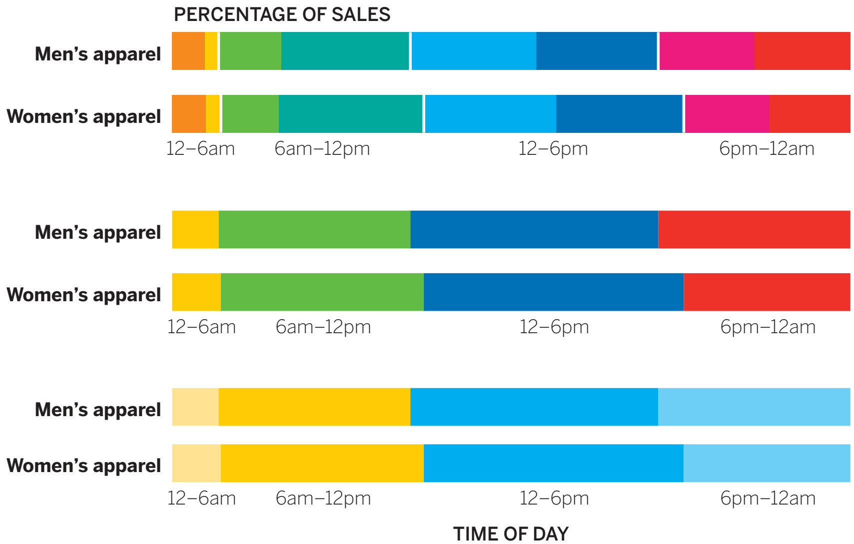
Most of the charts in this book use both size and weight to distinguish title type. Two distinguishing characteristics are quite common with titles. As a design choice, it’s fine. We present the examples with only one distinguishing characteristic to drive home the point that you don’t need to overemphasize elements to get them to do their job. This kind of discipline becomes even more important as you add elements to the data visualization. That sense of simplicity is lost as you create unique visual attributes for axes labels, captions, pointers, and other elements. Instead of giving each its own unique design features, it can help to create classes of information that share design attributes: Captions, legends, and labels can share a text style, for example. Lines, arrows, boxes, and other marks can use consistent weight and style. In many cases, they can be eliminated in favor of simple alignment, which achieves the same end without marks on the page.

Its use of color is restrained. Simplicity suffers when you make charts too colorful because you want them to be eye-catching or you have lots of data categories to plot. To make meaning from a chart, viewers can't help but focus on color differences and wonder what they mean. That's how the brain processes the information. It wants to assign meaning to each color. What's more, the brain can't hold many different distinct colors simultaneously as it evaluates information. More than, say, four or five colors at most and the brain wants to start grouping like colors together. Even if your blue line and teal line aren't related variables in your chart with 12 different line colors, the mind is trying to put them together.

The more color differences, the more mental work to figure out what the distinctions represent. Challenge each addition of a color to a chart: *Why do I need to make this distinction? Can it be combined with other information as a group with a single color?*

Think of color in your charts as a fraction that you need to reduce. A colorful chart is like the fraction four-sixteenths. That ratio is more clearly expressed as two-eighths, and most simply expressed as one-fourth. Find the lowest common denominator that still preserves the distinctions you need to convey your idea. For example, the first chart in this chapter, on page 126, included eight distinct three-hour time periods. I've shown iterations on that chart on the facing page to show how color reduction increases effectiveness. Eight unique colors result in a complicated looking chart with many elements fighting for our attention. Reducing that to four colors in six-hour chunks—enough categories to convey the idea well—helps, but the colors still fight with one another. We can push it even further. Clustering the data as just two colors: yellow for before noon and blue for afternoon, with the less common nonworking hours in paler hues, creates an obvious improvement in clarity.

WHEN DO PEOPLE BUY ON OUR WEBSITE?



Another note: Gray is your friend. It creates an information hierarchy. We typically think of gray information as background or secondary by comparison with information presented in color. It provides context without disrupting the main idea by fighting for too much attention.[5] Retaining axis lines but making them gray preserves their usefulness but lets them recede *behind* the important visual information. Background data that provides context also benefits from being made gray. The Mount Mansfield Snow Stake Depth chart earlier in this chapter is a masterful example of using color and gray to represent foreground and background information.

Color choice, too, should follow convention.[6] Contrasting data? Contrasting colors. Complementary data? Similar colors. Groups of data? Same or similar colors. Data ranges? "Empty" colors (low saturation, paler, whiter) for lower values and "full" colors (higher saturation, richer, darker) for higher values.

COURAGE

You've already heard most of the wisdom about simplicity: It's the ultimate sophistication (Da Vinci); style depends on it (Plato); less is more (Robert Browning via Ludwig Mies van der Rohe); simple is hard (variations attributed to hundreds of people). All that is true, of course. But for managers, here's a new aphorism: Simplicity is courageous.

A manager's impulse often is to show everything, which leads to dense, difficult-to-read charts that don't so much convey an idea as turn hundreds or thousands of spreadsheet cells into a visual. In part, this is the curse of knowledge—we think it's important to represent all the data that we know about and that we've produced. Dense, complex charts, we think, convey something about the person who created them: *I know my stuff. Look at all this data. Look how hard I'm working.*

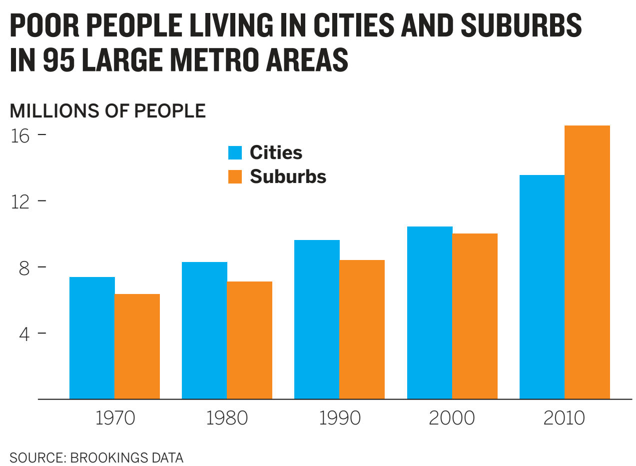
This deep-seated belief that more is better, that complex equals smart, must be eradicated. That's not what makes charts good.

Standing up at an important meeting to present a few clear, simple charts probably seems scary. Andrew Abela hears this when he's working with executives on their presentation skills. "When it comes to simplicity and clarity, there's a correct fear and a false fear," he says. "The correct fear is you do need to convey the right information, the right detail." That's what this book wants to help you do. "But then there's the false fear that if you don't show everything, they won't understand or they won't think you're working hard." In some ways, the first fear leads to the second: *I'm scared I might not show the right information, so I'll show all the information.* "I've been doing this a long time," Abela says, "and I will tell you now, nothing makes an executive happier than seeing someone show up with just a couple of excellent charts. They tell me, 'Finally, someone confident enough to just show me what I need and not bombard me with 60 slides.'"

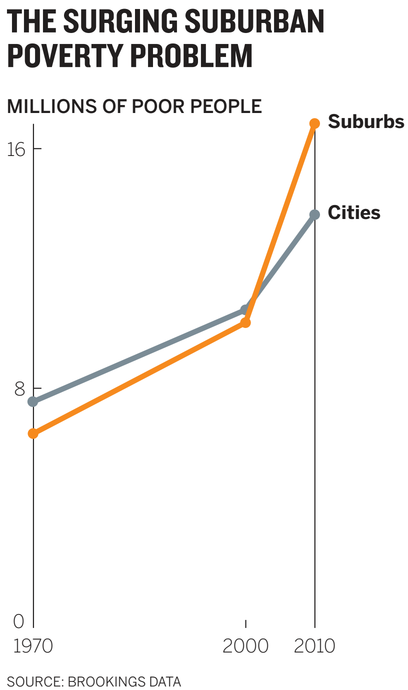
"Once," he continues, "I helped a manager prepare for a presentation to the CEO, and even though he was nervous about it, we decided he should make the entire presentation based on one great chart that he had created. The CEO was so impressed. They spent three hours talking about that one visualization."

THE ART OF PERSUASION

A manager at a not-for-profit is preparing to stand in front of 20 potential donors with deep pockets and many options for where to take their philanthropy. She's launching a program to fight suburban poverty, which she will tell them is a significant, growing problem. But she knows her audience will need more than that to be persuaded to back her initiative. She's already anticipating skeptical questions, such as "Why suburban poverty? It can't be as bad as urban poverty, can it?" These people will want to see evidence. She looks at a chart that will provide it:



A good effort. It's simple and well designed. All the information is there. Although poverty is growing in both cities and suburbs, it has grown more in the suburbs. Still, she's unsatisfied by her effort. The first thing she sees is that poverty is growing; it takes a minute to find the suburban poverty story. So, she tries to build a more persuasive visualization and comes up with this:



She's thrilled with this version, which is more accessible and far more convincing. The surge in suburban poverty comes through immediately, and almost directly after that, so does the idea that more poor people now live in suburbs than in cities. This will surprise and move her audience.

How did she get from her original, perfectly accurate but unsatisfying bar chart to something she's certain will help her line up donors for the program?

MAKING A CASE

It's often not enough to make a chart that's simply accurate. You're trying to reveal truths dormant in data; to make a case; compete for attention, resources, and money; make a pitch to clients; recruit new customers; sway an opinion or help to form one. You don't just want people to believe the chart is true—you want it to lead to action, suggest a way forward.

Persuasion science defines three strategies we use to influence behavior or thinking: *economic* (carrots and sticks), *social* (everybody else is doing it), and *environmental* (relaxing music at the dentist). Visualization falls for the most part into the third category. Steve J. Martin, a heavyweight in the field and a coauthor of several books on influence and persuasion, provides a legion of examples from his and others' research of how environmental persuasion strategies work.[7] For example, a professor doubled the number of people who were willing to participate in a survey by attaching a handwritten note to the request.[8] Hotels increased the reuse of towels by 25% when they changed the wording of placards next to the towels.[9] People serve themselves less food when the color of a plate contrasts with the color of the food.[10]

The mechanisms by which information visualizations persuade us are similarly subtle and equally powerful. “Whilst we’d like to think that our decisions are the result of effortful cognition, the reality is somewhat different,” Martin writes. “Much of our behaviour is driven by unconscious cues present in our environment.”

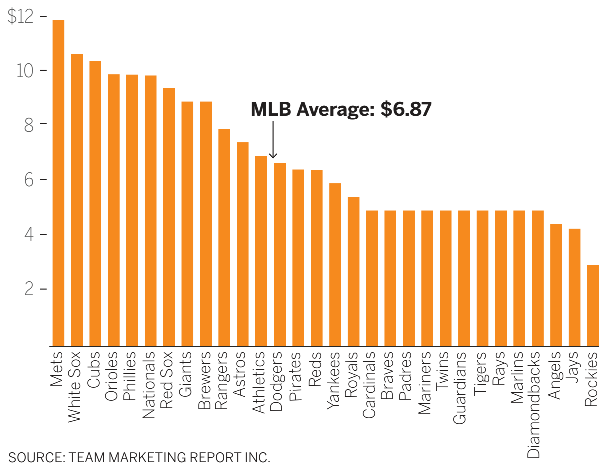
We’re veering away from the data scientists now. It’s often their job, or at least they see their own jobs as an effort to *show all the data*—to be as objective as possible and present everything that’s available for analysis. This makes sense when we’re doing exploratory visualization. It’s for fact-finding, hypothesis-testing, and analysis. This chapter focuses on those times when visual communication needs to sway an audience and effect change.

Even if we don’t think much about it, we recognize the distinction between conveying information and persuading, and we allow for both types of communication. A play-by-play announcer calls the action, describing mostly what’s actually happening on the field; a color commentator influences our sense of the game’s narrative. A house for sale can be accurately described as “2,400 square feet with 4 bedrooms and 2 baths on 1.2 acres” or, to make you want it more, as “a huge, open-concept Colonial with a brand-new modern kitchen, on a secluded, wooded lot with spectacular views.” What you may call a used car, the person hoping you’ll buy it calls pre-owned. Newspapers publish both reported stories and op-eds about the same topic. Compare the sentences below:

Reported story	Op-ed
The budget again seeks to retire the popular A-10 “Warthog” close air support aircraft for savings of $382 million, a move sure to anger Congress, which rejected a similar proposal last year.[11]	I appreciate the budget pressures that the Pentagon faces these days. But those arguments have serious flaws—and if we retire the A-10 before a replacement is developed, American troops will die.[12]

Is the reported story *better* than the op-ed? No, a qualitative comparison is impossible. One is informative, the other persuasive, and they use different rhetorical techniques.[13] The reported story describes facts, and speculation (Congress will get angry) is bolstered by evidence (it was rejected before). The editorial, though, uses the first person, joins the

COST OF ONE SMALL BEER AT EVERY MLB STADIUM



SOURCE: TEAM MARKETING REPORT INC.

If you wanted to persuade someone that beer is too expensive at baseball games, it's clear which chart you'd use.[14] But if the commissioner of baseball wanted to understand the costs associated with attending games, then such persuasion would be inappropriate. Admittedly, this transformation is extreme; it was conceived as an exercise in making data as persuasive as possible, a kind of op-ed experiment. But persuasion doesn't need to veer into blatant editorializing. Most of the time, managers just want to make a point more clearly and forcefully than an accurate, well-designed, but passive chart does.

audience ("we"), and feels more personal and conversational ("these days"). A significant claim ("troops will die") is stated without evidence. Neither text is better or worse than the other; each is good in its context (and, conversely, not good in the other's).

The same holds true for dataviz. When you have a point of view, you can employ techniques—manipulations—to heighten the effect. The unconscious cues—color, contrast, space, words, what you show and, as crucially, what you leave out—all work to make the idea more accessible and increase the chart's persuasiveness. This page shows the dataviz equivalent of the news story/op-ed comparison.

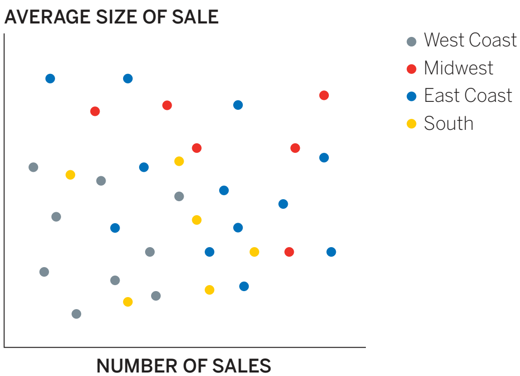
SOURCE: TEAM MARKETING REPORT INC.

THREE STEPS TO MORE-PERSUASIVE CHARTS

What often makes a chart persuasive is how easily people's attention goes to the main idea.[15] Persuasion scientists refer to this as the *availability* of salient information. If you make an idea easy to access, viewers will often find it more appealing and persuasive.[16]

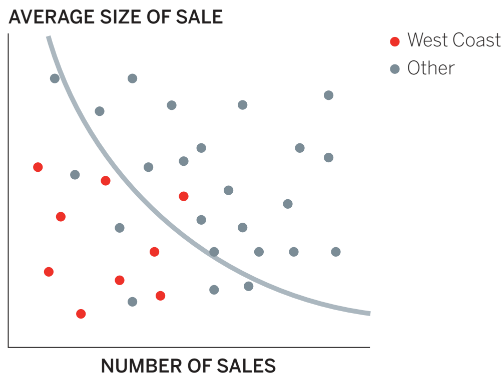
Which chart does a better job of persuading you that the West Coast sales team is a problem?

SALES PERFORMANCE BY REGION



SOURCE: COMPANY RESEARCH

WEST COAST SALES REPS UNDERPERFORM



SOURCE: COMPANY RESEARCH

The left chart may seem more informative because it includes more-detailed information. But persuasion is not about how detailed and precise you are; it's about how easy you make it to see the most important thing. The chart on the right is more persuasive.

The manager who made this chart employed many of the same techniques used by the manager at the not-for-profit who charted the growth of suburban poverty. When you're trying to increase persuasiveness, focus on these three things:

1. Hone the main idea.
2. Make it stand out.
3. Adjust what's around it.

Hone the main idea. The context-setting process outlined in chapter 4 for arriving at your defining statement will put you on the path to persuasion. Look again at the two urban/suburban poverty charts on pages 148–149 and try to imagine what statements might have been made during the talk and listen phase to inform the creation of those charts. They might be something like this:

Nonpersuasive	Persuasive
I want to compare suburban and urban poverty populations, decade by decade.	I need to convince people that suburban poverty is a huge and growing problem that has rapidly overtaken urban poverty.

To find your persuasive voice, you can go through a mini round of talk and listen with a counterpart. (If you're already at the talk and listen stage, add this in.) Change your prompt. Instead of asking *What am I trying to say or show?* try *I need to convince them that . . .* The former is still the best first prompt for your conversations (and for more-objective visualization projects). You may arrive at a more persuasive approach from that question alone. But if you don't, and your charts aren't having the persuasive effect you hoped for, the statement may help. Examples:

What am I trying to say or show?	I need to convince them that . . .
I am trying to show the distribution of costs of buying a beer at baseball stadiums.	I need to convince them that beer is unbelievably expensive at every single baseball stadium.
I am trying to show the relationship between increased automation in manufacturing and fewer jobs being available. Automation increases profits but creates a need for new jobs that are hard to fill.	I need to convince them that although profits are higher, robots are killing manufacturing jobs and creating a massive skills gap that offsets those short-term gains.
I am trying to show how increasing hours spent on work isn't increasing productivity and may be decreasing it.	I need to convince them that all this extra work we do is backfiring. It's hurting the company's productivity, not helping.
I am trying to show that getting vaccinated is safe and effective.	I need to convince them that vaccines save lives, not getting vaccinated leads to unnecessary deaths, and the risks from vaccines are vanishingly small.
I am trying to show that the gardening population is a large, growing, diverse, and underserved market.	I need to convince them that growth in the gardening market is real and comes from people who are hungry for apps, younger, and more technically savvy than they think.

Notice how in each case, the second prompt gives rise to more-emotional language. You've shifted from visualizing an idea (I want you to know something) to trying to persuade someone that the idea is good (I *need* you to believe something). Words that describe statistical trends (*increasing, declining, underserved*) naturally give way to words that describe feelings (*hurting, helping, hungry*).

One caveat: It's easy to slip into unhelpful editorializing when using the *I need to convince them that . . .* prompt. The manager looking at the gardening market, for example, may have arrived at *I need to convince them that they're wrong about gardeners and they're missing a major opportunity*. That's not a useful starting place for sketching and prototyping. It reflects his feelings about his audience and the results he foresees if he fails—not the ideas he wants to communicate in his charts.

Still, talking through his frustration with a colleague might help steer him toward a more useful statement of persuasion, especially if the colleague asks that pesky question "Why?"

> I need to convince them that they're wrong about gardeners and they're missing a major opportunity.
>
> Okay, why are they wrong?
>
> Because gardeners use apps and shop online. They're not these Luddites who can't work an iPad. Hell, 7 out of 10 gardeners are under 55.
>
> That's the missed opportunity?
>
> Yes. Most of the growth in gardening is with people under 35. Obviously they use apps. Even older gardeners are more tech-savvy than people assume. They shop online more than the average person. That's what I need to show.

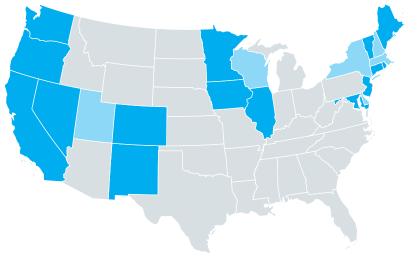
Now he has documented some usable information and found a revised, persuasive statement that he can begin to sketch.

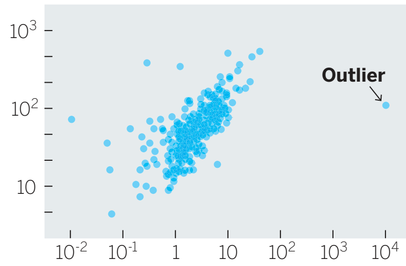
Make it stand out. With a sharper statement, sketching and prototyping will naturally veer toward more-persuasive forms. But you can amplify the persuasive effect even further with a couple of design decisions and techniques. Specifically, you can emphasize and isolate your main idea.

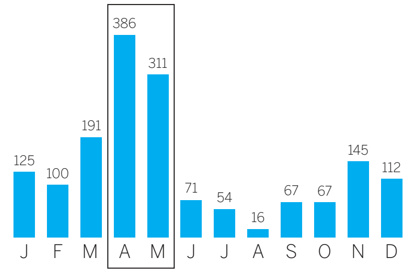
Emphasize. There, I just did it. Boldface and color are forms of visual emphasis. Did you say to yourself, *This bolded word is important; I should pay attention to it*? Probably not. But you did assign meaning to it, without even thinking about it. You treated it differently from the words you're reading now. You're more likely to remember it because I emphasized it.

Just as text allows for multiple forms of emphasis, such as **boldface,** *italics,* ALL CAPS, underline, color, and highlights, visuals use a variety of techniques to emphasize key information and ideas: Color. Highlights. Pointers. Labels. Tell me what I'm supposed to see. Make it easy for me to get it.

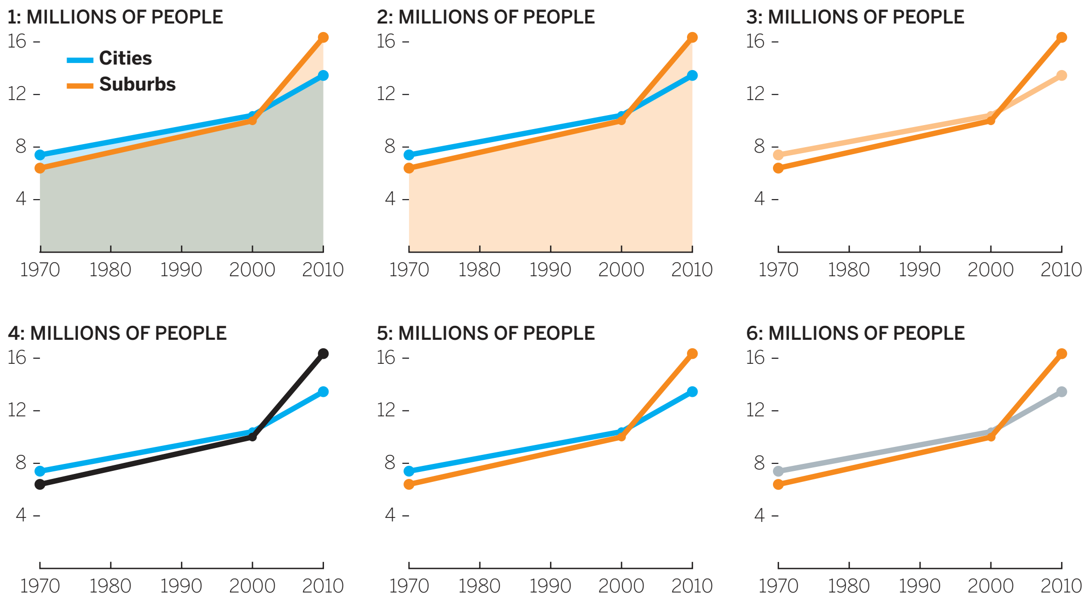
It doesn't take much to emphasize an idea. Color, simple pointers, or demarcations will draw the eye.







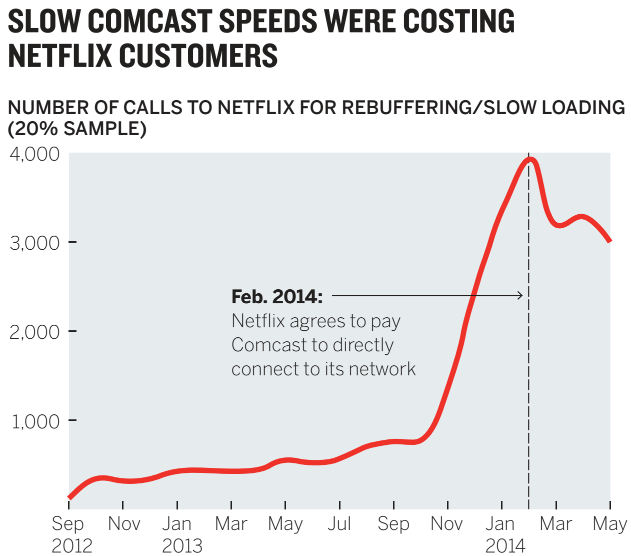
The most obvious and common form of emphasis is color. Use rich color to bring forward and diminish other information with lighter or contrasting colors. The not-for-profit manager went through several color iterations in trying to make her main idea the most accessible one. Each iteration attempts to make the surging suburban poverty trend the first thing we see and to use the comparative information, urban poverty, to support rather than compete with that idea. Here's why the manager rejected each previous iteration:



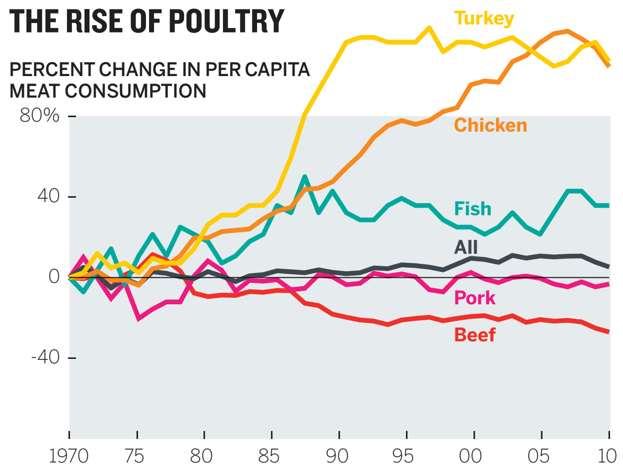
1. The overlay of semitransparent colors creates a third color that dominates the chart and draws attention to the filled area, not the lines.
2. This clearly highlights suburban poverty more, but why is one shaded and one not? The shaded area is still distracting.
3. Darker and lighter hues of the same color suggest two variables in a group, not a comparison. She wants to contrast, not complement.
4. Black on white provides the most contrast, but black and blue don't contrast so much that the black line pops.
5. Better! But the blue is still fighting for attention.
6. Final color choice.

Demarcations may seem almost unnecessarily simple, but they can be extremely influential. The curved gray line of demarcation on the chart that maps West Coast sales performance on page 152 makes it impossible to see the team as anything other than performing below expectations. Pointers can also nudge an audience toward the narrative we want to convey. Without the dotted line and label, it would be hard to understand what was happening in the Netflix Customers chart.

Demarcations can also be used to editorialize. By exceeding the border of the visual field, the author of the Rise of Poultry chart is making a value judgment about the reasonable limits within which the data should fall. The two lines that flout convention by going outside the border draw our eye immediately—they are meant to persuade us that the values represented by these lines are *too much* or



SOURCE: FCC REPORT, NETFLIX VS. COMCAST & TWC

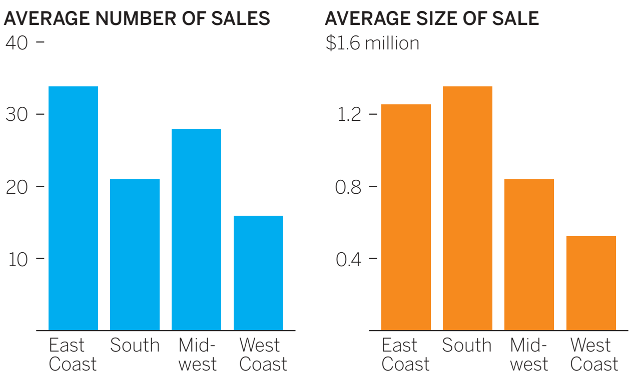


SOURCE: USDA/ECONOMIC RESEARCH SERVICE

above the expected range. (Similarly, in the editorial chart showing the cost of beer at MLB stadiums, the axis stops before it reaches the highest value: This suggests that the cost of beer at Giants and Red Sox games is, literally, *off the charts*.)

The West Coast sales-reps scatter plot uses another, less obvious way to make an idea more accessible. When charts are meant to represent some number of people or individual units, it's useful to show those units (or multiples of them) rather than a more abstract statistical representation of the whole set. In that chart, each dot represents a person. The same information could be conveyed more abstractly but would be less persuasive because it takes us further away from thinking about the individuals and their performance:

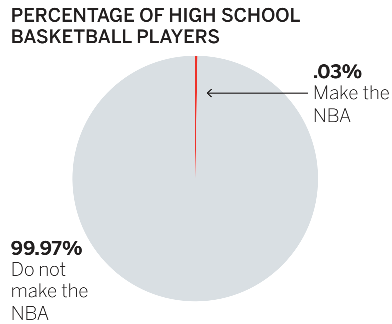
SALES REP PERFORMANCE BY REGION



SOURCE: COMPANY RESEARCH

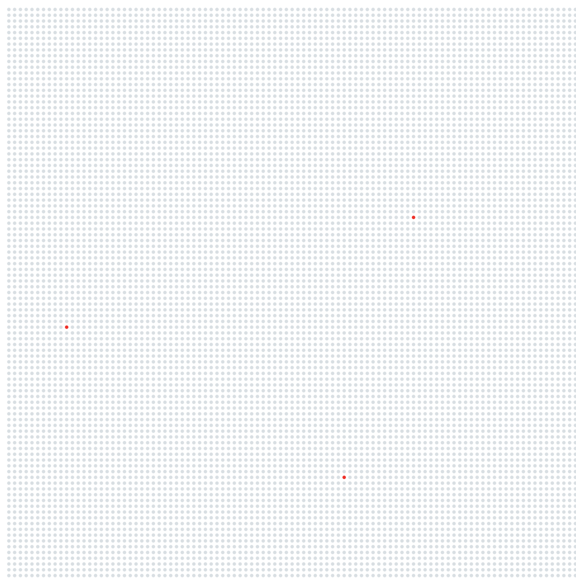
Even if the relative smallness of the West Coast bars were emphasized, this chart would be less persuasive than the chart that plots individuals' performance. That's because statistics are abstract things, and our minds would prefer to focus on more tangible, relatable things.[17] For example, which of these more convincingly shows the extreme unlikeliness that a high school basketball player will make it to the NBA, the pie or the unit chart?

HIGH SCHOOL BASKETBALL PLAYERS TO THE NBA



SOURCE: NCAA RESEARCH

FOR EVERY 10,000 HIGH SCHOOL BASKETBALL PLAYERS, HOW MANY MAKE IT TO THE NBA?



SOURCE: NCAA RESEARCH

The dots turn data into units that we can relate to—people—better than we can relate to a number like 0.03%. (It probably took you a moment to locate the three red dots. In this case, the random placement of them and the *lack* of accessibility *helps*, illustrating as it does that those individuals are so rare that you must work to find them in the crowd.)

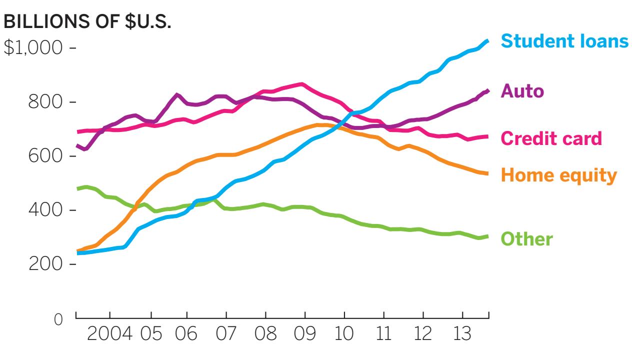
The way unit charts convey a sense of individuality have made them a popular way to communicate ideas about people. They're also effective when visualizing risk and probability (as in the NBA example, or in some other common examples, death rates).[18] Another potentially powerful use of unit charts is to represent money. We often show budgets and spending as proportional breakdowns. Showing individual units of money allocated to various groups might persuade us to think more carefully about where we put those literal units of money.

High-resolution displays have also helped popularize unit charts, because they can display tiny points as clearly as print can. How such a chart will play on a large screen in a presentation is worth considering beforehand.

Isolate. As much as we can emphasize the main idea, we can also isolate it by de-emphasizing other aspects of the visualization. Every element that earns a unique attribute, such as color, is fighting for attention with the main idea to which we want to draw people's eyes. The fewer the unique elements, the easier it is for viewers to know where to look and to understand what they see.

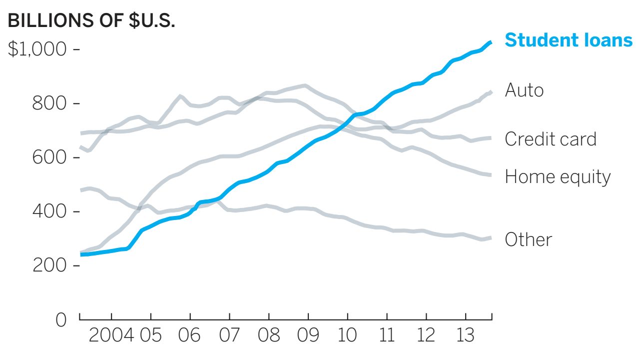
Software programs that generate charts don't automatically create influential emphasis. They tend to assign colors to every variable without accounting for which ones you want your audience to focus on first or most, or how color and categorization can be used to create primary and complementary information.

NONMORTGAGE DEBT OUTSTANDING



SOURCE: FEDERAL RESERVE BANK OF NEW YORK

NONMORTGAGE DEBT OUTSTANDING



SOURCE: FEDERAL RESERVE BANK OF NEW YORK

When every variable gets a bright color; no one variable stands out. Which idea is most available in the first Nonmortgage Debt Outstanding chart? Many people first see the green line because it's somewhat separate from the others. But this chart is in fact meant to persuade us that there's a student debt crisis. Now you may see it, but that idea was less available than it should have been. Isolating that variable creates a more persuasive chart.

For all the power of software programs and online services to generate reasonably good looking visualizations, they're not yet capable of injecting such persuasive design cues. That makes sense: Software renders data. People present ideas. It's still up to us to intervene with decisions and techniques that bring our ideas into high relief. The writing program I'm using right now can't anticipate which words I want bolded or italicized. It's up to me to decide which need emphasis and then apply the right kind at the right time.

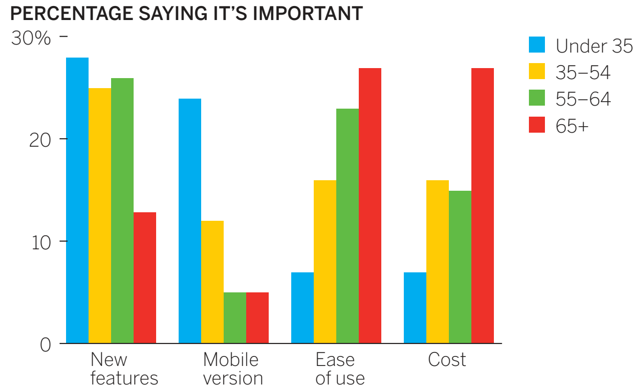
Adjust what's around it. The most aggressive way to make an idea pop is to change the reference points—the variables that complement or contrast with the main point. We can remove, add, or shift them.

Remove reference points. A chart similar to the one above right was tweeted with the message "The age divide in what people want from products."[19]

How available is the age divide in this chart? Do you see it? Are you persuaded there *is* an age divide? What about with the bottom chart?

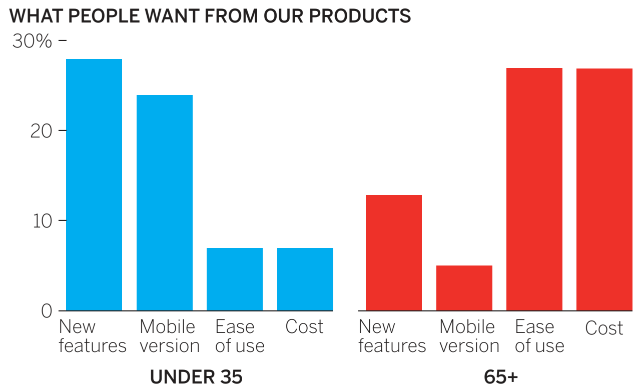
Removing reference points made the idea pop. Think of this as a more aggressive form of isolation. Instead of diminishing color or grouping elements together, you eliminate some information altogether. In the Opposing Desires chart, the middle two age groups have been removed because they don't help illustrate the idea of an age divide. This chart also groups bars by age rather than by feature requests. That makes sense because the main idea is an age divide; those are the categories we want to compare first.

WHAT ARE THE MOST IMPORTANT ASPECTS OF THIS PRODUCT THAT MAKE YOU WANT TO BUY IT?



SOURCE: COMPANY RESEARCH

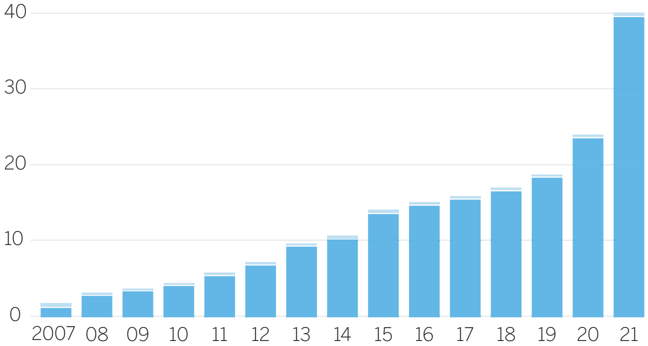
OPPOSING DESIRES OF THE YOUNGS AND THE OLDS



SOURCE: COMPANY RESEARCH

LONG LIVE VINYL

VINYL LPs AND VINYL 45s SOLD, IN MILLIONS



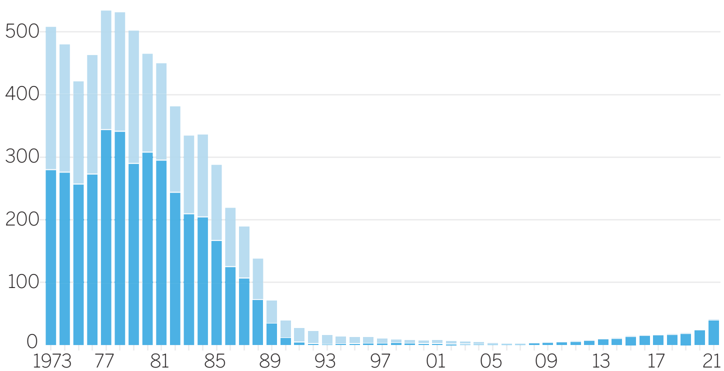
SOURCE: RIAA

Add reference points. It may seem that removing information will always make the main idea more available because it has less visual information fighting with it. But sometimes adding reference points works too. For example, a case can be made that vinyl LPs are making a major comeback. There's also a persuasive case to be made that vinyl LPs are *not* making a major comeback. New reference points incontrovertibly alter the persuasive message—in this case from one story to its opposite.

Shift reference points. Another way to change the narrative, and therefore the persuasive direction of the idea, is to shift a comparison entirely.

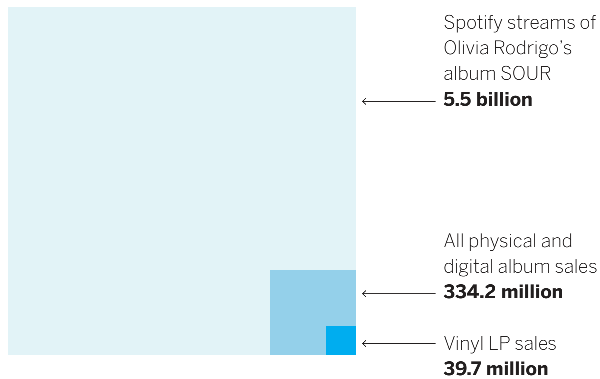
VINYL RECORDS ARE DEAD

VINYL LPs AND VINYL 45s SOLD, IN MILLIONS



SOURCE: RIAA

ALBUM SALES IN PERSPECTIVE, 2021



SOURCE: RIAA, SPOTIFY

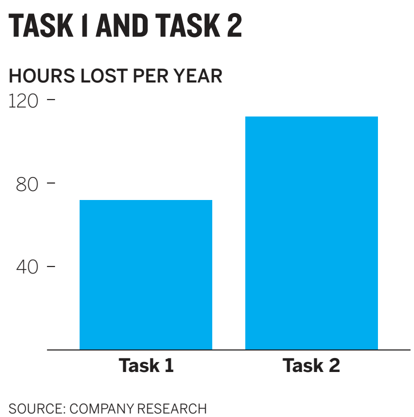
It may be true that vinyl is experiencing a minor resurgence within the context of vinyl albums. But when that trend is compared with a new reference point—total album sales in all formats—we can see right away that it's still only a tiny piece of the business.

This strategy is especially effective when the new reference points are familiar ones. The beer prices at MLB stadiums charts on page 151 compared the costs of one small beer at each ballpark. Unfortunately, a small beer is not the same size at all stadiums. To compare prices fairly, you'd have to calculate the cost per ounce. But how much is an ounce of beer? One sip? Two? The reference point is not easily accessible. We don't typically think about (or pay for) beer by the ounce. We do, however, pay for cases of beer. By shifting to roughly the amount we expect to pay for a case, something the audience will be able to easily access in their mind, the chart makes a faster, deeper connection with the audience.

Here's another example: A manager wants to make the case that the tech team should automate two menial processes. Each task takes only a few seconds, but both must be done constantly. He wants to show that performing the task dozens of times a day adds up over time. So he adds up all the time and plots it, as shown on the bottom left.

Hours per year is a respectable reference point, but it's not terribly dramatic—there are thousands and thousands of hours in a year, so about a hundred doesn't seem like that many. But if the manager shifts the reference point as on the bottom right, his boss may be persuaded to take action.

Workdays—now that's something the boss gets right away. Who wants to lose workdays to menial tasks? What's more, rather than focusing on hours lost to the tasks, the manager is focusing on *who* loses the hours. A new narrative forms: *Susan spends almost three weeks a year just on these menial tasks.* (Notice,



too, that the manager changed the bars into a unit chart, with five-day blocks composing a week. This creates another easily accessible unit—a work-week—to help persuade.)

YOU'RE NEVER NOT PERSUADING

We like to think that we're most persuasive when we provide comprehensive information and then lay out a detailed, accurate argument for our point of view. More content is more convincing.

But that's often not the case. Persuasion doesn't necessarily increase in lockstep with the volume of evidence or the breadth and depth of the data. In fact, some evidence suggests that providing too many supporting claims for your idea can have the opposite of a persuasive effect.[20] Persuasive charts tend to be simpler and to convey one or two ideas powerfully rather than many ideas equally—depending, as always, on context.

And no matter what kind of chart you need to create, you are never *not* persuading. A chart itself is a persuasion strategy—a manipulation that exploits the overwhelming power of the visual perception system to communicate something more convincingly than text can. Even a basic declarative chart is a form of persuasion, a deliberate attempt to not take a stance and persuade an audience you are impartial. People don't particularly like the idea that they're being persuaded all the time; they think that happens to others but not to them. Not true. That experiential part of the brain that relies on heuristics, metaphors, and experience to color interpretations of the world is a powerful influence, even when we look at data visualizations. If you internalize this fact, you can work with it, rather than fight it, and understand that your job is not to avoid persuasion—you can't do that—but to be responsible with your persuasion. That's next.

In the time I've spent with people talking about data visualization, the concern they express above all others is that they'll need to learn to be a designer to make good charts. They often say this as if they'll literally need to take design classes and learn about color theory and negative space and other "designy" concepts.

Not true. It can't be stressed enough that well-designed, persuasive charts are still primarily a function of good context. The simple concepts in this chapter merely provide ways to bring that context to its fullest realization.

It won't take long until some of these approaches to design and persuasion in visualization become second nature, and you're routinely turning out good charts that create good feelings behind our eyes.

RECAP

REFINE TO IMPRESS AND PERSUADE

The goal of good chart design isn't to make visualizations more attractive; it's to make them more effective and easier to understand. While most of us sense good design when we see it, we don't always know why. Here are some techniques to create that sense of good design in your charts:

1. **To make charts feel neat or clean, focus on design *structure*:**
   - Include three elements: title (and sometimes a subtitle), visual field, and source line. Within the visual field include axes, labels, and sometimes captions and legends.
   - Use consistent weights: title (about 12% of your visualization); subtitle (8%); visual field (75%); source line (5%).
   - Align elements: place them along as few horizontal and vertical lines as possible.

2. **For charts that just make sense or feel instantly understood, focus on design *clarity*.**
   - Remove extraneous elements. Be aggressive. Take away as much as possible while maintaining the meaning.
   - Use text to support the visual. Highlight the idea instead of describing the chart's structure.
   - Remove ambiguity. Make sure each element has a single purpose that can't be misinterpreted.
   - Use conventions and metaphors. Take advantage of ideas we don't need to think about to understand, such as red is "hot" and north is "up."

3. **To make charts that look elegant or beautiful, focus on design *simplicity*.**
   - Show only what's needed. Every element should be necessary, unique, and rendered as simply as possible.
   - Avoid belt-and-suspenders design. One form of emphasis per element is enough.
   - Minimize the number of colors you use. Gray works for contextual and second-level information and for structural elements such as grid lines.
   - Limit eye travel. Place labels and legends proximate to what they describe.

REFINE TO PERSUADE

It's often not enough to make a chart that's simply accurate. Managers may need to reveal truths that are dormant in the data to help make a case—to compete for attention, resources, and money; to pitch clients; to recruit new customers; to sway an opinion or help form one. To make charts more persuasive, use these three techniques:

1. Hone the main idea.

Adjust your prompt. Instead of asking *What am I trying to say or show?* start by saying *I need to convince them that . . .* This will expose where and how you can focus your energy on persuading an audience. For example:

What am I trying to say or show?	I need to convince them that . . .
I am trying to show the relationship between unbundling products and declining revenue.	I need to convince them that unbundling our software suite will devastate revenue streams.

2. Make it stand out.

Use simple design techniques to reinforce your main idea.

- **Emphasize** the main idea by adding visual information that calls attention to it. For example, use unique colors, pointers, labels, and markers to draw the audience's focus.
- **Isolate** the main idea by reducing the number of unique attributes for all other elements. For example, group them together; make them gray to bring the main idea into high relief.

3. Adjust what's around it.

Manipulate the variables that complement or contrast with the main point to make it pop.

- **Remove reference points.** Eliminate information and plotted data that distract or dilute the main idea.
- **Add reference points.** Add plotted data to the chart to expose otherwise hidden context.
- **Shift reference points.** Change the plotted data used in comparison with the main idea to create new context.

CHAPTER 6

FACTS AND TRUTH

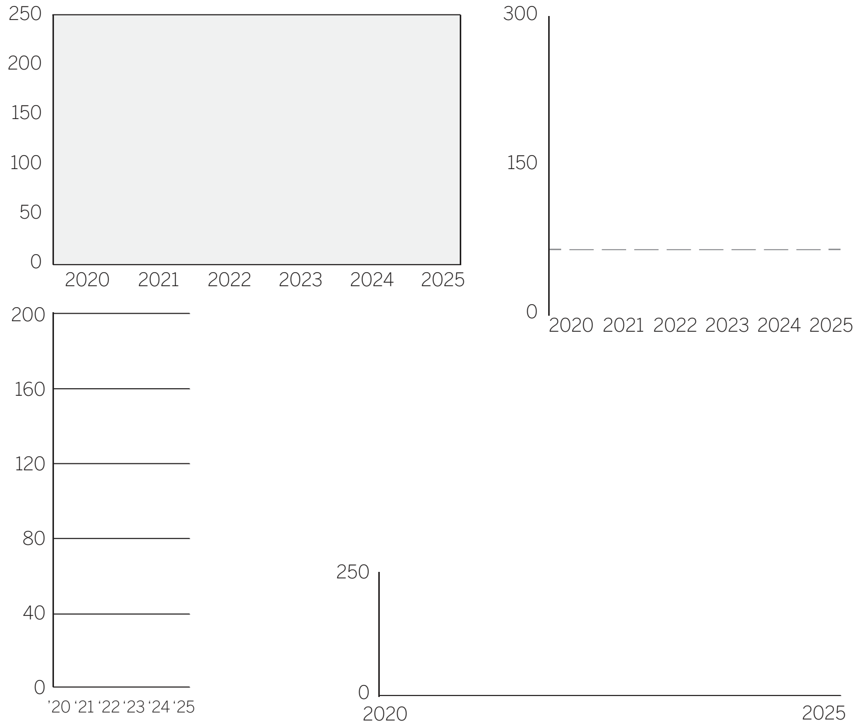
THE BLURRED EDGE OF PERSUASION AND DECEPTION

EVERY CHART is a manipulation.

Behind every chart are dozens of decisions, conscious and subconscious, that influence what someone sees and thinks about that chart.

This idea makes some people uncomfortable. Data visualization has what's called "high facticity"—that is, people feel like charts represent some reality accurately.[1] That data itself is dispassionate. That numbers don't lie. The whole point of data is that it's objective, right? And visualization is just a way to show data.

Well, yes. But also, no. Data visualization is not just a visualization of facts; it's the manipulation of them. Here's an exercise to reinforce the idea. I need to plot year-by-year data of my LDL or "bad cholesterol" level for five years.[2] Which set of axes should I use?



First, we need to know the proper distance to put between years on an x-axis. What is the correct amount of space between two years?

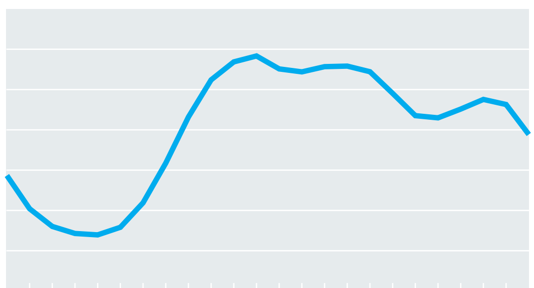
Obviously, it's an absurd question. The space between years on a two-dimensional surface is not a real thing; it's an arbitrary decision based on any number of factors that have nothing to do with time. If I wanted, I could put six meters between each year and I wouldn't be *wrong*. I'd be impractical, but not incorrect.

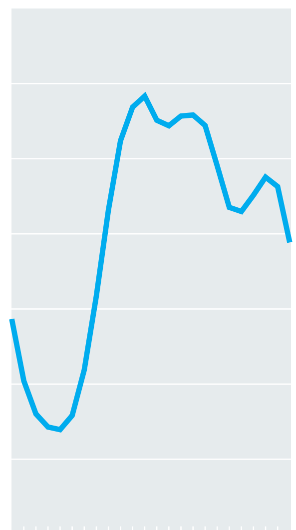
Even in this bare example, before I've plotted any data, I have several decisions to make about my axes: the length of the x- and y-axes; the range I use for the y-axis; the number of axis labels I use; the number of tick marks and where they're placed. I've probably at this point also made several decisions about my data: What years should I include or leave out? What other data will I include or leave out?

And once I make those choices, I face *dozens* more decisions. What chart type do I use? What colors? How many colors? Similar colors or different ones? How thick do I make my lines or how big do I make my dots or how much space do I put between my bars? What's the title and subtitle? What fonts do I use? Do I add a caption? Where's the key? Should I label specific numerical values? Which ones? And on and on.

Some of these questions I'll barely think about. I might just go with what the software gives me, or I might just do what I usually do. Others I'll consider more carefully. But in every case, I'm manipulating the visualization in ways that will affect what the user of the chart sees, feels, and understands about the data. I cannot avoid this. Every chart is a manipulation.

Sometimes, the manipulation is automated. Take these two curves:





They tell different stories. With the first, you notice a rolling trend, almost like a plane's trajectory. The second's sheer and bumpy—like a roller-coaster ride. It seems sharper, more volatile.

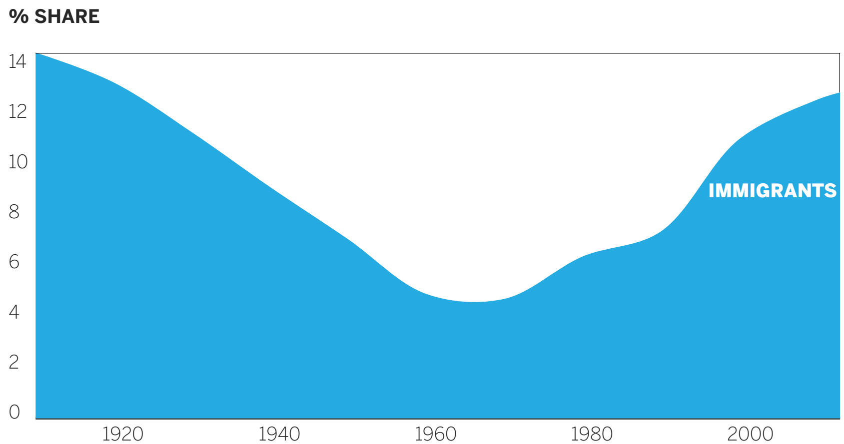
But these curves plot the same data on the same y-axis. The only difference between them is the length of the x-axis, a change that's merely the result of tilting a phone from landscape to portrait mode. So which chart is more "objective"? More "correct"?

Which is true?

A MAGIC TRICK

Let's be clear: The word *manipulation* as used above is reasonably neutral. Its connotative, nonpejorative meaning is just to work something with your hands. (Its Latin origins come from the words "hand" and "fill."[3])

IMMIGRANT SHARE OF POPULATION APPROACHES 1910 LEVELS



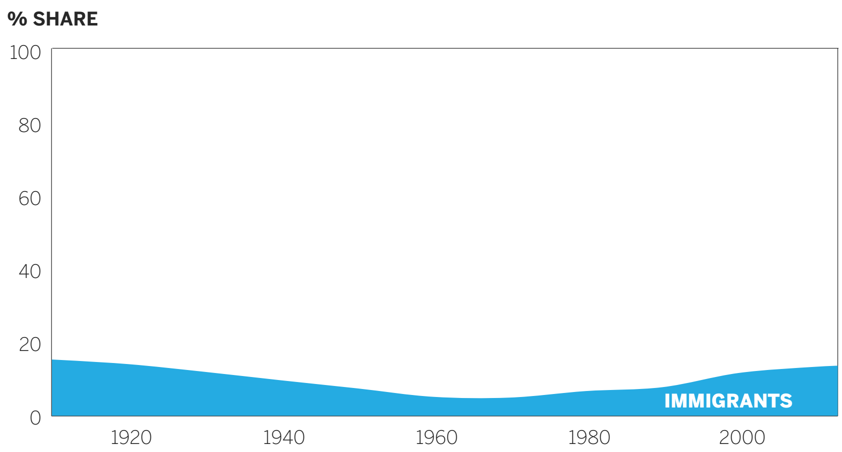
SOURCE: PEW RESEARCH

But it's also not a passive word. Some manipulation is the product of happenstance—decisions you don't even know you're making. But much is deliberate and skillful. And the more you understand how you control the truth you present to people, the more powerful your manipulations become.

Here's an example of manipulation that doesn't change a single data point but flips the meaning of the chart. We start with the chart on the left.

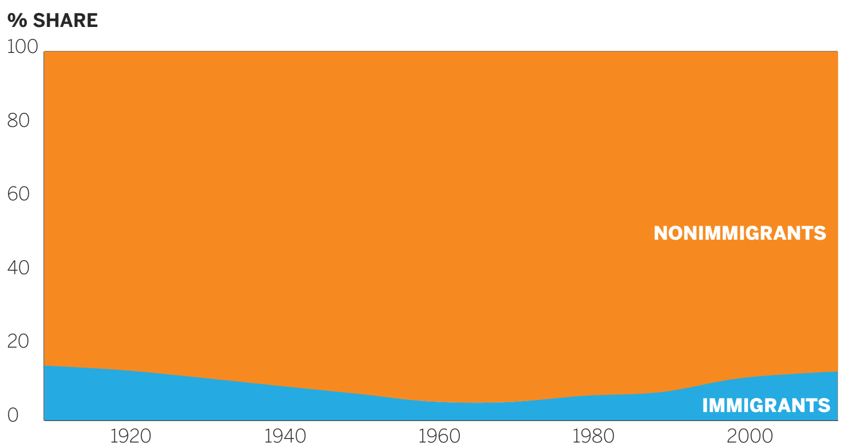
As a user of this chart, I see a clear message that the immigrant population is rising. In fact, it's almost filled up this bound box. It seems like it's making a point about this population reaching a high point it hasn't seen in 100 years. There are two variables in

IMMIGRANT SHARE OF POPULATION, 1910—2013



SOURCE: PEW RESEARCH

POPULATION, 1910—2013



SOURCE: PEW RESEARCH

my data—immigrants and nonimmigrants. I've only labeled one and I truncated my y-axis. This box only reaches "14% share," which is obviously out of 100%, but we don't see all that other data space, meaning you literally can't see some of the data that shows the nonimmigrant population (which again, I haven't labeled). I'm telling you what variable matters and making it hard for you to see or think about the other variable. Still I know it's 14%, so maybe I can imagine how much that is, or what the other 86% looks like. Try to imagine what 14% looks like on a full y-axis that shows all the data. When you do see it, the story feels different.

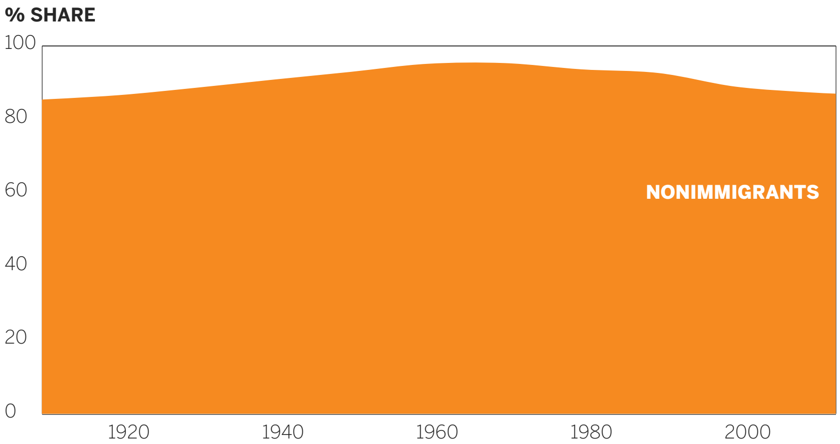
Now I've plotted both variables entirely, but I've still only labeled one. Also the second one is white—negative space in the visual, meaning I think of it as background information. I've made sure you focus on the variable I want you to see, as small as it now looks compared to the previous chart.

I could easily change your focus.

Now both variables are labeled and given strong colors. Now the massive size of the nonimmigrant variable draws the eye. I also changed the title to reinforce that I want you to think about both variables.

One more step: Let's do what the initial chart did, but in reverse. One variable shown. Title reinforcing what I want you to see. I could have truncated the y-axis but chose not to, as this variable comes so close to using the whole axis.

NONIMMIGRANT SHARE OF POPULATION, 1910–2013



SOURCE: PEW RESEARCH

Keep in mind this is the exact data used in the first chart, just manipulated to deliver a completely different truth. The decisions I made with color, axes, labels, and titles drive you to see what I want you to see or what I think you need to see.

THE BLURRED EDGE OF TRUTH

My transformation of the immigration data wasn't meant to be devious or manipulative in the pejorative sense. It was only to show the broad spectrum of truths you can create with simple alterations.

You've probably come across real-world examples of visualizations that are designed to deceive, hide, or otherwise alter the story in data in an unfair or unethical way. I'm often asked in seminars and workshops how to know where the line is between visual persuasion and visual dishonesty.

Even if it were a fine line, at least we could see it and stay on the ethical side of it. But, of course, no such line exists. Instead, we have to negotiate a blurred and shifting borderland between truthfulness and unfair manipulation.

On one side of this indefinite border are the persuasion techniques outlined in chapter 5 and on the other are the four types of deception:

Visual Persuasion Techniques	Visual Deception Techniques
Emphasis: Drawing the eye to main idea Example: Making one line in trend chart thicker and brighter-colored than others	***Exaggeration:*** Making an idea look more important or dramatic than it warrants Example: Truncating a y-axis to make an upward sales trend look like steeper gains
Isolation: Drawing the eye away from other ideas Example: Making all dots in a scatter plot gray except for the group you want to discuss	***Falsification:*** Changing or altering an idea in a way not supported by the data Example: Using two distinct y-axes to create a correlation where none exists
Added or removed reference points: Adjusting how much data is around the main idea to shift the context Example: Removing U.S. data from a bar chart to focus only on European data	***Omission:*** Leaving out data that would discount the viability of your idea Example: Removing U.S. data from a bar chart to make overall performance look better than it was
Shifted reference points: Adding new or different data to create a new context Example: Layering in stock index trend line to compare to your stock performance	***Equivocation:*** Using unnecessary elements to hide ideas or make them vague or unclear Example: Adding dozens of unnecessary stock performance trend lines to a chart so it's difficult to focus on one company's stock performance

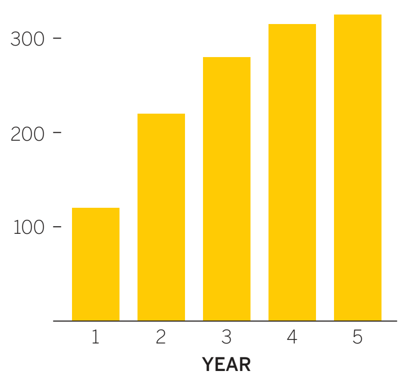
I won't dwell on falsification; the commandments should be obvious: Don't lie. Don't deliberately mislead. Don't create a chart like the one on the bottom left.

It looks like a positive revenue trend, but here each bar is *cumulative*, accounting for all previous years' revenue as well as new revenue. Year 1 is counted five times (see the middle chart), although that revenue was earned only once. This is continuous data, a trend line, hiding in a categorical form: We expect each bar to represent a discrete value. The breakdown shown bottom right is the more honest depiction of the revenue trend.

Of course, many charts don't fall neatly into one category or the other. One person's isolation is another's omission. It's easy to see how emphasis, applied too forcefully, might slip into exaggeration.

REVENUE GROWTH

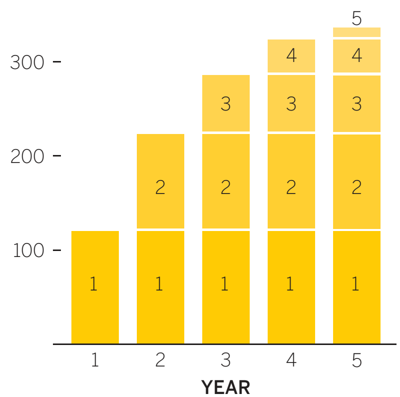
CUMULATIVE REVENUE
$400 million



SOURCE: COMPANY RESEARCH

REVENUE GROWTH

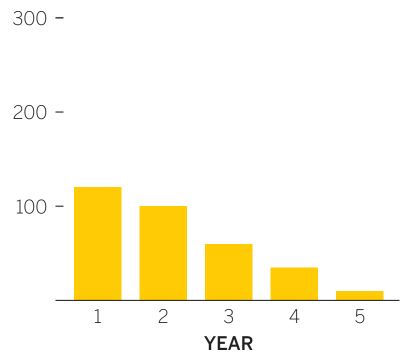
CUMULATIVE REVENUE
$400 million



SOURCE: COMPANY RESEARCH

FIVE-YEAR REVENUE TREND

ANNUAL REVENUE EARNED
$400 million

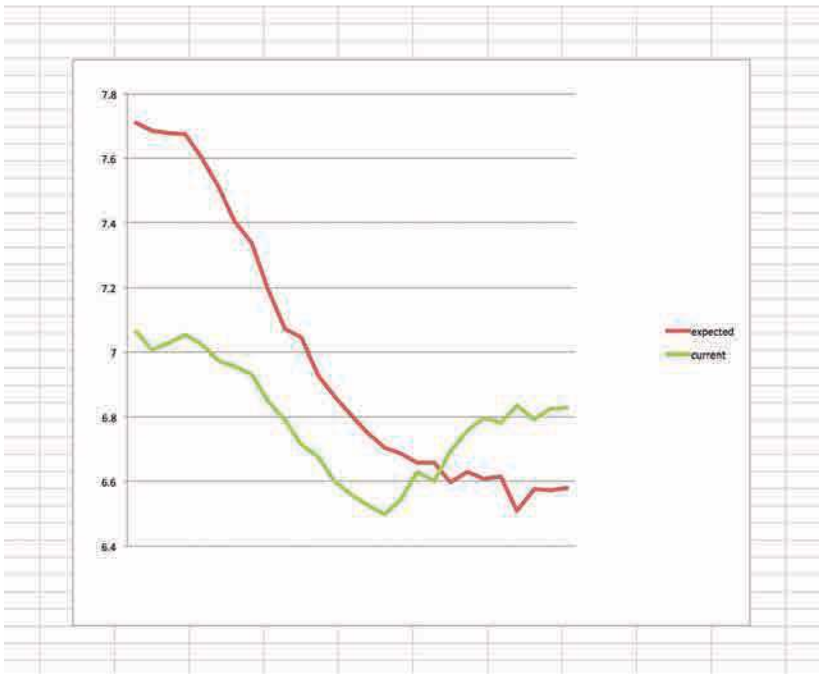


SOURCE: COMPANY RESEARCH

Imagine, for example, your boss asks you to prepare a presentation version of this chart that she quickly generated on current and expected job satisfaction across careers. She attached a note to it.

> Data and rough visual attached. For the board presentation, want to show the big change, the U-curve for current satisfaction and the huge gap in current vs. expected for young employees, which closes and flips in midcareer. Important to show where we need to address employee satisfaction issues before we propose funding for engagement programs.

You can see everything the boss is describing, but you also know that this satisfaction survey was scored on a 1 to 10 scale. This chart only shows from 6.4 to 7.8 on that scale,



14% of the actual range. When you reproduce this and compare it to a version with a full y-axis, you see a remarkable disparity.

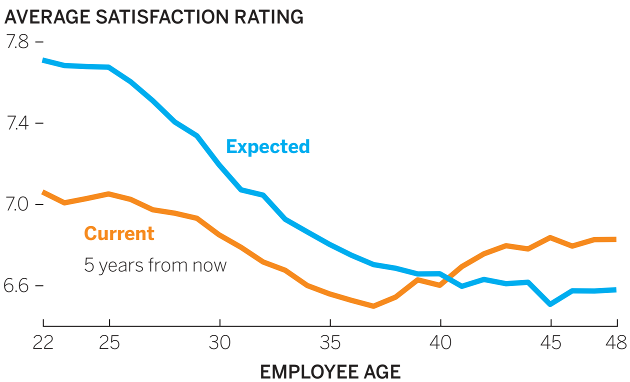
Remember the boss's keywords from her note: *big change, U-curve, huge gap, flips*. Those were clear in the original version, but the new version looks almost changeless—a small gap that converges in an unremarkable crossover.

What do you do? The boss thinks it's acceptable to "zoom in" like this, and indeed, we see this kind of truncation all the time. The boss insists that even though it's only about a point-and-a-half change, that is remarkably significant in this kind of data. By truncating, you're *emphasizing* what's important, not *exaggerating*. And it needs to be emphasized if the career programs are to get funded, which you agree are a good idea. No way they'll fund if you show the flat lines.

Which way do you go? Some of you will say show the whole range. You don't have to be a "y-axis fundamentalist" to see how dramatically truncation alters the idea that emerges from the data.[4] Others will say choose to truncate the axis. Though you think the changes look small on a full-axis chart, they *matter*, so they should be made to look less flat and more dynamic. If anything, some will argue, the full-axis version is deceptive because it makes something deemed significant look insignificant.

There's no easy answer here.

JOB SATISFACTION



SOURCE: COMPANY RESEARCH

JOB SATISFACTION

SOURCE: COMPANY RESEARCH

"HAPPY MANIPULATING!"

Deception in visualization is often unintentional, but there are plenty of examples of ill-intentioned people and organizations using charts—and their knowledge of how we see them—to lie. Partisan politics often fall to manipulating visuals to cater to a point of view in a way that belies the facts.

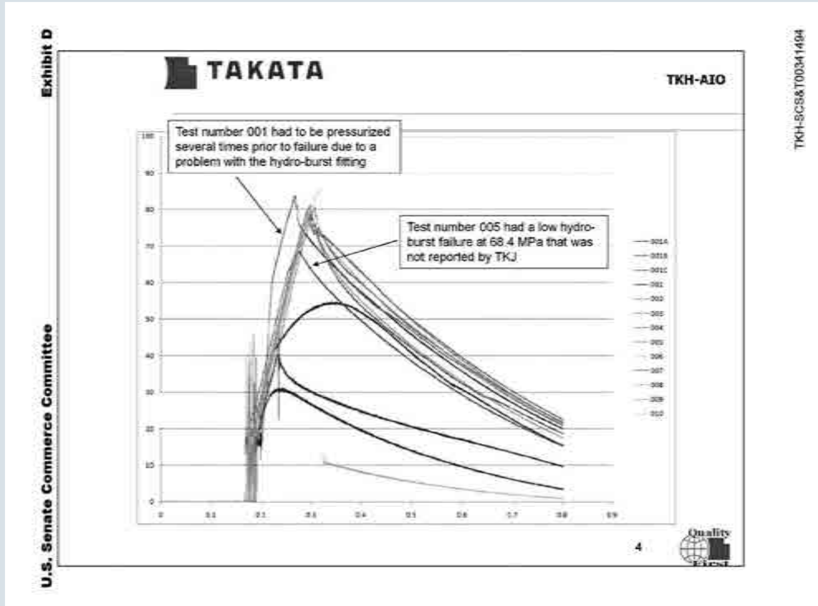
One of the worst cases of the past two decades was at Takata, an airbag manufacturer. Some airbags were failing safety tests, but they were not taken off the market or recalled. Over several years, many injuries and nearly two-dozen deaths have resulted from misfiring airbags. Eventually, a congressional investigation found that Takata had hidden the fact that there were technical problems with its product, both through omission of data and through manipulating charts.[5]

The company's airbag recall was one of the biggest in automotive history and eventually led to a multibillion-dollar class-action lawsuit settlement.

In one case a manager sent data to a team that was meant to clean up reports with the message "Happy manipulating!"

One of the emails uncovered in the investigation included an engineer saying, "I showed all the data together, which helped disguise the bimodal distribution. Nothing wrong with that. All the data is there. Every piece." He also suggested using "thick and thin lines to try and dress it up, or changing colors to divert attention."

The charts in question have not been identified, but they probably looked something like the one here, which was presented as evidence in the congressional investigation. It's easy to see how some light design manipulation of the kind the engineer cites could hide important information.



As telling and as troubling as the design manipulation was, the justification the person tries to use for doing it is equally disturbing. "All the data is there," he says, as if that absolves one from responsibly representing the ideas in the data.

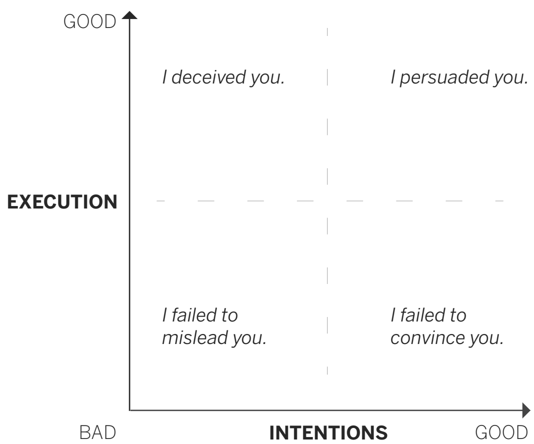
There could be no better way to illustrate the difference between facts and truth. Even when showing all the facts, they hid the truth. And people died.

EXPLORING THE GRAY AREA

Think of manipulating your dataviz like wielding a knife. Knives can be used in any number of ways: professionally by someone who's well trained; skillfully by a careful amateur; carelessly by someone not paying attention; recklessly by someone who isn't careful or considerate; even illicitly by a bad actor.

How you wield this knife really comes down to your intentions and your execution.

THE MANIPULATION MATRIX



You strive to reach the top right here, but that empty space in the middle is where we sometimes end up. Unpacking the ways in which charts slip into deception, even if we don't mean them to, is like learning to handle a knife so that you don't accidentally cut yourself or others.

These cases, like the earlier example of the career satisfaction chart, aren't cut and dried, so my advice isn't either. Rather than trying to create a doctrinaire list of dos and don'ts, I'll deconstruct four of the most common techniques that put charts in this gray area, explain why and when you might want to use them, and lay out why and when they may not be okay.

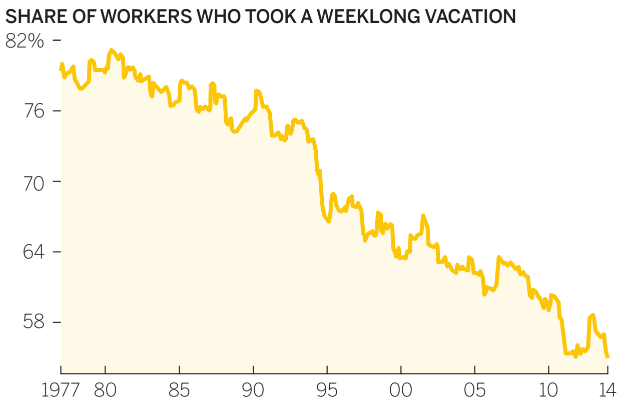
The truncated y-axis: exaggerating trends. The debate over the y-axis is visualization's version of grammarians arguing about ending a sentence with a preposition. Even if we think it's wrong, we do it because the proper alternative often feels awkward.

Why it may be effective. It emphasizes an idea. Cutting empty ranges out of an axis increases the physical distance between values, revealing more texture in the changes and making change look more dramatic, as shown in the career satisfaction example on page 176.

It's clearly true that not truncating makes it harder to see change and difference. The full-axis version uses 7% of the y-axis to show a 7% gap. The truncated version uses almost 50% of the chart's vertical space to represent a 7% gap. Truncation is a way of zooming in and isolating the main idea. It's not unlike looking through a magnifying glass.

It's also true that if a range of data is consistently far from zero, you'll need much more space to effectively unflatten the visual while maintaining a full y-axis.[6] You'll have to manipulate the height and width of the chart. This quickly becomes an impractical exercise: It yields strangely formatted charts that, although they preserve some detail of the curves, ultimately distract the viewer.

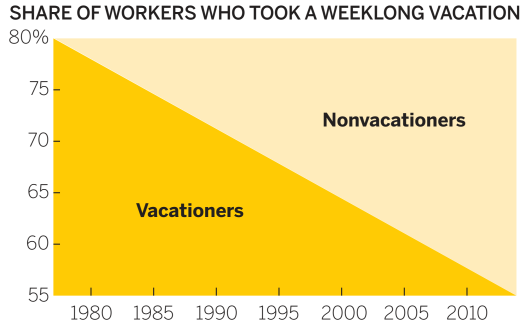
TAKING A VACATION



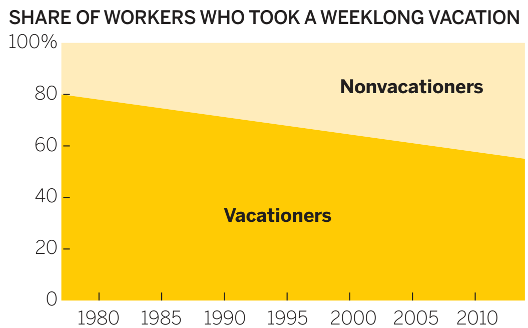
SOURCE: BUREAU OF LABOR STATISTICS, VOX

Why it may be deceptive. Some will argue that truncation acts less like a magnifying glass than like a fun house mirror, distorting reality by exaggerating select parts of it. The line on the Taking a Vacation chart above represents a drop of 25 percentage points, from 80% to 55%. But its physical descent covers almost the entire y-axis. In other words, the line descends 100% of the y-axis to represent a 25% decline. Truncation also hides representative space. The line here divides space that represents vacationers (below) and nonvacationers (above), but neither space accurately represents the proportions between the two at any given point. We can see this when we chart the *space* devoted to each variable, as shown below. In the truncated version, the proportions are simply inaccurate.

TAKING A VACATION



SOURCE: BUREAU OF LABOR STATISTICS, VOX



Another good way to understand the effect of truncation is to pluck three points from the data set and turn them into stacked bars, one group with a truncated y-axis and one that spans from zero to one hundred, as shown on the next page.

THE DATA

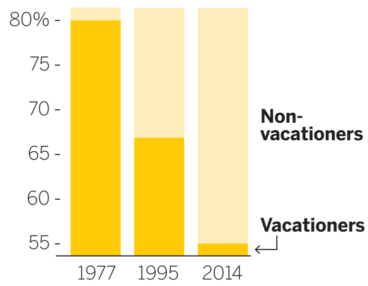
SHARE OF WORKERS

	Vacationers	Non-vacationers
1977	80%	20%
1995	67	33
2014	55	45

SOURCE: BUREAU OF LABOR STATISTICS, VOX

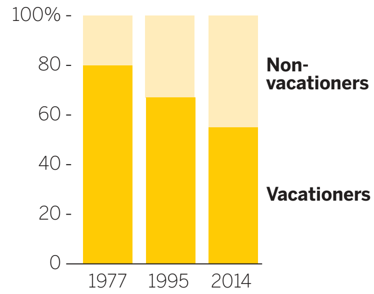
TRUNCATED AXIS

SHARE OF WORKERS



FULL AXIS

SHARE OF WORKERS



Rather than persuasive or even deceptive, the truncated-axis chart looks plain wrong, and it is. Its 1995 bar, for example, at 67%, should be two-thirds dark yellow and one-third pale yellow, but it's split about 50/50. Truncation with categorical data doesn't work. We see it used like this mostly when deception is the goal.[7] And yet the original line chart represents a similar dividing of space, except with many more data points along a continuum.

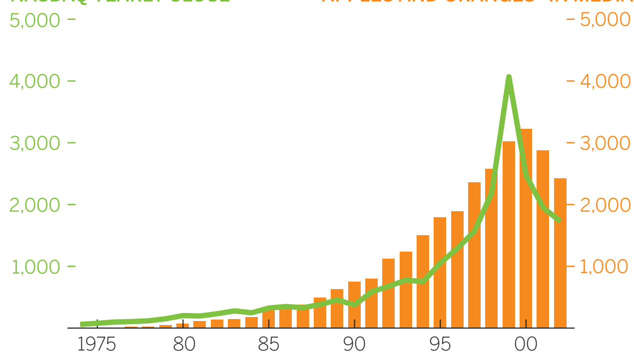
Sometimes people equate truncating the y-axis with not starting at zero. But even if it starts at zero, lopping off the top of an axis's true range also produces a distortionary effect, as it did in the immigration chart series. That kind of truncation is less often noticed and produces fewer outbursts from y-axis fundamentalists, but it can hide representative space in the same way.

The double y-axis: comparing apples and oranges. Compared with truncation, double-y-axis charts provoke little agitation. An internet search for "truncated y-axis" returns top results about lying with charts, but a search for "secondary y-axis" turns up mostly sites that teach you how to add one in Excel. Still, charts with two y-axes deserve similar scrutiny.

APPLES AND ORANGES

NASDAQ YEARLY CLOSE

NUMBER OF USES OF "APPLES AND ORANGES" IN MEDIA



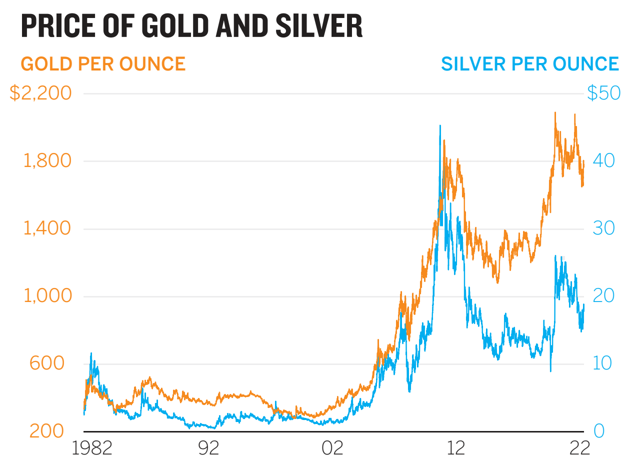
SOURCE: LEXIS-NEXIS RESEARCH, NASDAQ

Why it may be effective. It compels an audience to make comparisons. Instead of trying to convince people that there's a relationship between two variables, it creates a relationship by fiat. On the facing page is an example I created for a humorous essay on the use of the term "apples and oranges" in the media.

You can't look at this chart and consider each plot on its own merits. The fact that they're together forces you to think about them as *something*, not two things that happen to share a space. What does this chart say? You've probably formed the narrative I wanted you to: *Stock market gains lead to more people using the term "apples and oranges."*

Of course, that idea is absurd on its face—but it's almost impossible not to make the connection. I knew that (or at least I sensed it; this was created long before I had considered the mechanics of chart making) and leveraged it to send you down a path of trying to figure out *why* this relationship exists and to make a funny point. (This is one case where visual deception is allowed: in humor, when the audience knows you're being deceptive to make a funny point.) Two y-axes can shape a narrative that goes in the direction you want it to, and it is economical, using the space of one chart to plot two.

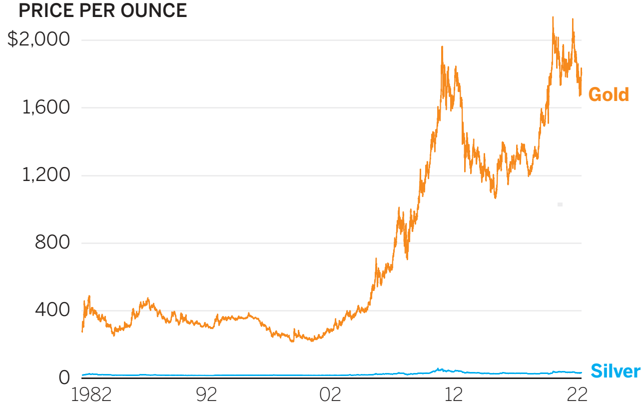
Why it may be deceptive. The relative sameness or difference in the shapes of lines or the heights of bars being measured on two different scales is much less meaningful than it appears to be. The simplest illustration is a chart that uses two axes representing the same type of value but in different ranges.



SOURCE: AURONOM.CO.UK

PRICE OF GOLD AND SILVER

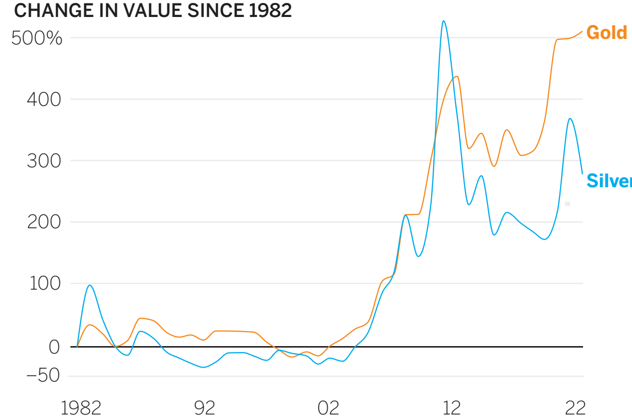
PRICE PER OUNCE



SOURCE: AURONOM.CO.UK

VALUE OF GOLD AND SILVER

CHANGE IN VALUE SINCE 1982



SOURCE: AURONOM.CO.UK

In the chart on the previous page, it appears that gold and silver are roughly the same price, and their prices move together. But the range of the secondary y-axis is two orders of magnitude lower than that of the primary y-axis. (In addition, they're truncated, so the closeness of the lines is artificial.) That means we're seeing lines that interact in fake ways. When the blue line is higher on the chart, the price of silver isn't higher than the price of gold. When the lines cross over, prices aren't crossing over. Both axes measure U.S. dollars, so why not use just one y-axis?

That's what the left gold and silver chart on this page shows, and it's simply less useful. We can't see what's happening to silver prices. One solution to this dilemma would be to show relative change in price rather than raw price, as the right-hand chart on this page shows.

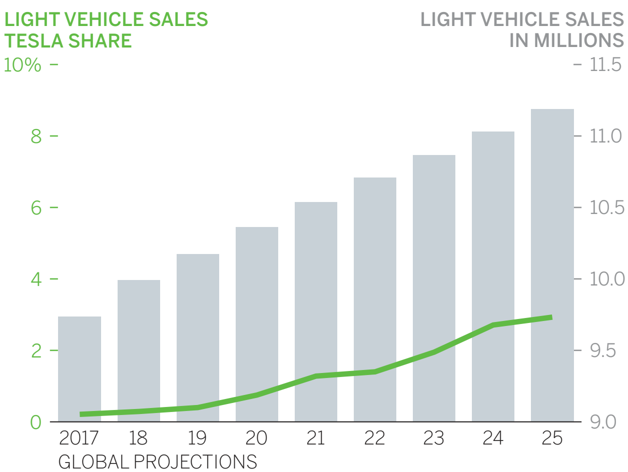
The price of silver, a flat line in the previous chart, is actually more volatile than the price of gold—an idea we don't see in the first chart. If anything, the price of gold looks more dynamic in that first chart, but the relative change from $1,300 to $1,200 is smaller than the change from $21 to $18, even though the slopes match when we use separate y-axes in the same space. Still, this new version creates new challenges. It shifts the main idea from the price of precious metals to the change in price—from value to volatility. Knowing the actual price of gold and silver at any given time is not possible in a percentage change chart.

Things get even murkier when the second y-axis uses a different value altogether. In the bottom left chart, it's hard to miss the narrative that Tesla's market share is going to come on strong in light vehicle sales. Its line reaches higher and higher into the bars that represent all light vehicle sales.

Unfortunately, that narrative is illusory. In 2025 the line reaches about a third of the way up the total light vehicle sales bar, which suggests Tesla will approach 10 million vehicle sales. Except that its y-axis is measured in percentage, not raw numbers. In 2025 it would have just a 3% market share—only 1/33rd of that year's plotted bar. The bottom right chart is an accurate portrayal of the scenario.

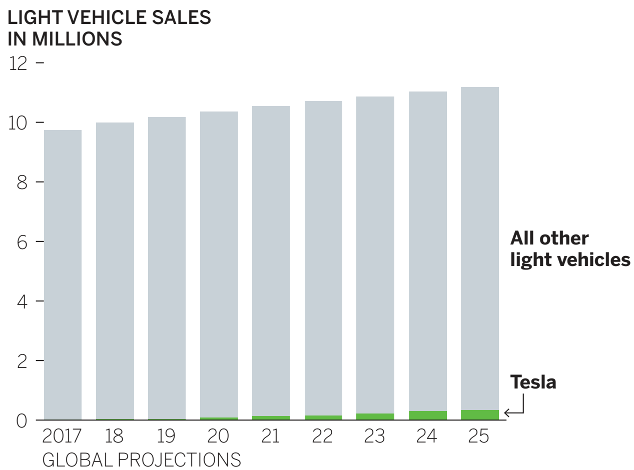
When two measures bear no relationship at all, things get truly weird, as with the chart top left on the next page.

GLOBAL LIGHT VEHICLE PENETRATION—ONE SCENARIO



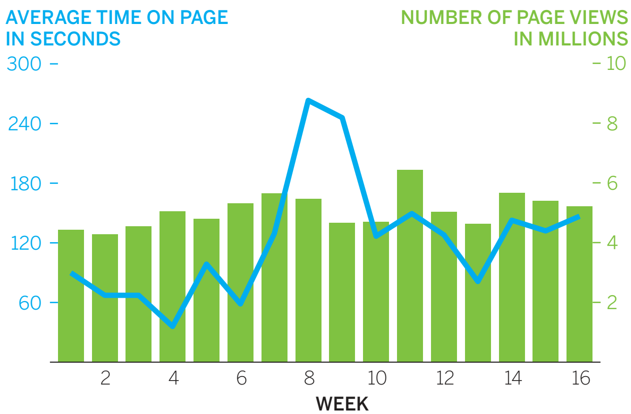
SOURCE: GOLDMAN SACHS GLOBAL INVESTMENT RESEARCH

GLOBAL LIGHT VEHICLE PENETRATION—ONE SCENARIO



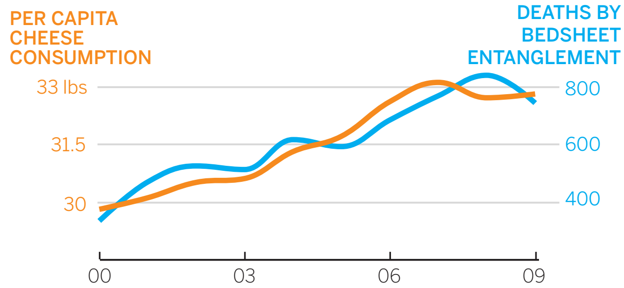
SOURCE: GOLDMAN SACHS GLOBAL INVESTMENT RESEARCH

PAGE VIEWS AND TIME ON PAGE



SOURCE: COMPANY RESEARCH

CHEESE AND BEDSHEETS



SOURCE: TYLERVIGEN.COM

We see events in physical space—crossovers, meeting points, divergences, convergences—that suggest a relationship that doesn't exist. Time on page didn't *cross over* or *go higher than* page views between the seventh and eighth weeks—and what would it even mean for seconds to be higher than page views? It's as if rugby and baseball are being played on the same field and we're trying to make sense of both as one game.

Nevertheless, when we see data charted together, our minds want to form a narrative around what we see. Charts can be concocted that combine truncation with dual y-axes to manipulate the curves into similar shapes to encourage that narrative-seeking, such as the chart to the right. The two variables here are correlated, but that's just an accident of statistics. The tempting if unlikely causal narrative is that eating more cheese increases the chances you'll suffocate in your bedsheets.[8] What happens when this visual parlor trick is applied to less silly examples? In an age of very big data sets and sophisticated tools for mining them, it becomes easy, as the Stanford professor of medicine John Ioannidis puts it, to "confer spurious precision status to noise."[9] Chart 1 in the series on the next page is a good example.

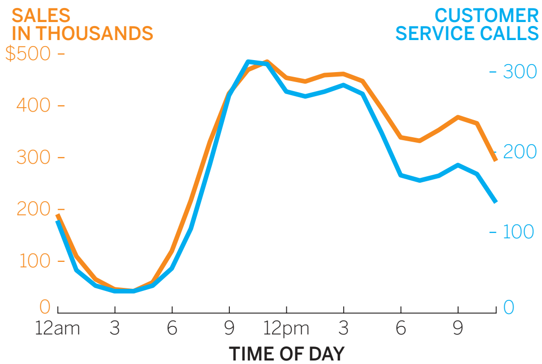
Sales and customer service calls map closely over the course of the day. The tight link might make a manager think that customer service should be staffed according to how much money the company expects to be bringing in at that time of day. More money, more reps. But the way these lines stick together, as much as we might want to believe it means something, is artificial. First, the lines stick together in part because they use separate grids.

Chart 2 in the series exposes the grid lines to show that the tight connection between lines is artificial.

It's almost as if each chart were on a semitransparent piece of paper and we slid one over the other until the curves aligned. In chart 3, when the axes are lined up to share a single grid, the picture changes.

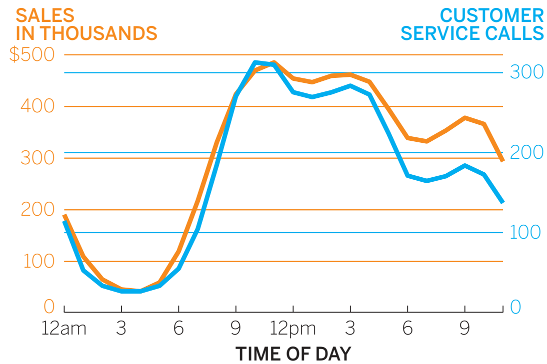
Similarity remains, but now calls are *always* lower than sales (keep in mind this is all still nonsensical since the values are completely different). Even so, we get the sense that sales and calls go up and down together. This chart still might persuade us that staffing should follow the day's sales trends.

1: SALES VS. CUSTOMER SERVICE CALLS



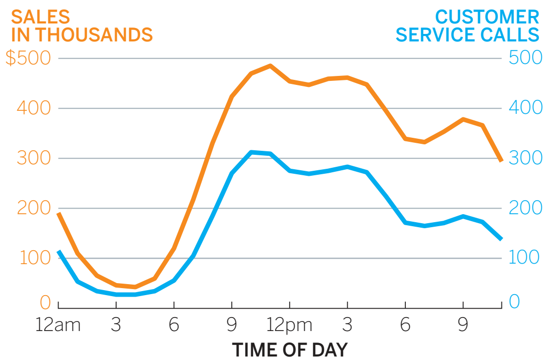
SOURCE: COMPANY RESEARCH

2: SALES VS. CUSTOMER SERVICE CALLS



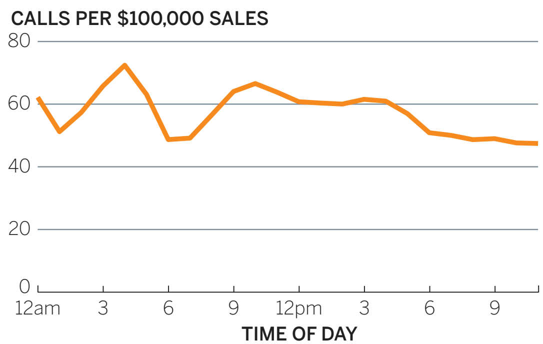
SOURCE: COMPANY RESEARCH

3: SALES VS. CUSTOMER SERVICE CALLS



SOURCE: COMPANY RESEARCH

4: DO FEWER CUSTOMER SERVICE CALLS MEAN MORE SALES?



SOURCE: COMPANY RESEARCH

But what if we take a view of the data that doesn't rely on an artificial similarity in the shape of curves? Using the same data, let's recalculate to compare sales per customer service call each hour as a ratio, shown in chart 4 in the series.

If sales and customer service calls really were as closely linked as the original chart suggests, this line would be essentially flat—as sales rise, calls rise. But this view tells a different, somewhat more nuanced story: The customer service team is handling 30% more calls for every $100,000 earned at 9 a.m. compared to 9 p.m. And the ratio bounces up and down all morning. In the first chart in this series, morning was when the lines were almost perfectly in sync, but that's when there's the most change in calls per sales.

Comparisons are one of the most basic and useful things we do with charts. They form a narrative, and narrative is persuasive. But it should be obvious by now that there are no easy ways to handle different ranges and measures in a single space. Pushing down one misleading problem can cause another to pop up. More-accurate portrayals, such as percentage change, may be less accessible or useful, or even alter the idea being conveyed.

The simplest way to fix this is to avoid it. Placing charts side by side rather than on top of each other and using presentation techniques that we'll talk about in chapter 7, can help create comparisons without creating false narratives.

The map: Misrepresenting Montana and Manhattan. Maps are themselves information visualizations, but they're also popular containers for dataviz. Assigning values from spreadsheets to geographic spaces has become essential practice in public policy circles and politics especially. The rise in popularity of color-coded maps, or choropleths," has spawned one of the toughest dataviz challenges in terms of toeing the line between effectiveness and deceptiveness.

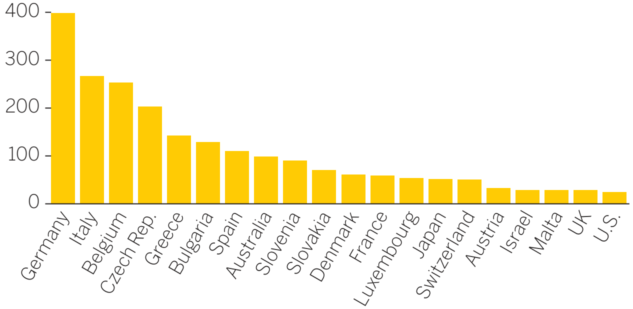
Why it may be effective. Maps make data based on geography more accessible by making it simple to find and compare reference points, because we are generally familiar with where places are. Comparing country data, for example, is easier when we embed values in maps, especially as the number of locations being measured increases. Looking at the Solar Capacity charts on the next page, see how long it takes you to complete the fairly simple task of comparing the United States with Japan, then Spain with France, and finally Germany with Australia on the bar chart. Then do the same on the map.

Choropleths also help us see regional trends that other forms of charts cannot. It's difficult, for example, to look at the bar chart and form ideas about, say, the European versus Asian deployment of solar capacity, but in the map we can make those assessments almost without thinking.

Why it may be deceptive. The size of geographical space usually over- or underrepresents the variable encoded within it. This is especially true with maps

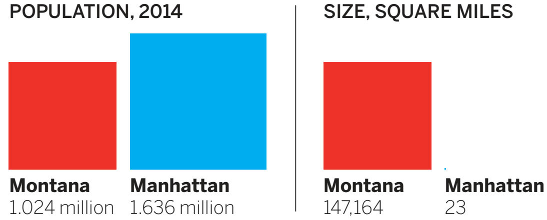
SOLAR CAPACITY

MILLIWATTS PER MILLIONS OF PEOPLE



SOURCE: CLEANTECHNICA.COM

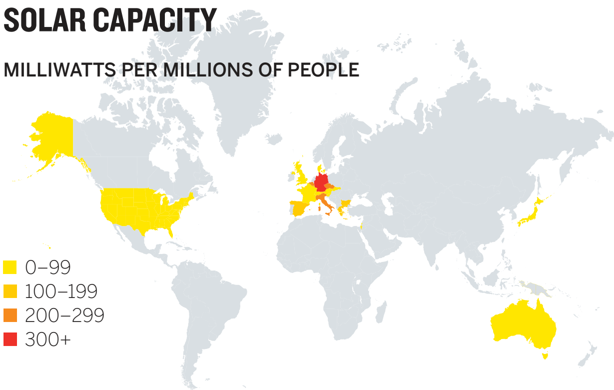
THE MONTANA-MANHATTAN PROBLEM



SOURCE: U.S. CENSUS

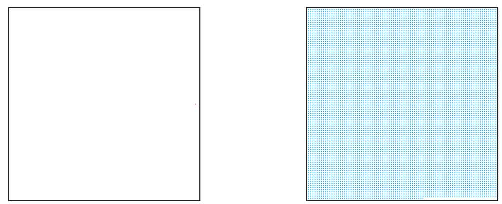
SOLAR CAPACITY

MILLIWATTS PER MILLIONS OF PEOPLE



SOURCE: CLEANTECHNICA.COM

POPULATION DENSITY: MONTANA VS. MANHATTAN



Montana
1.024 million people reside in 147,164 square miles

Manhattan
1.636 million people reside in 23 square miles

SOURCE: U.S. CENSUS

that represent populations, as we see during elections. You might call this the Montana-Manhattan problem.

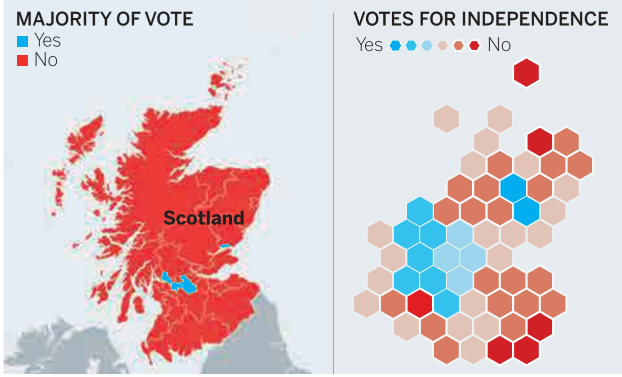
More people live in Manhattan, even though Montana is almost 6,400 times bigger. Another way to express this is to show how many people live in one square mile of each place. Each dot represents seven people.

It may be hard to see, but Montana's square mile contains one dot. So, when Montana votes one way during an election, the visual representation is of a colored-in area that's more than 6,400 times the size of the one for Manhattan, even though 60% more people live in Manhattan. This happens all over the world. On the next page are the election results for Scotland's referendum on independence plotted as a map and as a simple proportional bar chart.

Geographically it looks like about 95% of the country voted no. But what looks like an overwhelming victory isn't actually so one-sided. Less than 5% of the landmass on the map represents a yes vote; 38% of eligible voters voted yes. Consider that in the Highlands, that massive northernmost red region on Scotland's mainland, only about 166,000 people voted in total—fewer than the 195,000 who voted yes in Glasgow, one of the small blue wedges. The hexagon version of this map is an attempt to add some of the nuance back into the data, and it succeeds in downplaying the value of large spaces that have relatively small numbers of voters, but you do lose some sense of actual geographical navigation. For example, try to locate the Scottish highlands on this map.

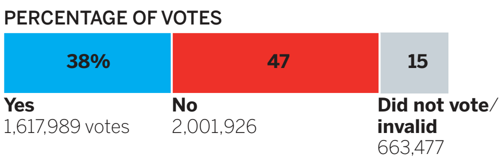
Moving away from maps, though, reintroduces the problems that maps are meant to solve by using our knowledge of where things are to make values more accessible. The proportional bar chart below, for example, makes it nearly impossible to

TWO VIEWS OF THE SCOTTISH REFERENDUM



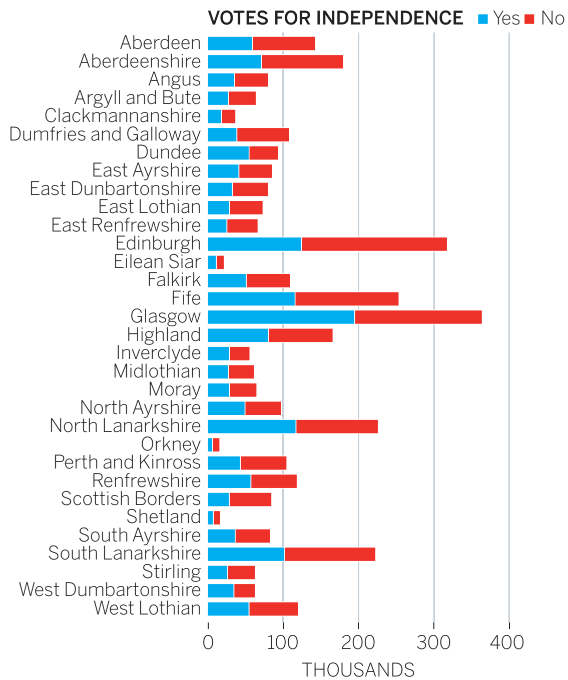
SOURCE: WIKIPEDIA

SCOTTISH REFERENDUM RESULTS



SOURCE: WIKIPEDIA

THE SCOTTISH REFERENDUM



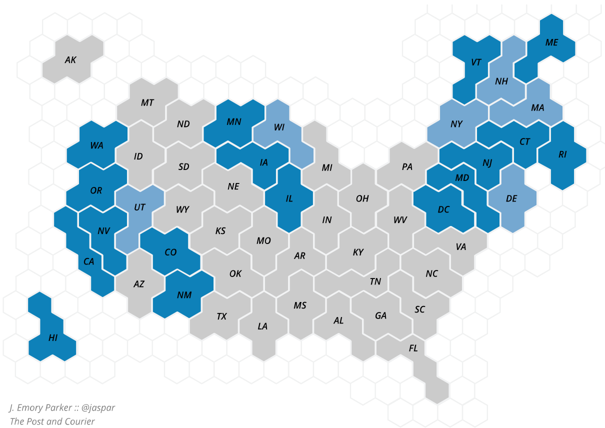
SOURCE: WIKIPEDIA

connect places to values quickly or to make regional estimations.

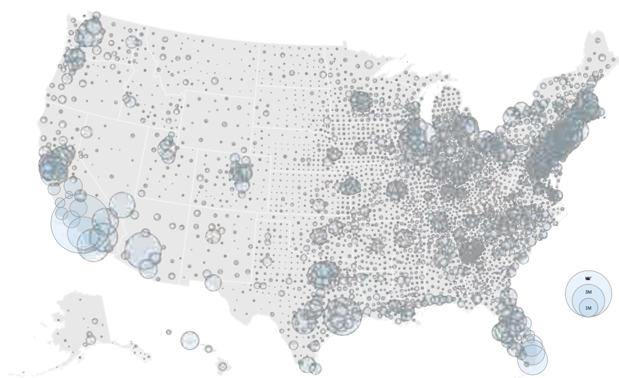
More-accurate representations of the data lead to less accessible geographic information. Conversely, good maps tend to misrepresent data values. This paradox has vexed designers, cartographers, and data scientists for some time, and they continue to look for solutions to this challenge; none has taken hold as a standard.

Grid maps provide an alternative solution. In a grid map, every region is assigned an equal size and placed roughly where we imagine it belongs on a regular geographical map. Some use squares, some hexagons, and some use compound hexagons that all have the same area but can change shape, as shown here.[10] It still takes more work to grab locations in these grids than it would in a regular map. Find New York in this grid map, for example. When I looked for Texas, I found Louisiana.

Other maps use proportional circles overlaying states, which can be striking, but it's hard to use if there are too many circles crashing into each other and it's still difficult to make comparisons between geographically disparate circles, say, Washington State and Maine (and if the values encoded in the circles have a wide range, they can become overwhelmingly disparate in size). Some use three-dimensional bars rising up from geographies. They also can be striking, but comparing values in this form is difficult; they tend to be best deployed when one geography is an outlying large value that draws the eye.



U.S. POPULATION BY COUNTY



SOURCE: MIKE BOSTOCK; HTTPS://BOST.OCKS.ORG/MIKE/BUBBLE-MAP/

These efforts are less misrepresentative than the ones that use real area to encode other variables, but they also flout a deeply ingrained convention in our heads—the shape of the world—and make us work harder, sometimes much harder, to find what we're looking for. That can be frustrating and therefore less persuasive.

Uncertainty: The paradox of showing potential futures. How do you show what hasn't happened, and might not happen? The paradox of charts showing uncertainty is that they force you to visually determine the undetermined. You must show what might be, but the act of showing it makes it appear to be. Humans struggle processing probability. Combine that with the high facticity of data visualization—when we see things charted, they seem to reflect a true reality—and you have a steep challenge of making uncertainty visible while not making it seem certain.

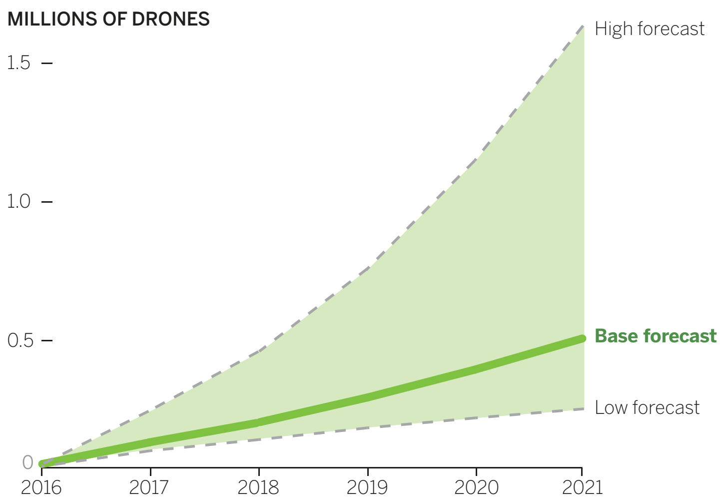
Why it may be effective. This is not to say we shouldn't visualize uncertainty. It's a highly valuable way to discuss multiple potential futures and ranges of possible outcomes. It's most effective when there is some certainty within the range of possibilities, and even more effective if you can assign probabilities to those potential outcomes. The classic academic approach is a box-and-whisker plot that shows a certain range as a bar and lines extending from either side to show the full potential range of outcomes. For statisticians and those used to using them, box-and-whisker plots are fine, but most audiences don't parse them so easily.

Another approach is to change a solid line to a dotted line, or a semitransparent line when real data becomes projected data, signaling that *we think this is going to happen* or *on the current trajectory, this is what will happen,* but it hasn't happened yet. Again, this may be rendered as multiple scenarios.

A most popular approach is what's sometimes called a "fan chart" because of the way the range of outcomes fans out, as with the drone chart on the facing page. The lighter hue band of data around a "most expected trend" line represents the range of possible outcomes. In some charts, the saturation of the uncertainty color deepens as probability increases and goes paler as it decreases—a smart way to signal likelihood using our brain heuristic that as color empties, so do values.

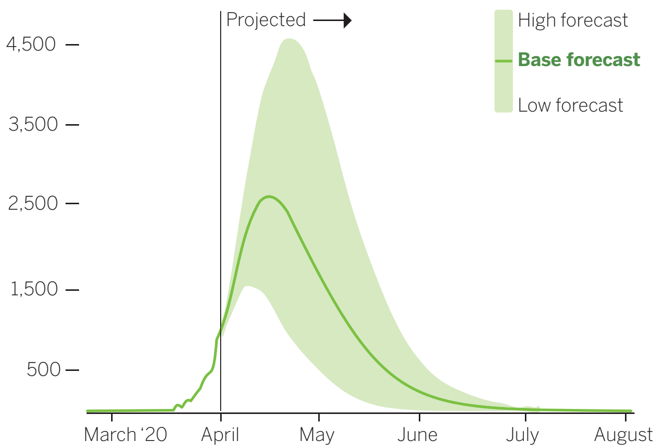
Why it may be deceptive. There's no getting around the fact that plotting uncertainty gives it a veneer of certainty. For example, look at the drone map again. The high forecast is extremely unlikely. Let's say it has a one-in-a-thousand chance of happening. And let's say the base forecast has a one-in-five chance of happening. Would you get the sense that the high forecast is 200 times less likely to exist than the base forecast from this visual? Can you even say what "200 times less likely" *should* look like?

COMMERCIAL DRONES ARE SET TO TAKE OFF



SOURCE: FAA

PROJECTED DAILY DEATHS FROM COVID-19



SOURCE: 2020 MODEL, INSTITUTE FOR HEALTH METRICS AND EVALUATION

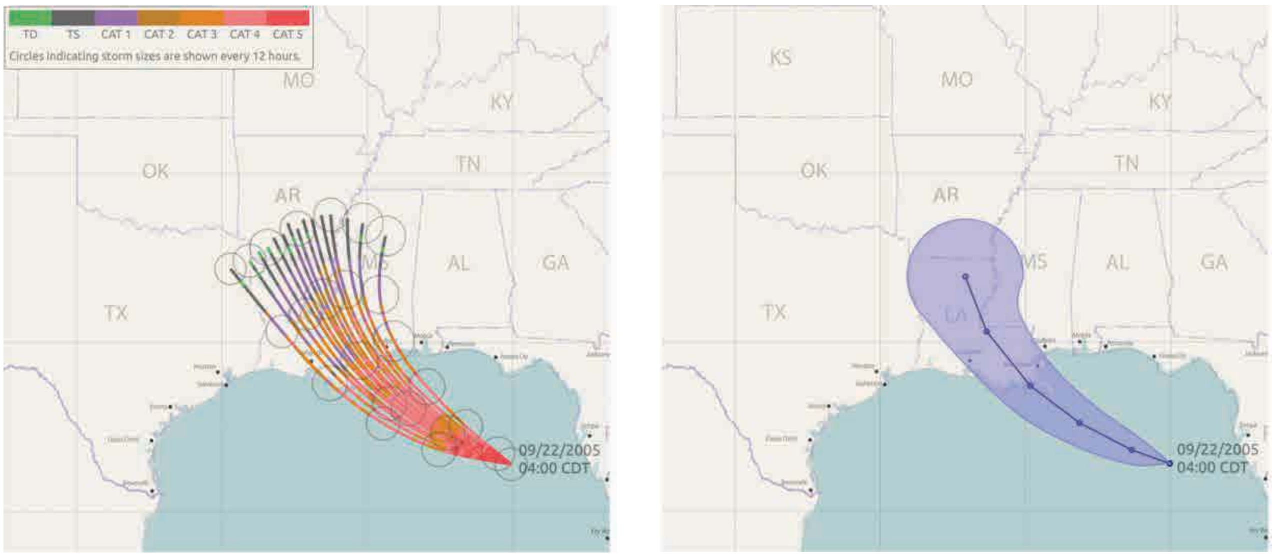
Such difficulty accurately representing uncertainty manifests in users of charts as anxiety and frustration. Famously during the 2016 election, the *New York Times* used a needle over vote percentages for each candidate, like a pressure gauge. The needle jittered as results came in to represent uncertainty in the outcome. The frenetic moves left and right were not representative of any real probability values, even though the needle hovered over real data values. It was just meant to metaphorically represent "uncertainty." The gauge only generated confusion, complaints, anger, and angst.

The pandemic became a master class in the problems with visualizing uncertainty. Charting potential deaths in such a fluid situation was a precarious task, even with probabilities attached to the outcomes. Experts were desperately hoping that visualizing the potential dire consequences would change people's actions to minimize poor outcomes.

But such charts also produce anxiety as it makes real "worst-case scenarios"—the mere act of visualizing such a thing affects the audience in potentially disproportionate ways to what the data suggests is *likely*.

This is also true in another classic uncertainty visualization, the hurricane projected path map. Such maps are widely deployed as storms barrel toward land. Often, they're animated. But they present a host of problems that can make them deceptive.[11] The ones shown below, for example, show two approaches. The sprayed lines give me no sense whatever of the likelihood of any one of these paths. What if the line that curls back into Mississippi is 90% likely to occur and all the others are a combined 10% likely? What's more, I don't get a good sense of the area that will be affected. The *path* of the storm is less important to me as a user of the visual than the *swath* it will pummel.

So, the second one fixes that, right? Actually no. It's a clearer depiction of probability—I see the most likely path and then the shaded area is all other paths, but that cone looks more like *the area the storm will affect*. I see one path and the swath of the storm getting bigger as it hits land when in fact that's not what it's meant to represent at all. These visual decisions have real consequences. When Hurricane Ian hit Florida, people were making decisions to stay in place or evacuate based on what they saw in a similar cone projection—mistaking the visual for showing what area would



CREDIT: LE LIU

be affected. But these charts show neither the affected area or secondary effects of a storm such as flooding and tide surges.[12]

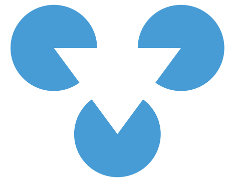
I don't mean these observations about deceptive charts to be criticism. They are not accusations that people are trying to deceive. Remember the manipulation matrix on page 178. Many visualizations fall into a gray area, and we're just trying to spot the reasons they might, and how to avoid that. Visualizing uncertainty remains one of the most difficult challenges for anyone. They're only offered here to help you avoid becoming deceptive when you visualize uncertainty and probability.

WHAT'S GOING ON HERE?

By now it should be clear that facts and truth are different. You can create multiple truths from one set of facts. That's strange when you think about it. Data is data. How can I present the same data to two people and get them to believe different, even opposite, truths?

The answer goes back to how we process visuals, and a concept called the Law of Prägnanz. This word translates roughly to "pithiness." Without getting too deep into gestalt psychology theory, all this means is that the simplest organization, requiring the least cognitive effort, will emerge as the figure. That is, our brains and visual perception systems do as little work as possible to find the easiest meaning.

This is true even if the figure that emerges in your mind isn't actually there. To the left is a famous example.



There are no circles or triangles in that figure. But your brain *can't not see them.* As gestalt psychologist Kurt Koffka put it, "The whole is other than the sum of the parts."

Applied to data visualization, that means we don't assemble the parts into a whole idea. We don't process all the data points and note their arrangement; compare their placement, their colors, and all the other marks on the page; and say to ourselves, "This all adds up to an idea."

We don't evaluate that picture and think, *There are three circular shapes, two on top, one in between the two on top below, each with 25-degree wedges removed, and the radius that makes up each edge of the wedge pointing to a corresponding radius on one of the others.*

We see it and think, *There's a triangle on top of three circles.*

The same is true for data visualization. When we look at the Scottish referendum map, we're not thinking about percentages of people who voted one way or another; we're thinking, *"No" won by a* *massive landslide*. When we look at a steep up-and-down curve, we think, *That's a volatile trend*. When there's an outlier on a scatter plot, we think, *That's different from the others*.

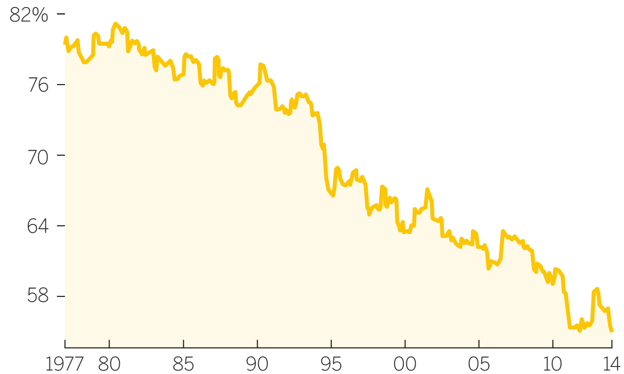
It's crucial to remember that, when you are creating persuasive visualization, and trying to avoid being deceptive, the audience is not reading your data. They are not parsing statistical information. They are seeing a whole and only afterward thinking about the parts.

First, they feel something. Then they try to relate to it, make sense of it. And *then* they think. But by that time, they've already formed the idea about it, and often those ideas are based on those heuristics and conventions we talked about in chapter 2. Those shortcuts in our mind we use to rapidly grab meaning so that we don't have to think much about something we see all the time. Up is positive, down is negative. Time goes left to right. So on.[13]

When the line in the vacation chart approaches the bottom—the "end" or the "floor"—of a chart, we take that as a cue that it's approaching zero, or nothing. This creates a false sense of termination. We *expect* the bottom to be zero, and our brains want to process it that way. When we realize it's not zero, we have to expend more mental energy trying to understand what we're actually looking at. Conversely, we see the top of the chart as the maximum, pinnacle, or ceiling. The truncated-axis vacation chart leads us toward the idea that *everybody used to go on vacation and now no one does*. But compare it to the full y-axis version below it.

TAKING A VACATION

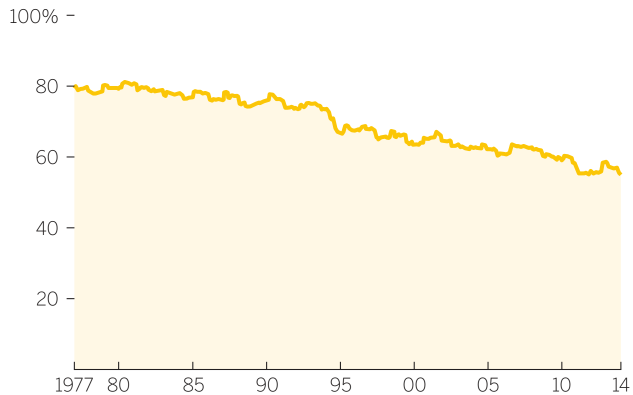
SHARE OF WORKERS WHO TOOK A WEEKLONG VACATION



SOURCE: BUREAU OF LABOR STATISTICS, VOX

TAKING A VACATION

SHARE OF WORKERS WHO TOOK A WEEKLONG VACATION

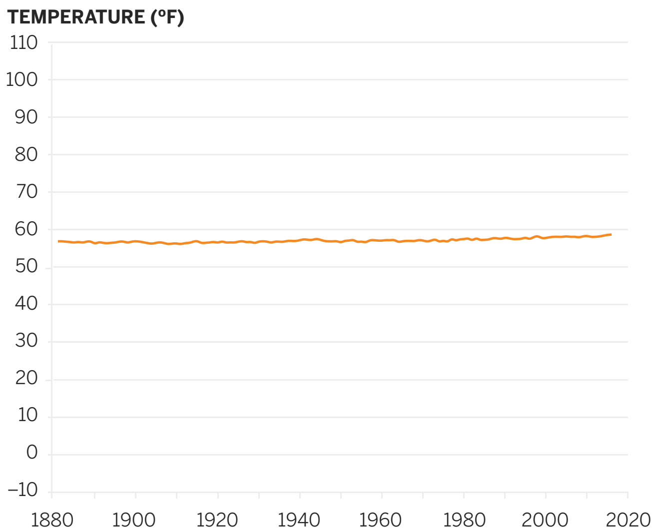


SOURCE: BUREAU OF LABOR STATISTICS, VOX

Okay, the number of vacationers is indeed declining, *but more people than not still take a vacation*. Did that idea come through from the truncated version? Did you see it first? Was it an accessible idea? Did you get the sense that on average, over nearly four decades, a vast majority of people took vacations and a majority still do?

You can see how someone might use how we process information to engineer persuasion or deception into a visualization. Look at this chart, based on one that made the rounds on Twitter.

AVERAGE GLOBAL TEMPERATURE, 1880–2020



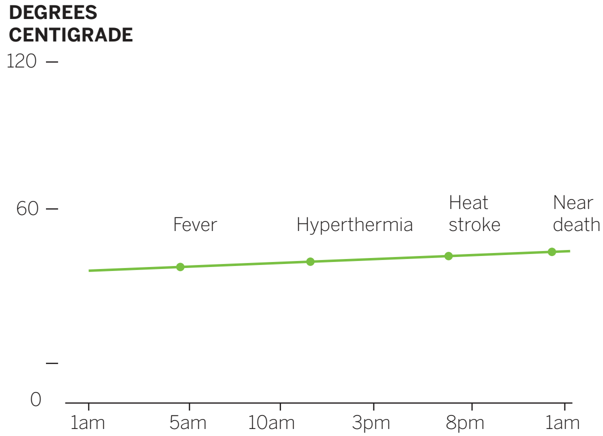
SOURCE: NASA

I love that this chart is accurately plotted. It's an inane plot, but it's not wrong. The chart maker has engineered this knowing that you see this and don't think about temperature changes and their significance. You look at this and see a flat line, and flat means no change, status quo, safe. The trick here is to drastically exaggerate the y-axis (a kind of anti-truncation). Global temperature averages will never range more than a few degrees, but this chart includes a possible range of 120 degrees. In truth, a half-degree change is significant, but you can't see that significance here, never mind feel it. A half-degree is only 0.4% of the y-axis. Significant changes have been designed out of the chart.

There are two ways to combat such deception. One is to show a different view that engineers the opposite feeling in the audience, and many critics of this chart did just that, truncating the y-axis and showing steeply rising curves and dark red zones of data that all looked dangerous and menacing. When you do this, you have to be prepared to be able to demonstrate how small changes are significant. Even if they look significant in a truncated-axis chart, for example, that's just what the chart shows. Can you demonstrate significance in another way? Can you, for example, show the correlation between a half-degree change and famine?

Another way is to overcome the this, to demonstrate to the audience that flat lines are sometimes very bad. Using the same techniques that the chart maker used for the global temperature

BODY TEMPERATURE



SOURCE: CDC

charts, I created a visualization that I hoped could change the conversation on flat lines.

This works, but it's a lot of work to constantly challenge how people's brains naturally process information. One way that deception can be overcome is through datavisual literacy. The more we know about how our brains process data visualizations, and the more we know the techniques that are used to persuade or deceive, many of which we've learned here, the more prepared we are to detect and disqualify deceptive visualizations, whether they're deceptive by accident or on purpose.

KNIFE SKILLS

I described the borderland between persuasion and deception as blurred. It should be obvious why. Most of the examples deconstructed here feel not perfectly right or wrong but, rather, endlessly debatable.

I also described the borderland as shifting, and in some ways that's the more difficult characteristic of persuasion techniques to reconcile. A truncated y-axis chart may be fine in one setting and violative in another. Even two colleagues in the same meeting might disagree about whether it's convincing or spurious.

Judging whether your visualization crosses that indefinite line will, like any other ethical consideration, come down to one of those difficult, honest conversations with yourself. Ask yourself:

- Does my chart make it easier to see the idea, or is it actively changing the idea?
- If it's changing the idea, does the new idea contradict or fight with the one in the less persuasive chart?
- Does eliminating information hide something that would rightfully challenge the idea I'm showing?
- Does the chart make me feel or see something I know doesn't reflect reality?
- Would I feel duped if someone else presented me with a chart like this?

If you find yourself answering yes to questions like these, you've probably entered deceptive territory. Another way to check yourself is to imagine someone challenging your chart as you present it. You might even recruit a colleague to practice. Do you have the supporting evidence to counter a challenge? Could you defend your chart and yourself against attacks on its and your credibility?

The line between visual persuasion and visual deception will never be completely clear. The most important thing you can do is to not think about the design techniques you use as right or wrong but rather make sure that the idea those techniques help you convey is defensible.

RECAP

FACTS AND TRUTH

Every chart is a manipulation. We make dozens of decisions, conscious and subconscious, about what we're showing and how we show it that affect the truth an audience will see.

One set of facts can lead to multiple truths represented in multiple data visualizations.

We use manipulations of visuals to persuade, but used too aggressively or recklessly, persuasion techniques—emphasis, isolation, adding or removing reference points—can become deceptive techniques: exaggeration, omission, equivocation. The line between persuasive and deceptive isn't always clear. The best way to negotiate it is to understand the most common techniques that put charts in the gray area, understand why you'd be tempted to use them, and realize why they might not be okay. Here are four:

1. THE TRUNCATED Y-AXIS

What it is:

A chart that removes valid value ranges from the y-axis, thereby removing data from the visual field. Most often it doesn't start the y-axis at zero.

Why it may be effective:
It emphasizes change, making curves curvier and distance from one point to another bigger. It acts as a magnifying glass, zooming in on the space where data occurs and avoiding empty space where data isn't plotted.

Why it may be deceptive:
It can exaggerate or misrepresent change, making modest increases or declines look "steep." It disrupts our expectation that the y-axis starts at zero, making it possible or even likely that the chart will be misread.

2. THE DOUBLE Y-AXIS

What it is:
A chart that includes two vertical scales for different data sets in the visual field—for example, one for a line that tracks revenues and one for a line that tracks share price.

Why it may be effective:
It compels the viewer to make a comparison between data sets that may not naturally go together. Plotting different values in the same space establishes a relationship between the two.

Why it may be deceptive:
Relationships between different values are artificial. Plotting those values in the same space creates crossovers, matching curves, or gaps that don't actually mean anything.

3. THE MAP

What it is:
A map that uses geographical boundaries to encode values related to that location, such as voting results by region.

Why it may be effective:
Geography is a convention that allows us to find data quickly on the basis of location rather than searching through a list of locations to match data. It also allows us to see trends at local, regional, and global levels simultaneously.

Why it may be deceptive:
The size of a region doesn't necessarily reflect the data encoded within it. A voting map, for example, may be 80% red but represent only 40% of the vote, because fewer people live in some larger spaces.

4. UNCERTAINTY

What it is:
A chart that depicts something that isn't real or only has a chance of becoming real.

Why it may be effective:
Modeling multiple futures can help drive better decision making, especially if those uncertain futures are weighted by probability.

Why it may be deceptive:
It makes tangible something that doesn't yet exist and may not ever exist. It may overweight highly

unlikely outcomes so that they look more possible than they are. It creates anxiety in an audience, especially when the data projects bad outcomes, like potential deaths or where a hurricane might hit.

THE LAW OF PRÄGNANZ

The reason we can see many truths from one data set is because our minds don't read data, they find the simplest, fastest explanation for a picture, a gestalt psychology principle called the Law of Prägnanz, or pithiness. The whole is not the sum of the parts; it's other than the sum of the parts. We see the whole to make sense of the parts.

Understanding this, we can design persuasion into our charts, but persuasion can slip into deception if we're not careful, and there are no hard-and-fast lines between the two. Situational context may make a chart persuasive in one setting and deceptive in another.

It's crucial to avoid accidentally, or intentionally, deceiving an audience with data visualization. Datavisual literacy helps combat visual deception.

Judging whether your visualization crosses that indefinite line between persuasion and manipulation will, like all other ethical considerations, come down to a difficult, honest conversation with yourself. Ask:

- Does my chart make it easier to see the idea, or is it actively changing the idea?
- If it's changing the idea, does the new idea contradict or fight with the one in the less persuasive chart?
- Does eliminating information hide something that would rightfully challenge the idea I'm showing?
- Would I feel duped if someone else presented me with a chart like this?

PART FOUR

THE LAST MILE

CHAPTER 7

PRESENT TO IMPRESS AND PERSUADE

GETTING A GOOD CHART TO THEIR EYES AND INTO THEIR MINDS

BY NOW YOU'RE CONCEIVING of and building smart, persuasive visualizations—good charts. So far, all your energy has gone into working out ways to develop and manipulate the charts themselves. Now, you can focus on taking that well-conceived object and helping people connect to it.

Typically, we aren't terribly good at that. We build a smart viz and hope that the chart itself—this clear, self-sufficient, persuasive little object of visual communication—will engage an audience. But the text of a brilliant speech doesn't compel an audience to action; the orator does. The score of a symphony doesn't move people; its performance does.

How you get a good chart to people's eyes and into their minds is what matters most. Effective presentation marks the difference between information visualizations that are merely adequate exposition and ones that move people.

Getting charts to eyes and into minds may sound figurative, but I mean these things literally. The twin challenges here are to help people when they first see the visual—how you present it to them—and to help them process it: how you get them to engage with it. I'll take these in turn.

GETTING IT TO THEIR EYES: PRESENTATION

At some point most managers learn how to give a presentation. They read books about it, take a class, or hire a coach.[1] The skills those tools offer are useful in presenting charts, but they may not cover specific techniques for presenting visualizations that can help make them easier to understand and more persuasive. Here are several techniques that will help you.

First show the chart and stop talking. Researchers estimate that about 55% of our brain activity is devoted to processing visual information. The visual system, crudely explained, includes a high road that handles spatial information and navigation, and a low road that recognizes and processes objects and shapes. No matter what the visual input is that hits your eyes, both roads teem with activity. Put a chart on a screen, and the entire ventral section of the brain fires up to suss out some meaning. As George Alvarez, a visual perceptions researcher at Harvard, puts it, “Mostly, vision is what the mind does.”

So if you present a chart and immediately start talking over it, you’ll make it harder for your viewers to understand the chart. Their brains really want to *look*, and you’re asking them to *listen*, too, and their brains are actually trying to shut down the listening stuff to put more processing power into vision. Visual processing is so intense that once we see something salient such as a color or a shape, we even start to tune out *other visual information* around what we’re focusing on—never mind sounds—in order to make sense of what we see.

Instead of talking over your new chart, display it and don’t talk for several seconds. If it helps, count five beats in your head. Let the viewers’ brains dial in on this new thing to look at. You’ve done the hard work of making the visualization clear and persuasive. You’ve made the salient information highly accessible. You’ve used the title and subtitle as confirming cues about the idea you want to convey. Don’t undercut your own hard work. Let the chart do what it was built to do.

The urge to start talking over a visualization is noble enough: You want to make sure people *get it*, and silence can be unnerving. Those five beats, I promise, will feel uncomfortably long at first. But inevitably, what happens during this initial pause is more useful than anything you might preemptively say. In education, such an extended silence is a well-established tactic called “wait time” or “think time.”[2] Teachers who allow three seconds or more to pass after they ask a question tend to have classes that are more engaged, think more critically, and come up with more-sophisticated answers to problems.

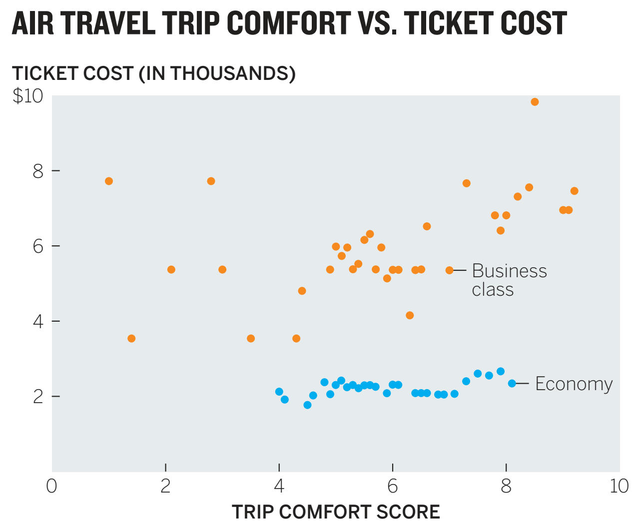
That’s what will happen if you pause after showing a chart. Eventually, someone in the audience will puncture the silence with a question, or offer analysis or an opinion. You may find that the chart spurs discussion without you saying a word. If you let people arrive at their own insights, the idea in the visual will be talked about more, and more deeply, than if you immediately tell them what they should see. Paradoxically, the silence creates a deeply interactive moment.

When it's time to talk, don't read the picture. The easiest way to lose your audience in any presentation is to read bullet points verbatim from a slide. Explaining the structure of a chart that you're presenting will disengage an audience just as badly.

Imagine presenting this map with the following script:

So, here's a map of China divided into its provinces. North is at the top of the map, and each province is distinguished by a light-yellow border outline and labeled with its name. Surrounding countries are labeled as gray, and the East and South China seas are shown, which are lighter gray. As you can see, distance is measured according to a key in the upper left corner.

Explaining how a map works toes the line of condescension, but more importantly, wastes time and disengages an audience. Here's a typical presentation of a chart and the script that might accompany it:



SOURCE: CARLSON WAGONLIT TRAVEL (CWT) SOLUTIONS GROUP, TRAVEL STRESS INDEX RESEARCH (2013)

So, here we are showing trip comfort versus how much a plane ticket costs. Comfort is 0 to 10 on the x-axis, and the cost of the ticket is on the y-axis. As you can see, economy-class tickets—the blue dots—don't vary much in cost, but comfort does. There seems to be a little more correlation between comfort and cost for business-class tickets, but only at the very high end, and even then, it's not a very strong effect.

Everything this presenter has said we can already see; he even says "as you can see," which is a clear tip-off that he's wasting time declaring the obvious. If they can see it, why say it?

Once it's time to talk, discuss *the idea*, not the object that shows the idea. Here's a new script for the presenter:

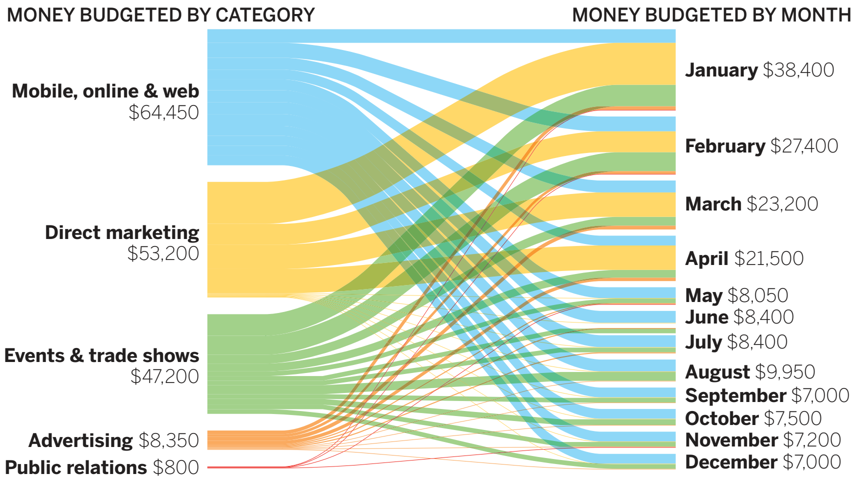
> [After five beats] Money doesn't seem to buy much comfort on plane trips, unless we pay the very top prices in both economy and business class. For most trips, comfort is average—in the middle—whether we pay $5,500 for a business-class ticket or $2,200 for an economy ticket. This suggests that only the most expensive tickets are worth the cost. Since we know it's not cost that determines comfort, we should explore what does so that we can ensure productive trips at the best cost.

There's no talk here of axes or color or how dots are clustered. Everything the presenter says is about the idea (money doesn't buy comfort), analysis of the idea (most business-class tickets aren't worth the cost), prompts for discussion (if cost doesn't affect comfort, what does?), and a reminder of the value of discussing the idea (happier employees at a reasonable cost).[3] Notice how discussing ideas instead of explaining the data and structure naturally leads to more human-centered language. Rather than explaining a price-to-comfort ratio, he's talking about comfortable employees and successful business trips. That's good. As the presentation guru Nancy Duarte put it to me, "Don't project the idea that you're showing a chart. Project that you're showing a reflection of human activity, of things people did to make a line go up or down. It's not 'Here's our Q3 financial results,' it's 'Here's where we missed our targets.'"

Reading a chart's structure during a presentation is often a sign that you lack confidence in the visualization. If you aren't sure the audience will get it, you probably haven't highlighted the main idea well enough. If you find yourself explaining the salient information, maybe you haven't emphasized and isolated it the way you could. Resist the urge to just read the chart, let those five beats of silence go by, and the questions and comments that come back will be a referendum on the chart's effectiveness. If people are asking about axes and labels and what they should be looking at, the visualization needs improvement.

With unusual forms and for added context, guide the audience. Mostly you should avoid talking about the chart itself, but there are exceptions. Unusual or complicated forms may require brief explanation prior to discussing ideas. Familiarity with forms does affect the ability to understand visualizations: You probably can't, for example, drop an alluvial diagram like the one on the next page on an audience without at least some explanation of how it works.

MARKETING COMMUNICATIONS PLAN BUDGET



SOURCE: COMPANY RESEARCH

This may elicit *oohs* and *aahs* when it first pops up, but if the viewers can't find meaning in it, they'll quickly write it off as a pretty picture or, worse, an attempt to show off that favors eye candy over insight.

That doesn't mean you should avoid unusual and complex forms: If they help frame ideas well, they can be powerful ways to engage people. But the time from *Gee whiz!* to *I see!* must be short. To make the transition, describe the function of the chart form before focusing on the idea:

> This alluvial diagram shows how our marketing communications dollars flow throughout the year. It helps us see three things: One, how our budget is distributed by program, represented by the thickness of the bars on the left. Two, how our budget is allocated by month, represented by the thickness of the bars on the right. And three, how each program's money flows over the course of the year, represented by how the lines move from left to right. Take a look. [Wait five beats] We seem to have two seasons for marketing communications: January to April, a shorter season of heavy, heavy spending. And May to December, a long season of spending a little bit on a lot of programs. Big direct marketing investments fall into that first time frame, which also happens to be when our events business needs heavy investment. Is this distribution okay? Do we need to rethink this?

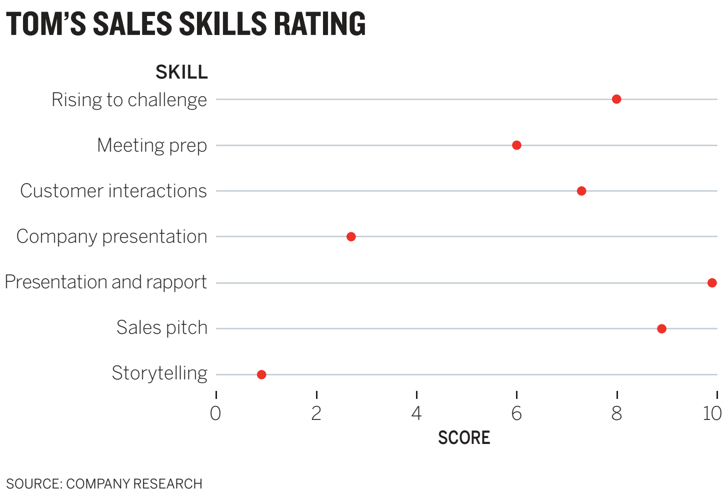
Notice that even in this case, while the speaker rightly explained the function and mechanics of an alluvial chart, she didn't fall into the trap of describing this particular example. She didn't say:

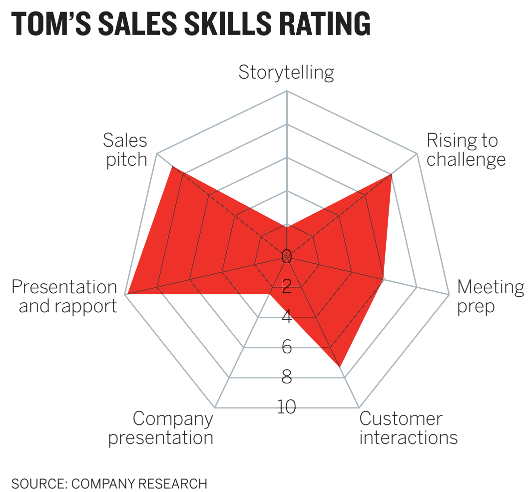
> The events business, in green here, represents a little more than 25% of our budget, and the spending skews slightly toward the beginning of the year, as you can see by the thicker bars flowing into January and February.

Keep explanations of forms brief, clear, and general, not specific to the data encoded in your chart.

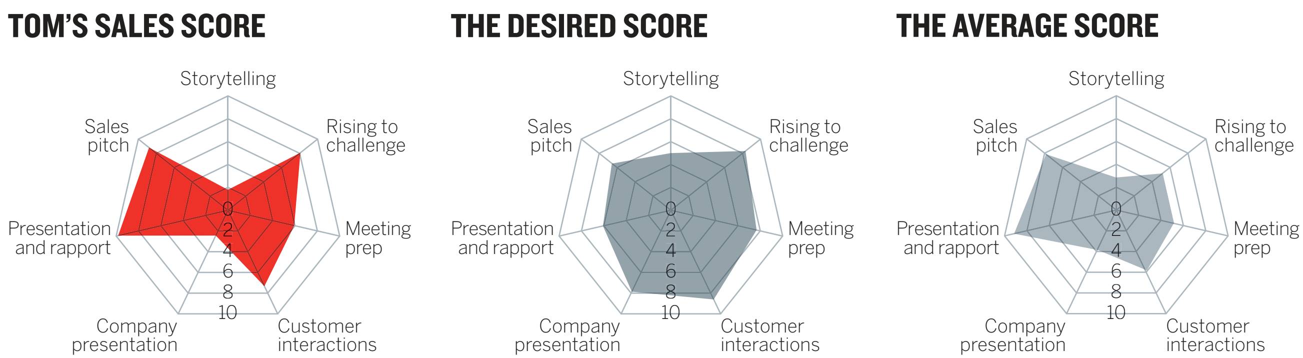
Use reference charts. Prototypical examples can also guide the audience, providing cognitive nudges toward clearer meaning. Presenting an average case, and ideal case, or other reference points works well even with basic charts, but it can be especially effective when presenting unusual forms. If you wanted to assess Tom's sales skills on seven different measures, you could use a dot plot like the one below. Or you could try the spider graph (also called a radar chart) next to it, which gives shape to multiple data points.

It's more difficult to assess Tom's *overall* performance in the dot plot, because we have to evaluate seven discrete data points and then intuit what they combine to mean. But with the spider graph, we see a whole thing: one shape to represent overall performance.[4]





Okay, but the shape is meaningless in itself. Presenting this chart on its own would create questions that aren't easily answered. Is this a *typical* shape? Is it *good*? The data, Tom's overall *score*, is clear. The main idea, Tom's overall *performance*, isn't nearly as accessible. Let's add two prototypical references, *average performance* and *desired performance*, along with an accompanying script for the spider graphs:



SOURCE: COMPANY RESEARCH

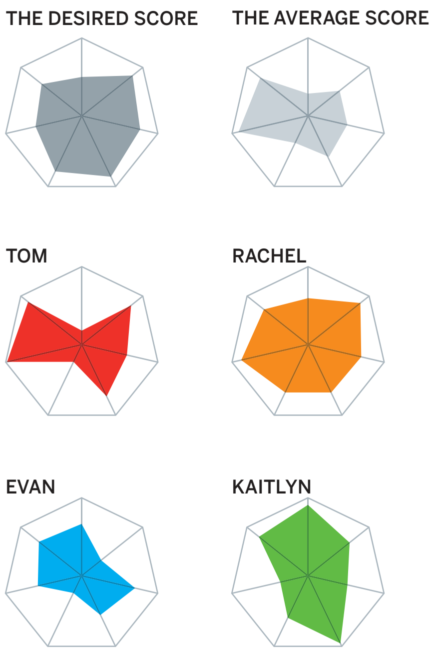
> [Wait five beats] The shape we're looking for skews out to the right. Those skills are more closely linked to sales success. But on average our sales team is strong on the left. Is that at the expense of developing the skills on the right? Tom's performance skews more right than average, but notice the bowtie shape. Those pinched points at the top and bottom are below average. Storytelling skills and company presentation skills have got to improve, but especially company presentation. We need to invest there.

Notice how the reference charts inject meaning into Tom's chart that isn't possible to access without the references. They help us set expectations and make sense of an otherwise arbitrary visual.

Also, since we're now evaluating a reasonably simple shape, the charts don't require much detail and can be scaled down. The entire sales staff could be presented in multiple small charts, with little additional explanation required. A team that had grown accustomed to these visualizations might not even need labels. Imagine a sales dashboard in which a sales manager could see the shape of team performance at a glance, such as the set shown on the facing page.

Now, without labels, and having looked at just one example previously, you can spot the best- and worst-performing salespeople.

COMPARING MULTIPLE SALESPEOPLE



SOURCE: COMPANY RESEARCH

When you have something important to say, turn off your chart. This presentation technique comes from George Alvarez, who had noticed in his Harvard lectures that when he kept a dataviz on the screen during class, students' eyes would be fixed on it. Even when Alvarez had moved on to another subject, he sensed that his students weren't fully with him as he tried to make important points.

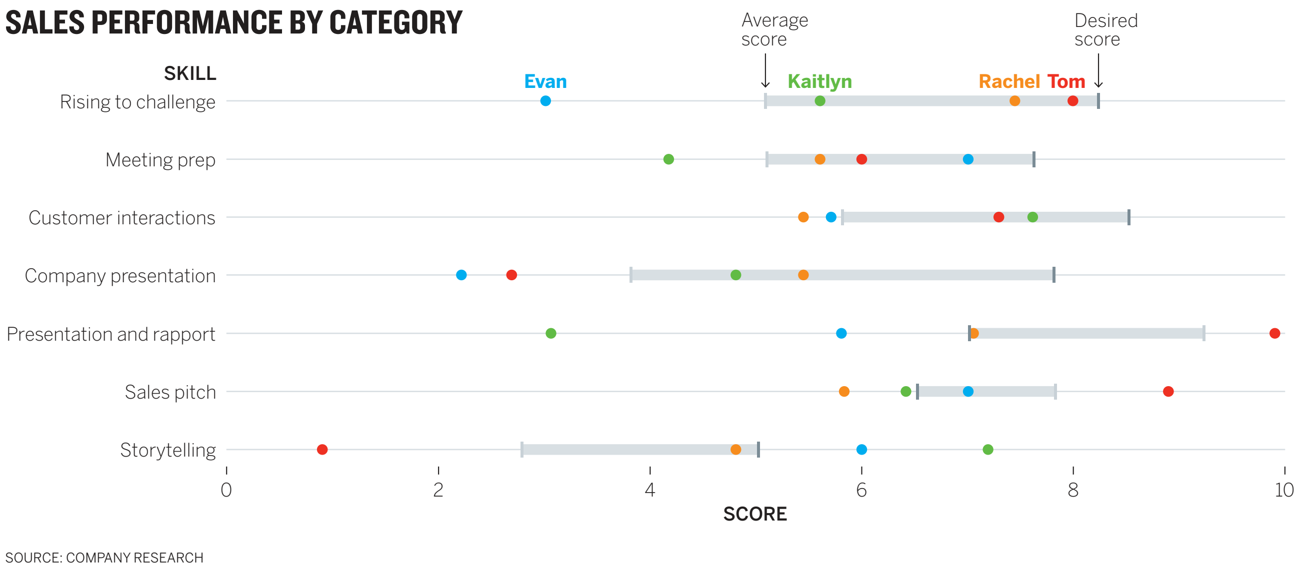
One day in class he showed his visualization and then, when he was ready to say something that the students needed to hear, he shut off the screen. The effect was stunning. Eyes that had been fixed on the picture darted to him and locked in. With nothing else to look at, the students listened intently.

I've adopted this technique in my presentations and have experienced the same dramatic effect. It's uncomfortably immediate and takes getting used to, but it works. I don't shut off my screen entirely; I put up a blank page, usually a solid color. Sometimes if you shut it off or make it black, it creates confusion, as if there was an A/V glitch.

There's a moment in every presentation when you want the audience to focus on what you have to say. It could be when you raise an alarm about performance, or explain the reason for a strategy shift, or ask for money. At those crucial times, the best thing to do with your visualizations is to put them away. Give the audience no choice about where to focus.

Show something simple. Leave behind something detailed. A good chart depends on context—we know this. The context of a presentation requires disciplined simplicity—you have a few seconds for the audience to get it. But nothing precludes your producing more-detailed versions of the visualization to leave behind with your audience so that they can explore the visual in more detail in their own time and at their own pace.

Compare the spider graphs of sales team performance from before—a good choice for a presentation or a dashboard—with a leave-behind chart that combines all that data in one space:



This plot wouldn't play well projected on a screen. It contains too many data points and offers too many places to focus. An audience may not know whether to focus vertically (comparing, say, average with desired scores across all seven categories) or horizontally (comparing everyone within one category). It doesn't steer us to any particular idea.

The spider graphs gave an at-a-glance sense of how individual salespeople were performing. This leave-behind visualization would allow a sales manager to spend time alone more deeply absorbing the information. Think of it as a bit of visual discovery—a category of dataviz in which we tolerate additional complexity for the sake of finding new things. The sales manager may want to confirm or refute a hypothesis he has about what skills his team needs to improve on. He may make notes about how to get certain dots moving in the right direction. He may notice, for example, the lack of dots near the desired score for "company presentation"; the whole sales team isn't even close to where it needs to be in that skill.

Finally, it's good practice to make data tables available as leave-behinds too. This mini-system of visualizations—the presentation version that requires a few seconds to understand, and the personal version that an individual can spend time looking at and thinking about, and a table that provides the raw material, and may allow someone to do some of their own visualizing—extends the usefulness of your presentation beyond the formal group setting.

GETTING IT INTO THEIR MINDS: STORYTELLING

The presentation techniques above are tactical and, frankly, somewhat defensive. Mainly they focus on preventing you from undermining your own charts and helping you to keep the audience from disengaging. Now let's focus on increasing engagement by getting the ideas in your charts into their minds.

We'll do that by telling stories.

Nothing's trendier than storytelling with data. An entire genre of journalism has emerged from it. Twitter is rife with links that promise to tell you "the story of [unemployment, climate change, the Roman Empire] in [1, 7, 50] charts."[5]

Data scientists, too, are latching on to narrative to communicate the complexities they pluck from big data sets, and software is trying to make it easier to string visuals together into a story.

In a way, visual storytelling is just a tributary feeding into a deeper, swifter river of business activities that use narrative as a catalyst—selling, persuading, leading. Much of it is born in design thinking and bolstered by neuroscience. As much as visual perception scientists might say that *vision* is what the mind does, many neuroscientists would argue that *stories* are what the mind does. They've shown that our brains react differently, and more positively, to stories than to bulleted lists or series of data points.[6] Many more parts of our brain are active when we're engaged with a narrative. Stories increase empathy, understanding, and recall. Storytelling is persuasive. The psychologist Robyn Dawes even argues that we can't make sense of statistics very well *without* narrative—that our "cognitive capacity shuts down in the absence of a story."[7]

CHARTS IN THE TIME OF COVID

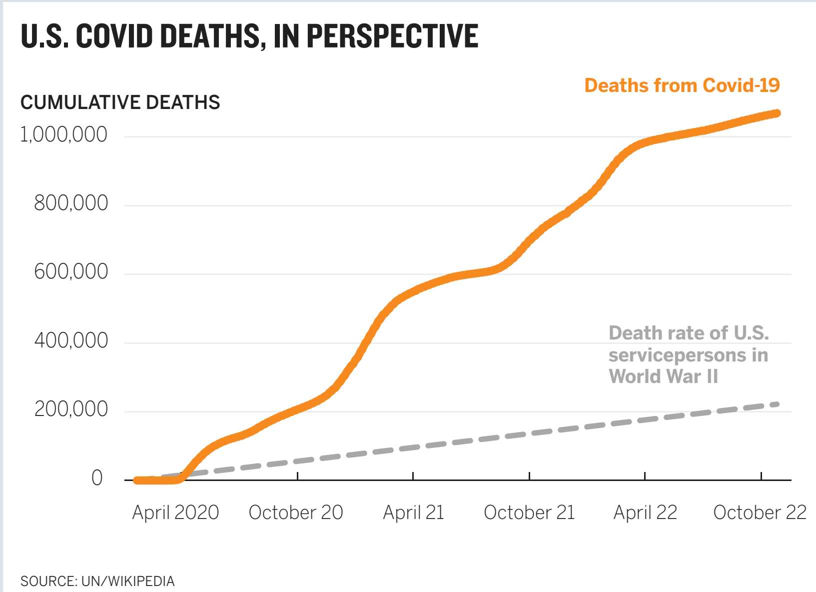
In 2020, good charts became an important weapon in the fight against the global pandemic. The stakes were obviously extreme.

Those who understood the emerging data were challenged to present increasingly complex and difficult ideas to an anxious, confused, and sometimes skeptical public. The pandemic pushed many into tackling the hardest challenges with data visualization—projecting uncertain futures responsibly, and dealing with extreme changes in data and very large numbers.

Websites popped up to recommend best practices for people visualizing Covid data. (Some of these were just good charting practices, such as being clear with labels; others were specific to the situation's data, such as the best ways to count positivity rates.) Simple procedures like heat maps had to be rethought as the data got "so hot, so fast" that maps of infection turned into fields of deep red. Others struggled to make y-axes that could accommodate the drastic and sudden changes in values. The *New York Times* famously used the full length of its page-one broadsheet to accommodate a y-axis showing a staggering rise in infections.

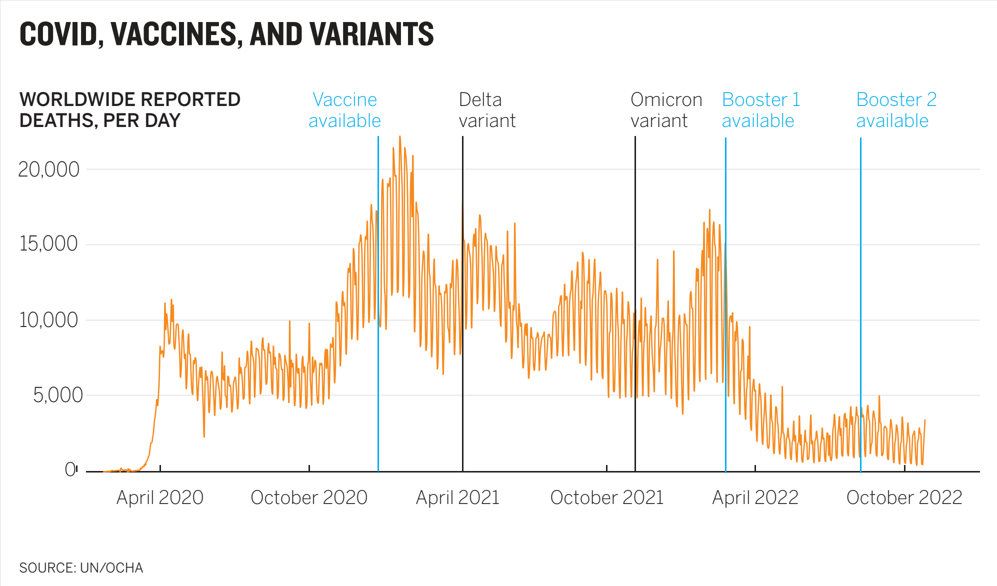
For me, the most heartening and interesting work was the flurry of charting activity aimed at presenting data visualization that moved people and spurred action.

Many worked hard to "personify" the data so that the staggering death toll didn't become a mere statistic. One effort measured the death toll in "9/11s"—the thinking being that that event was an unspeakable tragedy, and during the pandemic, there were some days when several "9/11s" happened. (Some people found this in poor taste, but the concept moved people to think about it.)



The cumulative death chart here uses a powerful reference point to put the scale of the pandemic into a new context—deaths during World War II—that may move people to see the toll of the pandemic in a new way.

Another important job for charts during the crisis was to show efficacy—that the actions we take to combat the virus have positive outcomes, as many charts show multiple models of what would happen



through different courses of action (lockdown, masks, no masks) and the chart above shows worldwide deaths before and after vaccines and variants.

Other positives came out of the work. Many media outlets upped their visualizations significantly. Small multiples—series of small charts with the same axis to map the same variable against different subsets—say, charting infection rates by country—emerged as a workhorse. Animations to represent models of how infections might unfold were put to powerful use.

Many charts that emerged during the crisis are destined to become models for future work, taught in classrooms and parsed by experts.

Here are two stories about two different topics. The first, on this page, is textual. The second, on the facing page, is visual.

I chose to tell different stories with the text and the chart because if I had used the same narrative, reading the text first would have made it hard for you to evaluate the chart's merits independently. Still, you can compare your experiences with each because their storylines are nearly identical—stable prices followed by sequential events that changed conditions and sent prices skyrocketing.

Notice how much more quickly you reach understanding when you look at the picture. The text feels like a transfer of information—something you have to work hard to understand and retain through reading and thinking. In the chart, you just see the idea. You don't have to hold specific values—prices, dates—in your mind, or calculate time frames for the change. You see a long period of stability followed by a quick spike. Comprehension feels almost instantaneous.

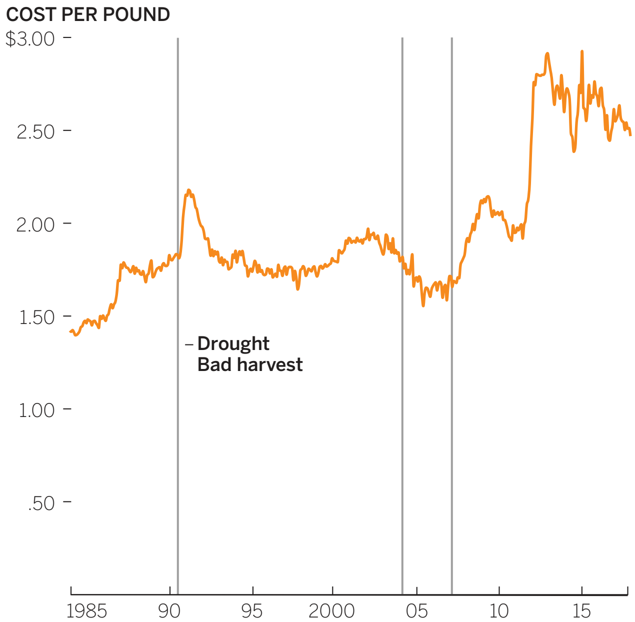
Narrative emerges much more quickly when it's visual. Thus, visual storytelling is an immensely powerful way to present ideas. If we define narrative broadly, as just a sequential presentation of related information, then even a simple chart can become a visual story. Intuitively, we know this.

> In the early 1990s the price of copper was low, under 50 cents a pound, but the shutting down of two mines because of political turmoil sent prices up to nearly 90 cents a pound. It stayed that way for several years and even dropped again to a stable 75 cents per pound. But then in 2003 a landslide at a mine sent prices over $1 per pound. Then, after a strike at a mine in Chile in 2004, prices passed $2 per pound. Because of these events and continued high demand, production fell below consumption, which caused prices to reach nearly $4 per pound by 2006.

We sometimes present charts by saying, "This chart tells the story of . . ."

But more practically, you need to know how to present visuals in a way that taps into the human need for narrative and exploits visualization's power to convey a story instantly. Here are a few techniques:

THE RISING PRICE OF PEANUT BUTTER



SOURCE: CPI

Create tension. Your boss likes to play games. He walks into a meeting and sings the familiar alphabet song: "A-B-C-D-E-F-G. H-I-J-K . . ." He stops. And waits. The room fills with real tension. Everyone feels beholden to the unresolved melody. It's captivating, in a literal sense. Nothing else can happen until it's been finished. You can't *not* finish it, and inevitably, someone will finally sing out, "LMNOP!"

If you think of a chart as having a melody—the shape of a line, or how dots are scattered on a plot—you can similarly captivate an audience by reaching points of tension and stopping. Until you reveal all the visual information, that melody is unresolved, and people will want to resolve it.

The easiest way to do this is exactly what your boss did with the alphabet song—pause before you get to the natural stopping point. "Here's how we scored with customers last quarter. And this quarter's scores [pause] . . ."

The short, unexpected silence generates anticipation, causes people to look up from their doodles, turn away from their screens, focus on the visual, and seek the ending.

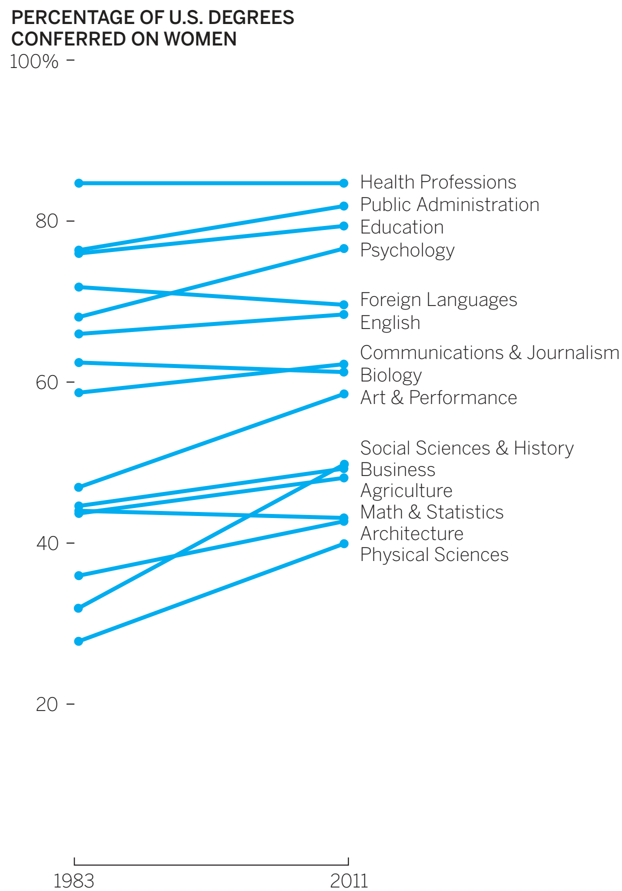
This technique invites interaction. Viewers are forced to think about how the melody will resolve. They'll try to fill in the blank space. Encourage this. Show three versions of your revenue chart and ask them to guess which one reflects reality before revealing the answer. Withhold labels from a bar chart that shows which products generate what portion of overall revenue, and ask them to match products to the values. Withhold key information, as with the slope graph to the right and its script below.

> [Wait 5 beats] There's not a college major charted here in which women earn fewer than 40% of the degrees given out. This shows great progress. But we haven't yet added two other degree categories. [Pause]

The presenter signals their intention to show more. The audience wants to know what the majors are, and where they fit in. Many people (including you) are already guessing. Encouraging speculation increases the tension. "What are they? Where will they fit?" And the longer they hold the moment, the more people will need an answer—the more they will want to resolve the melody before proceeding.

There are other ways to create tension. Using time and distance can help convey a sense of vastness or large values. A simple and effective example is distancetomars.com, an animated visualization that supposes that Earth is 100 pixels wide and then "travels" through space from Earth to Mars as stars fly by. A few seconds after you leave Earth, you arrive at the moon, 3,000 pixels away. Then you take off again (moving at the equivalent of three times the speed of light). After ten seconds or so, tension rises, because it's unclear when you'll finally "arrive" at Mars. Ten seconds becomes 20. Then 30. The longer it goes on, the more a sense of uncertainty overtakes you as you watch. Even though you've already grasped the main idea—Mars is really, really, really far away—you still want to get there.

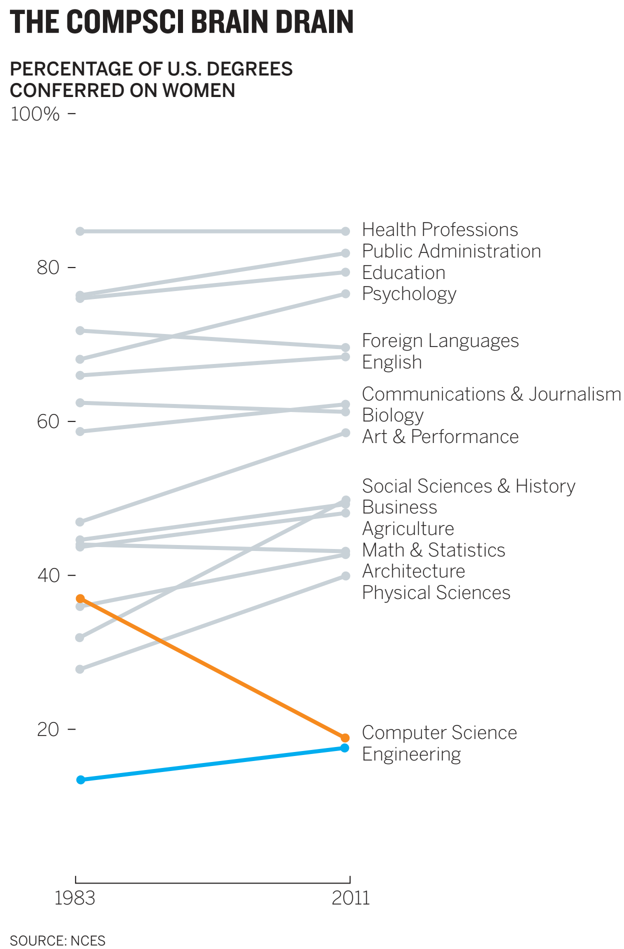
MORE WOMEN ARE EARNING DEGREES



SOURCE: NCES

Ultimately, it takes about one minute to get to Mars. It feels like a long time, but also just short enough that you don't become annoyed and start thinking, *Okay, I get the point*. That elicits the first of two caveats about creating tension: make sure you resolve it soon enough after you create it.

For example, are you annoyed that you don't yet know what percentage of computer science and engineering degrees are given to women? Did you forget about that chart? I probably ruined the effect of the tension by waiting too long to resolve it and distracting you with other things in the meantime. It won't be as effective now. Here it is anyway.



That would have been a powerful reveal had I timed it well. Second caveat: Use the reveal technique judiciously, when its effect will be felt because the idea being conveyed is somehow remarkable. A typical quarterly revenue chart that contains no surprises doesn't lend itself to creating tension. Pausing with every chart and inviting speculation on every chart you present would grow tiresome quickly.

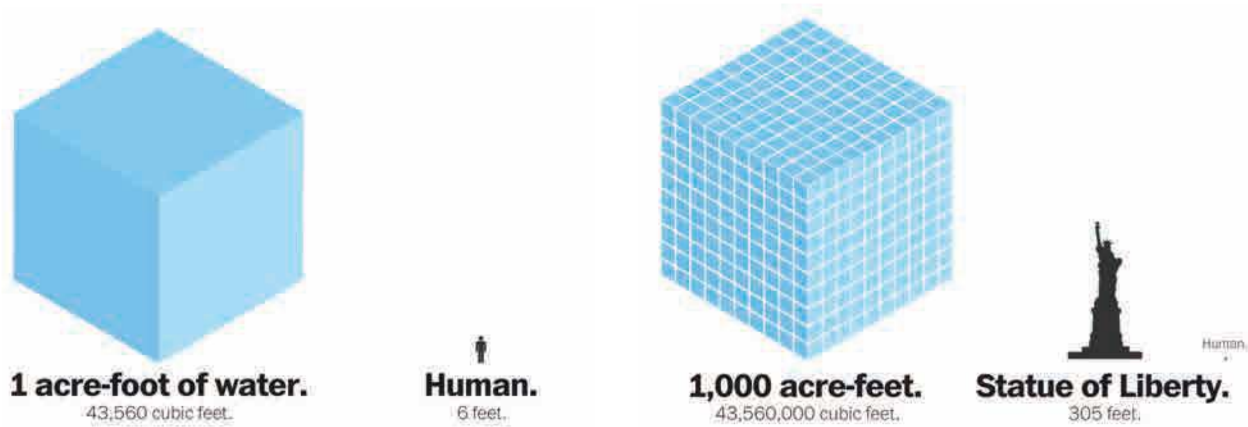
Creating tension works best when the reveal is dramatic. The reveal about women's degrees is unexpected—even if you were sure that computer science degrees would be lower, did you think they would be *that much* lower? Did you expect they would have fallen by *half*?

It also works when the information is overwhelming. Christopher Ingraham, a journalist at the *Washington Post*, used this kind of tension and reveal well when he wanted his audience to understand how much water had flowed into Houston's reservoirs during a series of severe storms. The amount is hard to comprehend, so Ingraham started by comparing two things we can relate to—one acre-foot of water (a standard measure) and a person—and then walked us through increasingly large comparisons.

"Quite a bit, isn't it?" Ingraham asks after the first in the series. After the second he says, "We're still not at the right scale."

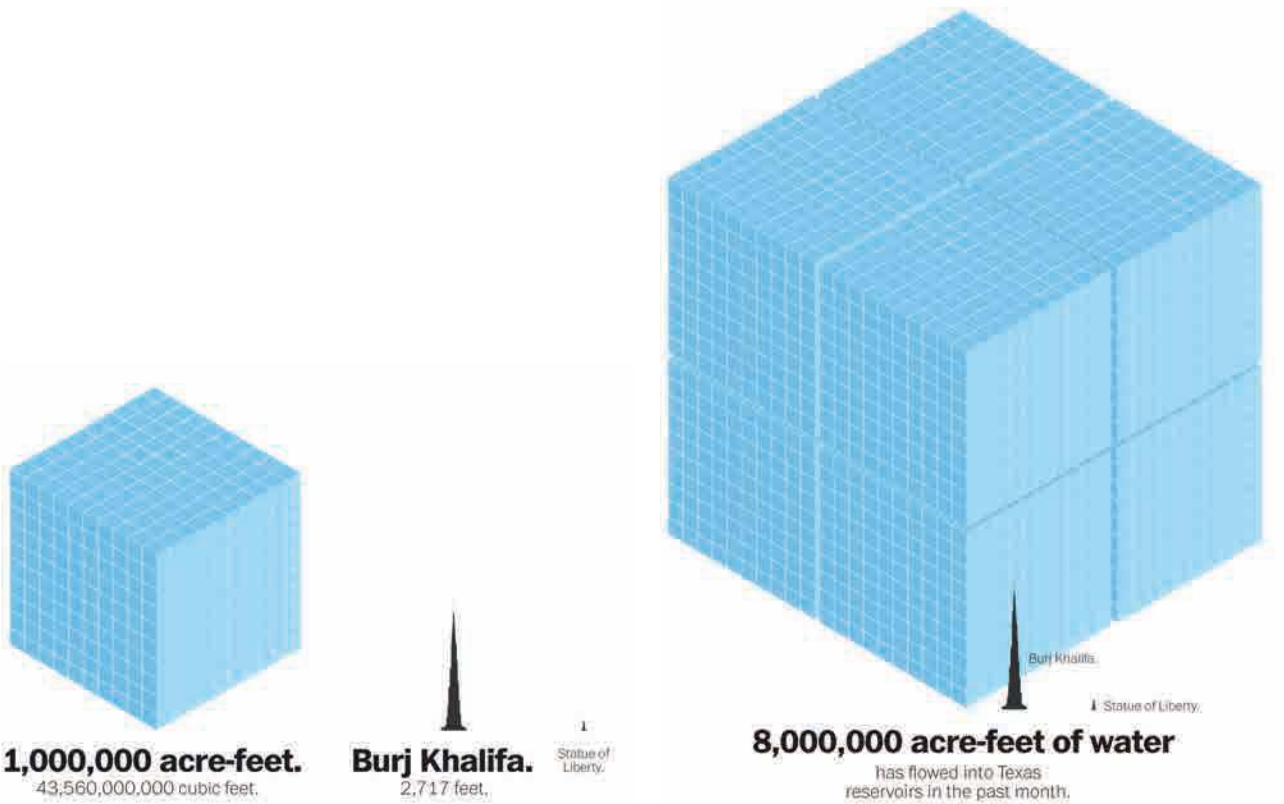
At each step the audience's tension increases a little, but so does its understanding of the vast volume of water we're talking about.[8] These intermittent reference points make us wonder just how "insane" (Ingraham's word) the amount of water was.

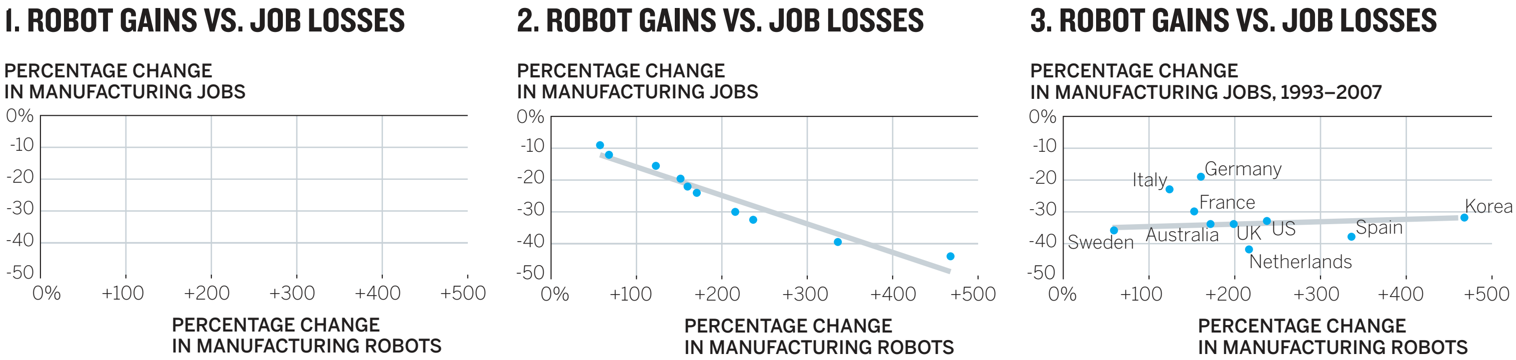
"Now we're getting somewhere," he writes after the third visual, and at this point we feel that he's just playing with us. We need this melody resolved. How much water flowed into Houston's reservoirs?



Finally, the reveal. It's enough water, he explains, to serve 64 million people's water needs for one year. The scale of the disaster is better understood because of how he brought us through the story.

Before-and-after charts are also effective at creating and resolving tension. Think of home-makeover shows. We stay tuned to see a bathroom transformed from something rundown into something astonishingly attractive.





SOURCE: GRAETZ AND MICHAELS, "ROBOTS AT WORK," AND BROOKINGS INSTITUTE, MARK MURO ANALYSIS OF BUREAU OF LABOR STATISTICS DATA

[Pause five beats] Robots are taking our jobs, right? Automated systems obviate the need for workers. We wanted to see the trend, so we mapped manufacturing job losses against number of robots deployed over the past 15 years in 10 countries. What do we expect to see on chart 1? [Pause, wait for answers] Right. As the deployment of robots increases, jobs decrease. Something like chart 2. [Pause five beats] Well, when we plotted the actual data in chart 3, this is what we saw: [Pause three beats] We were wrong. There's no correlation between more robots and fewer jobs. In fact, the UK and Sweden, two of the four countries that have lost the most manufacturing jobs, have deployed robots much more slowly than other countries.

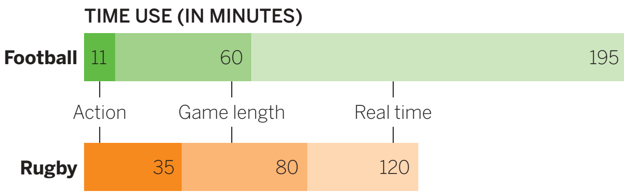
Lure. A bait and switch, or what scientists sometimes refer to, delightfully, as a "lure procedure," is also a powerful presentation technique.[9]

In this series, the center chart with the expected results lures people to commit to an idea. The reveal is so completely different, however, that it compels the audience to think through what just happened. *Why* isn't it what I thought it would be? Inconsistency creates internal anxiety that we feel compelled to fix.[10] And the greater the inconsistency, the more we want to reconcile the dissonance. In the face of visual evidence like this, it's hard to hold on to assumptions or even deeply held beliefs. It's a powerfully persuasive presentation technique that forces an audience to think about why things aren't the way they thought they were.

Be careful with this procedure, though, and make sure the bait is clearly labeled or designed so that it doesn't look like real data. It may include a title to signal this, something like "What We Expected to See" or "A Guess at What We'll See." In this case I also didn't label the plotted points with country names to prevent anyone from misinterpreting the bait as the real data.

Deconstruct and reconstruct. I'm fond of this chart that compares time use in American football games and rugby matches.

FOOTBALL VS. RUGBY



SOURCE: *WALL STREET JOURNAL*, THE ROAR

The point I want to make is that rugby is more exciting than football: It's a longer game that features more action in less real time. That idea comes through, eventually. In truth, as simple as this is, I could make it better in a presentation.

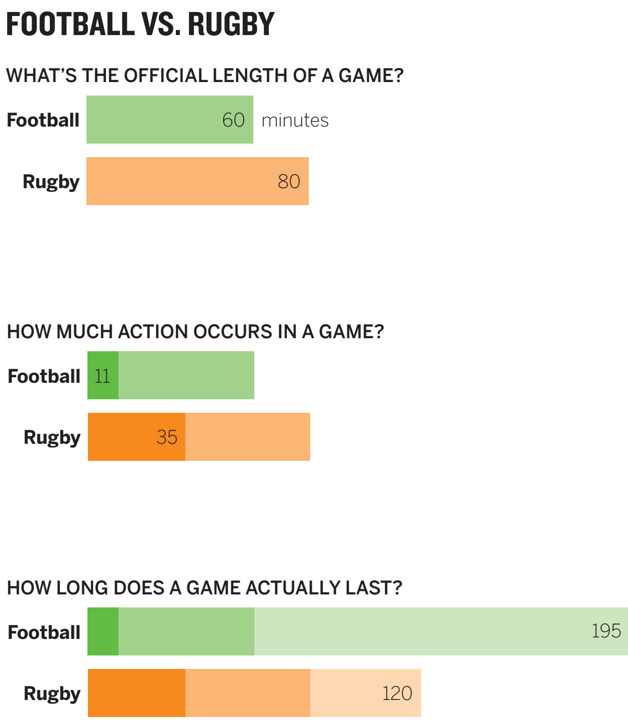
The challenge with what I've shown is that I've given you no fewer than 15 bits of information to look at here. The main idea, a simple one, doesn't pop as well as it could. If I want to make it work better, I could show one bar at a time:

> [Pause 5 beats, show only rugby bar] A rugby match contains a lot of action, and because there's very little stopping except for half-time, most of the time you're watching, you're watching the game itself. Compare that with American football. [Pause 3 beats before adding football chart to the screen]. The commercials make the game take up more time, even though it's a shorter game.

This is better; my viewers can focus on one sport at a time. But I'm still asking them to think about three things in relation to one another (action, game length, game time), then to do it again, and then to compare the two sets of relationships. In contrast, the women's degrees chart showed one thing—all other degrees—in the before state and just two new pieces of information in the reveal.

The thing about having options is that it slows us down. Here we borrow from Braess's paradox, a principle of traffic management developed by the mathematician Dietrich Braess, which states that adding route options (new roads, new lanes) to congested roadways can decrease traffic performance.[11] That's because when many people can switch routes (and switch again) for more-favorable personal outcomes, they slow the system down. Braess's paradox has been demonstrated in the real world many times when traffic improved after roads were removed. It has been applied to phenomena other than traffic, including power transmission (performance declined after systems were decentralized), protection of endangered species (the prospects for many species improve when one species goes extinct), and crowd control (multiple paths from a concourse to a seat make it take longer to get to seats).

What we experience with a complex chart isn't technically Braess's paradox, but it's similar. Think of all the places to focus on the Football vs. Rugby chart as route options. Should you start with the orange bar or the green? Should you compare the bars overall or the pieces? Do the specific values matter enough to focus on them? Which route will get you to understanding fastest? Options require choices, and choices take time. In a presentation, different people may choose to focus on different things.



By deconstructing a chart, you can remove all possible routes except one so that your presentation provides the fastest path to understanding. On this page is the Football vs. Rugby chart deconstructed for a presentation. Each chart would be shown one by one, starting with the top one.

The top chart is unambiguous. We've eliminated all but one route here: how long a game lasts. Viewers will grasp this immediately because it's a simple comparison and it's the only one available. The subtitle further prods, asking the question that the chart answers, in case there was any doubt.

In the second chart, we've added some new information now but, crucially, we've also *removed* some of the previous labeling and the first subtitle. We feel confident removing them because they were so clear and immediately understood. The bars of lighter color are all that remain from the previous chart, serving to put the new information in context. Because viewers aren't figuring out where to look, they can quickly assess that a rugby match has proportionally much more action.

One more time—add new information, remove old. Only one route to meaning. This time the reveal feels much more powerful. Viewers haven't once had to think about where to focus or decide what's important. Instead of spending mental energy figuring out a path through the chart, viewers are free to think about and discuss the idea. It's also more unlikely that they'll disagree about the meaning of this story, because it has been presented in such a way that they can't start from different places or focus on different things. Everyone can agree on what's shown here.

Some vanguard neuroscience suggests that might be important. Neuroscientist and marketing professor Moran Cerf, with Sam Barnett, published a paper suggesting that what makes a story memorable or engaging or vivid is how many brains respond similarly to it.[12] Put another way, what the authors call "cross-brain correlation," or CBC, predicts whether people will remember a story as well as or better than other measures, such as how they rate the story or how long they spend with it. To the extent that we can make our visual stories concise and unambiguous, they're likely to be far more engaging and memorable.

I'm going to pause here for a brief rant. Once, when I showed this rugby-football run to a group, someone challenged me. He said I created many more slides. Yes, I said, that's true. Well, he said, he had a slide quota. So, he couldn't do this. He overloads his presentations with dense slides and charts per slide not because he wants to, but because he has to fit everything into a limited number of permissible slides.

I felt for him. And I told him that slide quotas are facile and counterproductive (though I used more colorful language) and do the opposite of what they're meant to do. They neither save time nor deliver insight faster. If anything, they block insight by forcing everyone into parsing turgid slides rather than grasping ideas and talking about them.

I get it, though. It's an easy metric to set and monitor. "How many slides? 10? Too many!"

I told him, and I tell you, that the number of slides is an awful, misguided metric to use when building presentations. *Time to comprehension per slide* is a far superior metric. Look at the rugby run again. I can get through those three slides, each with one piece of information on it, in under 10 seconds. The full chart shown all at once usually takes me 20 or so. That's half the time to reach the same level of understanding. Imagine I can do that with every data set in my presentation. I might triple the number of slides while halving the length of the presentation.

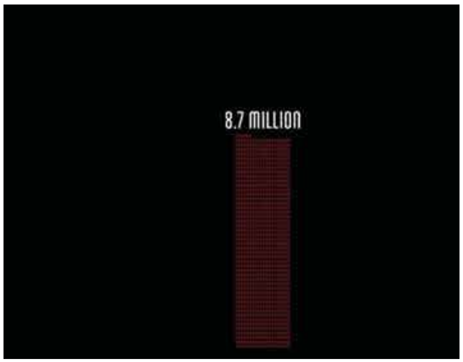
And that's a simple example. Think of the turgid slides of some poor presenter showing a high volume of data in such a cramped space because someone told them that's the rule.

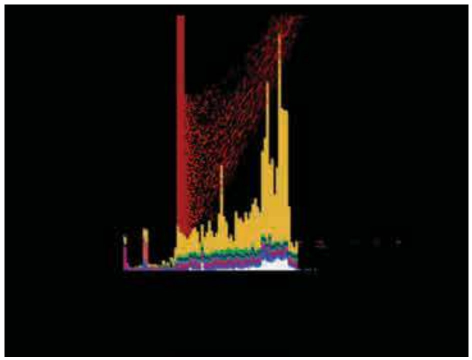
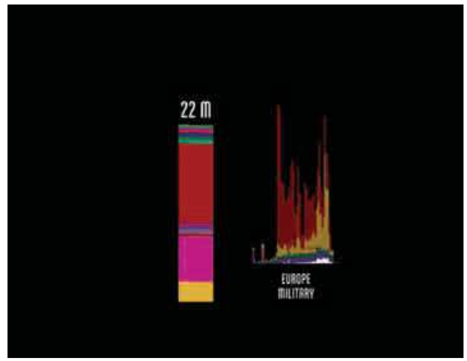
To bring the point home, I asked the person stuck with his slide quota how many slides he thought I had used during the presentation I was giving. I like to ask audiences this to see the range of answers. The highest number I've ever heard over several years was 80. Many people guess between 30 and 50. But in a standard 40-minute presentation I go through roughly 130 slides. That's under 20 seconds per slide.

The less you show at once, the shorter time to comprehension. Use that and forget about nonsense quotas. Okay, back to it.

Animate. Deconstruction and reconstruction lends itself to animation. Used skillfully—that means sparingly and functionally, not decoratively—animation can deepen understanding and engagement.

To show the massive scale of death in World War II in a way that conveys the tragic loss of human life rather than reporting statistics, for example, is difficult even with data visualization. Neil Halloran did it, though, in his interactive documentary *Fallen.io*. Halloran deftly uses movement (along with narration and sparse background music) to traverse a field of data points, zooming in and out to help viewers understand the vast scale of the war's carnage. In one powerful section he tallies deaths in the Soviet Union. The animation adds units of people killed—each icon added represents 1,000 deaths—for 45 harrowing seconds (using time to create tension). Dots are added until the finally tally of 8.7 million deaths is reached. A fast zoom out shows the total in comparison with other countries' columns of units before redistributing all the deaths over time as a stacked area chart. One person who commented on the dataviz illuminated the effectiveness of the narrative techniques we've discussed here in communicating statistics that are too absurd, too abstract, to grasp in other forms.



One million, six million, seventy million. Spoken or written, these numbers become a buzz. Incomprehensible. Presented graphically, they hit closer to the heart. As the Soviet losses climbed, I thought my browser had frozen. Surely the top of the column must have been reached by now, I thought.[13]

Note that animation is best used when it serves the idea. Showing change in data, for example, or animating a set of comparisons to parse and reparse a data set, are good uses of the technology. The *Washington Post* created a magnificent series of animated charts to show how various social distancing measures change the spread of Covid-19.[14] Merely decorating with animation—making bars grow out of an axis or spinning a pie chart onto a slide—may catch the eye, and that may serve a purpose to get someone's attention, but in general, the time it takes to achieve such animations probably isn't worth the limited benefit they may produce.

Tell stories. When you want to deeply impress an audience with dataviz, your impulse may be to show them uncommon and unusually beautiful forms. "Eye candy" is the perfect moniker for charts like that because they tend to give the audience a quick buzz that doesn't last. That sweet moment doesn't carry much nutrition.

Storytelling is the best, most powerful tool for making the kind of lasting impression that can create new understanding, change minds, or even effect policy change. Halloran's animation is visual storytelling at its most captivating. It moves us in a way that the text and static charts I've used to describe it won't ever capture. In a world in which it's said that people can't sit still for more than a minute or two, this 18-minute dataviz went viral.

It essentially consists of three basic chart types—unit charts, bar charts, and stacked area charts—deconstructed and reconstructed over and over. Powerful presentations that grab an audience don't have to rely on clever chart types. They can rely on your ability to craft your idea as drama.

Any story can be told in multiple ways, but a good way to start is to break the idea into three basic dramatic parts: setup, conflict, and resolution:[15]

Setup: Some reality.
Conflict: New information that affects reality.
Resolution: Some new reality.

In general, when we tell stories, the setup and resolution get about half of our attention. The other half is devoted to the conflict. That's where the action is. That's what makes narrative. No change, no story.

This formula is deeply entrenched in how humans experience stories; most successful narratives follow it. We can crudely map just about any story, or story archetype, onto it:

MAPPING STORIES

	MOBY-DICK	HARRY POTTER	WILE E. COYOTE AND THE ROAD RUNNER
Setup	Man goes on whaling voyage	Boy Wizard survives attack by Evil Wizard	Wile E. Coyote sets trap to catch speedy Road Runner
Conflict	Man's captain becomes unhinged seeking revenge on one whale	To defeat Evil Wizard, Boy Wizard must give up his life	Trap fails spectacularly
Resolution	Ship sinks, only man survives	Boy Wizard gives up life, Evil Wizard defeated	Road Runner escapes, Coyote injures self

As a narrative structure, this is obviously deeply reductive, but that's intentional. Obviously, crafting a great novel or making eight feature films involves much more than a few sentence fragments outlining three points on the story arc. But it's a useful way to practice deconstructing narratives (try it with your favorite stories) that will help make your presentations of charts more engaging.

Setup, conflict, resolution. Beginning, middle, end. You don't have to follow chronology, though usually you will; you only need to have your story proceed such that the setup makes sense on its own, the conflict affects the setup, and the resolution emerges from the conflict. Focus primarily on the conflict: That's what creates uncertainty, or introduces obstacles, or simply changes the status quo. It doesn't have to be negative. It could be the hiring of a star performer that increases productivity. Or it could be starting a new exercise regimen that helps you lose weight.

To find this rough story structure in your visualization, break down and refine the idea statement that you came up with in the process of talking and sketching. (It should be clear by now just how valuable arriving at some statement of your idea is to making good charts and presenting them well.) The story will be easiest to find in time-series data, which is inherently sequential. The peanut butter chart's story on page 217 would look like this:

Consecutive droughts and bad harvests have sent once-stable peanut butter prices to historical highs. That is:

Setup: Prices are stable
Conflict: Drought, bad harvest
Resolution: Prices rise sharply, then fall sharply
Setup: Prices are stable again for more than a decade
Conflict: Drought, bad harvest
Resolution: Prices hold
Conflict: Another drought, bad harvest
Resolution: Prices spike and then stay high.

This one happens to be a compound story with three conflicts. Even looking at the structure, you can imagine how you might roll out your visual using some of the techniques covered above. You could, for example, break it down and build it up, like with the Football vs. Rugby chart. You could tease your audience by stopping at the first drought line and asking what they think happened. You may even lure them—getting them to suggest that prices *must* have gone up after the second drought, when in fact they didn't.

Reserve dramas for your most complex ideas—explaining how multiple economic factors are affecting your business, for example—and your most important ideas, those for which you need to be especially convincing and persuasive.

PUTTING IT ALL TOGETHER

It can be useful to apply narrative principles to a chart, but it's far more powerful when, with multiple charts, you turn a presentation, or part of one, into a story.

Let's say you're a start-up pitching potential investors on a new type of coffee pod for single-serve coffee machines. The market for coffee pods is saturated, but yours is different. It's recyclable. You could just go in and say, "We have a recyclable pod that fixes a problem in the market." But will they understand the problem? Do they care? You want them to *feel* the problem so that when it comes time to show them your solution, they'll have no doubt that there's a need for it. Turn the beginning of your presentation into a short narrative.

INVESTOR PRESENTATION
—
RECYCLABLE POD

COUPLE OF CONCEPT CHARTS → X | X BASIC CHARTS

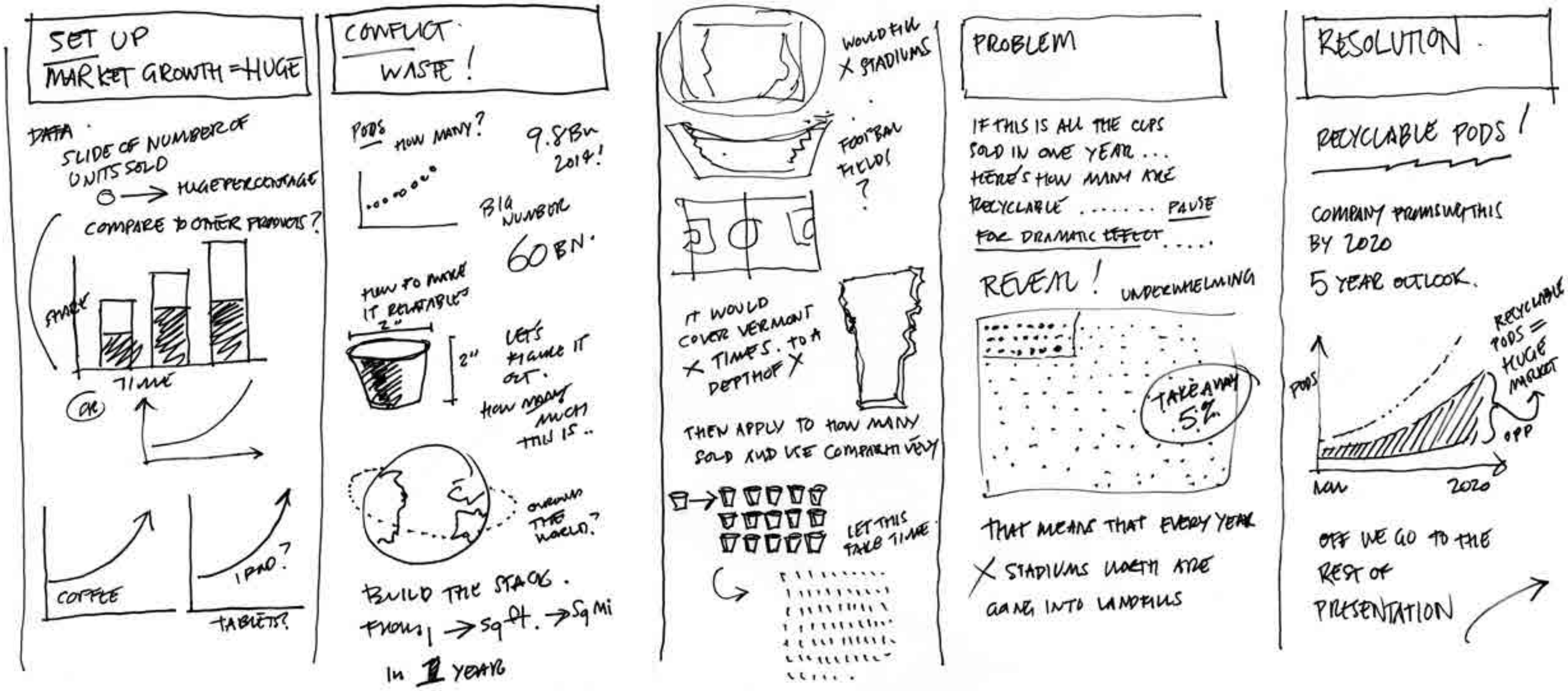
USE MAPS/SPACE
"FILL-UP"

SET UP: SINGLE SERVE IS HUGE!
CONFLICT: WASTE PRODUCED. HUGE!
RESOLUTION: WE HAVE A RECYCLING OPPORTUNITY. HUGE!

STATEMENT:
THE WASTE GENERATED BY PODS IS MASSIVE
EVEN A SMALL GROWTH IN RECYCLED PODS
WILL MAKE A BIG DIFFERENCE.

MAKE THEIR JAWS DROP!

VISCERAL CONNECTION TO
THE VOLUME OF WASTE

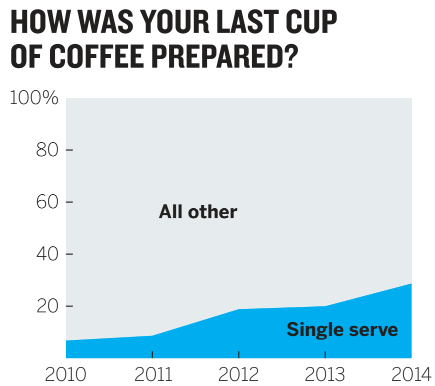


First sketch out the three main parts of the drama, in words and literal sketches:

Setup: Single-serve coffee machines are dominating the consumer coffee market.
Conflict: Nonrecyclable coffee pods have produced a staggering and growing amount of waste.
Resolution: Recyclable pods will help solve this problem.

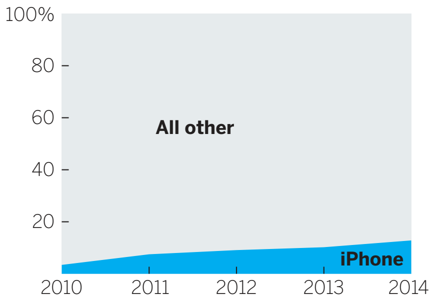
You've mapped out a story. It's a good sign that most of the time and space has gone into the conflict section, where drama has the greatest effect. Another good sign: You're already thinking about the presentation of the idea, making notes about using tension, time, and reveals to increase the persuasive effect of what you're showing.

Now you have to build those charts. Each chart will still go through the talk-sketch-prototype process; some may go through it together. But each needs to be well conceived and convey its idea effectively so that the audience can focus on the story rather than on making sense of the visuals. For brevity's sake, I'll skip to final charts and presenter's notes. Notice how they pull together everything discussed in this chapter, from not reading the picture, to using silence, to creating tension and reveals, to telling a story.



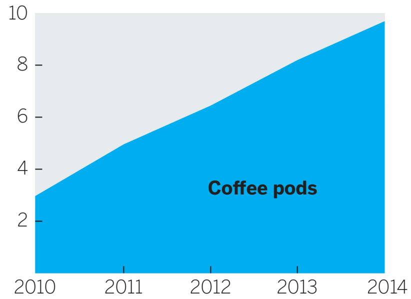
Setup:
[Show first chart, pause five beats] We all know that single-serve coffee is a growing phenomenon, but just how intense its surge is can't be understated. Its share has quadrupled in the past four years. In less than a decade, it's gone from zero share to almost one in three people saying their last cup came from a single-serve machine.

IPHONE MARKET SHARE



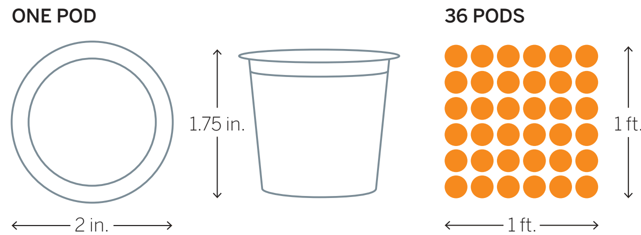
For perspective, here's the growth of the iPhone's share of the mobile market over the same period. [Display next chart next to first]

BILLIONS OF PODS SOLD



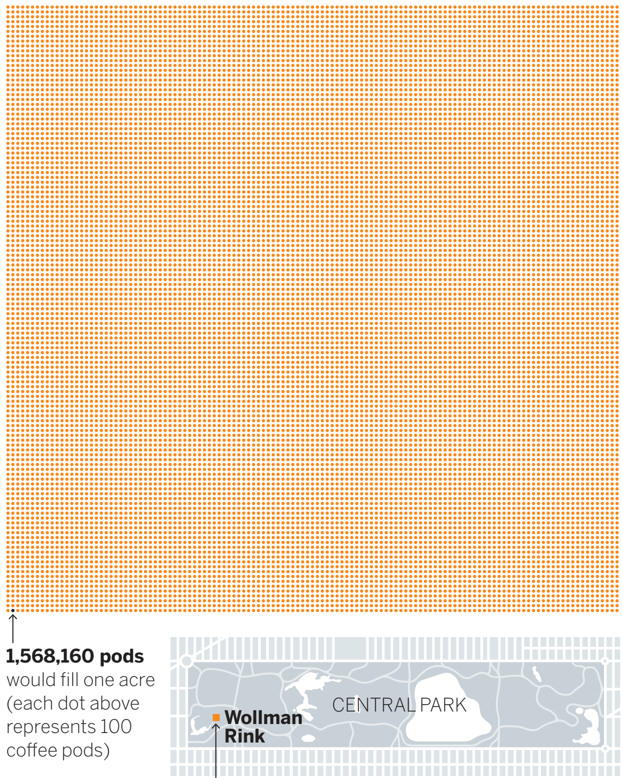
[Pause 3 beats] Every single-serve cup brewed requires a pod. Sales numbers on pods are notoriously difficult to pin down, but we know that the leading vendor alone is approaching 10 billion pods sold in one year—six times as many as five years ago. [Pause three seconds] During my pause right there, almost 1,000 pods were sold.

A TYPICAL COFFEE POD



Conflict:
And most of them aren't recyclable, which has created a significant waste problem. The dominant vendor sold 18 billion pods in the past two years. But it's hard to fathom how much waste that really is, so let's try to break it down. If we lined up the pods, 36 would fill a square foot.

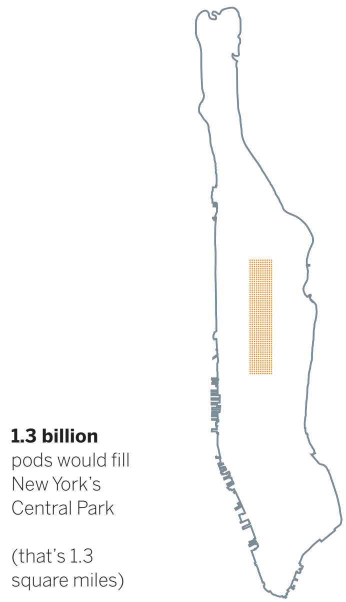
HOW MANY COFFEE PODS WOULD FILL AN ACRE?



One acre is about the size of New York's Central Park ice skating rink

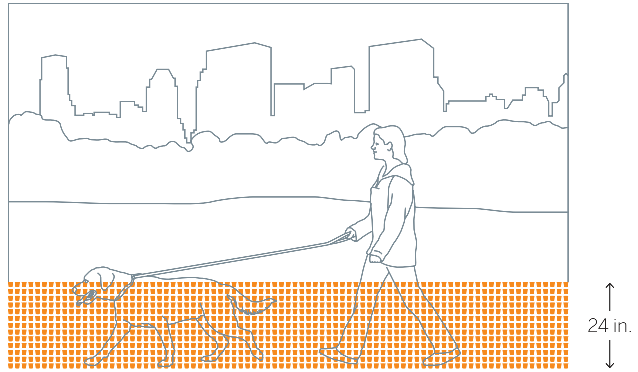
[Pause 3 beats] Think of an acre: like the skating rink in Central Park. Covering that with pods would account for eight one-thousandths of 1% of the pods sold by just the leading vendor in the past two years.

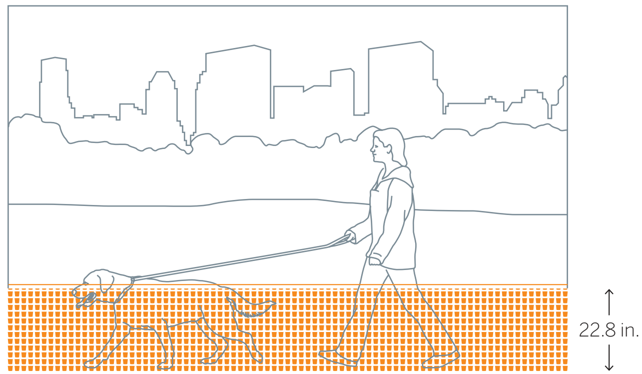
PODS TAKE CENTRAL PARK



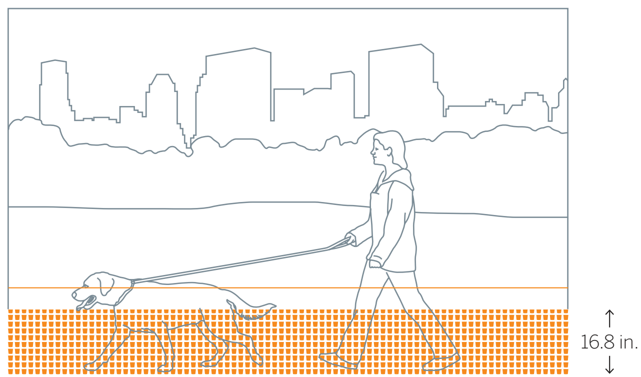
We have to go far bigger to see how much space 18 billion pods would take up. Would covering Central Park in pods use up all those sold in the past two years? Would it at least take half the pods to cover Central Park? [Pause] No. It would only take 7% of those pods to cover Central Park.

We're going to have to start stacking them to account for the rest. If we did that in Central Park, the entire park would be just over two feet deep in the coffee pods sold by the leading vendor in the past two years. We'd be thigh-high in unrecyclable used coffee pods. But to be fair, we should subtract the pods that are already recyclable. If we did that, how much lower would the pile be in Central Park? [Pause, remove recyclables on same image, change label from 24 in. to 22.8 in.] About 1.2 inches lower. Not even a single pod's height.





Resolution:
The reason there are so few recyclable pods is that it's been a technically difficult design to achieve. We have a design that we believe solves that problem. And if our design can gain 15% of the market in two years, we can reduce this pile of trash in Central Park by almost half a foot. [Again tweak image to remove pods and change label to 16.8 in.] That's a big start.



This story doesn't guarantee success, but it's a lot more engaging and impressive than projecting standard spreadsheet charts and reading the data in them—or, worse, reading bullet points verbatim. Notice how little you've said about the data itself; instead, you focus incessantly on helping the audience understand the idea. Even early, you're using a form of the lure procedure, showing the adoption of single-serve, which looks modest until you compare it to the iPhone. You don't overwhelm them with statistics. The area chart shows that massive growth in sales of pods well enough.

And then you pivot to the conflict, which is not simply that 18 billion coffee pods were thrown away. The idea is that *the popularity of single-serve coffee creates a major waste problem that current recycling efforts can't begin to address—but we can.* The number 18 billion is big and abstract enough that simply stating it can't possibly convey its meaning in terms of objects. Imagery, relatable points of reference, and a narrative arc conspire to make that abstract number tangible. You can see a dog nearly *drowning* in unrecyclable waste. We use reference points the audience will understand—the size of Central Park—to get them to grasp very big numbers.

And you resolve by showing how you can make this sea of waste *go down*, significantly.

You're sparking viewers' brains so that they'll understand the idea better, engage with it more, and remember it in a way they otherwise wouldn't. Even when charts are perfectly executed, to truly engage an audience, the play's the thing.

RECAP

PRESENT TO IMPRESS AND PERSUADE

Beyond manipulating charts themselves, you can make visualizations more effective by improving your presentation skills. The twin challenges here are to help viewers when they first see the visual (how you present it to them) and to help them process it (how you get them to engage with it).

PRESENTATION TIPS

- **Show the chart and stop talking.**
  A good chart will speak for itself. Let the viewers' active visual systems work without distractions.

- **Don't read the picture.**
  Talk about the ideas in the chart, not its structure.

- **For unusual visual forms, guide the audience.**
  Don't read the picture, but do provide some brief explanation of how the form works.

- **Use reference charts.**
  Companion visuals that show "ideal" or "average" cases can add context and make your chart easier to understand.

- **When you have something important to say, turn off your chart.**
  As long as a visual is displayed, viewers will look more than listen. If you want them to hear you, turn off the screen for a moment to refocus them.

- **Show something simple. Leave behind something more detailed.**
  Use the simplest forms possible in presentations, but create versions with more information that audience members can spend time with on their own.

ENGAGEMENT TIPS

- **Create tension.**
  Before revealing a full visual, show parts of it and ask the audience to speculate on what it will ultimately show.

- **Use time.**
  To make an audience grasp large values, reveal them gradually.

- **Zoom in or out.**
  To give viewers a sense of scale, start with a relatable value and then increase or decrease the scale step-by-step to show the value you want them to understand.

- **Lure.**
  Lure viewers in with a visual they may expect to see and then show them the actual version, which contradicts expectations.

- **Deconstruct and reconstruct.**
  Break down a visualization into multiple, simpler charts and then put it back together for the audience.

- **Tell stories.**
  Use the dramatic structure of setup, conflict, and resolution to make a chart or several charts tell a short story.

CHAPTER 8

A RETURN TO TEAMWORK

ON UNICORNS AND CATHEDRALS

WHEN I SPEAK TO A GROUP ABOUT DATA VISUALIZATION, a skeptic inevitably will challenge me. Sure, they say, this all looks great, but it won't fly *here*.

Why not? I ask. But I already know what they're going to say, which depends on the skeptic. There are two types: the people with the data (data scientists, managers, business analysts), and the people who need the data (executives, subject-matter experts, decision makers).

They've both arrived at the same conclusion, but for completely different reasons.

Why won't improving data visualization work in your organization?

Data scientists, business analysts, and managers say . . .	Executives, subject-matter experts, and decision makers say . . .
"They don't give us the time, tools, or money we need."	"We've invested [thousands, millions, billions] in a data operation and haven't seen the returns."
"We don't even know what they want from the data."	"It's their job to find the trends and show them to us."
"They don't understand the data."	"They don't know how to communicate."
"They don't use it to make decisions anyway."	"They don't make compelling cases with the data."

Both sides vent their frustration to me. Data teams know they're sitting on valuable insights but can't sell them. They say decision makers misunderstand or oversimplify their analysis and expect them to do magic, to provide the right answers to all their questions. Executives, meanwhile, complain about how much money they invest in data operations that don't provide the guidance they hoped for. They don't see tangible results because the results aren't communicated in their language.

The skepticism is more than anecdotal. In a question on Kaggle's 2017 survey of data scientists, to which more than 7,000 people responded, four of the top seven "barriers faced at work" were related nontechnical ones: "lack of management/financial support," "lack of clear questions to answer," "results not used by decision makers," and "explaining data science to others." Those results are consistent with what the data scientist Hugo Bowne-Anderson found interviewing 35 data scientists for his podcast; as he wrote, "The vast majority of my guests tell [me] that the key skills for data scientists are . . . the abilities to learn on the fly and to communicate well in order to answer business questions, explaining complex results to nontechnical stakeholders."[1]

I call this the last-mile problem. Despite massive investments to get the talented data scientists to set up shop, amass zettabytes of material, and run it through their deduction machines to find signals in the unfathomable volume of noise, many companies aren't getting the value they could from data science. Even well-run operations that generate strong analysis fail to capitalize on their insights, because, right at the end, they lack the ability to present findings in a way that bridges the gap between those with the data and those who manage the company. They've done the hard work of analysis and insight, but are neither prepared nor equipped to deliver it effectively to those who want to use it.

In general, the last-mile problem follows one of these scenarios. See if you recognize any of them.

The statistician's curse. A data scientist with vanguard algorithms and great data develops a suite of insights and presents them to decision makers in great detail. She believes that her analysis is objective and unassailable. Her charts are "click and viz" with some bullet points added to the slides—in her view, design isn't something that serious statisticians should spend time on. The language she uses in her presentation is unfamiliar to her listeners, who become confused and frustrated. Her analysis is dead-on, but her recommendation is not adopted.

The top-down demand. A business stakeholder wants to push through a pet project but has no data to back up his hypothesis. He asks the data team to produce analysis and charts for his presentation. The team knows that his hypothesis is ill-formed and offers helpful ideas about a better way to approach the analysis, but he wants only charts and speaking notes. One of two things will happen: His meeting will be upended when someone asks about the data analysis and he can't provide answers, or his project will go through and then fail because the analysis was unsound.

The beautiful mirage. A top-notch information designer is inspired by some analysis from company data and offers to help create a beautiful presentation for the board, with on-brand colors and typography and engaging, easily accessible stories. But the scientists get nervous when the executives start to extract wrong ideas from the analysis. The clear, simple, captivating charts make certain relationships look like direct cause and effect when they're not, and they remove the uncertainty that's inherent in the analysis. The data team is in a quandary: Finally, top decision makers are excited about their work, but what they're excited about isn't a good representation of it.

This divide runs deep. But despite the finger-pointing, it's neither side's fault. This is neither a technology failure nor a management failure. It's a system failure.

And not a new one, either. Consider that 105 years ago, before spreadsheets and analytics and Chart Wizard and Tableau, Willard Brinton began *Graphic Methods for Presenting Facts* with this:

> Time after time it happens that some ignorant or presumptuous member of a committee or a board of directors will upset the carefully-thought-out plan of a man who knows the facts, simply because the man with the facts cannot present his facts readily enough to overcome the opposition . . . As the cathedral is to its foundation so is an effective presentation of facts to the data.

Brinton nailed the last-mile problem, more than a century ago. How could this song remain the same for more than a century? Like anything else this deeply rooted, the last-mile problem's origins are multiple.

For one, visualization capabilities, the ones used to create charts, are usually welded onto the tools used to do the data science, like spreadsheets or analytics software. This reinforces the notion that it's the responsibility of the data person to be the communicator. After all, it's part of their toolset. Systems and processes are a function of the tools, not vice versa.

But the tools' default output can't match well-conceived, well-designed good charts, and the people who use those tools aren't inclined to improve on the default output. They're trained deeply in data manipulation, not data communication. They haven't been trained in making good charts, and many don't want to be the ones doing the communicating. Many data scientists have told me they're wary of visualization because it can dumb down their work and spur executives to draw conclusions that belie the nuance and uncertainty inherent in any analysis. (Executives, for their part, tell me the data scientists make it impossible to understand ideas because of their insistence on focusing on all the data and their statistical methods.) In the rush to grab in-demand data scientists, organizations have been hiring the most technically oriented people they can find, ignoring their ability or desire (or lack thereof) to communicate with a lay audience.

That would be fine if those organizations also hired other people to close the gap—but they don't. They still expect one person to wrangle data, analyze it in the context of the business and its strategy, make good charts, and present them to a lay audience. That breadth of skills and expertise is unreasonable to ask from any one job. As talent goes, that's unicorn stuff.

To begin solving the last-mile problem, companies must stop looking for unicorns. They don't exist, for the most part. And data teams must stop mistaking data for a cathedral. Data's the foundation. Insight's the church.

The key to solving the last-mile problem isn't fixing the data people's capabilities, as management would have it. And it's not changing management's expectations and statistical literacy, as the data people would have it.

The key is to dismantle the system most companies use in data operations and rebuild it in a new way. To free data scientists from unreasonable expectations and introduce new types of workers to the mix. It relies on cross-disciplinary teams composed of members with varying talents who work in proximity. Empathy, developed through exposure to others' work, facilitates collaboration among the types of talent. Work is no longer passed between groups; it's shared among a group.

A team approach—hardly new, but newly applied—can get data operations across the last mile, delivering good charts that provide invaluable insight into opportunities, threats, outliers, and trends.

This chapter is a bit different from the previous ones. You may be on either side of the last-mile divide, nodding your head about the characterization of the other side. But no matter which side you're on, I hope you might find this material valuable, and that you'll share and discuss it with people on the other side.

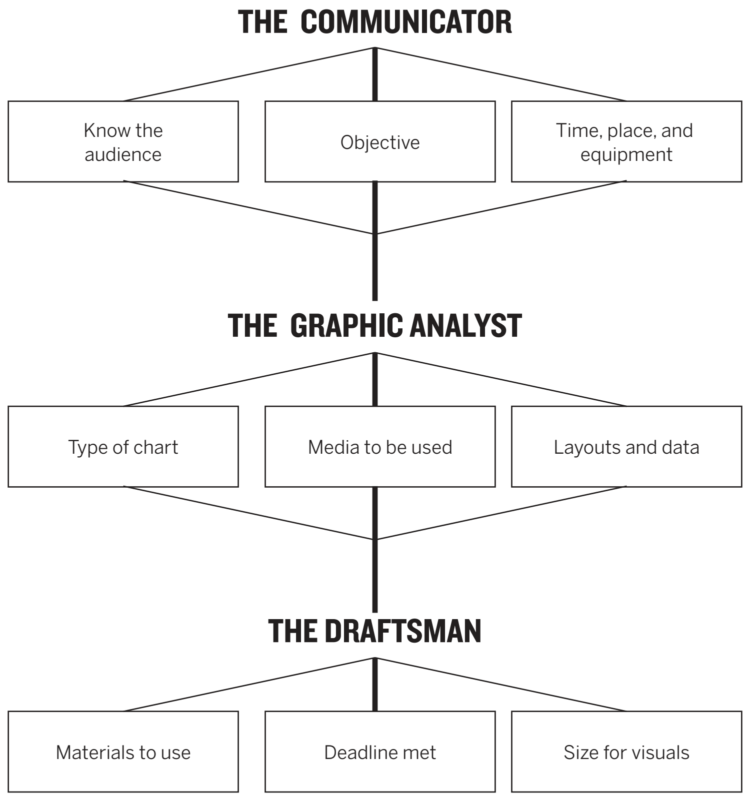
WHY ARE THINGS LIKE THIS?

In the early twentieth century, when Brinton was talking about cathedrals, pioneers of modern management ran sophisticated operations for turning data into decisions through visual communication, and they did it with teams. Brinton describes deeply cross-disciplinary efforts that included gang punch operators, card sorters, data collectors, managers, and draftsmen (they were nearly always men), among others. Examples of the results of this collaboration are legion in Brinton's book (see chapter 1). Railroad companies and large manufacturers were especially adept, learning through good charts the most efficient routes to send materials through factories, achieving targets for regional sales performances, and even optimizing vacation schedules.

Notably, Brinton never advocates for building teams as he describes the roles of the many people involved in creating "records for the executive" and the tools they would use. It wouldn't have occurred to him that he needed to advocate for teams; it would have been impossible for him to imagine any other approach to visualization. Arguing for a team doing visualization would have been like arguing for breathing. Teams weren't a choice; they were an inevitable prerequisite to crafting even a single chart.

The team approach persisted through most of the century. In her 1969 book *Practical Charting Techniques*, Mary Eleanor Spear details the ideal team—a communicator, a graphic analyst, and a draftsman (still mostly men)—and its responsibilities. "It is advisable," Spear writes, "that [all three] collaborate."

In the 1970s, things started to split. Scientists flocked to new technology that allowed them to visualize data in the same space (a computer program) where they manipulated it. Visuals were crude but available



A recreation of the matrix developed by Mary Eleanor Spears in 1969 to outline who's on the data visualization team and what they are responsible for. Though some of the responsibilities have changed, this basic structure remains a good foundation for building a team. Today, you might describe these roles as a subject-matter expert, a data analyst, and an information designer.

fast and required no help from anyone else. A crack opened in the dataviz world between computer-driven visualization and the more classic design-driven visualization produced by draftspeople (finally).

In the early 1980s, Chart Wizard, Microsoft's innovation in Excel, introduced "click and viz" for the rest of us. Suddenly anyone with a few cells of data could instantly create a chart along with overwrought variations on it that made bars three-dimensional, turned a pie into a doughnut, or laid a line over bars.

The profoundness of this shift can't be overstated. It helped make charts a lingua franca for business. It fueled the use of data in operations and eventually allowed data science to exist, because it overcame the low limit on how much data human designers can process into visual communication in a reasonable time. It opened modeling and hypothesis testing as it allowed one to quickly chart and rechart data based on different variables (a task the tools are still best suited to).

Most crucially, it changed the structure of work. Designers—draftspeople—were devalued and eventually fell out of the process. Visualization became the job of those who managed data, most of whom were neither trained to visualize nor inclined to learn. The speed and convenience of pasting a Chart Wizard graphic into a presentation prevailed over slower, more resource-intensive, design-driven visuals, even if the latter were demonstrably more effective.

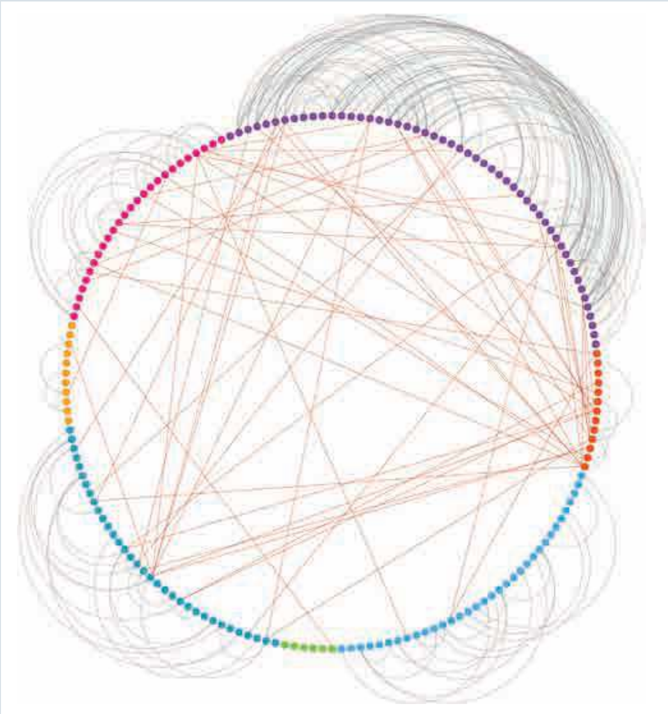
With the advent of data science, the expectations put on data scientists have remained the same—do the work and communicate it—even as the requisite skills have broadened to include coding, statistics, and algorithmic modeling. Indeed, in *Harvard Business Review*'s landmark 2012 article on data scientist as the sexiest job of the twenty-first century, the role is described in explicitly unicornish terms: "What abilities make a data scientist successful? Think of him or her as a hybrid of data hacker, analyst, communicator, and trusted adviser. The combination is extremely powerful—and rare."[2]

Right on both counts. That combination of talents is wildly unlikely for one person. But it's also an extremely powerful combination. So why not make a team that combines those talents?

REBUILDING THE SYSTEM

The sheer volume of data and the investments companies have made to extract value from it means the pendulum is swinging back to teams. Today, Spear's graphic analyst may be a business analyst or data scientist, and the draftsman may be a programmer, but the collaborative approach is similar. Many companies now contract with "data designers" and programmers who specialize in visualization. Investing in specialized talent to collaborate on making great data visualization is starting to make sense again, no matter how

INTERLOCKING BEAUTY



Okay, you can't figure out what's going on here immediately, but you can with just one more sentence of explanation: Each dot is a company, and each line is a person who sits on the board of the companies their line connects.

You're looking at the global corporate power structure. And the more you look, the more intriguing it becomes. But it's not here for the insights it delivers, per se. It's here to demonstrate what a team approach to data visualization can accomplish. Watch how every talent described in this chapter is reflected in the story of this visual—and several accompanying ones.

The data for this diagram was wrangled by researcher Eelke Heemskerk, who is also a subject-matter expert. He also did some analysis, hypothesizing that the nature of these interlocking boards had changed over time.

He worked with me to conceive of an approach to show this information, and we prototyped several approaches: force-directed network diagrams, geographical maps, heat maps.

We couldn't have arrived at the approach we took without another team member, a coder who quickly shot out prototypes and helped us arrive at this approach. Once we arrived at this, we did more analysis of what we saw, asking the coder to create alternate views, such as zoomed-in versions and adjusting our geographical groupings. It was during this collaboration we came up with the idea to put international connections inside the circle and intranational ones outside.

Content with our approach, we gave the prototype to a designer who worked with the coder (while consulting us) to clean up the rough versions and make them presentation-ready. Meanwhile, Heemskerk and I developed the story that we would tell with these charts (which I encourage you to read, if for no other reason than to see how storytelling with data can really bring ideas to life; the link is in the endnote[3]), which included zooming in to certain geographies and even individual company's board interlocks.

None of the four of us could have come up with this on our own, but as a team, we were able to make something out of complex data that is captivating, yes, but also informative, helping you see ideas that would be nearly impossible to glean any other way.

convenient some of the tools are. Routine projects may not call for rigorously structured teamwork, but all charts benefit from collaboration, even if it's just talking through the ideas to set context or consulting a design friend on a particular design challenge (chapter 4).

Complex data sets, large projects, and visualizations for which you want to go beyond standard chart forms will benefit from a team and free you up to focus on the ideas. You'll use the team approach when you're set on finding profound new insights, or you want people to see something in a powerful new way.

An effective data operation that produces visual insight for decision makers must be based on a team approach. It can borrow from Brinton's and Spear's basic models, but will account for the modern context, including the volume of data being processed, the automation of systems, and advances in visualization tools and techniques. It will also account for a wide range of project types, from the reasonably simple reporting of standard analytics data (say, financial results) to the most sophisticated big data efforts that use cutting-edge machine learning algorithms.

You can start with these four steps for setting up a more effective team to bridge the last mile and make good charts that become cathedrals atop data's foundation:

1. Define talents, not team members. It might seem natural that the first step toward dismantling unicorn thinking is to assign various people to the roles the "perfect" data scientist now fills: data manipulator, data analyst, designer, and communicator.

Not quite. Rather than assign people to roles, define the talents you need to be successful. A talent is not a person; it's a skill that one or more people possess. One person may have several talents; three people may be able to handle five talents. It's a subtle distinction but an important one for keeping teams nimble enough to configure and reconfigure during various stages of a project. (We'll come back to this.)

Any company's list of talents will vary, but a good core set includes these six:

Project management. Because your team is going to be agile and will shift according to the type of project and how far along it is, strong project management employing scrumlike methodologies will run under every facet of the operation. A good project manager will have great organizational abilities and strong diplomacy skills, helping to bridge cultural gaps by bringing disparate talents together at meetings and getting all team members to speak the same language.

Data wrangling. Skills that compose this talent include building systems; finding, cleaning, and structuring data; coding; and creating and

CORE TALENTS FOR COMMUNICATING DATA

PROJECT MANAGEMENT

Tasks: Manage creation of team, timeline, and schedules; marshal resources; troubleshoot

Skills: Organization; methodology (such as scrum); people management

Tools: Project management software; collaboration software

Leads: During creation of a data science operation; during creation and execution of a project

Supports: Ongoing data science operations

DATA WRANGLING

Tasks: Find, clean, and structure data; develop and implement data and visualization systems, algorithms, and models; develop templates and systems for repeatable processes

Skills: Coding; statistics; systems architecture

Tools: Data prep, cleaning, and governance software; programming languages; database software and tools; analytics software with visualization capabilities

Leads: Early in a data team's existence; early in a project's development

Supports: During routine data analysis, hypothesis testing, and visual exploration of data

maintaining algorithms and other statistical engines. People with wrangling talent will look for opportunities to streamline operations—for example, by building repeatable processes for multiple projects and templates for solid, predictable visual output that will jump-start the information-design process.

Data analysis. The ability to set hypotheses and test them, find meaning in data, and apply that to a specific business context is crucial—and, surprisingly, not as well represented in many data operations as one might think. Some organizations are heavy on wranglers and rely on them to do the analysis as well. But good data analysis is distinct from coding and statistics. Often this talent emerges not from computer science but from the liberal arts. The software company Tableau ranked the infusion of liberal arts into data analysis as one of the biggest trends in analytics in 2018.[4] Critical thinking, context setting, and other aspects of learning in the humanities also happen to be core skills for analysis, data or otherwise. In an online lecture about the topic, the Tableau research scientist Michael Correll explained why he thinks infusing data science with liberal arts is crucial. "It's impossible to consider data divorced from people," he says. "Liberal arts is good at helping us step in and see context. It makes people visible in a way they maybe aren't in the technology."

Subject expertise. It's time to retire the trope that data teams are stuck in the basement to do their arcane work and surface only when the business needs something from them. Data science shouldn't be thought of as a service unit; it should have management talent on the team. People with knowledge of the business and the strategy will inform project design and data analysis and keep the team focused on business outcomes, not just on building the best statistical models. Joaquin Candela, who ran applied machine learning at Facebook, worked hard to focus his team on business outcomes and to reward decisions that favored those outcomes over improving data science. "You might look for the shiniest algorithm or the people who are telling you they have the most advanced algorithm. And you really should be looking for people who are most obsessed with getting any algorithm to do a job."[5]

When you focus on outcomes, you will naturally gravitate toward engineering clear communication and human design into your data. Subject expertise bridges the data team and the stakeholders that need insight from the data.

Design. This talent is widely misunderstood. Good design isn't just choosing colors and fonts or coming up with an aesthetic for charts. That's styling—part of design, but by no means the most important part. Rather, people with design talent develop and execute systems for effective visual communication. They understand how to create and edit visuals to focus an audience and distill ideas. Information-design talent—which emphasizes understanding and manipulating data visualization—is ideal for a data team.

CORE TALENTS FOR COMMUNICATING DATA (CONT.)

DATA ANALYSIS

Tasks: Develop and test hypotheses on data and data models; find patterns and useful trends to inform business decisions

Skills: Statistics; scientific method; critical thinking; technical and nontechnical communication

Tools: Analytics software systems; spreadsheets; visualization software

Leads: During routine data analysis, project design, hypothesis testing, and visual exploration of data

Supports: Early in a data team's existence; early in project development; during visual communication development and presentations to lay audiences

SUBJECT EXPERTISE

Tasks: Define business goals; develop and test hypotheses; develop nontechnical communication

Skills: Functional knowledge; critical thinking; strategy development; nontechnical communication

Tools: Spreadsheets; collaboration software; sketching materials; visualization software; presentation software

Leads: During project design, hypothesis testing, and visual exploration of data; during communication to nontechnical audiences

Supports: Early in a data team's existence; during visualization and design process

CORE TALENTS FOR COMMUNICATING DATA (CONT.)

DESIGN

Tasks: Develop visual communication and presentations; create templates and styles for repeatable visualization

Skills: Information design; presentation design; design thinking; persuasive communication

Tools: Spreadsheets; sketching materials; visualization software; design suite software; presentation software

Leads: During data visualization and the creation of presentations and visual systems (templating)

Supports: During data visualization and the creation of presentations and visual systems (templating)

STORYTELLING

Tasks: Develop stories from data and visuals; help construct presentations in story format; present to nontechnical audiences

Skills: Information design; writing and editing; presenting; persuasive communication

Tools: Presentation software; visualization software; sketching materials

Leads: During creation of data visualization and presentations; during presentation to nontechnical audiences

Supports: During visual iteration and prototyping

And it's what you've been learning to do throughout this book. All that work around setting context as deeply as you can, with understanding some basic design principles, with understanding persuasion and manipulation and storytelling, is helping you to develop this talent.

Storytelling. Narrative is an extremely powerful human contrivance and one of the most underutilized in data science. The ability to present data insights as a story will, more than anything else, help close the communication gap between algorithms and executives.

2. Create a portfolio of necessary talents. Once you've identified the talents that you need, free yourself from the idea that these are roles you should hire people to fill. Instead focus on making sure these talents are available to form a team. Some of them naturally tend to go together: Design and storytelling, for example, or data wrangling and data analysis, may exist in one person.

Sometimes the talent will be found not in employees but in contractors. For my work, I keep a kitchen cabinet of people who have talents in areas where I'm weak who are well worth investing in to get projects done on time and with good charts. You may want to engage an information-design firm, or contract with some data wranglers to clean and structure new data streams.

Thinking of talents as separate from people frees you from trying to find that rare person who can do the data science, make good charts, and communicate ideas to a general audience. It also allows each person on this team to focus on their strengths and thus be more effective. Nabbing some people who have superior design skills, for example, will free data analysts from trying to figure out how to make their charts presentation-ready so they can spend their time on analysis. Designers and storytellers won't lose time trying to manipulate data, time that's better spent on creating a series of powerfully persuasive visualizations.

The talent portfolio also creates opportunities for people who previously might have been overlooked as generalists. An average coder who also has good design skills, for example, might be very useful in a system that prioritizes talents, as they can serve both roles. Their ability to speak both languages makes them a good bridge.

Randal Olson, the lead data scientist at Life Epigenetics and curator of the Reddit channel Data Is Beautiful (devoted to sharing and discussing good dataviz), used to focus solely on how well someone did the technical part of data science. "I know, when I started, I had zero appreciation for the communication part of it," he says. "I think that's common." Now, in some cases, he has changed the hiring process. "You know, they come in and we immediately start white-boarding models and math," he told me. "It's data scientists talking to data scientists. Now I will sometimes bring in a nontechnical person and say to the candidate, 'Explain your model to this person.'"

3. Expose team members to talents they don't have. Helping people focus on what they do best doesn't mean preventing them from learning other skills to develop more talents. Overcoming culture clashes begins with understanding others' experiences. Design talent often has little exposure to statistics or algorithms. Its focus is on aesthetic refinement, simplicity, clarity, and narrative. The depth and complexity of data work could be hard for designers to reconcile. On the other side, data-centric people focus on objectivity, statistical rigor, and comprehensiveness; the communication part is not only foreign to them but distracting. "It goes against their ethos," says one manager of a data science operation at a large tech company. "I was the same way, working in data science for 10 years, but it was eye-opening for me when I had to build a team. I saw that if we just learned a little more about the communication part of it, we could champion so much more for the business."

There are many ways to expose team members to the value of others' talents. Designers can learn some basic statistics—take an introductory course, for example—while data scientists learn basic design principles. Neither must become experts in their counterparts' field—they just need to learn enough to appreciate each other, because in this new system, they'll be working on a single team.

Project managers should make sure that stand-ups and other meetings always include a mix of talents. A scrum stand-up geared mostly to updating on tech progress can still include a marketer who makes presentations, as happens at Olson's company. Subject-matter experts should bring data wrangling and analysis talent to strategy meetings. Special sessions at which stakeholders answer questions from the data team and vice versa also help to bridge the gap. The former chief algorithms officer at Stitch Fix, Eric Colson (who is something close to a unicorn, having brought both statistical and communication talents to a company where data science is intrinsic), asks his team members to make one-minute presentations to nontechnical audiences, forcing them to frame problems in smart ways that everyone can understand. "To this day," Colson says, "if you say 'coconuts' here, people will know that was part of a metaphor one person used to describe a particular statistical problem he was tackling. We focus on framing it in ways everyone understands because the business won't do what it doesn't understand." Another manager of a data team created a glossary of terms used by technical talent and design talent to help employees become familiar with one another's language.

If your organization contains some of those rare people who, like Colson, have both data talents and communication and design talents, it helps to have them mentor one another. People who express interest in developing talents that they don't have but that you need should be encouraged, even if those strengths (design skills, say) are far afield from the ones they already have (data wrangling). Indeed, in my workshops I hear from data scientists who would love to develop their design or storytelling talent but don't have time to commit to it and don't know where to look for resources. Others would love to see that talent added to their teams, but their project management focuses primarily on technical outcomes, not business ones.

All this exposure is meant to create empathy among team members with differing talents. Empathy in turn creates trust, a necessary basis for effective teamwork. Colson recalls a time he used storytelling talent to help explain something coming from the data analysis talent: "I remember doing a presentation on a merchandising problem, where I thought we were approaching it the wrong way. I had to get merchandising to buy in." Instead of explaining beta-binomial distribution and other statistical concepts to bolster his point of view, he told a story about someone pulling balls from an urn and what happened over time to the number and type of balls in the urn—and showing what happened with simple visuals as he proceeded. "People loved it," he says. "You watched the room and how it clicked with them and gave them confidence so that at that point the math behind it wasn't even necessary to explain. They trusted us."

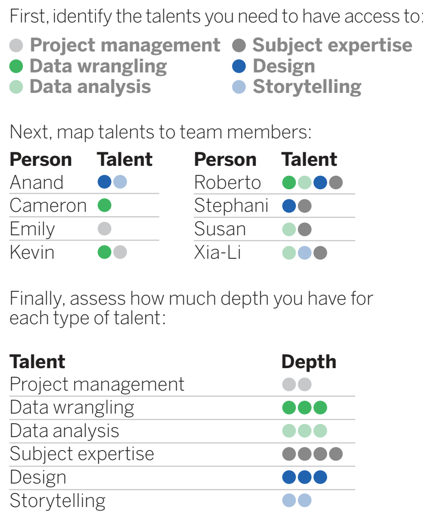
Notice in that simple example how nearly every talent in the portfolio is represented. The data wranglers set up a system to generate probability distributions based on company data. The analyst hypothesized and tested the model against a real business problem in merchandising—which he had been discussing with a subject-matter talent who knew about and understood merchandising's problem. Recognizing he had found an insight worth sharing, he then leaned on design and storytelling talents to visualize and explain the model to the lay audience whose buy-in he needed.

Now try to imagine a similar outcome if one person in the data operation was expected to create that successful chain of events. It's unlikely they could have.

4. Structure projects around talents. With a portfolio of talents in place, it's time to use it to accomplish your goals. The shifting nature of what talents are needed and when can make projects unwieldy. Strong project management skills and experience in agile methodologies will help in planning the configuration and reconfiguration of talents, marshaling resources as needed, and keeping schedules from overwhelming any part of the process.

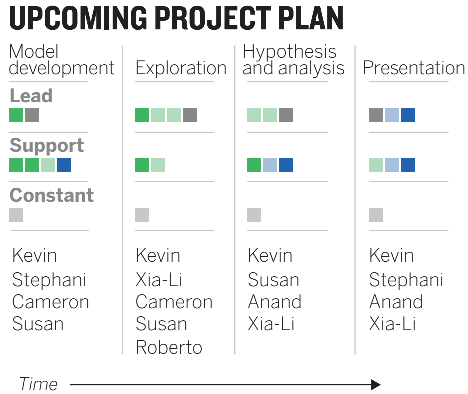
Build a talent dashboard. Performing a talent audit helps managers do a better job of planning for projects and configuring teams.

Map a project. Using the same unit measures, you can map any project to the talents you'll need across the project. You can assign some estimated value to each unit of talent. For example, a box for data analysis could represent 20 hours of work per week. So, four such boxes would equal two full-time employees' worth of time for data analysis on the project.



Try structuring this as a time-based map that shifts talent deployment as the project progresses. So, a project team isn't so much a collection of people who stay in the same role on the project from beginning to end but rather a collection of talent that shifts into different roles at different points in the project. Typically, one cluster of talents will take the lead early in a project, and different clusters will do so in the later stages.

By cross-referencing talents needed to the team members, you can identify places where your portfolio of talent falls short and will need shoring up. As you can probably tell by now, the constant in this model is project management talent, which is responsible for managing the configuration of the rest of the talent and adjusting as needed.



Even in this simple example, it becomes clear how powerful this approach can be. In the first part of the project, talents are mapped to people perfectly. Later on, some talents go unused. Kevin, for example, is needed for project management during the presentation phase, but his data wrangling talent goes unused. You could pull him off the project and put Emily in his place, but at this point, that would probably disrupt the team dynamic. So, Kevin has spare talent—data wrangling—he can apply to another project if need be and time permits. Or perhaps he can do some mentoring of team members who want to develop that talent.

Over time, trends will emerge for the talent you consistently find yourself scrambling to find, say, design talent, but you seem to always have excess data analysis talent.

PUTTING IT ALL TOGETHER

Once you build such a system, there's still more you can do to ensure successful visual communication. Consider these steps to make projects successful:

Assign a single, empowered stakeholder. It's possible, or even likely, that not all the people whose talents you need will report to the same manager or even be in the same department. Design talent may report to marketing; subject-matter experts may be executives reporting to the CEO. Nevertheless, it's important to give the team as much decision-making power as possible. Stakeholders will most often be people with business expertise who are closely connected to or responsible for business goals; the aim of the work, after all, is better business outcomes. Those people can create shared goals and incentives for the team. Ideally you can avoid the responsibility-without-authority trap, in which the team is dealing with several stakeholders who may not all be aligned.

Assign leading talent and support talent. Who leads and who supports will depend on what kind of project it is and what phase it's in. For example, in a deeply exploratory project, in which large volumes of data are being processed and visualized just to find patterns, data wrangling and analysis take the lead, with support from subject expertise; design talent may not participate at all, since no external communication is required. Conversely, to prepare a report for the board on evidence for a recommended strategy adjustment, storytelling and design lead with support from data talent.

Colocate. Have all team members work in the same physical space during a project. Also set up a shared virtual space for communication and collaboration. It would be undesirable to have those with design and storytelling talent using a Slack channel while the tech team is using GitHub and the business experts are collaborating over email. Use "paired analysis" techniques, whereby team members literally sit next to each other and work on one screen in a scrumlike iterative process (see chapter 4). They may be people with data wrangling and analysis talent refining data models and testing hypotheses, or a pair with both subject expertise and storytelling ability who are working together to polish a presentation, calling in design when they have to adapt a chart.

Colocation is trickier now with the emergence of permanent hybrid work, so it's smart to think about the best collaboration platforms to make sure you can at least colocate some members virtually. The key is you're a team, so you want to use the same tools as much as you can and limit the asynchronous act of throwing chunks of work over a proverbial wall. The whole value here comes from people with different skills interacting, whether that be in person or on screen.

Make it a real team. The crucial conceit in colocation is that it's one empowered team. At Stitch Fix, "our rule is no handoffs," Colson says. "We don't want to have to coordinate three people across departments." To this end he has made it a priority to ensure that his teams have all the talents they need to accomplish their goals with limited external support. He also tries to hire people many would consider generalists who cross the tech-communication gap. He augments this model with regular feedback for, say, a data person who needs help with storytelling, or a subject expert who needs to understand some statistical principle.

Reuse and template. One of the most powerful forms of resistance I encounter when showing people how they can transform their data into good charts is protests over *time*. They don't have it, they tell me. It takes too much time to make the charts better. A casual cost-benefit analysis in their minds tells them a suboptimal chart that's easy to get into the presentation is *good enough*, and making charts and visual communication better *isn't worth it*.

Of course, the executives I talk to don't see it that way.

But this goes back to the core problem of expecting one person to possess such disparate skills. The team approach makes good output come faster. And it can be made faster still if you devote time to creating templates.

When people complain about how much time it takes to make good charts, they're usually talking about the time it takes to take default output and design it up. But many software systems, like the analytics packages and the online dataviz tools such as Tableau and Flourish and Plotly, allow you to create templates with standard colors, typefaces, and other standard elements so your default output is much better. When you are not spending time finding the right Hex color value for your bars or deciding how thick they should be, or how much space should go between them because all that's templated, then you can spend your time focusing on making the ideas clear, simple, persuasive, and powerful with more-specific design refinements.

Colson also created an "algo UI team"—a group of people who combine their design talents and data wrangling talents to create reusable code sets for producing good dataviz for the project teams. Such templates are invaluable for getting a team operating efficiently. For example, conversations that someone with design talent would have with a person who has data analyst talent about best practices in visualizing, say, histograms, become hard-coded in the tools. Graham MacDonald, the chief data scientist at the Urban Institute, has successfully fostered this kind of cooperation on templating. His group produces data by county for many U.S. counties. By getting data wrangling and subject expertise together to understand what they need to communicate to stakeholders, the group built a reusable template that could customize the output for any county. Such an outcome would have been difficult without the integration of those talents on the team.

A powerful exercise to consider is to take the talent mapping system outlined in the previous section and apply it to a chart templating project as a first proof of concept. That way you're gaining experience operating this way while building valuable, reusable visual assets for the future.

The presentation of data science to lay audiences—the last mile—hasn't evolved as rapidly or as fully as the science's technical part. It must catch up, and that means rethinking how data and data visualization teams are put together, how they're managed, and who's involved at every point in the process, from the first data stream to the final, good chart shown to the board. Until companies can successfully traverse that last mile, data will underdeliver. It will provide, in Willard Brinton's words, foundations without cathedrals.

RECAP

A RETURN TO TEAMWORK

For most of the twentieth century, data visualization and good charts emerged from teamwork between subject-matter experts, data manipulators, and draftspeople, or designers. But late in the twentieth century, data manipulation programs added click-and-viz tools that made creating crude charts extremely easy and convenient. The teamwork ethos fell away. But with the advent of data science, and the massive investments in building data operations, leaders are demanding a fix to "the last-mile problem"—when stakeholders fail to communicate insights from data, and leaders fail to heed their advice because the case hasn't been made well with good visual representations.

The team approach is reemerging because of this. Here is a four-step process to building your own:

1. Define talents you need.
This is not identifying people but rather the talents people have. There are six you will draw on:

- Project management
- Data wrangling
- Data analysis
- Subject expertise
- Design
- Storytelling

2. Create a portfolio of these talents.
Map your people to the talents they have. Many people have more than one, for example, design and storytelling. Also map outside resources and organizations you can draw on to fill out your talent portfolio when you need, for example, a data wrangling company that will clean data for you.

3. Expose team members to talents they don't have.
Cross-disciplinary training and mentoring will overcome cultural barriers between people on the team, such as designers and data scientists, and will make collaboration more effective.

4. Structure projects around talents.

First, build a talent dashboard that shows the talents each team member possesses and the overall depth for each talent (the number of people who possess that talent). Then, map a project to the talents you need at each stage of the project, assigning the talent as either having a leading role or supporting role at each step.

Also consider these best practices:

Assign a single, empowered stakeholder.
This person serves as a bridge between team members who may well come from different business units and have different reporting structures.

Assign leading talent and support talent.
Team members should know if they are driving progress or just supporting others at each stage of a project.

Colocoate.
Have all team members work in the same physical space during a project. Also set up a shared virtual space for communication and collaboration.

Make it a real team.
Discourage handoffs and ad hoc collaboration. Set goals and incentives for the entire team.

Template.
Build visual output in a way that creates good default data visualizations and can be reused in future projects.

CONCLUSION

THE CRAFT IS IN THE THINKING

IN SOME WAYS, *data visualization* is a terrible term. It reduces the idea of good charts to a mechanical procedure. It evokes the tools and methodology required to create when it should evoke the creation itself. It's like calling *Moby-Dick* a "word sequentialization" or *Starry Night* a "pigment distribution."

It also reflects an ongoing obsession in the dataviz world with process over outcomes. Even now, most of the energy poured into teaching dataviz focuses on making sure you do it the "right" way or judging you if you do it the "wrong" way; on picking the right form; on when to use what colors. Chart crit is all about technique, how the thing was built, what it looks like.

Enough of all that. Forget right charts and wrong charts. Data is only a middleman between phenomena and your ideas about them.[1] And visualization is merely a procedure, a way of using that middleman to communicate ideas that convey much more than just pictures of statistics. What we do, really, when we make good charts is get at some truth and move people to feel that truth: To see what couldn't be seen before. To change minds. To cause action. It's not data visualization so much as visual rhetoric: the art of graphical discourse.

A common understanding of some basic grammar is necessary for that, of course. We all need to use subjects and verbs in roughly the same way if we're to communicate. But letting them govern our communication would be paralyzing and counterproductive. When you obsess on the minutiae of visualization rules—or, worse, when you judge a chart according to its relative adherence to those rules—you become one of Emerson's little statesmen, adoring foolish consistencies, which as he noted, are the hobgoblins of little minds.

Besides, software is beginning to take care of all that for you. Tools are evolving to manage some of the grammar.[2] They're getting their own versions of document templates, spell-check, and grammar check to guide formatting decisions and correct common missteps. Decisions about color, labels, grid lines, even what chart type to use—decisions to which entire books and courses have been devoted—are being encoded into visualization software so that the output in its default state is at least pretty good.

Interactivity helps too. The number and type of labels to include in a visualization, for example, is a decision that we're used to making as we construct charts, and it can be difficult. Too many labels create clutter, making it hard to know where to focus; not enough confuse viewers and, likewise, make choosing the proper focus a challenge. But hover states help solve the problem. Toggles manage complexity by showing or hiding variables as needed. A simple Next button can control the pace at which information is added or removed from a visualization.

More intelligence is being built into software. It's early days still, but some programs aim to look at your data and be able to suggest a chart type to you—not unlike the way the streaming platform Netflix will now just play a show it thinks you'll like. If it works, this could be a powerful prototyping advance. Software is also trying to build story into its structure, helping you to find and build multiple charts that work together.

Visualization is becoming fundamentally more interactive. I look forward to the day when we'll take for granted that decisions about what to show or where to focus—decisions you once had to make ahead of time and commit to—can be handled in medias res, often by the user. And those decisions will be alterable. Users will control the pace of the storytelling. Depth and complexity will become on-demand services. *Show me more. Show me less. Show me just this. Show me only that.* In a presentation, a manager will display a good chart and then filter and adjust it when the CEO asks, "What does that curve look like if we exclude the younger demographic?" A new, good chart will immediately appear on the screen. "Now just show me how women responded." Presentations will become conversations, exploratory dataviz in the boardroom.

We're starting to see such functionality in a tool like Flourish, which is evolving to make good visualization with presentation-worthy design into deeply interactive storytelling. First, I see an overall picture; then that picture breaks apart into a series of small multiples that represent each variable as a separate chart (animated well). Then I zoom in on one of those small multiples to talk about that one variable. Then I add another variable to compare two. On and on. I can do on-demand prototyping, exploratory visualization, and declarative visualization in one space. All I have to do is find the idea I want to convey, the story I want to tell, and iterate until I have it.

In short, visualization tools are evolving to make everything *available* but not always *visible*. That cracks things wide open. It changes a visualization's essential nature from *imparted* to *shared*; from a transaction—something you present or hand over—to a collaboration, which you work on and adjust with others.

It's not near perfect yet. And doing it well, as ever, requires training and time (which you've started with this book), but it's time well spent for anyone who wants to be a good visual communicator.

Charles Hooper is a dataviz consultant who works mostly with Tableau these days, but he used to work in Excel and remembers using Lotus 1-2-3, Harvard Graphics, and a program called Brio. Before that, he hand-drew his visualizations, transferred them to acetate, and displayed them with an overhead projector. "I'm turning 70 next week," he declares. "And right now, I'm telling you, this is the most exciting time, because it's getting easy to try things. When it's not easy, people just follow the specs. But you make it easy, put it in the hands of the masses, give it to businesspeople and not just specialists like me, and they come up with really innovative ways of looking at things. I learn something new every day from people trying out visualization."

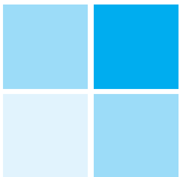
Software will continue to improve, in ways we can already see and in ways we can't yet imagine. But what it won't do—what it can't do—is intuit your specific context. And context, still, is everything. Visual thinking and visual communication will become no less relevant no matter what features are added to software programs. If anything, the better the software gets, and the less you need to stress over the number of ticks you put on your x-axis, or even what chart type to choose, the freer you will be to focus on bringing into high relief the ideas you want to communicate. The process of setting your context, finding your main idea, and visualizing it persuasively—that is, the guts of this book—will still be the most critical skills you can develop.

And still—despite Hooper's (and my own) excitement about the tools and where they're going—I insist that you can develop those visual thinking and communication skills with little more than paper, some pencils, and someone to talk to. I believe as strongly as ever that the craft is in the thinking. That the feelings behind their eyes which your good charts create don't come from software.

They come from you.

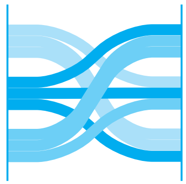
Good luck.

GLOSSARY OF CHART TYPES



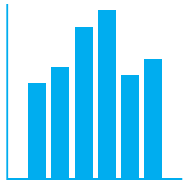
2 × 2 matrix: Box bisected horizontally and vertically to create four quadrants. Often used to illustrate a typology based on two variables. (Also called a matrix.)

+ Easy-to-use organizing principle for categorizing elements and creating "zones"
– Plotting items within quadrants at different spatial intervals suggests a statistical relationship that likely doesn't exist



Alluvial diagram: Nodes and streams show how values move from one point to another. Often used to show changes over time or details in how values are organized, such as how budget allocations are spent month by month. (Also called a flow diagram.)

+ Exposes detail in value changes or exposes detailed breakdowns in broad categories of data
– Many values and changes in flow make for complex, crisscrossed visuals that, while pretty, may be difficult to interpret



Bar chart: Height or length of bars shows relationship between categories ("categorical data"). Often used to compare discrete groups on the same measure, such as salaries of ten different CEOs. (Also called a column chart when bars are vertical.)

+ Familiar form that's universally understood; great for simple comparisons between categories
– Many bars may create the impression of a trend line rather than highlight discrete values; multiple groups of bars may become difficult to parse

Bubble chart: Dots scattered along two measures that add a third (size of bubble) and sometimes fourth (color of bubble) dimension to the data to show distributions of several variables. Often used to show complex relationships, such as multiple pieces of demographic data plotted by country. (Also called, erroneously, a scatter plot.)

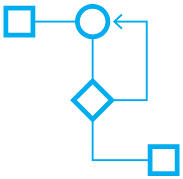
+ One of the simplest ways to incorporate a "z-axis"; bubble sizes can add crucial context to distribution visuals
– Sizing bubbles proportionally is tricky (area is not proportional to radius); by their nature, three- and four-axis charts require more time to parse, so are less ideal for at-a-glance presentation

Bump chart: Lines show change in ordinal rank over time. Often used to show popularity, such as box office rankings week to week. (Also called a bumps chart.)

+ Simple way to express popularity, winners, and losers
– Changes aren't statistically significant (values are ordinal, not cardinal); many levels and more change make for eye-catching skeins but may make it difficult to follow rankings

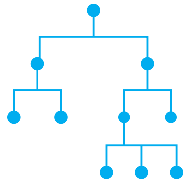
Dot plot: Shows several measures along a single axis. Often used in place of a bar chart when the comparison that matters is not the height of each bar but the difference in height between bars.

+ Compact form that works vertically or horizontally in a small space; makes comparison much easier than the traditional form (bar chart) along a single measure
– With many dots to plot, can be difficult to label effectively; removes any sense of trend across categories if that's important



Flow chart: Polygons and arrows arranged to show a process or workflow. Often used to map out decision making, how data moves through a system, or how people interact with systems, such as the process a user goes through to buy a product on a website. (Also called a decision tree, which is one type of flow chart.)

+ Formalized system, universally accepted, for representing a process with many decision points
– Must understand established syntax (e.g., diamonds represent decision points; parallelograms represent input/output, etc.)

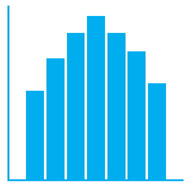


Hierarchical chart: Lines and points used to show the relationship and relative rank of a collection of elements. Often used to show how an organization is structured, such as a family or a company. (Also called an org chart, a family tree, or a tree chart, all of which are types of hierarchies.)

+ Easily understood method for documenting and illustrating relationships and complex structures
– Line-and-box approach limited in the amount of complexity it can show; harder to show less formal relationships such as how people work together outside the bounds of a corporate hierarchy

Geographical chart: Maps used to represent values attributed to locations in the physical world. Often used to compare values between countries or regions, such as a map showing political affiliations. (Also called a map.)

+ Familiarity with geography makes it easy to find values and compare them at multiple levels (i.e., comparing data by country and region simultaneously)
– Using the size of places to represent other values can over- or underrepresent the value encoded in those places

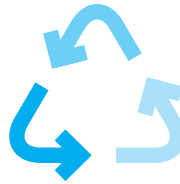


Histogram: Bars show distribution based on the frequency of occurrences for each value in a range. Often used to show probability, such as the results of a risk-analysis simulation. (Also called, erroneously, a bar chart, which compares values between categories, whereas a histogram shows the distribution of values for one variable.)

+ A fundamental chart type used to show statistical distribution and probability
– Audiences sometimes mistake a histogram for a bar chart

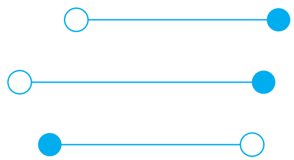
Line chart: Connected points show how values change, usually over time (continuous data). Often used to compare trends by plotting multiple lines together, such as revenues for several companies. (Also called a fever chart or a trend line.)

+ Familiar form that's universally understood; great for at-a-glance representation of trends
– Focusing on the trend line makes it harder to see and talk about discrete data points; too many trend lines make it difficult to see any individual line



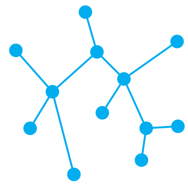
Metaphorical chart: Arrows, pyramids, circles, and other well-recognized figures used to show a nonstatistical concept. Often used to represent abstract ideas and processes, such as business cycles.

+ Can simplify complex ideas; universal recognition of metaphors makes understanding feel innate
– Easy to mix metaphors, misapply them, or overdesign them



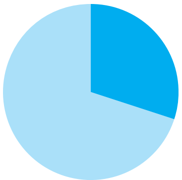
Lollipop chart: Similar to a dot plot, but plots two points on a single measure connected by a line to show some relationship between the two values. Plotting several lollipops can create an effect similar to a floating bar chart, in which values aren't all anchored to the same point. (Also called a double lollipop chart.)

+ Compact form that works horizontally and vertically; great for making multiple comparisons between two variables when the difference between the two is what matters most
– When variables "flip" (the high value was the low value in a previous lollipop), it can be confusing to read across multiple lollipops; multiple lollipops of similar value make it hard to evaluate individual items in the chart



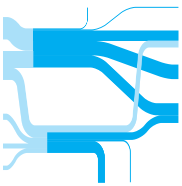
Network diagram: Nodes and lines connected to show the relationship between elements within a group. Often used to show interconnectedness of physical things, such as computers or people.

+ Helps illustrate relationships between nodes that might otherwise be hard to see; highlights clusters and outliers
– Networks tend to get complex quickly. Some network diagrams, while beautiful, can become difficult to interpret



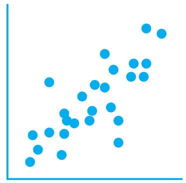
Pie chart: A circle divided into sections that each represent some variable's proportion of the whole value. Often used to show simple breakdowns of totals, such as population demographics. (Also called a donut chart, a variation shown as a ring.)

+ Ubiquitous chart type; shows dominant versus nondominant shares well
– People don't estimate the area of pie wedges very well; more than a few slices makes values hard to distinguish and quantify



Sankey diagram: Arrows or bars show how values are distributed and transferred. Often used to show the flow of physical quantities, such as energy or people. (Also called a flow diagram.)

+ Exposes detail in system flows; helps identify dominant components and inefficiencies
– Complex systems with many components and flow paths make for complex diagrams



Scatter plot: Dots plotted against two variables show the relationship between those two variables for a particular set of data. Often used to detect and show correlation, such as a plot of people's ages against their incomes. (Also called a scatter diagram, scatter chart, or scatter.)

+ A basic chart type that most people are familiar with; spatial approach makes it easy to see correlation, negative correlation, clusters, and outliers
– Shows correlation so well that people may make a causal leap even though correlation doesn't imply causation

Slope chart: Lines show a simple change in values. Often used to show dramatic change or outliers that run counter to most of the slopes, such as revenues falling in one region while rising in all others. (Also called a line chart.)

+ Creates a simple before-and-after narrative that's easy to see and grasp either for individual values or as an aggregate trend for many values
– Excludes all detail of what happened to the values between the two states; too many crisscrossing lines may make it hard to see changes in individual values

Small multiples: A series of small charts, usually line charts, that show different categories measured on the same scale. Often used to show simple trends dozens of times over, such as GDP trends by country. (Also called grid charts or trellis charts.)

- \+ Makes simple comparisons across multiple, even dozens, of categories more accessible than if all the lines were stacked in one chart
- – Without dramatic change or difference, can be hard to find meaning in the comparison; some "events" you'd see in a single chart, such as crossover points between variables, are lost

Stacked area chart: Lines plot a particular variable over time, and the area between lines is filled with color to emphasize volume or cumulative totals. Often used to show multiple values proportionally over time, such as product sales volume for several products over the course of a year. (Also called an area chart.)

- \+ Shows changing proportions over time well; emphasizes a sense of volume or accumulation
- – Too many "layers" create slices so thin it's hard to see changes or differences or track values over time

Stacked bar chart: Rectangles divided into sections that each represent some variable's proportion to the whole. Often used to show simple breakdowns of totals, such as sales by region. (Also called a proportional bar chart.)

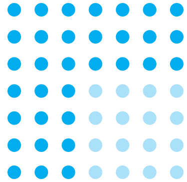
- \+ Some consider it a superior alternative to a pie chart; shows dominant versus nondominant shares well; may effectively handle more categories than a pie chart; works horizontally and vertically
- – Including too many categories or grouping multiple stacked bars together may make it difficult to see differences and changes

Table: Information arranged in columns and rows. Often used to show individual values over time across multiple categories, such as quarterly financial performance.

- \+ Makes every individual value available; easier to read and compare values than a prose version of the same information
- – Difficult to get an at-a-glance sense of trends or to make quick comparisons between groups of values

Treemap: A rectangle divided into smaller rectangles that each represent some variable's proportion to the whole value. Often used to show hierarchical proportions, such as a budget divided into categories and subcategories.

+ Compact form for showing detailed proportional breakdowns; overcomes some limitations of pie charts with many slices
– Detail-oriented form not optimal for at-a-glance understanding; too many categories make for a stunning but harder-to-parse visual; usually requires software capable of accurately arranging the squares



Unit chart: Dots or icons arranged to represent collections of individual values associated with categorical variables. Often used to show tallies of physical items, such as dollars spent or people stricken in an epidemic. (Also called a dot chart or dot plot.)

+ Represents values in a way that feels more concrete, less abstract than some statistical representations
– Too many unit categories may make it hard to focus on central meaning; strong design skills needed to make arrangement of units most effective

NOTES

Introduction

1. The socialization of marketing and the consumerization of technology, two ideas that can be applied to what's happening to data visualizations, come from the work of Josh Bernoff. See Charlene Li and Josh Bernoff, *Groundswell* (Harvard Business Review Press, 2008, rev. ed. 2011); and Josh Bernoff and Ted Schadler, *Empowered* (Harvard Business Review Press, 2010).

2. See hotshotcharts.com. Basketball analytics are a hotbed of advanced visualization because basketball has become a hotbed of advanced statistical analysis, as have all sports.

3. Edward Tufte's books are considered canonical in terms of data visualization best practices. Stephen Few has published similarly smart textbooks on best practices in charting and information dashboard design. Dona M. Wong's compact, unambiguous *The Wall Street Journal Guide to Information Graphics* (W.W. Norton, 2010) is a rule book for quick reference.

4. Joseph M. Williams, *Style: Toward Clarity and Grace* (University of Chicago Press, 1990), 1.

5. Wong, *The Wall Street Journal Guide to Information Graphics*, 90.

6. See "Terabyte" at http://www.whatsabyte.com/.

7. Mary Bells, "The First Spreadsheet—VisiCalc—Dan Bricklin and Bob Frankston," About.com Inventors, http://inventors.about.com/library/weekly/aa010199.htm.

8. For an excellent summary of the research on visual versus verbal learning styles, listen to the podcast "Visual, Verbal, or Auditory? The Truth behind the Myth of Learning Styles," part of a podcast series called "Learning about Teaching Physics" (http://www.compadre.org/per/items/detail.cfm?ID=11566). In it, Hal Pashler, of the University of California, San Diego, and Richard Mayer, of the University of California, Santa Barbara, review their separate work, all of which points to a muddy picture about inherent learning biases. In a meta-analysis, Pashler couldn't find many studies that were even constructed to test learning styles effectively. Mayer found that people do tend to sense that they prefer to learn one way or the other—and their brains actually respond differently—but also found that whether or not people identified as visual or verbal learners, they found visually oriented information more valuable. The podcast cohost, Michael Fuchs, says: "Our intuition of how we learn sometimes doesn't match how we actually learn." Pashler adds: "We should be very distrustful of our casual intuition about what works best for us . . . without having evidence of it." Ultimately, Mayer concludes that

“multimedia” information that combines pictures and words is what leads to “deeper understanding.”

9. For the smartest discussion of the state of visualization and critique, see Fernanda Viégas and Martin Wattenberg, “Design and Redesign in Data Visualization,” https://medium.com/@hint_fm/design-and-redesign-4ab77206cf9.

Chapter 1

1. Though it’s popularly reported that more than 80% of brain activity is devoted to what we see, the Harvard visual perception scientist George Alvarez says the number is probably closer to 55%—still far more than for any other perceptual activity.

2. Willard C. Brinton, *Graphic Methods for Presenting Facts* (The Engineering Magazine Company, 1914), 61, 82, https://archive.org/details/graphicmethodsfo00brinrich.

3. Naveen Srivatsav, “Insights for Visualizations—Jacques Bertin & Jock Mackinlay,” hastac.org blog post, February 16, 2014, https://www.hastac.org/blogs/nsrivatsav/2014/02/16/insights-visualizations-jacques-bertin-jock-mackinlay.

4. Jock Mackinlay, “Automating the Design of Graphical Presentations of Relational Information,” *ACM Transactions on Graphics* 5 (1986), http://dl.acm.org/citation.cfm?id=22950.

5. One computer scientist and visualization expert, who asked not to be named, has described Tufte as “basically a Bauhaus designer with an understanding of statistics.”

6. William S. Cleveland and Robert McGill, “Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods,” *Journal of the American Statistical Association* 79 (1984); “Graphical Perception and Graphical Methods for Analyzing Scientific Data,” *Science* 229 (1985); and William S. Cleveland, Charles S. Harris, and Robert McGill, “Experiments on Quantitative Judgments of Graphs and Maps,” *Bell System Technical Journal* 62 (1983).

7. In order to get through this history quickly so that we can move on to the practical lessons, I’m skimming right over important researchers such as Stephen Kosslyn and Barbara Tversky, among others. Suffice to say that dozens of important people and papers were influential during this time.

8. For better or worse, pie charts became anathema, while treemaps and other new procedures gained purchase.

9. I’m also speeding past the development of visualization software. It started in the 1970s, but in the past ten years the number of tools has exploded, and their ease of use is one of their core selling points. Strangely, Excel, among business’s core data tools, remains in the estimation of many frustratingly behind the curve in its visualization capabilities and default settings. Most visualization software mitigates this disconnect by allowing easy imports of data from the Excel spreadsheets that businesses will no doubt continue to use.

10. See davidmccandless.com and Carey Dunne, “How Designers Turn Data into Beautiful Infographics,” Fast Company Design, January 6, 2015, http://www.fastcodesign.com/3040415/how-designers-turn-data-into-beautiful-infographics.

11. See Manuel Lima's website, visualcomplexity.com.

12. An excellent example is "A Visual Guide to Machine Learning," R2D3, http://www.r2d3.us/visual-intro-to-machine-learning-part-1/.

13. See Alex Lundry, "Chart Wars: The Political Power of Data Visualization," YouTube video, April 28, 2015, https://www.youtube.com/watch?v=tZl-1OHw9MM.

14. M. A. Borkin, et al., "What Makes a Visualization Memorable?," *IEEE Transactions on Visualization and Computer Graphics* (Proceedings of InfoVis 2013). This research is still highly controversial. Memorability is a useful quality in a chart, but the research doesn't test the effectiveness of communicating the idea in the data, or whether the chartjunk skews attitudes toward it. Still, that the authors merely call into question the long-held belief that chartjunk is verboten indicates the provocative tenor of the new generation of research, which doesn't assume anything about tenets that *feel* true.

15. The research also suggests that pies work well when proportions are recognizable, such as 25% or 75%. J. G. Hollands and Ian Spence, "Judging Proportion with Graphs: The Summation Model," *Applied Cognitive Psychology* 12 (1998); and Ian Spence, "No Humble Pie: The Origins and Usage of a Statistical Chart," *Journal of Educational and Behavioral Statistics* 30 (2005).

16. Alvitta Ottley, Huahai Yang, and Remco Chang, "Personality as a Predictor of User Strategy: How Locus of Control Affects Search Strategies on Tree Visualizations," *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, 2015; Caroline Ziemkiewicz, Alvitta Ottley, R. Jordan Crouser, Ashley Rye Yauilla, Sara L. Su, William Ribarsky, and Remco Chang, "How Visualization Layout Relates to Locus of Control and Other Personality Factors," *IEEE Transactions on Visualization & Computer Graphics* 19 (2013); Evan M. Peck, Beste F. Yuksel, Lane Harrison, Alvitta Ottley, and Remco Chang, "Towards a 3-Dimensional Model of Individual Cognitive Differences," *Proceedings of the 2012 BELIV Workshop: Beyond Time and Errors—Novel Evaluation Methods for Visualization* (2012).

17. Anshul Vikram Pandey et al., "The Persuasive Power of Data Visualization," *New York University Public Law and Legal Theory Working Papers*, paper 474 (2014), http://lsr.nellco.org/cgi/viewcontent.cgi?article=1476&context=nyu_plltwp.

18. Brendan Nyhan and Jason Reifler, "The Roles of Information Deficits and Identity Threat in the Prevalence of Misperceptions," June 22, 2015, http://www.dartmouth.edu/~nyhan/opening-political-mind.pdf.

19. Michael Greicher et al., "Perception of Average Value in Multiclass Scatterplots," http://viscog.psych.northwestern.edu/publications/GleicherCorellNothelferFranconeri_inpress.pdf; Michael Correll et al., "Comparing Averages in Time Series Data," http://viscog.psych.northwestern.edu/publications/CorrellAlbersFranconeriGleicher2012.pdf.

20. Jeremy Boy et al., "A Principled Way of Assessing Visualization Literacy," *IEEE Transactions on Visualization and Computer Graphics* 20 (2014).

21. Alberto Cairo asks these and other good questions in the foreword to *Data Visualization in Society* (Amsterdam University Press, 2020).

Chapter 2

1. Gestalt psychology principles are often used to describe how we see charts. For example, the law of similarity suggests that like objects, such as data categories, should share values, such as color. Throughout this chapter and in others, I offer principles that borrow from Gestalt psychology but also go beyond it to other science.

2. See "Writing Direction Index," Omniglot.com, http://www.omniglot.com/writing/direction.htm#ltr.

3. Dereck Toker, Cristina Conati, Ben Steichen, and Giuseppe Carenini, "Individual User Characteristics and Information Visualization: Connecting the Dots through Eye Tracking," *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (2013); Dereck Toker and Cristina Conati, "Eye Tracking to Understand User Differences in Visualization Processing with Highlighting Interventions," *Proceedings of UMAP 2014, the 22nd International Conference on User Modeling, Adaptation, and Personalization* (2014).

4. No magic number exists as the threshold for the number of variables we can handle before they become "too much." I chose eight colors as a maximum on the basis of a conversation with the visualization researcher and author Tamara Munzer, who said, "There are fewer distinguishable categorical colors than you'd like. You don't get more than eight."

5. Display media limits this visualization as well. We can't zoom in to discrete points here, but all the data points are plotted, and the creator of this chart, Alex "Sandy" Pentland of MIT, had a version from which he could zoom into subsets to see all the points.

6. Researcher Steven Franconeri used this term to distinguish how we process information at two levels. The "blurry level" is fast, almost subconscious and helps us quickly pick out patterns. More deliberate parsing, which evaluates single values and compares values, is a slower process. Franconeri's point was that the blurry level, which is often disregarded when talking about making good charts, shouldn't be. He said: "Heat maps are disparaged because it's hard to pick out a single value from them. But take a year's worth of sales data, typically shown as a line graph, then imagine it as a heat map. It's hard in the heat map to read off absolute values, but ask someone what is the month with highest average sales and it turns out that the heat map is way better because you're not obsessed with the peaks and shape recognition as you would be with a line chart." George Alvarez of Harvard University described perception similarly as happening on a "low road" and a "high road."

7. Viola S. Störmer and George A. Alvarez, "Feature-Based Attention Elicits Surround Suppression in Feature Space," *Current Biology* 24 (2014); and Steven B. Most, Brian Scholl, Erin R. Clifford, and Daniel J. Simons, "What You See Is What You Set: Sustained Inattentional Blindness and the Capture of Awareness," *Psychological Review* 112 (2005).

8. Jon Lieff, "How Does Expectation Affect Perception," Searching for the Mind blog, April 12, 2015, http://jonlieffmd.com/blog/how-does-expectation-affect-perception.

9. Scott Berinato, "In Marketing, South Beats North," *Harvard Business Review*, June 22, 2010, https://hbr.org/2010/06/in-marketing-south-beats-north/.

10. Ludovic Trinquart, David Merritt Johns, and Sandro Galea, "Why Do We Think We Know What

We Know? A Metaknowledge Analysis of the Salt Controversy," *International Journal of Epidemiology* 45, no. 1 (February 2016): 251–260, https://doi.org/10.1093/ije/dyv184.

11. I wish I could give more proper credit to the creator of this rather elegant chart. In trying to track down its provenance, no one seemed to want to take credit for it. The author of the story, Sandro Galea, said it was the work of *Fortune,* but the head of information design there said it came from the research paper, though a thorough search of those papers did not turn up any such chart. Whoever created it, nice work! Even if it doesn't represent virtuous chaos, it's a smart solution to visualizing a complex data set.

12. I've changed the title, subject, and data points to protect the innocent, but the structure and conventions they used remain the same.

13. *Encyclopedia Britannica Online*, s.v. "Weber's law," http://www.britannica.com/science/Webers-law.

14. Ronald A. Rensink and Gideon Baldridge, "The Perception of Correlation in Scatterplots," *Computer Graphics Forum* 29 (2010).

15. In statistics, correlation is referred to with "r" where r = –1 is negative correlation, r = 0 is no correlation, and r = 1 is correlation.

16. Lane Harrison et al., "Ranking Visualizations of Correlation Using Weber's Law," *IEEE Transactions on Visualization and Computer Graphics* 20 (2014); Matthew Kay and Jeffrey Heer, "Beyond Weber's Law: A Second Look at Ranking Visualizations of Correlation," *IEEE Transactions on Visualization and Computer Graphics* 22 (2016).

17. Helen Kennedy and Rosemary Lucy Hill, "The Feeling of Numbers: Emotions in Everyday Engagements with Data and Their Visualisation," *Sociology* 52, no. 4 (2018): 830–848.

18. *Data Stories* (podcast), "Data Visualization Literacy with Jeremey Boy, Helen Kennedy, and Andy Kirk," episode 69, March 9, 2016.

19. Kennedy and Hill, "The Feeling of Numbers."

20. Daniel M. Oppenheimer and Michael C. Frank, "A Rose in Any Other Font Wouldn't Smell as Sweet: Effects of Perceptual Fluency on Categorization," *Cognition* 106 (2008).

Chapter 3

1. For thoughtful and entertaining examinations of "crap circles," see Gardiner Morse, "Crap Circles," *Harvard Business Review*, November 2005, https://hbr.org/2005/11/crap-circles; and Gardiner Morse, "It's Time to Retire 'Crap Circles,'" *Harvard Business Review*, March 19, 2013, https://hbr.org/2013/03/its-time-to-retire-crap-circle.

2. An idea pioneered by Eric von Hippel, as cited in Marion Poetz and Reinhard Prügl, "Find the Right Expert for Any Problem," *Harvard Business Review*, June 2015, https://hbr.org/2014/12/find-the-right-expert-for-any-problem.

3. The process described here is inspired by the process used by a data analysis company called Quid. The network diagram is inspired by one of Quid's examples. See Sean Gourley, "Vision Statement: Locating Your Next Strategic Opportunity," *Harvard Business Review*, March 2011,

https://hbr.org/2011/03/vision-statement-locating-your-next-strategic-opportunity.

Chapter 4

1. Abela's best-known book is *Advanced Presentations by Design: Creating Communication That Drives Action*, 2nd ed. (Wiley, 2013).

2. The sketches in this book look neat and reasonably orderly. A highly skilled designer created them to be readable. You should not expect or aim to sketch as neatly as what appears here. It's only necessary that you can interpret your sketches. Value speed over aesthetics.

3. Andrew Wade and Roger Nicholson, "Improving Airplane Safety: Tableau and Bird Strikes," http://de2010.cpsc.ucalgary.ca/uploads/Entries/Wade_2010_InfoVisDE_final.pdf.

4. See Richard Arias-Hernandez, Linda T. Kaastra, Tera M. Green, and Brian Fisher, "Pair Analytics: Capturing Reasoning Processes in Collaborative Analytics," *Proceedings of Hawai'i International Conference on System Sciences 44*, International Conference on System Sciences 44, January 2011, Kauai, Hawai'i.

5. Bart deLanghe, Stefano Puntoni, and Richard Larrick, "Linear Thinking in a Nonlinear World," *Harvard Business Review*, May–June 2017, https://hbr.org/2017/05/linear-thinking-in-a-nonlinear-world.

6. Roger Nicholson and Andrew Wade, "A Cognitive and Visual Analytic Assessment of Pilot Response to a Bird Strike," International Bird Strike Committee Annual Meeting, 2009, http://www.int-birdstrike.org/Cairns%202010%20Presentations/IBSC%202010%20Presentation%20-%20R%20Nicholson.pdf.

7. David McCandless, "If Twitter Was 100 People . . ." information is beautiful, July 10, 2009, http://www.informationisbeautiful.net/2009/if-twitter-was-100-people/.

Chapter 5

1. Williams, *Style*, 17.

2. Sometimes a title more like the former is not only okay but desirable. If you're striving for total objectivity, a literal transfer of facts and a straight description of the chart's structure may work fine as a headline. By using more-descriptive supporting elements, you may be shaping the audience's thinking.

3. Like Twain, Einstein is too often cited as the source of quotations. As Quote Investigator shows, we can't be sure that he said this first, but he seems to have said something like it. http://quoteinvestigator.com/2011/05/13/einstein-simple/.

4. Edward Tufte, *The Visual Display of Quantitative Information*, 2nd ed. (Graphic Press, 2001).

5. Remember, though, that the medium of presentation matters. Some grays that appear "quiet" but readable on a page disappear when projected on a large screen or in a light room. Light colors, too, may fade or disappear, or their fidelity may be low; oranges may become indistinguishable from reds. Know your equipment and choose colors that work with it.

6. The web is full of sites that help create color schemes. My favorite is paletton.com, which lets you switch easily between complementary and contrasting color schemes.

7. Most recently, Steve J. Martin, Noah J. Goldstein, and Robert B. Cialdini, *The Small Big: Small Changes That Spark Big Influence* (Grand Central Publishing, 2014), about how small persuasions can lead to massive change. Cialdini is the author of several seminal works on persuasion science.

8. Steve J. Martin, from the April 2015 issue of *High Life*, the British Airways in-flight magazine.

9. Noah J. Goldstein, Steve J. Martin, and Robert B. Cialdini, *Yes!: 50 Scientifically Proven Ways to Be Persuasive* (Free Press, 2008).

10. Koert van Ittersum and Brian Wansink, "Plate Size and Color Suggestibility: The Delboeuf Illusion's Bias on Serving and Eating Behavior," *Journal of Consumer Research* 39 (2012).

11. "U.S. Budget Boosts Funding for Weapons, Research, in New Areas," Reuters, February 2, 2015, http://www.reuters.com/article/2015/02/02/us-usa-budget-arms-idUSKBN0L625Q20150202.

12. Martha McSally, "Saving a Plane That Saves Lives," *New York Times*, April 20, 2015, http://www.nytimes.com/2015/04/20/opinion/saving-a-plane-that-saves-lives.html.

13. I recognize that in the modern, blogging world, this line has smudged to near imperceptibility, a trend some rue. The point stands that reporters report, don't insert opinion without evidence, and present both sides of an argument, whereas editorials are well-structured arguments that proffer a point of view.

14. I've updated this data since I first wrote *Good Charts*. Just a few years ago, the average price of a beer was $5.98, the price of an average "Ballpark Case" was $115, and the most expensive was the Red Sox's, at $186 per case.

15. Daniel Kahneman and Richard Thaler, "Anomalies: Utility Maximization and Experienced Utility," *Journal of Economic Perspectives* 20 (2006); Amos Tversky and Daniel Kahneman, "Availability: A Heuristic for Judging Frequency and Probability," *Cognitive Psychology* 5 (1973).

16. Petia K. Petrova and Robert B. Cialdini, "Evoking the Imagination as a Strategy of Influence," *Handbook of Consumer Psychology* (Routledge, 2008), 505–524.

17. We tend to react more viscerally to the unit chart than to a statistically driven chart. This is related to a phenomenon known as *imaging the numerator*. In a notable study that demonstrates this effect, experienced psychiatrists were given the responsibility of deciding whether or not to discharge a psychiatric patient. All the doctors were given an expert analysis, but some were told by the expert that *20%* of patients like this one were likely to commit an act of violence upon release. Other doctors were told that *20 out of every 100* patients like this one were likely to commit an act of violence.

In the group that was told "20%," about 80% of the doctors decided to release the patient. In the group that was told "20 out of every 100," only about 60% suggested releasing him. The likelihood of recidivism was the same for both groups, so why the great disparity? The latter group was imaging the numerator. In the minds of those doctors, 20 out of 100 turned into 20 people committing acts of violence. The former group didn't react the same way because percentages don't commit acts of violence.

This phenomenon occurs because the experiential part of the brain—the part that relies on metaphor and narrative to create feelings—quickly and powerfully overrides the rational part that analyzes statistics. Unit charts take advantage of this. See Veronica Denes-Raj and Seymour Epstein, "Conflict between Intuitive and Rational Processing: When People Behave against Their Better Judgment," *Journal of Personality and Social Psychology* 66 (1994); and Paul Slovic, John Monahan, and Donald G. MacGregor, "Violence Risk Assessment and Risk Communication: The Effects of Using Actual Cases, Providing Instruction, and Employing Probability versus Frequency Formats," *Law and Human Behavior* 24 (2000): 271–296.

18. I should note that imaging the numerator in evaluating risk is considered a negative phenomenon. For example, in the original study Denes-Raj and Epstein showed that when people were offered a chance to win money by picking red beans from a jar, they chose to pick from a jar that had more red beans even if red beans were proportionally fewer in that jar. Thus they were picking from a jar in which their odds of getting a red bean were lower. Imaging the numerator can also make us inflate risks. Paul Slovic noted in one study that when trying to communicate how infinitesimal parts per billion were, researchers told people to imagine one crouton in a 1,000-ton salad. Unfortunately, although the numerator (the crouton) was an easily understood concept, the massive salad was not. People ended up thinking that risks stated in parts per billion were more significant than they actually are. So although unit charts can persuasively convey individuality and help connect us to values by making statistics less abstract, they can also backfire or artificially exaggerate the data.

19. I kept the design and the data but changed the subject.

20. Suzanne B. Shu and Kurt A. Carlson, "When Three Charms but Four Alarms: Identifying the Optimal Number of Claims in Persuasion Settings," *Journal of Marketing* 78 (2014).

Chapter 6

1. The term "facticity" carries several meanings, including some related to philosophy. This use of it isn't necessarily the most common one, though it is increasingly common as a way to describe something that *feels* like an objective reflection of data, facts, and reality.

2. This is not real data.

3. "manipulate," *Merriam-Webster*, https://www.merriam-webster.com/dictionary/manipulate.

4. A term coined by Matthew Zeitlin as part of a discussion with my former colleague Justin Fox, who had the temerity to tweet positively about a chart with a truncated y-axis. Read the entertaining and thoughtful account here: Justin Fox, "The Rise of the Y-Axis-Zero Fundamentalists," byjustinfox.com, December 14, 2014, http://byjustinfox.com/2014/12/14/the-rise-of-the-y-axis-zero-fundamentalists/.

5. Danielle Ivory and Hiroko Tabuchi, "About Data Tampering," *New York Times*, January 4, 2016, https://www.nytimes.com/2016/01/05/business/takata-emails-show-brash-exchanges-about-data-tampering.html.

6. This was the case Tufte cited when arguing for truncation. You might suspect he'd be a y-axis-zero fundamentalist, but in fact he was open to the idea of truncation and cited its common use in scientific and academic circles as support for his view: "The scientists want to show their data, not zero." See the bulletin board conversation

"Baseline for Amount Scale" at http://www.edwardtufte.com/bboard/q-and-a-fetch-msg?msg_id=00003q.

7. Hannah Groch-Begley and David Shere, "A History of Dishonest Fox Charts," *Media Matters*, October 1, 2012, http://mediamatters.org/research/2012/10/01/a-history-of-dishonest-fox-charts/190225.

8. This comes from tylervigen.com, whose owner, Tyler Vigen, is a JD student at Harvard Law School. He wrote a script that finds statistical correlations in unrelated data sets and then charted them. Vigen's examples are usually silly; he has collected them in an entertaining book, *Spurious Correlations* (Hachette Books, 2015).

9. Ioannidis was writing about data, not visualizations—specifically, how research into the effects of nutrients on the human body is notoriously dodgy: "Almost every single nutrient imaginable has peer reviewed publications associating it with almost any outcome." We can apply what he says about big data sets to the visualization of such sets. John P. A. Ioannidis, "Implausible Results in Human Nutrition Research," *BMJ*, November 14, 2013, http://www.bmj.com/content/347/bmj.f6698.

10. For an excellent discussion of this trend, see Nathan Yau, "The Great Grid Map Debate of 2015," FlowingData, May 12, 2015, https://flowingdata.com/2015/05/12/the-great-grid-map-debate-of-2015/; and Danny DeBelius, "Let's Tesselate: Hexagons for Tile Grid Maps," NPR Visuals Team, May 11, 2015, http://blog.apps.npr.org/2015/05/11/hex-tile-maps.html.

11. An excellent discussion of these hurricane charts and other visualizations of uncertainty can be found in Jessica Hullman's excellent article, "How to Get Better at Embracing Unknowns," *Scientific American*, September 1, 2019, https://www.scientificamerican.com/article/how-to-get-better-at-embracing-unknowns/.

12. Scott Dance and Amudalat Ajasa, "Cone of Confusion: Why Some Say Iconic Hurricane Map Misled Floridians," *Washington Post*, October 4, 2022, https://www.washingtonpost.com/climate-environment/2022/10/04/hurricane-cone-map-confusion/.

13. Scott Berinato, "In Marketing, South Beats North," *Harvard Business Review*, June 22, 2010, https://hbr.org/2010/06/in-marketing-south-beats-north.

Chapter 7

1. I recommend Nancy Duarte, *HBR Guide to Persuasive Presentations* (Harvard Business Review Press, 2012); Duarte's work at Duarte.com; and Andrew Abela, *Advanced Presentations by Design: Creating Communication That Drives Action* (Wiley, 2013).

2. Mary Budd Rowe is generally considered the inventor of this educational technique, and multiple studies have confirmed its positive effects. See Mary Budd Rowe, "Wait Time: Slowing Down May Be a Way of Speeding Up!" *Journal of Teacher Education* 37 (January–February 1986), http://www.sagepub.com/eis2study/articles/Budd%20Rowe.pdf.

3. You might suggest that this presenter change the title of the chart to something that reflects the idea, such as "Money Doesn't Buy Comfort in Air Travel (Unless You Spend a Lot)."

4. Some may take exception to connecting discrete categorical data like this. For example, if I rolled this radial

chart out flat, it would essentially be a line chart whose area was filled in with color. And connecting would make categorical data look like a continuous trend line, which is one of the few absolute no-nos in charting, because there is no inherent connection between categories of sales skills rankings, but a trend would suggest that they are connected. That's a fair argument, and I'd understand if you chose to forgo using radar charts because of it. But I still believe they're useful, because connecting the points radially doesn't spark the trend line convention in our minds. Instead, it makes us see a shape to which we can assign meaning.

5. Two of my favorites: Gregor Aisch et al., "Where We Came From and Where We Went, State by State," *New York Times* Upshot, August 14, 2014, http://www.nytimes.com/interactive/2014/08/13/upshot/where-people-in-each-state-were-born.html; and Timothy B. Lee, "40 Maps That Explain the Roman Empire," Vox, August 19, 2014, http://www.vox.com/2014/8/19/5942585/40-maps-that-explain-the-roman-empire.

6. Ho Ming Chow, Raymond A. Mar, Yisheng Xu, Siyuan Liu, Suraji Wagage, and Allen R. Braun, "Personal Experience with Narrated Events Modulates Functional Connectivity within Visual and Motor Systems during Story Comprehension," *Human Brain Mapping* 36 (2015).

7. Robyn M. Dawes, "A Message from Psychologists to Economists," *Journal of Economic Behavior & Organization* 39 (May 1999), http://www.sciencedirect.com/science/article/pii/S0167268199000244.

8. Ingraham's story was an online article, not a live presentation. Smartly, he broke up the page so that the visualizations were separated by enough text that the audience could see only one at a time, as if they were presentation slides. This maximizes the effect of the final reveal. Each block of text that follows its visualization could actually serve as a smart script for a live presentation, because it adds context and understanding about the amount of water we're looking at and doesn't simply repeat what we see. Christopher Ingraham, "Visualized: How the Insane Amount of Rain in Texas Could Turn Rhode Island into a Lake," *Washington Post Wonkblog*, May 27, 2015, http://www.washingtonpost.com/blogs/wonkblog/wp/2015/05/27/the-insane-amount-of-rain-thats-fallen-in-texas-visualized/.

9. See "Bait and Switch," changingminds.org, http://changingminds.org/techniques/general/sequential/bait_switch.html; and Robert V. Joule, Fabienne Gouilloux, and Florent Weber, "The Lure: A New Compliance Procedure," *Journal of Social Psychology* 129 (1989). This work refers more to people's commitment to a menial task when they thought they'd be doing a fun one, but the mechanism is similar: if you get someone to commit to one way of seeing things, the inconsistency upon reveal of a new way of seeing things creates tension that the person feels compelled to resolve. The greater the inconsistency, the more they will feel compelled to understand and resolve the dissonance.

10. See "Consistency," changingminds.org, http://changingminds.org/principles/consistency.htm.

11. Dietrich Braess, Anna Nagurney, and Tina Wakolbinger, "On a Paradox of Traffic Planning," *Transportation Science* 39 (November 2005), http://homepage.rub.de/Dietrich.Braess/Paradox-BNW.pdf.

12. Moran Cerf and Samuel Barnett, "Engaged Minds Think Alike: Measures of Neural Similarity Predict Content Engagement," *Journal of Consumer Research*, in review.

13. writzter, comment on "The Fallen of World War II," http://www.fallen.io/ww2/#comment-2044710701.

14. This is a masterful use of animation and data. Harry Stevens, "Why Outbreaks like Coronavirus Spread Exponentially, and How to 'Flatten the Curve,'" *Washington Post*, March 14, 2020, https://www.washingtonpost.com/graphics/2020/world/corona-simulator/.

15. For a fuller exploration of storytelling with data, you can purchase my "Storytelling with Data Toolkit" with the *Good Charts Workbook*. Both include deep dives on this topic. https://store.hbr.org/product/good-charts-workbook-storytelling-with-data-toolkit/10310.

Chapter 8

1. Hugo Bowne-Anderson, "What Data Scientists Really Do, According to 35 Data Scientists," *Harvard Business Review*, August 15, 2018, https://hbr.org/2018/08/what-data-scientists-really-do-according-to-35-data-scientists.

2. Thomas H. Davenport and DJ Patil, "Data Scientist: The Sexiest Job of the 21st Century," *Harvard Business Review*, October 2012, https://hbr.org/2012/10/data-scientist-the-sexiest-job-of-the-21st-century.

3. Eelke Heemskerk, "How Corporate Boards Connect, in Charts," *Harvard Business Review*, April 21, 2016, https://hbr.org/2016/04/how-corporate-boards-connect-in-charts.

4. Ben Jones and Michael Correll, "BI Trend #2: Liberal Arts Impact," n.d., https://www.tableau.com/learn/webinars/bi-trend-2-liberal-arts-impact.

5. Scott Berinato, "Inside Facebook's AI Workshop," *Harvard Business Review*, July 19, 2017, https://hbr.org/2017/07/inside-facebooks-ai-workshop.

Conclusion

1. This sentence is paraphrased from Kirk Goldsberry.

2. Some visualization pros marvel at Microsoft's missed opportunity with charts and graphs in Excel, where a lot of corporate data sits. Excel wasn't originally terrible at generating charts, says Leland Wilkinson, a dataviz veteran and the author of *The Grammar of Graphics*, 2nd ed. (Springer, 2005), who recently joined Tableau. "Its first charts were rather nice," he said to me. "Then they got nervous because people were out there doing chartjunk"—3-D charts and gradient fills; cones instead of flat bars; exploded pies. There's a certain look to Excel charts from the 1990s and the early 2000s that is closely identified with the prototypical business presentation: gray background, heavy horizontal grid lines, blue line with large square dots as data points. "Bad software leads people to do bad graphics," Wilkinson says. "I'm delighted by PowerPoint. If you use it right, it's wonderful. I think almost the opposite of charting in Excel." At any rate, other software and online services have filled the void left by Excel, and the ease of importing and exporting spreadsheet data has obviated the need for good charting in the spreadsheet program itself.

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